# THE SEPARATION BETWEEN KNESER-TYPE TETRATION AND REGULAR ITERATION FOR $1 < b < \mathrm{e}^{1/\mathrm{e}}$ AND THE INVERSE HORN MAP OF $\mathrm{e}^u - 1$

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ABSTRACT. For bases $1 < b < \eta = \mathrm{e}^{1/\mathrm{e}}$, the equation $F(z+1) = b^{F(z)}$, $F(0) = 1$, has the regular iteration $R_b$ at the attracting fixed point. We construct a second, Kneser-type solution $K_b^W$ by sewing the two regular charts and characterise it by the position of its seam. Its difference from $R_b$ is exponentially small and is governed, mode by mode, by the inverse Écalle–Voronin horn map of the parabolic germ $u \mapsto \mathrm{e}^u - 1$.

Write $K_b^W = R_b \circ (\mathrm{id} + P_b)$ with $P_b = \sum_{n\geq 0} c_n \mathrm{e}^{2\pi \mathrm{i} n z}$, and let $\Lambda = \exp(4\pi^2/\log\lambda)$, where $\lambda$ is the attracting multiplier. We prove that $c_n/\Lambda^n \to B_n \mathrm{e}^{2\pi \mathrm{i} n a}$ as $b \to \eta^-$ for every $n \geq 1$. Here $B_n$ are the Fourier coefficients of the inverse horn map in the normalisation fixed by the formal Fatou coordinate, and $a$ is the attracting Fatou coordinate of the base point. The proof runs in three steps.

(1) The separation is expressed, up to an additive $O(\Lambda)$ with explicit constants, through the transition map between the two regular solutions.
(2) The limit of that transition map is established by a direct confluence argument for the saddle-node unfolding.
(3) The sewn solution is identified by a sewing construction and a uniqueness theorem.

The coefficient limits and sewing theorem hold for the characterised solution. A parabolic-renormalisation theorem for maps with one singular value proves $B_1 \neq 0$, so the first-mode ratios and limiting phase follow. The sewn family varies holomorphically near every real $b \in (1, \eta)$; near $\eta$ its imaginary part at real heights has an explicit leading $\Lambda$-profile and is not identically zero on any interval about 0. At every sufficiently small nonzero real height, this nonzero, exponentially flat splitting rules out meromorphic continuation through the cusp after any finite ramification of the base parameter. We prove that Kneser's solution for $b > \eta$ continues analytically in the base through the upper half-disc about $\eta$, across the Shell–Thron boundary irrespective of the arithmetic of the neutral multiplier, and that along this continuation, followed near the real axis, its boundary values on $(1, \eta)$ are exactly $K_b^W$. The key is a family of tilted initial regions whose quotient ends are punctures for every multiplier. Hence all these results hold for the boundary values of the continued Kneser family; continuation through the lower half-plane gives the complex-conjugate branch. The invariants $\tau_n$, and hence $c_n/\Lambda^n$ up to an exponentially small $O(\Lambda)$, are differentiable from the side $b < \eta$ in the unfolding parameter $p = |\log\lambda| \log\lambda_2$. The first-order coefficient is given by an absolutely convergent series over parabolic orbits. This series is obtained by adding to the model time a correction that is holomorphic in the parameter and makes the one-step error vanish to high order at both fixed points. The numerically evaluated first-mode deficit is $0.010199006345897\,p$, and it matches fits from the hyperbolic side to eleven digits. The same method gives an asymptotic expansion to all orders in $p$, with every coefficient built from convergent parabolic series. Independent ladder computations of a two-sided numerical solution agree with the transition-map prediction to eight digits, arguments included. An independent implementation of a two-sided regular construction (`fatou.gp`) coincides with the sewing solution to 40 digits at four complex bases, two of them near $(1, \eta)$. That pointwise comparison is independent of the continuation argument. An interval-certified quasiconformal argument separately extends the exterior Kneser family across a non-cusp boundary subarc on a common complex neighbourhood of the full height segment $[0, 1]$.

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## 1. Introduction

The tetration equation

$$F(z+1) = b^{F(z)}, \qquad F(0) = 1, \tag{1}$$

has, for real $b > \eta := \mathrm{e}^{1/\mathrm{e}}$, a distinguished real-analytic solution constructed by Kneser [11]. It tends to the two complex-conjugate fixed points of $w \mapsto b^w$ as $\operatorname{Im} z \to \pm\infty$. Its conformal uniformisation is uniquely characterised by the initial-region criterion of [30]. For $1 < b < \eta$ the fixed points are real: $L = \mathrm{e}^{\lambda}$ is attracting, with $0 < \lambda < 1$, and $L_2 = \mathrm{e}^{\lambda_2}$ is repelling, with $\lambda_2 > 1$. The Koenigs construction at $L$ yields the regular solution $R_b$. This is the solution usually meant in that range [13]. Paulsen [24] proposed continuing Kneser's two-fixed-point construction through the upper half of the Shell–Thron region. If this base continuation $\kappa$ has a boundary value, denote it by

$$K_b = \lim_{y \downarrow 0} \kappa_{b+\mathrm{i}y}. \tag{2}$$

High-precision algorithms for the relevant pointwise constructions are developed in [12, 13, 14]. We construct a two-sided sewing solution $K_b^W$ independently and determine its difference from $R_b$. We then prove that the continuation of Kneser's solution through the upper half-disc about $\eta$ has boundary values $K_b = K_b^W$ (Theorem E), so every result about $K_b^W$ holds for these boundary values.

**Notation.** Let $h = 2\pi/|\log\lambda|$ be the imaginary period of $R_b$, and put

$$\Lambda = \mathrm{e}^{-2\pi h} = \exp(4\pi^2/\log\lambda).$$

Both $R$ and $K^W$ are normalised by $F(0) = 1$. The map $P = R^{-1} \circ K^W - \mathrm{id}$ is 1-periodic with $P(0) = 0$. In the relevant region it has only non-negative modes, $P = \sum_{n\ge 0} c_n \mathrm{e}^{2\pi\mathrm{i}nz}$, and the separation is

$$K_b^W(z) - R_b(z) = c_1 R_b'(z)(\mathrm{e}^{2\pi\mathrm{i}z} - 1) + O(c_2) + O(c_1^2).$$

At $b = \eta$ the map becomes the parabolic germ $f(u) = \mathrm{e}^u - 1$ in $w = \mathrm{e}(1+u)$. Let $\Phi_{\mathrm{att}}, \Phi_{\mathrm{rep}}$ be its Fatou coordinates, both normalised by the formal Abel function $\alpha(u) = -2/u + \frac{1}{3}\log(-u) + O(u)$. Let

$$\Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1}(z) = z + \sum_{n\ge 1} B_n \mathrm{e}^{2\pi\mathrm{i}nz}$$

be the inverse horn map on the upper gate [8, 31], and $a = \Phi_{\mathrm{att}}(1/\mathrm{e} - 1)$ the attracting Fatou coordinate of the base point $w = 1$. Numerically

$$|B_1| = 0.0890584364122133\ldots, \quad \frac{B_2}{B_1^2} = -0.0253204931 + 1.5742190652\,\mathrm{i},$$
$$a = 3.0292972144180\ldots$$

(Section 2). The transition map between the two regular solutions is $T_b = R_b^{-1} \circ S_b$, where $S_b$ is the regular solution at the repelling point. Write $T_b(z) - z = \sum t_n \mathrm{e}^{2\pi\mathrm{i}nz}$ and $\tau_n = t_n \mathrm{e}^{-2\pi\mathrm{i}nt_0}$ (Section 3).

**Main results.**

**Theorem A** (Confluence)**.** *For every $n \ge 1$, $\tau_n(b) \to B_n \mathrm{e}^{2\pi\mathrm{i}na}$ as $b \to \eta^-$. Moreover, $B_1 \ne 0$ and $t_n/t_1^n \to B_n/B_1^n$ for every $n \ge 1$.*

**Theorem B** (The sewing solution)**.** *There is $b_1 < \eta$ such that for $b \in (b_1, \eta)$ the following holds. There is exactly one solution $K_b^W$ of (1) with the following three properties:*

- *it has a representation $R_b \circ (\mathrm{id} + P)$ on an upper half-plane;*

- *it has a representation* $S_b \circ (\mathrm{id} + Q)$ *on a lower half-plane;*
- $P$ *and* $Q$ *are bounded and* 1*-periodic, and the seam lies half a period below the real axis.*

*The precise statement is Definition 8.1. Its coefficients satisfy*

$$\left|\frac{c_n}{\Lambda^n} - \tau_n(b)\right| \le C_n \Lambda$$

*with explicit constants* $C_n$ *independent of* $b$*.*

**Corollary 1.1.** *For* $K_b^W$*,* $\hat{c}_n := c_n/\Lambda^n \to B_n e^{2\pi i n a}$ *for every* $n \ge 1$*. In particular*

$$\lim_{b \to \eta^-} \frac{|c_1|}{\Lambda} = |B_1|.$$

*The limiting phase is* $\arg B_1 + 2\pi a$*, and* $c_n/c_1^n \to B_n/B_1^n$ *for every* $n \ge 1$*.*

In the coordinate $u = (w - e)/e$ the map $\eta^w$ becomes $e^u - 1$, the germ used by Trappmann and Kouznetsov [14] at $b = \eta$. Their Abel function $\alpha$ has the normalisation of $\Phi_{\mathrm{att}}$, so $a = \alpha(\zeta(1))$ with $\zeta(z) = (z - e)/e$; the two agree to 40 digits. Paulsen [**?**, Prop. 1] shows that $\mathcal{K}_b(z)$ tends to their regular superexponential through 1 as $b \to \eta$ off $(-\infty, \eta]$. Corollary 1.1 describes the non-regular part that is lost in this limit: it is of size $\Lambda$, with the constants $B_n e^{2\pi i n a}$.

**Theorem C** (First-order term)**.** *Let* $s = 1 - e \log b$ *and* $p = |\log \lambda| \log \lambda_2 = 2s + O(s^2)$*. For every* $n \ge 1$ *with* $B_n \ne 0$ *(in particular* $n = 1$ *by Proposition 2.4),*

$$\log \frac{\tau_n(b)}{B_n e^{2\pi i n a}} = \kappa^{(n)} p + O(p^2) \qquad (b \to \eta^-),$$

*where* $\kappa^{(n)}$ *is given by an absolutely convergent series over orbits of the parabolic germ* $e^u - 1$ *(Theorem 5.6; the remainder* $O(p^2)$ *is Theorem 5.8 with* $m = 1$*). More generally,* $\log(\tau_n / B_n e^{2\pi i n a})$ *has an asymptotic expansion* $\sum_{j \ge 1} \kappa_j^{(n)} p^j$ *to all orders, and every coefficient is built from convergent parabolic series of the same kind (Theorem 5.8).*

**Theorem D** (Boundary values)**.** *Suppose the base-continued Kneser-type family* $\kappa$ *belongs to the class of Definition 8.1 for all bases in some nonempty open set near* $(b_1, \eta)$ *in the upper half-plane. Then* $\kappa = K^W$ *there. Its representations are bounded uniformly along every ladder, the boundary value (2) exists and equals* $K_b^W$*, and Corollary 1.1 holds for* $K_b$*.*

The hypothesis of Theorem D is qualitative. The next theorem removes it for the continuation that passes near $\eta$.

**Theorem E** (Continuation around the cusp)**.** *Kneser's real-normalised solution for real* $b > \eta$ *close to* $\eta$ *continues analytically in the base through the upper half-disc about* $\eta$*. The continuation crosses the Shell–Thron boundary arc in that half-disc, whatever the arithmetic of the neutral multiplier, and extends holomorphically across a real interval* $(\eta - r, \eta)$*,* $r > 0$*, where it equals* $K_b^W$*. Continued further near the real axis, it equals* $K_b^W$ *on all of* $(1, \eta)$*. Consequently, for this continuation, the boundary value (2) exists,* $K_b = K_b^W$*, and Corollary 1.1 holds for* $K_b$*:* $c_n/\Lambda^n \to B_n e^{2\pi i n a}$ *for every* $n \ge 1$*, and* $|c_1|/\Lambda \to |B_1| > 0$*. Continuation through the lower half-disc gives the complex-conjugate branch instead.*

The precise statement is Theorem 9.13. The proof (Section 9) tilts Kneser's initial region, the chord between the two fixed points and its image, continuously along the path. In the coordinate $\log \frac{u - u_1}{u - u_2}$ the region becomes a thin strip about a line through $i\pi$ whose slope turns from 0 to $\pi/4$. Its quotient ends are punctures for every multiplier, so the neutral boundary is no obstacle. At real $b < \eta$ the tilted region is a fundamental domain of the sewn surface of Proposition 3.4. The path from the base orbit to the seam then passes below the attracting fixed point, which is

exactly the seam condition of Definition 8.1. An independent implementation of the two-sided regular solution coincides with the sewing solution to 40 digits at four bases, two of them near $(1, \eta)$ (Section 10.4). The family of [24] is defined by analytic continuation in the base. There the continuation across the Shell–Thron boundary is argued only numerically, and the uniqueness statement needs an additional root-selection condition (Remark 1.2). Theorem E proves the continuation near the cusp; it agrees with Paulsen's family if continuation in the upper half of the Shell–Thron region is path-independent (Remark 9.15). Corollary 10.21 independently continues the exterior family across one non-cusp boundary subarc; whether continuation across such arcs far from $\eta$ agrees with the one near the cusp is not needed here and is not addressed. By Lemma 2.1 and Proposition 2.2, $\Lambda$ has the saddle-node form

$$\Lambda = \exp(-C_\eta(\eta - b)^{-1/2}(1 + o(1))), \qquad C_\eta = \frac{4\pi^2}{\sqrt{2\mathrm{e}^{1-1/\mathrm{e}}}} = 20.3508\ldots.$$

Thus the coefficients of the separation have the stated exponentially small upper bounds. Proposition 2.4 makes the first-mode asymptotic sharp with a strictly positive constant; the quoted numerical value of that constant is not a certified enclosure.

**Numerical results.** Theorems A and B are confirmed and extended numerically in Section 10.

(1) Ladder computations of $K_b$ reproduce $\tau_1$ and $t_2/t_1^2$ to their full precision, arguments included; for example, eight digits at $b = 1.1$.
(2) At $b = \sqrt{2}$, far from $\eta$, the sewing equation gives $\operatorname{Im} K^W_{\sqrt{2}}(\frac{1}{2}) = -1.18899697185 \cdot 10^{-48}$, which agrees with Paulsen's 120-digit value $-1.18899697 \cdot 10^{-48}$ for his base-continued family [24, §6] in all nine quoted digits, sign included. This is strong numerical evidence that his continuation path gives the same branch as Theorem E (Open problem 1).
(3) $\tau_1$, $t_2/t_1^2$ and $t_3/t_1^3$ have been followed up to $\eta - b = 3 \cdot 10^{-5}$.
(4) The parabolic series of Theorem C gives

$$\log \frac{\tau_1}{B_1 \mathrm{e}^{2\pi i a}} = (-0.010199006345897402441 - 0.013250956182506712250\,\mathrm{i})\,p + O(p^2),$$

in agreement with fits from the side $b < \eta$ to their eleven digits. The second coefficient, $\kappa_2^{(1)} = -7.88992074991511334 \cdot 10^{-5} + 3.6554727325799459 \cdot 10^{-4}\,\mathrm{i}$, agrees with the fits to within one unit in their sixth digit.

**Method.** Three observations drive the proofs.

(1) The separation is a property of the transition map $T_b$ between two classical Koenigs coordinates. A welding identity expresses $c_n$ through the Fourier modes of $T_b$ (Theorem 3.2). The scale $\Lambda$ appears because the seam sits half a period below the real axis, and the gate carrying the positive modes a further half period below it.
(2) The limit of $T_b$ is a confluence problem for the saddle-node unfolding in the direction where both fixed points are real (the "Glutsyuk" direction [10, 21, 6]). In the complementary Lavaurs direction, the corresponding statements are the classical results of parabolic implosion [16, 29, 22]. For vector fields, see [27]. We prove the needed statement directly (Section 4), using the model time $A_1 \log(u - u_1) + A_2 \log(u - u_2)$, whose one-step error vanishes at both fixed points.
(3) The Kneser-type solution is the conformal sewing of two half-cylinders along $T_b$. A contraction in a weighted Wiener algebra gives it with explicit bounds (Theorem 6.3). A proper-map argument proves uniqueness in the class of Definition 8.1 (Theorem 8.2).

**Organisation.** Section 2 collects the parametrisation and the parabolic data. Section 3 introduces $T_b$, the welding identity and the sewing construction. Section 4 proves Theorem A, and Section 5 proves Theorem C. Sections 6–8 prove Theorems B and D, and Section 9 proves Theorem E. Section 10 reports the numerics, and Section 11 lists open problems.

**Remark 1.2** (On the definition and uniqueness in [24])**.** In [24], $\kappa_b$ is defined by analytic continuation in $b$ of Kneser's solution $\kappa_b(z_0)$, $\operatorname{Re} z_0 > -2$, from $(\eta, \infty)$ (Prop. 3 there). That proposition asserts real-analytic dependence of Kneser's conformal uniformisation on the real base without a separate parameter-dependence proof; Proposition 10.1 supplies such a proof locally along the real exterior ray, and Corollary 10.27 gives uniform vertical tails there. Near the cusp, Theorem E carries the family through the Shell–Thron boundary by a different route; transporting those estimates to boundary arcs far from $\eta$ is not needed here and is not addressed. Off that boundary, Proposition 1 of that paper asserts two regular representations $\psi_1^{-1}(\rho_1)$ and $\psi_2^{-1}(\rho_2)$ from the limits at $\pm i\infty$. Its proof starts from $\psi_1(F(z+1)) = \psi_1(F(z)) + 1$; since $\psi_1 = \log \sigma_1 / \log s_1$ is multivalued, continuity on $\operatorname{Im} z > M$ only gives this up to a constant $2\pi i m / \log s_1$ with $m \in \mathbb{Z}$. We impose their degree-one form explicitly in Definition 8.1; Lemma 10.25 records the integer end label that a fixed-point limit alone leaves open. The continuation across the Shell–Thron boundary is argued in [24, §5] by the continuity of a numerical scheme; it is not proved there.

The uniqueness statement [24, Prop. 2] requires only analyticity on $\operatorname{Re} z > -2$, $F(0) = 1$ and the two limits. As stated, it is false, already at $b = e$, by the following exact existence argument. Let $K = \operatorname{sexp}_e$ be Kneser's real-normalised solution and $D_0 = \{|z| < \frac{1}{2}\}$. Since $K(0) = 1$ and $K$ is nonconstant and holomorphic, the open-mapping theorem gives $\delta > 0$ such that $B(1, \delta) \subset K(D_0)$. Decrease $\delta$ so that $e(1 - e^{-\delta}) > \delta$. For every real $t$,

$$\exp(B(t, \delta)) \supset B\big(e^t, e^t(1 - e^{-\delta})\big),$$

as follows by applying the principal logarithm and $|\log(1+v)| \le -\log(1-|v|)$. Put $u_n = K(n)$; then $u_0 = 1$ and $u_{n+1} = e^{u_n}$. Induction using the displayed inclusion gives $B(u_n, \delta) \subset K(D_0 + n)$ for every $n$, and indeed

$$B\big(u_{n+1}, u_{n+1}(1 - e^{-\delta})\big) \subset K(D_0 + n + 1).$$

Choose $n$ so large that the last radius exceeds $1 + \pi/2$, and let $k = \lceil e^{u_{n+1}}/(2\pi) \rceil$. For large $n$, $0 \le \log(2\pi k) - u_{n+1} < 1$. Hence $w = \log(2\pi k) + i\pi/2$ lies in $K(D_0 + n + 1)$. Choose $z_* \in D_0 + n + 1$ with $K(z_*) = w$ and set $c = z_* + 2$. Then $\operatorname{Re} c > 0$ and, exactly,

$$K(c - 1) = e^w = 2\pi i k, \qquad K(c) = e^{2\pi i k} = 1.$$

Thus $F(z) = K(z + c)$ is analytic on $\operatorname{Re} z > -2$, obeys $F(z+1) = e^{F(z)}$, has $F(0) = 1$, and has the same two vertical limits as $K$. Yet $F(-1) = 2\pi i k \neq 0 = K(-1)$. The construction works for every real $b > \eta$. Put $a = \log b > 1/e$ and $u_n = K_b(n) \to \infty$, since $e^{ax} > x$ for $x > 0$. Choose $N$ with $a u_{N+1} > 1$ and then $\delta > 0$ so that $B(u_N, \delta) \subset K_b(B(N, \frac{1}{2}))$ and $u_{N+1}(1 - e^{-a\delta}) > \delta$. The inclusion

$$E_b(B(t, \delta)) \supset B\big(e^{at}, e^{at}(1 - e^{-a\delta})\big)$$

propagates the fixed $\delta$-disks and yields image disks of radii $u_{n+1}(1 - e^{-a\delta}) \to \infty$. For large $n \ge N$ choose $k = \lceil a \exp(a u_{n+1})/(2\pi) \rceil$. The target $w_k = a^{-1}(\log(2\pi k/a) + i\pi/2)$ lies in the resulting image disk. Then $E_b^2(w_k) = 1$ and $E_b(w_k) = 2\pi i k/a \neq K_b(-1) = 0$. Choose $z_k \in B(n+1, \frac{1}{2})$ with $K_b(z_k) = w_k$, put $c_k = z_k + 2$, and set $F_k(z) = K_b(z + c_k)$. Since $\operatorname{Re} c_k > 0$, each $F_k$ meets the hypotheses of Proposition 2, but $F_k(-1) = 2\pi i k/a$. The values $k$ are unbounded, giving infinitely many distinct solutions. The gap in [24, Prop. 2] is the last step of its proof, which infers $C = 0$ from $F_1(C) = 1$. For real $b > \eta$ the uniqueness theorem of [23] is not affected: it

also assumes analyticity on $\mathbb{C}\setminus(-\infty,-2]$ and $F(\bar z)=\overline{F(z)}$, both of which this $F$ violates. Indeed $c$ is nonreal, and $K$ cannot extend holomorphically through $-2$, since $K(-1)=0$ whereas an exponential never vanishes; hence $F$ is singular at $-2-c$ off the standard cut. For complex bases, some root-selection condition must be added to [24, Prop. 2]. The seam-height condition of Definition 8.1 is such a condition, and Theorem 8.2 shows that it suffices near $(1,\eta)$.

Replacing $F(0)=1$ by $F(-1)=0$ excludes the solutions $F_k$, and part of the argument then goes through. Suppose $F_2=F_1(\cdot+C)$ on $\operatorname{Re} z>-2$, both with $F(-1)=0$; exchanging $F_1$ and $F_2$ if necessary, $\operatorname{Re} C\ge 0$. A solution on $\operatorname{Re} z>-2$ can vanish only where $\operatorname{Re} z\le -1$, since otherwise $F(z)=b^{F(z-1)}\neq 0$. Hence $F_1(C-1)=F_2(-1)=0$ forces $\operatorname{Re} C=0$. To conclude $C=0$ one still needs $F_1(-1+\mathrm{i}t)\neq 0$ for real $t\neq 0$. This holds if $F_1$ continues to $\mathbb{C}\setminus(-\infty,-2]$ with the functional equation, as Kneser's solution does for real $b>\eta$: then $F_1(-1+\mathrm{i}t)=b^{F_1(-2+\mathrm{i}t)}$. For complex bases this continuation is not available in general. For $b=\mathrm{i}$, [24] reports zeros of $\mathcal{K}_{\mathrm{i}}$ near $-3.810+0.146\mathrm{i}$ and $-6.591+0.273\mathrm{i}$, hence further branch points off $(-\infty,-2]$. So with $F(-1)=0$ the extra condition needed is $F(-1+\mathrm{i}t)\neq 0$ for $t\neq 0$, together with the degree-one form above.

## 2. Setup and parabolic data

Throughout, $1<b<\eta$ and $E(w)=b^w$.

**Lemma 2.1** (Parametrisation)**.** *Let $L$ be a fixed point of $E$ with multiplier $\lambda=L\log b$. Then $L=\mathrm{e}^{\lambda}$ and $\log b=\lambda\mathrm{e}^{-\lambda}$. The map $\lambda\mapsto\lambda\mathrm{e}^{-\lambda}$ increases on $(0,1)$ and decreases on $(1,\infty)$, with maximum $\mathrm{e}^{-1}=\log\eta$. Hence $\lambda\in(0,1)$ and $\lambda_2\in(1,\infty)$ parametrise $(1,\eta)$ through the attracting and the repelling fixed point, and the two branches meet at $b=\eta$.*

*Proof.* $L=b^L$ gives $\log L=L\log b=\lambda$. □

**Proposition 2.2.** *With $\lambda=1-\varepsilon$ one has $\varepsilon=\sqrt{2\mathrm{e}(\log\eta-\log b)}\,(1+O(\varepsilon))$, and as $b\to\eta^-$*

$$\Lambda=\exp\Big(-\frac{C_\eta}{\sqrt{\eta-b}}(1+o(1))\Big),\qquad C_\eta=\frac{4\pi^2}{\sqrt{2\mathrm{e}^{1-1/\mathrm{e}}}}.$$

*Proof.* Expand $\lambda\mathrm{e}^{-\lambda}=\mathrm{e}^{-1}(1-\varepsilon^2/2-\varepsilon^3/3+O(\varepsilon^4))$ and use $|\log\lambda|=\varepsilon+O(\varepsilon^2)$ and $\log\eta-\log b\sim(\eta-b)/\eta$. □

**Remark 2.3** (Translation invariance)**.** A simultaneous shift $K(\cdot+\delta)$, $R(\cdot+\delta)$ sends $c_n\mapsto c_n\mathrm{e}^{2\pi\mathrm{i}n\delta}$. Thus $\kappa_n=c_n/c_1^n$, when defined, is invariant under complex translations, while $|c_1|$ is invariant only under real ones. The normalisation $F(0)=1$ fixes the imaginary origin. This is why normalised first-mode limits require a specified base point, and why the constant $a$ enters Corollary 1.1.

2.1. **The parabolic germ.** At $b=\eta$, put $w=\mathrm{e}(1+u)$. Then $E$ becomes $f(u)=\mathrm{e}^u-1$, with attracting petal on $\operatorname{Re} u<0$ and formal Abel function

$$\alpha(u)=-\frac{2}{u}+\frac{\log(-u)}{3}+\sum_{k\ge1}c_k^{\alpha}u^k,\qquad \alpha(f(u))=\alpha(u)+1. \tag{3}$$

The Fatou coordinates are $\Phi_{\mathrm{att}}(u)=\lim[\alpha(f^n u)-n]$ and $\Phi_{\mathrm{rep}}(u)=\lim[\alpha(f^{-n}u)+n]$, with $\log(-u)$ continued along orbits. On the upper gate, the horn map $\Phi_{\mathrm{rep}}\circ\Phi_{\mathrm{att}}^{-1}(z)=z+\sum_{n\ge1}A_n\mathrm{e}^{2\pi\mathrm{i}nz}$ and its inverse $\Phi_{\mathrm{att}}\circ\Phi_{\mathrm{rep}}^{-1}(z)=z+\sum_{n\ge1}B_n\mathrm{e}^{2\pi\mathrm{i}nz}$ have no constant mode (Section 4, item (d)). Formal inversion gives

$$B_1=-A_1,\qquad B_2=-A_2+2\pi\mathrm{i}A_1^2. \tag{4}$$

**Proposition 2.4.** *The first coefficient of the inverse upper horn map satisfies $B_1\neq 0$.*

*Proof.* Let $A$ be the immediate parabolic basin of $f(u) = e^u - 1$. The negative real ray is contained in $A$, since $u < f(u) < 0$ for $u < 0$. The entire map $f$ has no critical points and exactly one finite singular value, $-1$, its asymptotic value along that ray. By [5, Lemma 87], the singular values of $f|_A : A \to A$ are contained in $A \cap \{-1\}$; the negative ray shows that $-1$ does occur. Hence $f$ belongs to the one-petal unisingular class $S_\infty$ of [5, Definition 8]. The parabolic-renormalisation theorem [5, Theorem 9 and Corollary 11] gives a renormalisation $R[f]$ with exactly one attracting petal at its parabolic fixed point. The upper renormalisation is the exponential projection of $\Phi_{\text{att}} \circ \Phi_{\text{rep}}^{-1}$, up to nonzero linear scalings of source and target [5, Appendix A]. In our normalisation, writing $\zeta = e^{2\pi i z}$, this projection is $H(\zeta) = \zeta + 2\pi i B_1 \zeta^2 + O(\zeta^3)$. Since $H'(0) = 1$, nonzero linear scalings that again give multiplier one preserve whether the quadratic coefficient vanishes. One attracting petal means that this coefficient is nonzero. Therefore $B_1 \neq 0$. □

**Remark 2.5** (The mechanism)**.** The underlying reason is the classical one. The map $f$ is of finite type in the sense of Epstein: it has no critical points and a single finite asymptotic value. Its horn maps are again of finite type, with singular values in $\mathbb{C}^*$ given by the projections of the single singular orbit [4]. So the upper horn map has exactly one singular value in $\mathbb{C}^*$. By the Fatou-type theorem for finite-type maps [4], the immediate basin of each attracting petal of a parabolic fixed point contains a singular value other than the fixed point. If $B_1 = 0$, then $H(\zeta) - \zeta = O(\zeta^3)$, and $H$ would have at least two attracting petals at 0, each needing its own singular value.

Computing both maps directly on horizontal lines gives

$$\begin{aligned} |B_1| &= 0.089058436412213331563, \qquad |B_2| = 0.012487384111564700811, \\ \kappa_2^\infty &:= \frac{B_2}{B_1^2} = -0.02532049309975755123 + 1.57421906523195162 67\,\mathrm{i}, \\ \frac{B_3}{B_1^3} &= -2.254799124523803111 - 0.024408714296883171\,\mathrm{i}. \end{aligned} \tag{5}$$

Also $a = \Phi_{\text{att}}(1/e - 1) = 3.0292972144180360989\ldots$, so the limiting phase in Corollary 1.1 is $\arg B_1 + 2\pi a = 1.5865330757\ldots \pmod{2\pi}$. These values are stable to all digits shown under changes of working precision (40–60 digits), transform size ($N = 16, 24, 32$) and sampling height ($\operatorname{Im} z = 1, 1.5, 2$). They are numerical values, not certified enclosures; nonvanishing of $B_1$ follows independently from Proposition 2.4.

**Remark 2.6** (Formal invariants)**.** Put $\zeta = e^{2\pi i z}$. The inverse horn map becomes the cylinder germ $F(\zeta) = \zeta \exp(2\pi i \sum B_n \zeta^n) = \zeta + a_2\zeta^2 + a_3\zeta^3 + \cdots$. Since $B_1 \neq 0$, its formal invariants under tangent-to-identity conjugacy include $a_2 = 2\pi i B_1$ and the iterative residue $\rho = 1 - a_3/a_2^2$ [3]. One has

$$\kappa_2^\infty = 2\pi i(\tfrac{1}{2} - \rho), \qquad \rho = 0.249455254258832 - 0.004029881638351\,\mathrm{i}. \tag{6}$$

When defined, the higher ratios $B_n/B_1^n$ are not formal invariants of $F$. They are, however, invariants under the translations that preserve the normalisation of Fatou coordinates, and Theorem A compares them for every $n$.

## 3. The separation is a transition-map invariant

The separation problem compares a Kneser-type solution $K_b$ with $R_b$. This section removes $K_b$ from the problem; Section 9 identifies it with the sewn solution constructed below. The separation coefficients turn out to be determined, up to a relative error of order $\Lambda$, by the transition map between the two *regular* solutions at the two real fixed points — an object built from two convergent Schröder series in a fraction of a second, at any $b < \eta$.

3.1. **The transition map.** Let $\tau$ be the Koenigs coordinate at the repelling point $L_2$, $\tau(E(w)) = \lambda_2\tau(w)$, $\tau'(L_2) = 1$, and put

$$S(z) = \tau^{-1}(-\lambda_2^z),$$

continued to an entire function by $S(z+1) = E(S(z))$. It maps $\mathbb{R}$ decreasingly onto the segment $(L, L_2)$ and has period $2\pi\mathrm{i}/\log\lambda_2$. Let $R^{-1}(w) = \log(\sigma(w)/\sigma(1))/\log\lambda$ be the attracting Abel coordinate, $R^{-1}(1) = 0$, which is multivalued with period $\mathrm{i}h = 2\pi\mathrm{i}/|\log\lambda|$. On the horizontal strip where $S$ takes values in the immediate basin of $L$ and a continuous branch is chosen, put

$$T(z) := R^{-1}(S(z)), \qquad T(z+1) = T(z) + 1, \qquad T(z) - z = \sum_{n\in\mathbb{Z}} t_n \mathrm{e}^{2\pi\mathrm{i}nz}.$$

**Lemma 3.1** (Real structure)**.** *For real $b \in (1, \eta)$, $\operatorname{Im} T(x) \equiv h/2 \pmod h$ for every real $x$. Consequently, on the branch with $\operatorname{Im} t_0 = h/2$, $T(x) - x - \mathrm{i}h/2$ is real, $t_{-n} = \overline{t_n}$, and the line $\operatorname{Im} w = -h/2$ of attracting time is the image of $\mathbb{R}$ under the branch $T - \mathrm{i}h$.*

*Proof.* $S(x) \in (L, L_2)$, where $\sigma > 0$, while $1 < L$ gives $\sigma(1) < 0$. Hence $\sigma(S(x))/\sigma(1)$ is negative and $\operatorname{Im}\log(\sigma(S(x))/\sigma(1)) \in \pi + 2\pi\mathbb{Z}$; divide by $\log\lambda = -2\pi/h$. □

Translations of $S$ ($S \mapsto S(\cdot + \delta)$, $\delta \in \mathbb{C}$) send $t_0 \mapsto t_0 + \delta$, $t_n \mapsto t_n \mathrm{e}^{2\pi\mathrm{i}n\delta}$, so

$$\tau_n(b) := t_n\, \mathrm{e}^{-2\pi\mathrm{i}nt_0} \qquad (\operatorname{Im} t_0 = h/2) \tag{7}$$

does not depend on the normalisation of $S$; it depends on that of $R$ only through the real translation fixed by $R(0) = 1$. The condition $\operatorname{Im} t_0 = h/2$ refers to the real normalisation of $S$ and fixes the branch of $R^{-1}$; a branch shift by $\mathrm{i}kh$ would multiply $\tau_n$ by $\Lambda^{-nk}$. The branch is fixed once in the real normalisation and then carried along.

3.2. **The welding identity.**

**Theorem 3.2.** *Let $K$ be a solution of (1), holomorphic on the sets below where it is represented, and suppose there are 1-periodic holomorphic maps $\varphi_\pm = \mathrm{id} + (\text{periodic})$ with*

1. *$K = R \circ \varphi_+$ on an upper half-plane $U_+$, where $P = \varphi_+ - \mathrm{id} = \sum_{n\ge 0} c_n \mathrm{e}^{2\pi\mathrm{i}nz}$ converges on $U_+$;*
2. *$K = S \circ \varphi_-$ on a lower half-plane $U_-$, where $\varphi_-$ is univalent, $G := \varphi_-^{-1} - \mathrm{id}$ is bounded on $V_- = \varphi_-(U_-)$, which contains a lower half-plane, and $G(w) \to g_0$ as $\operatorname{Im} w \to -\infty$;*
3. *$U_+ \cap U_-$ contains a horizontal line $\ell$ with $\varphi_-(\ell)$ inside the strip of $T$.*

*Then, with the branch of $T$ fixed by continuity from $\varphi_+ \circ \varphi_-^{-1}$, for every $n \ge 1$*

$$t_n = \sum_{m\ge n} c_m \big[\mathrm{e}^{2\pi\mathrm{i}mG}\big]_{n-m}, \qquad t_0 = g_0 + \sum_{m\ge 0} c_m \big[\mathrm{e}^{2\pi\mathrm{i}mG}\big]_{-m}, \tag{8}$$

*where $[F]_k$ is the $k$-th Fourier coefficient; in particular $[\mathrm{e}^{2\pi\mathrm{i}mG}]_0 = \mathrm{e}^{2\pi\mathrm{i}mg_0}$ and $[\mathrm{e}^{2\pi\mathrm{i}mG}]_k = 0$ for $k > 0$.*

*Proof.* On $\ell$, $R \circ \varphi_+ = S \circ \varphi_- = R \circ T \circ \varphi_-$, so $\varphi_+ - T \circ \varphi_-$ takes values in the discrete period lattice of $R$ and is constant; absorb it into the branch of $T$. Hence $T = \varphi_+ \circ \varphi_-^{-1}$ on the curve $\varphi_-(\ell)$, i.e. $T(w) - w = G(w) + P(w + G(w))$ there. The Fourier coefficients are computed by Cauchy's theorem on this curve. The region below it lies in $V_-$, and $\sum c_m \mathrm{e}^{2\pi\mathrm{i}m\varphi_-^{-1}(w)}$ converges uniformly on the curve $\varphi_-(\ell)$ itself, because $\ell \subset U_+$. Since $G$ is bounded and holomorphic on $V_-$, with limit $g_0$ at $-\mathrm{i}\infty$, so is $\mathrm{e}^{2\pi\mathrm{i}mG}$, whose Fourier series therefore contains only modes $k \le 0$ and has constant term $\mathrm{e}^{2\pi\mathrm{i}mg_0}$. Expanding $P(w + G) = \sum_m c_m \mathrm{e}^{2\pi\mathrm{i}mw} \mathrm{e}^{2\pi\mathrm{i}mG(w)}$ and collecting modes gives (8); the Fourier coefficients of $T - \mathrm{id}$ do not depend on the line inside its strip of holomorphy. □

**Corollary 3.3.** *Under the hypotheses of Theorem 3.2, for every $n \geq 1$,*

$$t_n\, \mathrm{e}^{-2\pi \mathrm{i} n g_0} = c_n + \mathrm{e}^{-2\pi \mathrm{i} n g_0} \sum_{m>n} c_m [\mathrm{e}^{2\pi \mathrm{i} m G}]_{n-m}.$$

*If $c_n \neq 0$, this may be written as*

$$t_n\, \mathrm{e}^{-2\pi \mathrm{i} n g_0} = c_n(1+\rho_n'), \qquad |\rho_n'| \leq \sum_{m>n} \frac{|c_m|}{|c_n|}\, |[\mathrm{e}^{2\pi \mathrm{i} m G}]_{n-m}|\, |\mathrm{e}^{-2\pi \mathrm{i} n g_0}|,$$

*and $g_0 = t_0 - c_0 - \sum_{m\geq 1} c_m [\mathrm{e}^{2\pi \mathrm{i} m G}]_{-m}$. Heuristically, if the $c_m$ decay like $\Lambda^m$ and $G$ is bounded on a line at height comparable to $h$ above the gluing line, both corrections are $O(\Lambda)$. The precise statement for the sewing solution is Theorem 6.3(4). On the branch of Lemma 3.1, $g_0 = t_0 - \mathrm{i}h + O(\Lambda)$, so that for the sewing solution, by Theorem 6.3(4),*

$$\hat{c}_n := \frac{c_n}{\Lambda^n} = \tau_n(b) + O(\Lambda). \tag{9}$$

*Since $B_1 \neq 0$, also $c_1 \neq 0$ for $b$ sufficiently close to $\eta$ and $\kappa_n := c_n/c_1^n = t_n/t_1^n + O(\Lambda)$ for each fixed $n$.*

The identity $\mathrm{e}^{-2\pi \mathrm{i} n(t_0 - \mathrm{i}h)} = \Lambda^n \mathrm{e}^{-2\pi \mathrm{i} n t_0}$ is where the scale $\Lambda$ enters: *the $\Lambda$ law is the statement that the gluing line sits at attracting time $-h/2$, half a period below the real axis, while the gate carrying the positive modes of $T$ sits a further half period below it.* Nothing in the argument uses Kneser's construction.

### 3.3. **Welding construction.**

**Proposition 3.4.** *For $b \in (1,\eta)$ glue the half-cylinders $C_- = \{\mathrm{Im}\, w' \leq 0\}/\mathbb{Z}$ (repelling time) and $C_+ = \{\mathrm{Im}\, w \geq -h/2\}/\mathbb{Z}$ (attracting time) along their boundaries by $w = T(w') - \mathrm{i}h$. The result is conformally $\mathbb{C}^*$; its universal-cover coordinate $z$ defines a solution*

$$K^W(z) = R(w(z)) \text{ on } C_+, \qquad K^W(z) = S(w'(z)) \text{ on } C_-,$$

*unique up to a translation in $z$, which satisfies the hypotheses of Theorem 3.2. It is holomorphic except where $w(z)$ meets the singular set $\bigcup_k((-\infty,-2] + \mathrm{i}kh)$ of $R$.*

Unlike $S$, which is entire (the Poincaré function of a repelling point), $R$ is not: its inverse Koenigs map extends only through inverse branches of $E$. So $K^W$ carries, besides the cut near $(-\infty,-2]$, the copies near $(-\infty,-2] + \mathrm{i}kh$, $k \geq 1$, lying in its $R$-represented region. The same holds for the Kneser-type solutions at complex bases inside the Shell–Thron region, whose upper representation is through the attracting regular superfunction.

*Proof.* By Lemma 3.1 the gluing map is a real-analytic diffeomorphism between the boundary circles that extends holomorphically to a neighbourhood, so the glued surface $X$ is a Riemann surface; each end is a punctured disc, hence $X \cup \{\text{two points}\}$ is a simply connected compact Riemann surface and $X \cong \mathbb{C}^*$. The definitions agree on the seam because $R \circ (T - \mathrm{i}h) = S$. The generator of $\pi_1(X)$ lifts to $z \mapsto z+1$ and acts as $w \mapsto w+1$, $w' \mapsto w'+1$, whence $K^W(z+1) = E(K^W(z))$. Near each puncture the change of coordinate is a holomorphic map of punctured discs fixing the puncture, which is the statement that $w - z$ (resp. $w' - z$) has only non-negative (resp. non-positive) Fourier modes. Uniqueness is the uniqueness of the uniformisation. □

**Proposition 3.5** (Local parameter dependence of the sewn family)**.** *For every $b_0 \in (1,\eta)$, the normalised geometric sewing of Proposition 3.4 extends to a holomorphic family over a complex disc $\Delta_{b_0}$ about $b_0$. On every common regular domain of the universal-cover coordinate, $K_b^W(z)$ is jointly holomorphic in $(b,z)$. These local families agree on overlaps and therefore define the inner sewn family on an open complex neighbourhood $N_W$ of the entire real interval $(1,\eta)$, the union of the discs $\Delta_{b_0}$. No uniform width of that neighbourhood is asserted.*

*Proof.* Fix $b_0$. Both fixed points and their Koenigs coordinates vary holomorphically near $b_0$. The compact set $S_{b_0}([0,1]) \subset (L, L_2)$ lies in the attracting basin and avoids its fixed point. Thus the chosen branch of $T_b = R_b^{-1} \circ S_b$ is jointly holomorphic on a parameter disc times a narrow strip about $[0,1]$; the relation $T_b(w'+1) = T_b(w')+1$ extends it to the whole strip. The real circle map is orientation preserving: $\tau_b'(S_b(x)) > 0$ by inverse iteration to $L_2$, $\sigma_b'(S_b(x)) > 0$ by forward iteration to $L$, and hence $S_b'(x) < 0$ and $(R_b^{-1})'(S_b(x)) = \sigma_b'(S_b(x))/(\sigma_b(S_b(x)) \log \lambda_b) < 0$. Consequently $T_{b_0}'(x) > 0$. Compactness of the circle and local injectivity give a smaller strip on which $T_{b_0}$ is injective modulo 1; shrinking the parameter disc preserves this property and makes the image of the real circle an analytic Jordan curve in the cylinder.

Put $U_b = T_b + \varpi_b$, where $\varpi_b = 2\pi\mathrm{i}/\log \lambda_b$; at $b_0$ this is $T_{b_0} - \mathrm{i}h$. Use $U_b$ to identify a thin open annular collar in the repelling cylinder with its image in the attracting cylinder. More explicitly, for a sufficiently small $\delta > 0$ take the repelling chart $\{\operatorname{Im} w' < \delta/2\}/\mathbb{Z}$ and the attracting chart above the curve $U_b(\{\operatorname{Im} w' = -\delta/2\})$; identify them on $|\operatorname{Im} w'| < \delta/2$. After shrinking the parameter disc, these charts and the transition $(b, w') \mapsto (b, U_b(w'))$ form a Hausdorff holomorphic family of cylinders. At $b = b_0$ it is the analytic-collar version of the boundary welding in Proposition 3.4. Adjoin the two ends using the parameter independent local coordinates $\mathrm{e}^{2\pi\mathrm{i}w}$ and $\mathrm{e}^{-2\pi\mathrm{i}w'}$. This yields a proper holomorphic submersion whose fibres are compact Riemann surfaces of genus zero. The two added points and the upper-chart point $w = 0$ are three disjoint holomorphic sections (the last remains above the seam after shrinking the disc).

Local holomorphic triviality of a proper family with mutually biholomorphic compact fibres [9], followed by the unique fibrewise Möbius map sending these three sections to $0, \infty, 1$, gives a jointly holomorphic coordinate $q_b$ on the compactified fibres. Choose the local lift $z = (2\pi\mathrm{i})^{-1} \log q_b$ for which the marked upper point $w = 0$ has $z = 0$. On the lift of the overlap, $R_b(U_b(w')) = S_b(w')$ because $\varpi_b$ is a period of $R_b$. The two regular superfunctions therefore define $K_b^W$, jointly holomorphic wherever their represented branches are regular, and $K_b^W(0) = R_b(0) = 1$. On the real parameter slice this is exactly the geometric welding. Its marked coordinate makes the local families unique, so the identity theorem patches them on overlapping discs. □

**Corollary 3.6** (Non-reality at fractional real heights)**.** *For every real $b < \eta$ sufficiently close to $\eta$, and every $\delta > 0$ sufficiently small, some real $x$ with $0 < |x| < \delta$ satisfies $\operatorname{Im} K_b^W(x) \neq 0$. In particular the sewn solution is not real-valued on any real interval about $z = 0$.*

*Proof.* Suppose that $K_b^W(x)$ were real for all real $x$ in a neighbourhood of 0. Shrink that neighbourhood so its values lie in a real interval about 1 on which the attracting Koenigs coordinate $\sigma_b$ is holomorphic and real, with $\sigma_b(K_b^W(x))/\sigma_b(1) > 0$. The upper representation of Theorem 6.3 gives

$$\frac{\sigma_b(K_b^W(x))}{\sigma_b(1)} = \lambda_b^{x+P_b(x)}, \qquad P_b(0) = 0.$$

Since $\log \lambda_b$ is real and nonzero, continuity at $x = 0$ makes $\operatorname{Im} P_b(x) = 0$ on this interval. The function $P_b$ is holomorphic and periodic on a strip containing the whole real line, so reflection and the identity theorem make it real on that line. But its Fourier expansion has only non-negative modes. A real-valued periodic function has conjugate positive and negative coefficients; therefore all its nonconstant coefficients vanish. This contradicts $c_1(b) \neq 0$ for $b$ sufficiently close to $\eta$, established in Corollary 3.3 using Proposition 2.4. □

This analytic non-reality statement concerns the sewn family near $\eta$. The tiny non-real part observed numerically by Paulsen at $b = \sqrt{2}$ [24, §6] is reproduced by $K^W$ in all nine quoted digits (Section 10.1); this is numerical, not a proof that his continuation path gives $K^W$.

**Corollary 3.7** (Complex cusp profile and its imaginary trace)**.** *Let $A = B_1 e^{2\pi i a} \neq 0$ and put $f = F_\eta^{\mathrm{att}}$. On some fixed complex disk about $z = 0$, as $b \to \eta^-$ through real bases,*

$$\frac{K_b^W(z) - R_b(z)}{\Lambda(b)} \longrightarrow f'(z) A(e^{2\pi i z} - 1)$$

*locally uniformly, hence with all $z$-derivatives. In particular, on a fixed real interval about $0$,*

$$\frac{\operatorname{Im} K_b^W(x)}{\Lambda(b)} \longrightarrow f'(x) \operatorname{Im}\Big[A(e^{2\pi i x} - 1)\Big]$$

*uniformly on compact subintervals. The right-hand side is not identically zero on any interval about $0$.*

*Proof.* By Theorem 6.3(4), with $B = e^{2\pi(Y_0+D+2)}$, there is a constant $C$ independent of $b$ and $n$ such that $|c_n(b)| \le C(\Lambda B)^n$ for all $n \ge 1$ and $b$ sufficiently close to $\eta$. Here we used $|\tau_n| \le M e^{2\pi n(Y_0+1)}$ and absorbed the factor $\Lambda$ in the error bound into $C$. Since $P_b(0) = 0$, its constant mode is $-\sum_{n\ge1} c_n$. Choose a disk in the common upper representation supplied by Corollary 6.5. The Fourier tail is geometric on every smaller disk once $b$ is close enough to $\eta$, and there

$$P_b(z) = c_1(b)(e^{2\pi i z} - 1) + O(\Lambda^2)$$

locally uniformly. Theorem 6.3(4) and Corollary 4.5 give $c_1/\Lambda \to A$. On a fixed disk about $0$, $R_b \to F_\eta^{\mathrm{att}}$ with its derivatives by Corollary 6.5 and its proof. Taylor's formula gives $K_b^W(z) = R_b(z) + R_b'(z)P_b(z) + O(\Lambda^2)$ on a smaller disk. This proves the complex limit, with derivative convergence by Cauchy's formula. For real $x$, $R_b(x)$ and $f'(x)$ are real, giving the imaginary trace. If its last factor vanished on an interval, both the first and second derivatives at $x = 0$ would vanish, forcing $\operatorname{Re} A = \operatorname{Im} A = 0$, contrary to $A \neq 0$. □

**Corollary 3.8** (Obstruction at the real cusp)**.** *For every sufficiently small complex neighbourhood $V$ of $0$, one can choose a real $x_* \in V \setminus \{0\}$ and constants $c, \delta > 0$ such that*

$$|\operatorname{Im} K_b^W(x_*)| \ge c\Lambda(b) \qquad (\eta - \delta < b < \eta,\ b \in \mathbb{R}).$$

*There is therefore no jointly holomorphic family $F(b, z)$ on a product of a disc about $b = \eta$ and $V$ which is real at real heights for real $b > \eta$ and agrees with $K_b^W$ for all real $b < \eta$ sufficiently close to $\eta$. In particular, direct holomorphic continuation of a real-normalised Kneser family through the real cusp cannot identify it with the inner sewn branch.*

*Proof.* The limiting profile in Corollary 3.7 is not identically zero on any real interval about $0$. Choose $x_*$ where it is nonzero. Since $\Lambda(b) > 0$ for real $b < \eta$, the uniform limit gives the stated lower bound after reducing $\delta$. For fixed real $x_*$, Schwarz reflection in the parameter says that a holomorphic function $b \mapsto F(b, x_*)$ which is real on a real interval above $\eta$ is also real on the real interval below $\eta$ in the same disc. This contradicts the lower bound if $F = K^W$ there. □

**Corollary 3.9** (Conjugate inner branches and their jump)**.** *For real $b \in (1, \eta)$ put $K_b^{W,\#}(z) = \overline{K_b^W(\bar z)}$ on the reflected regular domain. This reflected family is locally jointly holomorphic in $(b, z)$ near every such real base. Both branches obey the tetration equation, are normalised at $z = 0$, agree at every nonnegative integer height where defined, and have a simple zero at $z = -1$ with the same logarithmic monodromy $2\pi i/\log b$ at $z = -2$. For $x_*$ as in Corollary 3.8,*

$$|K_b^W(x_*) - K_b^{W,\#}(x_*)| \ge 2c\Lambda(b) \qquad (\eta - \delta < b < \eta,\ b \in \mathbb{R}).$$

*More precisely, on a fixed real interval about $0$ the quotient of their difference by $\Lambda(b)$ tends locally uniformly to $2i(F_\eta^{\mathrm{att}})'(x) \operatorname{Im}[A(e^{2\pi i x} - 1)]$.*

*Proof.* For complex $b$ near a real base, define the reflected family by $K_b^{W,\#}(z) = \overline{K_{\bar b}^W(\bar z)}$. Proposition 3.5 makes it jointly holomorphic. Reflection preserves the equation for real $b$ and therefore, by the identity theorem, on its local complex parameter disc. The integer values follow from the equation and normalisation. The zero at $-1$ and its simplicity follow from Proposition 10.41, as does the monodromy. On real heights the difference is exactly $2\mathrm{i}\,\mathrm{Im}\,K_b^W(x)$, so the lower bound and the profile follow from Corollaries 3.8 and 3.7. □

**Proposition 3.10** (The cusp splitting is beyond all algebraic orders)**.** *On a sufficiently small real interval $J$ about $0$, put*

$$M(x) = (F_\eta^{\mathrm{att}})'(x)\,\mathrm{Im}[A(\mathrm{e}^{2\pi\mathrm{i}x} - 1)].$$

*The interval $J$ may be chosen so that $M(x) \neq 0$ for every $x \in J \setminus \{0\}$. For every such $x$ and every integer $q \geq 1$, the germ along $t > 0$ of*

$$t \longmapsto K^W_{\eta - t^q}(x)$$

*has no meromorphic extension through $t = 0$. In particular, no finite ramification of the base parameter makes this value meromorphic at the real cusp; it has no convergent Puiseux–Laurent germ there. If it admits a single-valued holomorphic continuation on a full punctured $t$-disc, the singularity at $t = 0$ is essential.*

*Proof.* The profile $M$ is real analytic, vanishes at 0, and, by Corollary 3.7, is not identically zero on any interval about 0. Thus 0 is an isolated zero, and we shrink $J$ accordingly. Suppose such a meromorphic germ $G(t)$ existed. Its reflection $G^\#(t) = \overline{G(\bar t)}$ would also be meromorphic, so $D(t) = G(t) - G^\#(t)$ would be meromorphic near 0. For positive real $t$, the imaginary profile and the saddle-node scale of $\Lambda$ give

$$D(t) = 2\mathrm{i}\,\mathrm{Im}\,K^W_{\eta - t^q}(x) = O\Big(\exp(-c_0 t^{-q/2})\Big) = o(t^N) \qquad \text{for every } N \geq 0$$

with some $c_0 > 0$. A meromorphic Laurent germ with this decay on the positive axis is identically zero. This contradicts the nonzero limit $D(t)/\Lambda(\eta - t^q) \to 2\mathrm{i}M(x)$. □

**Corollary 3.11** (A two-jet witness at the normalisation height)**.** *Put $f = F_\eta^{\mathrm{att}}$ and $A = B_1\mathrm{e}^{2\pi\mathrm{i}a} \neq 0$. As $b \to \eta^-$ through real bases,*

$$\frac{1}{\Lambda(b)}\begin{pmatrix}\mathrm{Im}[(K_b^W)'(0)]\\ \mathrm{Im}[(K_b^W)''(0)]\end{pmatrix} \longrightarrow \begin{pmatrix}2\pi f'(0)\,\mathrm{Re}\,A\\ 4\pi f''(0)\,\mathrm{Re}\,A - 4\pi^2 f'(0)\,\mathrm{Im}\,A\end{pmatrix} \neq \begin{pmatrix}0\\0\end{pmatrix}.$$

*Consequently at least one of the first two $z$-derivatives at $0$, with the same choice of derivative for every $q \geq 1$, has no meromorphic base germ through $t = 0$ after $b = \eta - t^q$. The corresponding derivative of the reflected inner branch differs from that of $K_b^W$ by at least $c\Lambda(b)$ in modulus for some $c > 0$ and all sufficiently close real $b < \eta$.*

*Proof.* The complex limit of Corollary 3.7 and Cauchy's formula give convergence of the first two derivatives. The derivatives of $R_b$ at 0 are real for real $b$, giving the displayed vector. Since $f'(0) \neq 0$ by Corollary 6.5, vanishing of its first entry forces $\mathrm{Re}\,A = 0$, and then vanishing of its second forces $\mathrm{Im}\,A = 0$, a contradiction. For a nonzero entry, the reflection argument in the proof of Proposition 3.10 applies to that derivative. Reflection doubles its imaginary part, which also gives the lower bound. □

**Remark 3.12** (Conjugate bypasses of the cusp)**.** Suppose the real-normalised Kneser germ admits analytic continuation along two conjugate parameter paths bypassing $\eta$ and reaching a common real interval below $\eta$. If the continuation along one path agrees there with $K^W$, uniqueness of analytic continuation and Schwarz reflection make the continuation along the

reflected path $\overline{K_b^W(\bar{z})}$. At the fixed real height $x_*$ of Corollary 3.8, the two values differ in modulus by at least $2c\Lambda(b)$. The existence of such continuations is not asserted here.

For non-real $b$ sufficiently close to any fixed real $b_0 \in (1, \eta)$, the sewn solution thus depends holomorphically on $b$. Theorem 7.1 gives the stronger quantitative Wiener bounds near $\eta$. For $\operatorname{Im} b > 0$ it tends to $L$ and $L_2$ as $\operatorname{Im} z \to \pm\infty$, respectively. Those limits alone do not characterise it: Proposition 2 of [24] has the counterexample in Remark 1.2. Identifying the base-continued $\kappa$ with $K^W$ requires the stronger two-chart and seam condition of Theorem D. Table 1 gives numerical evidence: $\tau_1$ and $t_2/t_1^2$ reproduce the ladder values of $\hat{c}_1$ and $\kappa_2$ — moduli *and* arguments — to the precision of the ladder extrapolation, e.g. eight digits at $b = 1.10$.

TABLE 1. Kneser ladders (Sections 10.1, 10.2, extrapolated to $\operatorname{Im} b = 0$) against the Kneser-free invariants $\tau_1 = t_1 e^{-2\pi i t_0}$ and $t_2/t_1^2$ (script `transition_map.py`).

| $b$ | $\hat{c}_1$ (ladder) | $\tau_1$ | $\kappa_2$ (ladder) | $t_2/t_1^2$ |
|---|---|---|---|---|
| 1.05 | $0.0850235\,e^{1.534136i}$ | $0.0850226\,e^{1.534165i}$ | $1.374049\,e^{1.413420i}$ | $1.374336\,e^{1.413645i}$ |
| 1.10 | $0.08642720\,e^{1.55109214i}$ | $0.08642720\,e^{1.55109214i}$ | $1.440851\,e^{1.477690i}$ | $1.440851\,e^{1.477690i}$ |
| 1.15 | $0.0872053\,e^{1.561012i}$ | $0.0872052\,e^{1.561013i}$ | $1.479111\,e^{1.511255i}$ | $1.479107\,e^{1.511258i}$ |
| 1.30 | 0.088434 | $0.0884321\,e^{1.577580i}$ | $1.541516\,e^{1.561918i}$ | $1.541515\,e^{1.561919i}$ |

3.4. **Reduction to confluence.** Write $a := \Phi_{\mathrm{att}}(1/e - 1) = 3.0292972144180360989\ldots$ for the $\alpha$-normalised attracting Fatou coordinate of the base point $w = 1$ at $b = \eta$ (real, since the orbit of $u = 1/e - 1$ stays on the negative axis).

**Theorem 3.13.** *Assume the following confluence statement* (C)*: there are constants $s(b) \in \mathbb{C}$ such that, as $b \to \eta^-$,*

$$R_b^{-1}(e(1+u)) \to \Phi_{\mathrm{att}}(u) - a, \qquad S_b^{-1}(e(1+u)) + s(b) \to \Phi_{\mathrm{rep}}(u),$$

*uniformly on compact subsets of the upper gate, with the branches continued as in Section 2.1. Then for every $n \geq 1$*

$$\tau_n(b) \longrightarrow B_n\, e^{2\pi i n a}.$$

*Consequently, also $t_n/t_1^n \to B_n/B_1^n$ for every $n \geq 1$. Together with Corollary 3.3 and the identification of $K_b$ with $K_b^W$ in Theorem 9.13, this gives $\hat{c}_n \to B_n e^{2\pi i n a}$ for all $n$. The limiting phase $\arg \hat{c}_1 \to \arg B_1 + 2\pi a = 1.5865330757\ldots \pmod{2\pi}$.*

*Proof.* Replace $S_b$ by $S_b(\cdot - s(b))$, which leaves $\tau_n$ unchanged. By (C), $T_b \to \Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1} - a = h^{-1} - a$ uniformly on $[0,1] + iY_g$ for fixed $Y_g$ in the gate, so the Fourier coefficients on that line converge: $t_n \to B_n$ $(n \geq 1)$ and $t_0 \to -a$, the latter because $B_0 = 0$ in the $\alpha$ normalisation. Hence $\tau_n = t_n e^{-2\pi i n t_0} \to B_n e^{2\pi i n a}$. □

Statement (C) is a confluence theorem for the saddle-node unfolding $u \mapsto e^{\lambda e^{-\lambda} e(1+u)}/e - 1$ in the direction where the two fixed points are real (the "Glutsyuk" direction): the Koenigs coordinates at the two hyperbolic points converge, on the gate, to the sectorial Fatou coordinates of the parabolic germ. Results of this type are due to Glutsyuk [10] and, for the full unfolded modulus, Mardešić–Roussarie–Rousseau [21] and Christopher–Rousseau [6]. Rather than matching their normalisations, we prove (C) directly for this family in Section 4, so Theorem 3.13 holds unconditionally.

## 4. Proof of Theorem A: confluence

We prove (C) directly for the family at hand; no normalisation from the literature is needed. In $u$-coordinates ($w = \mathrm{e}(1+u)$, $\mu = \mathrm{e}\log b$)

$$f_b(u) = \mathrm{e}^{\mu-1+\mu u} - 1, \qquad f_\eta(u) = \mathrm{e}^u - 1.$$

For $b < \eta$ close to $\eta$ the map has, in a fixed disc $D_r$, exactly the two fixed points $u_1 = \mathrm{e}^{\lambda-1} - 1 < 0 < u_2 = \mathrm{e}^{\lambda_2-1} - 1$ (Rouché), and

$$f_b(u) - u = (u-u_1)(u-u_2)\, k_b(u), \qquad k_b \to k_\eta, \quad k_\eta(u) = \tfrac{1}{2} + \tfrac{u}{6} + \cdots, \tag{10}$$

uniformly on $D_r$, with $k_b$ zero-free there. Put $A_j = 1/\log\lambda_j$, so $A_1 < 0 < A_2$.

**Lemma 4.1** (Residues)**.** *As $b \to \eta^-$: $A_1(u_1-u_2) \to 2$, $A_1 + A_2 \to \nu := 1 - 4k'_\eta(0) = \frac{1}{3}$, hence $A_1u_1 + A_2u_2 \to 2$ and $A_ju_j^2 \to 0$.*

*Proof.* $\lambda_1 = 1 + (u_1-u_2)k_b(u_1)$ and $\lambda_2 = 1 + (u_2-u_1)k_b(u_2)$. With $d = u_1 - u_2 \to 0$ and $1/\log(1+x) = 1/x + \frac{1}{2} + O(x)$, $A_1 + A_2 = \frac{k_b(u_2)-k_b(u_1)}{d\,k_b(u_1)k_b(u_2)} + 1 + O(d) \to 1 - 4k'_\eta(0)$, and $A_1 d \to 1/k_\eta(0) = 2$. □

Numerically $A_1 + A_2 = 0.3333331469$ and $A_1(u_1-u_2) = 1.99676$ at $\lambda = 0.99$. Define the *model time*

$$\Psi_b(u) = A_1 \log(u-u_1) + A_2 \log(u-u_2),$$

which is exactly the time of the vector field $(u-u_1)(u-u_2)/(\dots)$ whose time-one map has multipliers $\lambda_1, \lambda_2$, and let $\alpha_2(u) = -2/u + \frac{1}{3}\log(-u)$. By Lemma 4.1, for $u, p$ in a simply connected region at positive distance from 0, with logarithms continued along the same path,

$$\Psi_b(u) - \Psi_b(p) \longrightarrow \alpha_2(u) - \alpha_2(p). \tag{11}$$

**Lemma 4.2** (Uniform one-step error)**.** *$F_b := \Psi_b \circ f_b - \Psi_b - 1$ is single-valued and holomorphic on a fixed disc $D_{r'}$, $F_b \to F_\eta := \alpha_2 \circ f_\eta - \alpha_2 - 1$ uniformly there, and*

$$|F_b(u)| \le M\,|u-u_1|\,|u-u_2| \qquad (u \in D_{r'/2})$$

*with $M$ independent of $b$.*

*Proof.* By (10), $(f(u)-u_1)/(u-u_1) = 1+(u-u_2)k(u)$ and $(f(u)-u_2)/(u-u_2) = 1+(u-u_1)k(u)$, so

$$F = A_1 \log\frac{1+(u-u_2)k}{1+(u-u_1)k} + (A_1+A_2)\log(1+(u-u_1)k) - 1,$$

holomorphic for $|u| < r'$ small. The first logarithm is $O(|u_1-u_2|)$ uniformly, and $A_1(u_1-u_2)$, $A_1+A_2$ are bounded (Lemma 4.1); so $|F| \le M_0$ on $|u| = r'$ uniformly, and the displayed formula converges. At $u = u_1$ the first term is $A_1\log\lambda_1 = 1$ and the second vanishes, so $F(u_1) = 0$; likewise $F(u_2) = 0$. The maximum principle applied to $F/((u-u_1)(u-u_2))$ on $D_{r'}$ gives the bound, and letting $b \to \eta$ gives $F_\eta = O(u^2)$. □

**Lemma 4.3** (Uniform petals and summability)**.** *Put $Z_b(u) = A_1 \log\frac{u-u_1}{u-u_2}$ and $\Omega_{R,b} = \{u : \operatorname{Re} Z_b(u) > R\}$ (a disc of Apollonius about $u_1$). There is $R_0$ such that for $R \ge R_0$ and all $b$ near $\eta$: $\Omega_{R,b} \subset D_{r'/2}$, $f_b(\Omega_{R,b}) \subset \Omega_{R,b}$, $\operatorname{Re} Z_b(f_b^k u) \ge \operatorname{Re} Z_b(u) + \frac{3}{4}k$, and*

$$\sum_{k\ge N} |f_b^k u - u_1|\,|f_b^k u - u_2| \le \frac{C}{R + \frac{3}{4}N} \qquad (u \in \Omega_{R,b}),$$

*with $C$ independent of $b$ and $u$. As $b \to \eta$, $\Omega_{R,b}$ converges on compacta to the petal $\{\operatorname{Re}(-2/u) > R\}$.*

*Proof.* $Z(f(u)) - Z(u) = A_1 \log \frac{1+(u-u_2)k}{1+(u-u_1)k} = 1 + O(|u| + |u_1| + |u_2|)$ uniformly, by the expansion in the proof of Lemma 4.2; on $\Omega_{R,b} \subset D_{C/R+|u_1|}$ this is within $\frac{1}{4}$ of 1 once $R$ is large, which gives invariance and the growth of $\operatorname{Re} Z$. With $q = (u-u_1)/(u-u_2)$, $|q| = \mathrm{e}^{-\sigma}$, $\sigma = |\log \lambda_1| \operatorname{Re} Z$, one has $u - u_2 = (u_2 - u_1)/(q-1)$, whence

$$|u-u_1||u-u_2| = \frac{|q|\,|u_2-u_1|^2}{|1-q|^2} \le \Big(\frac{|u_2-u_1|}{|\log \lambda_1|}\Big)^2 \frac{(1+\sigma)^2 \mathrm{e}^{-\sigma}}{(\operatorname{Re} Z)^2},$$

using $1-\mathrm{e}^{-\sigma} \ge \sigma/(1+\sigma)$. The first factor is bounded (Lemma 4.1) and $(1+\sigma)^2\mathrm{e}^{-\sigma} \le 4$; summing $(R + \frac{3}{4}k)^{-2}$ over $k \ge N$ gives the bound. The last assertion follows from $\operatorname{Re} Z_b(u) \to \operatorname{Re}(-2/u)$, i.e. $A_1(u_2 - u_1) \to -2$. □

**Theorem 4.4** (Confluence)**.** *Statement* (C) *holds. More precisely, with branches continued along paths in the respective parabolic basins:*

1. $R_b^{-1}(\mathrm{e}(1+u)) \to \Phi_{\mathrm{att}}(u) - a$ *uniformly on compact subsets of the parabolic basin of* $f_\eta$*;*
2. *for a fixed* $p'$ *in the gate (below), putting* $s(b) = \Phi_{\mathrm{rep}}(p') - \mathcal{S}_b(p')$, $\mathcal{S}_b(u) + s(b) \to \Phi_{\mathrm{rep}}(u)$ *uniformly on compact subsets of the repelling basin (backward orbits tending to* 0*). Here* $\mathcal{S}_b$ *is the real-normalised repelling time, i.e. the branch of* $S_b^{-1}$ *that is real on the segment* $(u_1, u_2)$ *and is continued through the band of Lemma 6.1, then along backward-invariant paths.*

*The* gate *of height* $Y_*$ *is* $\mathcal{G}(Y_*) = \Phi_{\mathrm{rep}}^{-1}(\{Y_* \le \operatorname{Im} z \le Y_* + 1,\ -1 \le \operatorname{Re} z \le 2\})$*. For* $Y_*$ *large it is a compact subset of both parabolic basins of* $f_\eta$*. For* $b$ *near* $\eta$ *it lies in the band* $\mathcal{B}_b(Y_* - 2)$*, because* $Z_b \to -2/u$ *uniformly on it (Lemma 4.1) and* $\operatorname{Im}(-2/u) = \operatorname{Im} \alpha(u) + O(1) = \operatorname{Im} \Phi_{\mathrm{rep}}(u) + O(1)$ *there.*

*Proof.* (1) On $\Omega_{R,b}$ put $\mathcal{A}_b(u) = \lim_n [\Psi_b(f_b^n u) - n] = \Psi_b(u) + \sum_{k\ge 0} F_b(f_b^k u)$; the series converges absolutely by Lemmas 4.2 and 4.3. Since $f^n u \to u_1$ and the Koenigs map satisfies $\sigma(f^n u) = \lambda_1^n \sigma(u)$, $\Psi_b(f^n u) - n = A_1 \log \sigma(u) + A_2 \log(u_1 - u_2) + o(1)$, because $A_1 \log \lambda_1 = 1$. Thus $\mathcal{A}_b$ is the Koenigs Abel coordinate up to an additive constant, and $R_b^{-1}(\mathrm{e}(1+u)) = \mathcal{A}_b(u) - \mathcal{A}_b(u^*)$, $u^* = 1/\mathrm{e} - 1$. Let $K$ be a compact subset of the petal $P_R = \{\operatorname{Re}(-2/u) > R\}$ and $U \supset K \cup \{f_\eta^{N_0} u^*\}$ a simply connected region in $P_R$ at positive distance from 0. For $b$ near $\eta$, $U \subset \Omega_{R,b}$ and $u_1, u_2 \notin U$. For $u, p \in U$,

$$\mathcal{A}_b(u) - \mathcal{A}_b(p) = \Psi_b(u) - \Psi_b(p) + \sum_{k\ge 0} [F_b(f_b^k u) - F_b(f_b^k p)].$$

The first term converges by (11). In the sum, each term converges because $f_b^k \to f_\eta^k$ and $F_b \to F_\eta$ uniformly. The tails beyond $N$ are $\le 2MC/(R + \frac{3}{4}N)$ uniformly in $b$ (Lemma 4.3). Hence the difference converges uniformly to $\alpha_2(u) - \alpha_2(p) + \sum_k [F_\eta(f_\eta^k u) - F_\eta(f_\eta^k p)] = \Phi_{\mathrm{att}}(u) - \Phi_{\mathrm{att}}(p)$. The last identity holds because $\Phi_{\mathrm{att}} = \lim[\alpha(f^n u) - n]$ and $\alpha - \alpha_2 = O(u) \to 0$ along orbits. A general compact $K$ of the basin satisfies $f_\eta^N(K) \Subset P_R$ for some $N$. Then $f_b^N(K) \Subset P_R$ for $b$ near $\eta$, and $\mathcal{A}_b = \mathcal{A}_b \circ f_b^N - N$. Taking $p = f_b^{N_0} u^*$, which tends to $f_\eta^{N_0} u^* \in U$, and using uniform convergence on a neighbourhood gives (1). (On $\Omega_{R,b}$ the function $\mathcal{A}_b$ is multivalued, since this punctured Apollonius disc surrounds $u_1$. It is single-valued on $U$, which avoids $u_1$. On $K$, $\sigma_b \ne 0$ because $f_b^N(K)$ avoids $u_1$. On non-immediate components of the basin the branch is defined by $\mathcal{A}_b \circ f_b^N - N$. The Koenigs map is normalised by $\sigma_b'(u_1) = 1$.)

(2) Apply (1)'s argument to $G_b(v) = -f_b^{-1}(-v)$, the principal inverse branch reflected. Then $G_\eta(v) = -\log(1-v) = v + \frac{v^2}{2} + \frac{v^3}{3} + \cdots$, which is of the same prepared form with $k_\eta'(0) = \frac{1}{3}$ and residue $-\frac{1}{3}$. The attracting fixed point of $G_b$ is $-u_2$, with multiplier $1/\lambda_2$. Its Koenigs Abel

coordinate is $-S_b^{-1}(\mathrm{e}(1-v))$ up to a constant, and its parabolic Fatou coordinate is $-\Phi_{\mathrm{rep}}(-v)$ up to a constant (since $\alpha_f(-v) = -\alpha_G(v) + \text{const}$). The free constant is fixed by $p'$. □

With Theorem 4.4, Theorem 3.13 becomes unconditional. Four points need checking; each is settled here. The proof of Lemma 6.1 uses only Lemmas 4.1–4.3, so there is no circularity.

*(a) Inversion on the gate line.* Fix $Y_* \geq Y_B + 6$ and the line $\ell = [0,1] + \mathrm{i}Y_*$. By Rouché, the solution $v_b(z)$ of $\mathcal{S}_b(v) + s(b) = z$ near $\Phi_{\mathrm{rep}}^{-1}(z)$ exists and converges uniformly on $\ell$. Moreover $S_b(z - s(b)) = \mathrm{e}(1 + v_b(z))$, because $\mathcal{S}_b$ is a branch of $S_b^{-1}$.

*(b) The gate line lies in the strip of holomorphy of $T$.* The points $v_b(z)$, $z \in \ell$, lie within $o(1)$ of the gate, hence in the band $\mathcal{B}_b(Y_B + 3)$. By Lemma 6.1(ii) their $S$-times $z - s(b)$ satisfy $|\operatorname{Im}(z - s(b))| \leq h/2 - Y_B - 3 + \pi/3 + C/Y_B + C\varepsilon < h/2 - Y_B - 2$. So the shifted line $\ell - s(b)$ lies in the strip $\Sigma$ of Lemma 6.1(iii), where $T$ is holomorphic. Therefore the Fourier coefficients of $\tilde{T} = T(\cdot - s)$ on $\ell$ are $t_n \mathrm{e}^{-2\pi \mathrm{i} n s}$ and $\tilde{t}_0 = t_0 - s$, with the $t_n$ those of $T$ on $\mathbb{R}$. No prior knowledge of $\operatorname{Im} s$ is used.

*(c) The branch of $R_b^{-1}$.* Continue $R_b^{-1}$ from $u^*$ along the negative axis to $f_\eta^{N_0} u^*$, where it is real. Then go inside $P_R \setminus D_\delta(0)$ through the upper half-plane to the gate, and then inside the band down to the segment $(u_1, u_2)$. This path passes above $u_1$ and does not wind around $u_2$. Along it $R_b^{-1} = \Psi_b + \sum_k F_b \circ f_b^k + c_R$, so the change of $\operatorname{Im} R_b^{-1}$ is $A_1 \Delta \arg(u - u_1) + A_2 \Delta \arg(u - u_2) + O(C/Y + \varepsilon) = h/2 + 0 + \text{small}$. Lemma 3.1 forces the exact value, so on the segment $\operatorname{Im} R_b^{-1} = +h/2$. The branch in Theorem 4.4(1) is therefore the one with $\operatorname{Im} t_0 = h/2$ used in (7).

*(d) $B_0 = 0$.* At $b = \eta$, on the gate, $\Phi_{\mathrm{att}} - \alpha_2 = \sum_{k \geq 0} F_\eta(f^k u)$ and $\Phi_{\mathrm{rep}} - \alpha_2 = -\sum_{k \geq 1} F_\eta(f^{-k} u)$, with the same principal $\log(-u)$ in $\alpha_2$. That logarithm stays continuous along both orbits, since $-f^{\pm k} u$ remains in the lower half-plane. By Lemma 4.2 and $f^{\pm k} u \approx -2/(x \pm k + \mathrm{i}Y)$, both sums are $O(1/Y)$. Hence $\Phi_{\mathrm{att}} \circ \Phi_{\mathrm{rep}}^{-1}(z) - z \to 0$ as $\operatorname{Im} z \to +\infty$, so its constant Fourier mode vanishes: $B_0 = 0$. Numerically $|B_0| < 10^{-59}$.

By (a)–(d) and Theorem 4.4, $\tilde{T} \to h^{-1} - a$ uniformly on $\ell$, where $h^{-1}$ denotes the inverse horn map (not the period $h$). So $\tilde{t}_n \to B_n$ and $\tilde{t}_0 \to -a$, and $\tau_n = \tilde{t}_n \mathrm{e}^{-2\pi \mathrm{i} n \tilde{t}_0}$ gives:

**Corollary 4.5.** *For every $n \geq 1$, $\tau_n(b) \to B_n \mathrm{e}^{2\pi \mathrm{i} n a}$ as $b \to \eta^-$. Since $B_1 \neq 0$, also $t_n / t_1^n \to B_n / B_1^n$ for every $n \geq 1$.*

A direct pointwise check of Theorem 4.4 on two gate points (`confluence_check.py`) gives errors 0.086, 0.021, 0.0050, 0.00079 (attracting side) at $\lambda = 0.8, 0.9, 0.95, 0.98$, decreasing like $\varepsilon^2$.

## 5. The first-order term: a parabolic series

Theorem A gives the limit of $\tau_n$. Here we prove that $\tau_n$ is differentiable at $b = \eta$ from the side $b < \eta$. The derivative is given by an absolutely convergent series computed entirely at the parabolic germ. Put

$$s = 1 - \mathrm{e} \log b = 1 - \mu, \qquad f_s(u) = \mathrm{e}^{-s + (1-s)u} - 1,$$

so $s \in (0, s_0]$ corresponds to $b < \eta$ near $\eta$ and $f_0(u) = \mathrm{e}^u - 1$. The map $(s, u) \mapsto f_s(u)$ is entire.

5.1. **Why termwise differentiation fails, and the remedy.** With the model time $\Psi_b$ of Section 4, the one-step error is only $O(|u - u_1||u - u_2|)$ (Lemma 4.2). Along a parabolic orbit $u_k \approx -2/k$ one has $\partial_s u_k \asymp k$. Hence both $\partial_s F(u_k)$ and $F'(u_k) \partial_s u_k$ are of order one, and the differentiated series diverges. The remedy is the idea behind the prepared form of Mardešić–Roussarie–Rousseau [21]. We add to the model time a polynomial correction, holomorphic in $s$, that makes the one-step error vanish to order $N$ at both fixed points simultaneously.

By the Weierstrass preparation theorem,

$$f_s(u) - u = q_s(u)\, k_s(u), \qquad q_s(u) = u^2 - \sigma_1(s)u + \sigma_2(s), \tag{12}$$

on a polydisc $|s| < s_0$, $|u| < r$. Here $\sigma_1, \sigma_2$ are holomorphic, $k_s$ is holomorphic and zero-free, and $k_0(0) = \frac{1}{2}$. From $f_s(0) = -s + O(s^2)$ and $f_s'(0) - 1 = -2s + O(s^2)$ we get $\sigma_2 = -2s + O(s^2)$, $\sigma_1 = O(s)$, and the discriminant

$$D(s) = \sigma_1^2 - 4\sigma_2 = 8s + O(s^2)$$

is biholomorphic near 0. For real $s > 0$ the roots $u_1 < 0 < u_2$ are the fixed points of Section 4. Put $t = \sqrt{D(s)}$. Then $u_{1,2} = \frac{1}{2}(\sigma_1 \mp t)$ are holomorphic functions of $t$, interchanged by $t \mapsto -t$. Since $f_s'(u_1) = 1 + (u_1 - u_2)k_s(u_1)$, the residues are $A_1 = a_s(u_1, u_2)$ and $A_2 = a_s(u_2, u_1)$, where

$$a_s(x, y) = \frac{1}{\log(1 + (x - y)k_s(x))} = \frac{1}{(x - y)k_s(x)} + \frac{1}{2} + (x - y)\, h_s(x, y)$$

with $h_s$ holomorphic.

**Lemma 5.1** (Uniform model)**.** *Fix $N \geq 1$. After shrinking $s_0$ and $r$, there are holomorphic functions $e_1, \dots, e_{2N-2}$ on $\{|s| < s_0\}$ with the following property. Put $P_s(u) = \sum_i e_i(s)u^i$ and, for $s \neq 0$,*

$$\Psi_s(u) = A_1 \log(u - u_1) + A_2 \log(u - u_2) + P_s(u).$$

*Then:*

1. *$\nu(s) = A_1 + A_2$ extends holomorphically to $|s| < s_0$. For every $\delta \in (0, r)$ there is $s_\delta > 0$ such that $\Psi_s(u) - \nu(s) \log u$ extends to a holomorphic function of $(s, u) \in \{|s| < s_\delta\} \times \{\delta < |u| < r\}$. Here, for $|u| > \delta$, the branches are fixed by $\log(u - u_j) := \log u + \operatorname{Log}(1 - u_j/u)$ with one common branch of $\log u$. Individually $A_j \sim \pm 1/\sqrt{2s}$, so any other convention would introduce $2\pi\mathrm{i}A_j$-jumps that are not holomorphic in $s$.*
2. *The one-step error*

$$F_s(u) = A_1 \log \frac{f_s(u) - u_1}{u - u_1} + A_2 \log \frac{f_s(u) - u_2}{u - u_2} + P_s(f_s(u)) - P_s(u) - 1$$

*(principal logarithms) extends to a holomorphic function on $\{|s| < s_0\} \times \{|u| < r\}$, and $F_s = q_s^N\, \Gamma_s$ with $\Gamma$ holomorphic and $|\Gamma| \leq M$ there.*

*Proof. Removability.* Let $\Theta(s, x, y, u) = a_s(x, y)\ell(y) + a_s(y, x)\ell(x)$, where $\ell$ is holomorphic in $(s, x, y, u)$. By the expansion of $a_s$, its singular part is

$$\frac{\ell(y)/k_s(x) - \ell(x)/k_s(y)}{x - y}.$$

The numerator vanishes on $x = y$, so $\Theta$ extends holomorphically across the diagonal. Evaluate $\Theta$ at $(x, y) = (u_1, u_2)$. The result is a holomorphic function of $t$, even in $t$, hence a holomorphic function of $t^2 = D(s)$, hence of $s$. The same holds for the other pairing $\Theta'(s, x, y, u) = a_s(x, y)\ell(x) + a_s(y, x)\ell(y)$, whose singular part $[\ell(x)/k_s(x) - \ell(y)/k_s(y)]/(x - y)$ is again removable and symmetric. These arguments apply to the following choices of $\ell$:

- $\ell \equiv 1$ in $\Theta'$, which gives $\nu$;
- $\ell(x) = \log(1 - x/u)$ in $\Theta'$, which gives $\Psi_s - \nu \log u - P_s$ for $|u| > \delta$ once $|u_j| < \delta/2$;
- $\ell(x) = \log(1 + (u - x)k_s(u))$ in $\Theta$, which gives $F_s^{\log} := A_1 \log \frac{f - u_1}{u - u_1} + A_2 \log \frac{f - u_2}{u - u_2} - 1$, because $(f_s(u) - u_1)/(u - u_1) = 1 + (u - u_2)k_s(u)$.

This proves (1) and the holomorphy of $F^{\log}$.

*Divisibility.* For $s \neq 0$, $F_s^{\log}(u_j) = A_j \log f_s'(u_j) - 1 = 0$ (as in Lemma 4.2). Weierstrass division by $q_s$ has a remainder $r_0(s) + r_1(s)u$ that is holomorphic in $s$ and vanishes at two

distinct points for $s \neq 0$. So the remainder is 0, and $F^{\log} = q_s\varphi$ with $\varphi$ holomorphic. Likewise $D_i := (f_s(u))^i - u^i = q_s d_i$ (powers, not iterates). At $s = 0$, $D_i = u^i((1 + \frac{u}{2} + \cdots)^i - 1)$, so $d_i = \frac{i}{2}u^{i-1} + O(u^i)$.

*The linear system.* Divide $\varphi + \sum e_i d_i$ by the Weierstrass polynomial $q_s^{N-1}$ of degree $2N-2$. The $2N-2$ coefficients of the remainder depend holomorphically on $s$ and affinely on $e$, with matrix $M(s)$. At $s = 0$ the remainder is the Taylor polynomial modulo $u^{2N-2}$, so $M(0)$ is lower triangular with diagonal entries $i/2 \neq 0$. Hence $e(s) = -M(s)^{-1}\operatorname{rem}(\varphi)$ is holomorphic near 0. Then $F_s = F_s^{\log} + \sum e_i D_i = q_s(\varphi + \sum e_i d_i)$ is divisible by $q_s \cdot q_s^{N-1}$. The bound on $\Gamma$ follows after shrinking the polydisc. □

Only the one-step error enters the estimates below, and it is single-valued. The logarithms in $\Psi_s$ are continued along orbits exactly as in the proof of Theorem 4.4. Since $P_s$ is single-valued, replacing $\Psi_b$ by $\Psi_s$ changes no branch. The path of item (c) in Section 4 runs outside $D_\delta(0)$ and does not pass between $u_1$ and $u_2$. So along it the convention of Lemma 5.1(1) agrees with the continuation used there.

5.2. **Orbits for complex parameters.** Let $P_R = \{\operatorname{Re}(-2/u) > R\}$ and put $\zeta = -2/u$.

**Lemma 5.2** (Finite orbits)**.** *Let $K \subset P_R$ be compact, with $R$ large, and let $Z = \max_K |2/u|$. There is $c > 0$ with the following property. For every integer $J \geq 1$, every $\sigma \in \mathbb{C}$ with $|\sigma| \leq c/J^2$, every $u \in K$ and every $0 \leq k \leq J$, the iterate $u_k = f_\sigma^k(u)$ is defined and $\zeta_k = -2/u_k$ satisfies*

$$\operatorname{Re}\zeta_k \geq \operatorname{Re}\zeta_0 + k/2, \qquad |\zeta_k| \leq |\zeta_0| + 2k.$$

*In particular $|u_k| \leq 2/(R + k/2)$. For $\sigma = 0$ this holds for all $k$.*

*Proof.* For $|u| \leq r$ we have $-2/f_0(u) = -2/u + 1 + \epsilon(u)$ with $|\epsilon(u)| \leq C_0|u|$, and $|f_0(u)| \geq |u|/2$. Write $f_\sigma = f_0 + \sigma g$ with $|g| \leq G$. Suppose the claim holds up to $k < J$. Then $|u_k| \leq 2/R \leq r$ and $w := \sigma g(u_k)/f_0(u_k)$ satisfies

$$|w| \leq G|\sigma||\zeta_k| \leq Gc(Z+2) \leq \tfrac{1}{2}.$$

Hence

$$\zeta_{k+1} = \frac{-2}{f_0(u_k)}(1+w)^{-1} = \zeta_k + 1 + e_k, \qquad |e_k| \leq \frac{2C_0}{R} + 4G|\sigma||\zeta_k|^2 \leq \frac{2C_0}{R} + 4Gc(Z+2)^2.$$

This is at most $\frac{1}{2}$ for $R \geq 8C_0$ and $c$ small. So $\operatorname{Re}\zeta_{k+1} \geq \operatorname{Re}\zeta_k + \frac{1}{2}$ and $|\zeta_{k+1}| \leq |\zeta_k| + \frac{3}{2}$. □

**Lemma 5.3.** *Let $u \in K$, $u_k = f_0^k(u)$ and $\dot{u}_k = \partial_\sigma f_\sigma^k(u)|_{\sigma=0}$. Then*

$$|\dot{u}_k| \leq C(k+1), \qquad |\partial_\sigma F_\sigma(v)|_{\sigma=0}| \leq M'|v|^{2N-2}, \qquad |F_0'(v)| \leq M'|v|^{2N-1} \quad (|v| \leq r/2).$$

*Proof.* For the first bound, apply Cauchy's estimate on $|\sigma| = c/(k+1)^2$ together with Lemma 5.2 (with $J = k+1$). This gives $|\dot{u}_k| \leq \frac{2}{R+k/2} \cdot \frac{(k+1)^2}{c}$. For the other two, write $F_\sigma = q_\sigma^N \Gamma_\sigma$ with $q_0(v) = v^2$ and $\partial_\sigma q_\sigma$ bounded, and bound $\partial_\sigma\Gamma$ and $\Gamma_0'$ by Cauchy's estimates. □

5.3. **One-sided differentiability.** For $u$ in a compact $K \subset P_R$ and real $s \in [0, s_0]$ put

$$\mathcal{A}_s(u) = \Psi_s(u) + \sum_{k \geq 0} F_s(f_s^k u),$$

with a fixed branch of $\log u$ on a simply connected neighbourhood $U \subset P_R$ of $K$. For $s > 0$ this is the attracting Koenigs Abel coordinate up to an additive constant, by the argument of Theorem 4.4(1). For $s = 0$, $\mathcal{A}_0 = \Phi_{\text{att}} + \text{const}$, because $\Psi_0 - \alpha = \text{const} + O(u)$.

**Proposition 5.4.** *Let $N \geq 3$. Uniformly for $u \in K$, as $s \to 0^+$,*

$$\mathcal{A}_s(u) = \mathcal{A}_0(u) + s\,\dot{\mathcal{A}}(u) + O(s^{10/9}),$$

*where*

$$\dot{\mathcal{A}}(u) = \partial_s \Psi_s(u)\big|_{s=0} + \sum_{k\geq 0}\Big[\partial_\sigma F_\sigma(u_k)\big|_{\sigma=0} + F_0'(u_k)\,\dot{u}_k\Big], \qquad u_k = f_0^k(u). \tag{13}$$

*The series converges absolutely and uniformly on $K$, with terms $O(k^{2-2N})$. Hence $\dot{\mathcal{A}}$ is holomorphic on the interior of $K$.*

*Proof.* The $\Psi$-part is holomorphic in $s$ by Lemma 5.1(1), so it contributes $O(s^2)$. Fix $J \geq 1$ and split the sum at $J$.

*Head.* $H_J(\sigma) = \sum_{k<J} F_\sigma(f_\sigma^k u)$ is holomorphic on $|\sigma| < c/J^2$. There $|u_j(\sigma)| \leq C\sqrt{|\sigma|} \leq C\sqrt{c}/J$. Since $k < J$ gives $1/J \leq (R+1)/(R+k/2)$, Lemma 5.2 yields $|q_\sigma(u_k)| \leq C_3(R+k/2)^{-2}$. Therefore

$$|H_J(\sigma)| \leq MC_3^N \sum_k (R + k/2)^{-2N} =: M_1,$$

uniformly in $J$. Cauchy's estimate gives $|H_J''| \leq 8M_1J^4/c^2$ on $[0, c/(2J^2)]$, so

$$|H_J(s) - H_J(0) - sH_J'(0)| \leq 4M_1c^{-2}s^2J^4.$$

*Tails.* We may assume $K \subset P_{R+1}$. Since $Z_s \to -2/u$ uniformly on $K$ (Lemma 4.1), $K \subset \Omega_{R,s}$ for all small $s > 0$. If $r$ was shrunk in Lemma 5.1, increase $R$ so that $\Omega_{R,s} \subset D_{r/2}$, and reach a general $K$ by $f^m$ as below. For real $s \in (0, s_0]$ the proof of Lemma 4.3 gives $|q_s(f_s^k u)| \leq C_1(R+\frac{3}{4}k)^{-2}$. For $s = 0$, Lemma 5.2 gives the analogous bound $|u_k|^2 \leq 4(R+k/2)^{-2}$. Hence $\big|\sum_{k\geq J} F_s(f_s^k u)\big| \leq C_4 J^{1-2N}$ for all $s \in [0, s_0]$. By Lemma 5.3, the terms of (13) are $O((R + k/2)^{2-2N})$. So $H_J'(0)$ differs from the series by at most $C_5J^{3-2N}$.

*Balance.* For $s \leq c/(2J^2)$,

$$\big|\mathcal{A}_s - \mathcal{A}_0 - s\dot{\mathcal{A}}\big| \leq Cs^2 + 4M_1c^{-2}s^2J^4 + 2C_4J^{1-2N} + C_5sJ^{3-2N}.$$

Take $J = \lceil s^{-2/9} \rceil$, which is admissible for small $s$. With $N \geq 3$ every term is $O(s^{10/9})$. □

**Remark 5.5.** Taking $J = s^{-2/(2N+3)}$ instead gives the error $O(s^{2-8/(2N+3)})$, so $O(s^{2-\epsilon})$ for any $\epsilon > 0$ with $N$ large. The proof needs $N \geq 3$ only because of the crude bound on the head ($N = 2$ would give $s^{6/7}$). The series converges for $N \geq 2$.

The derivative does not depend on $N$. The condition $F_0 = O(u^{2N})$ forces $\Psi_0$ to agree with the formal Abel function $\alpha$ up to a constant and $O(u^{2N-1})$, so the $e_i(0)$ are the coefficients of $\alpha$ and do not depend on $N$. Hence $\mathcal{A}_s^{(N)} - \mathcal{A}_s^{(N')}$, which equals the difference of the two polynomials evaluated at $u_1(s) = O(\sqrt{s})$, is a constant plus $O(s^{3/2})$. In any case additive constants cancel in $\dot{\mathcal{A}}(v) - \dot{\mathcal{A}}(u^*)$. Numerically, $N = 3, 4, 6, 8$ give the same $\kappa^{(1)}$ to 16–20 digits.

5.4. **The derivative of $\tau_n$.** Compact subsets of the basin are handled by $\mathcal{A}_s = \mathcal{A}_s \circ f_s^m - m$, where $m$ is chosen so that $f_0^m(K) \Subset P_R$. The iterate $f_s^m$ is jointly holomorphic, so the expansion of Proposition 5.4 persists, with $\dot{\mathcal{A}} = \dot{\mathcal{A}} \circ f_0^m + \mathcal{A}_0' \circ f_0^m \cdot \partial_s f_s^m$.

For the repelling side, apply everything to the reflected family $G_s(v) = -f_s^{-1}(-v) = -(\log(1-v) + s)/(1-s)$, exactly as in Theorem 4.4(2). This family is again of the form (12), with fixed points $-u_2, -u_1$. Let $\mathcal{S}_s(u) = -\mathcal{A}_s^G(-u)$. By Remark 2.3 and (7), $\tau_n$ does not depend on the additive normalisation of the repelling coordinate. We use $\mathcal{S}_s^\sharp = \mathcal{S}_s - \mathcal{S}_s(p') + \Phi_{\mathrm{rep}}(p')$. This differs from the real normalisation by the constant called $s(b)$ in Theorem 4.4(2), which we denote $s_*(b)$ here to avoid a clash with the parameter $s$, and it is right-differentiable with derivative $\dot{\mathcal{S}}^\sharp$. On the attracting side, $R_b^{-1}(\mathrm{e}(1+u)) = \mathcal{A}_s(u) - \mathcal{A}_s(u^*)$, with the branch fixed in item (c) of Section 4.

**Theorem 5.6.** *Let $\ell = [0,1] + \mathrm{i}Y_*$ be the gate line of Section 4, and put $v(z) = \Phi_{\mathrm{rep}}^{-1}(z)$ and $\tilde{T}_0 = h^{-1} - a$. Define*

$$\delta T(z) = \dot{\mathcal{A}}(v(z)) - \dot{\mathcal{A}}(u^*) - \tilde{T}_0'(z)\,\dot{\mathcal{S}}^\sharp(v(z)), \qquad \delta t_m = \int_\ell \delta T(z)\,\mathrm{e}^{-2\pi\mathrm{i}mz}\,dz.$$

*Then for every $n \geq 1$, as $s \to 0^+$,*

$$\tau_n = B_n \mathrm{e}^{2\pi\mathrm{i}na} + s\,\dot{\tau}_n + O(s^{10/9}), \qquad \dot{\tau}_n = \mathrm{e}^{2\pi\mathrm{i}na}\big(\delta t_n - 2\pi\mathrm{i}n\,B_n\,\delta t_0\big).$$

*Moreover $p = -\log\lambda_1 \log\lambda_2 = 2s + \frac{10}{9}s^2 + O(s^3)$ is holomorphic in $s$. Consequently, if $B_n \neq 0$,*

$$\log\frac{\tau_n}{B_n \mathrm{e}^{2\pi\mathrm{i}na}} = \kappa^{(n)} p + O(p^{10/9}), \qquad \kappa^{(n)} = \frac{\delta t_n}{2B_n} - \pi\mathrm{i}n\,\delta t_0.$$

*Proof.* By Proposition 5.4 and Cauchy's estimates, both $\mathcal{A}_s$ and $\mathcal{S}_s^\sharp$ have expansions $X_0 + s\dot{X} + O(s^{10/9})$ in the $C^1$ topology on a compact neighbourhood of $v(\ell)$. By Rouché, the inverse $v_s = (\mathcal{S}_s^\sharp)^{-1}$ exists near $v(\ell)$ and satisfies

$$v_s = v - s\,\dot{\mathcal{S}}^\sharp(v)/\Phi_{\mathrm{rep}}'(v) + O(s^{10/9}).$$

Hence $\tilde{T}_s := R_b^{-1} \circ \mathrm{e}(1 + v_s)$ satisfies $\tilde{T}_s = \tilde{T}_0 + s\,\delta T + O(s^{10/9})$ uniformly on $\ell$, where we used $\mathcal{A}_0'(v)/\Phi_{\mathrm{rep}}'(v) = \tilde{T}_0'$. By items (a)–(c) of Section 4, the Fourier coefficients $\tilde{t}_m(s)$ of $\tilde{T}_s - \mathrm{id}$ on $\ell$ satisfy $\tau_n = \tilde{t}_n \mathrm{e}^{-2\pi\mathrm{i}n\tilde{t}_0}$. Differentiate, using $\tilde{t}_n(0) = B_n$ and $\tilde{t}_0(0) = -a$. The expansion of $p$ follows from the removability argument of Lemma 5.1 applied to $\log\lambda_1 \log\lambda_2$, together with a direct Taylor expansion. Finally, $\log(\tau_n/\tau_n(0)) = s\,\dot{\tau}_n/\tau_n(0) + O(s^{10/9})$ and $s = p/2 + O(p^2)$. □

5.5. **Higher orders.** The argument of Proposition 5.4 extends to every order, at the price of a larger $N$.

**Proposition 5.7.** *Let $m \geq 1$ and $N \geq m^2 + m + 1$. For $u \in K$ and $j \geq 0$ put*

$$\mathcal{A}^{(j)}(u) = \partial_s^j \Psi_s(u)\big|_{s=0} + \sum_{k \geq 0} \partial_\sigma^j \big[F_\sigma(f_\sigma^k u)\big]_{\sigma=0}. \tag{14}$$

*For $j \leq m$ the series converges absolutely and uniformly on $K$, and its terms are $O(k^{2j-2N})$. Uniformly for $u \in K$, as $s \to 0^+$,*

$$\mathcal{A}_s(u) = \sum_{j=0}^{m} \frac{s^j}{j!}\mathcal{A}^{(j)}(u) + O(s^{m+\gamma}), \qquad \gamma = 1 - \frac{2(m+1)^2}{2N + 2m + 1} > 0.$$

*Proof. Term bounds.* Fix $k \geq 0$ and apply Lemma 5.2 with $J = k + 1$. For $|\sigma| \leq c/(k+1)^2$ the head estimate in the proof of Proposition 5.4 gives $|q_\sigma(f_\sigma^k u)| \leq C_3(R + k/2)^{-2}$, hence $|F_\sigma(f_\sigma^k u)| \leq M C_3^N (R + k/2)^{-2N}$. By Cauchy's estimate on this disc,

$$\big|\partial_\sigma^j [F_\sigma(f_\sigma^k u)]_{\sigma=0}\big| \leq j!\, M C_3^N\, c^{-j} (k+1)^{2j} (R + k/2)^{-2N} = O(k^{2j-2N}).$$

This is summable for $j \leq m < N - \frac{1}{2}$. For $m = 1$ this recovers the bounds of Lemma 5.3, without computing $\dot{u}_k$.

*Head.* $H_J(\sigma) = \sum_{k<J} F_\sigma(f_\sigma^k u)$ is holomorphic and bounded by $M_1$ on $|\sigma| < \rho := c/J^2$. For real $0 \leq s \leq \rho/2$, the Taylor remainder of order $m$ satisfies

$$\Big|H_J(s) - \sum_{j \leq m} \frac{s^j}{j!} H_J^{(j)}(0)\Big| \leq M_1 \sum_{j>m} (s/\rho)^j \leq 2M_1 c^{-m-1}\, s^{m+1} J^{2m+2}.$$

*Tails.* As before, $\big|\sum_{k \geq J} F_s(f_s^k u)\big| \leq C_4 J^{1-2N}$ for $s \in [0, s_0]$. By the term bounds, the tails of the $j$-th derivative series from $J$ on are $O(J^{2j+1-2N})$. Since $sJ^2 \leq c/2$, the tails contribute $\sum_{j \leq m} s^j J^{2j+1-2N} = O(J^{1-2N})$.

*Balance.* Write $\mathcal{T}_J^{(j)}$ for the tail from $k = J$ on of the $j$-th derivative series. The identity

$$\mathcal{A}_s - \sum_{j\le m} \frac{s^j}{j!}\mathcal{A}^{(j)} = \Big[\Psi_s - \sum_{j\le m} \frac{s^j}{j!}\partial_s^j \Psi\Big] + \Big[H_J(s) - \sum_{j\le m} \frac{s^j}{j!} H_J^{(j)}(0)\Big] + \sum_{k\ge J} F_s(f_s^k u) - \sum_{j\le m} \frac{s^j}{j!}\mathcal{T}_J^{(j)}$$

holds exactly. The $\Psi$-part is holomorphic in $s$. Altogether the error is $O(s^{m+1}J^{2m+2} + J^{1-2N})$. Take $J = \lceil s^{-\beta}\rceil$ with $\beta = (m+1)/(2N+2m+1) < \frac{1}{2}$, which is admissible. Both terms are then $O(s^{m+1-2(m+1)\beta}) = O(s^{m+\gamma})$, and $\gamma > 0$ exactly when $N \ge m^2 + m + 1$. □

For $m = 1$ this is Proposition 5.4; the error there is $O(s^{10/9})$ for $N = 3$.

**Theorem 5.8.** *For every $n \ge 1$ with $B_n \ne 0$, the function $\log(\tau_n/\tau_n(0))$ has an asymptotic expansion to all orders in $p$ as $b \to \eta^-$:*

$$\log \frac{\tau_n}{B_n e^{2\pi i n a}} \sim \sum_{j\ge 1} \kappa_j^{(n)} p^j, \qquad \kappa_1^{(n)} = \kappa^{(n)}.$$

*For each order $m$, the remainder after the $p^m$ term is $O(p^{m+1})$. Each coefficient $\kappa_j^{(n)}$ is a polynomial in the following data:*

- *$B_n^{-1}$ and $a$;*
- *the Taylor coefficients of $s(p)$;*
- *finitely many Fourier coefficients on $\ell$ of explicit polynomial expressions in $\mathcal{A}^{(i)} \circ v$, $\mathcal{S}^{\sharp(i)} \circ v$ $(i \le j)$, their $u$-derivatives, and $1/\Phi'_{\mathrm{rep}} \circ v$.*

*Here $\mathcal{A}^{(i)}$ and $\mathcal{S}^{\sharp(i)}$ are given by the convergent series (14) over parabolic orbits of $f_0$ and $G_0$. The coefficients do not depend on $N$.*

*Proof.* Fix $m$ and $N \ge m^2 + m + 1$. By Proposition 5.7 and the reduction $\mathcal{A}_s = \mathcal{A}_s \circ f_s^M - M$ (with $f_s^M$ jointly holomorphic), $\mathcal{A}_s$ and $\mathcal{S}_s^\sharp$ have expansions $\sum_{j\le m} s^j X_j/j! + O(s^{m+\gamma})$. These hold uniformly on a compact neighbourhood of $v(\ell) \cup \{u^*\}$ and are holomorphic in $u$, so by Cauchy's estimates they also hold in the $C^r$ topology on a smaller neighbourhood, for every $r$.

*Inverse.* Formal inversion of $\sum_j s^j \mathcal{S}_j^\sharp/j!$ gives a polynomial approximant $\tilde v_s = v + \sum_{1\le j\le m} s^j V_j$ with $\mathcal{S}_s^\sharp(\tilde v_s) = z + O(s^{m+\gamma})$. Since $\mathcal{S}_s^\sharp \to \Phi_{\mathrm{rep}}$ in $C^1$ and $\Phi'_{\mathrm{rep}} \ne 0$ near $v(\ell)$, the maps $\mathcal{S}_s^\sharp$ are uniformly bi-Lipschitz there, and the true inverse satisfies $v_s = \tilde v_s + O(s^{m+\gamma})$.

*Composition.* Taylor-expand $\mathcal{A}_s$ about $v$, using the bounded $u$-derivatives. This gives $\tilde T_s = \sum_{j\le m} s^j T_j/j! + O(s^{m+\gamma})$ uniformly on $\ell$. Hence the Fourier coefficients $\tilde t_i(s)$ and $\tau_n = \tilde t_n e^{-2\pi i n \tilde t_0}$ have expansions to order $m$. Since $\tau_n(0) \ne 0$, so does $\log(\tau_n/\tau_n(0))$.

*Change of parameter.* Finally, $p$ is holomorphic in $s$ with $p'(0) = 2$, so $s$ is a holomorphic function of $p$.

*Independence of $N$.* An asymptotic expansion of a fixed function is unique, so the coefficients do not depend on the choice of $N$.

*Remainder.* Applying the argument at order $m+1$ and absorbing the $p^{m+1}$ term gives the remainder $O(p^{m+1})$. □

**Remark 5.9.** In particular, Theorem 5.8 with $m = 2$ improves the remainder in Theorem 5.6 from $O(p^{10/9})$ to $O(p^2)$. Theorem 5.8 gives explicit asymptotic coefficients in the normalisation of $\tau_n$, but says nothing about summability. The $\frac{1}{2}$-summability of [6], in their normalisation, suggests Gevrey-2 growth $|\kappa_j^{(n)}| \lesssim C^j (j!)^2$ in $p \propto \epsilon$ (Gevrey-1 in $\sqrt{p}$). This matches the scale $\Lambda = \exp(-C/\sqrt{p})$ of the beyond-all-orders terms. The constants in our proof grow faster, because $N$ must grow like $m^2$.

**Remark 5.10** (Relation to the literature)**.** Christopher and Rousseau show that the unfolded modulus, in a particular normalisation, is $\frac{1}{2}$-summable in the canonical parameter $\epsilon = (A_2 - A_1)^{-2}$, with the Glutsyuk direction as the singular direction [6, Cor. 4.9]. One checks the exact identity $p = 4\epsilon/(1-\nu^2\epsilon)$. Their result gives the existence of an asymptotic expansion. Mardešić–Roussarie–Rousseau record that at $\epsilon = 0$ only continuity was known [21, §11]. Theorems 5.6 and 5.8 give the coefficients, in the normalisation of $\tau_n$, as explicit convergent series over parabolic orbits. We know of no closed form. An integer-relation search on $\kappa^{(1)}$ against $\pi, \pi^2, \log 2, \gamma$ returns only spurious relations.

5.6. **Numerics.** The series (13) is evaluated with $N = 3, \dots, 8$ on 24 points of the line $\operatorname{Im} z = 1$ or 1.5, with orbits of length 300–3000. The $s$-derivative of each (finite, holomorphic) truncation is taken by a central difference at $s = \pm 10^{-20}$ to $\pm 10^{-30}$ in 60–90 digit arithmetic (`mrr_derivative.py`). The results are stable to all printed digits:

$$\kappa^{(1)} = -0.010199006345897402441 - 0.013250956182506712250\,\mathrm{i},$$
$$\kappa^{(2)} = -0.05105655620132933 - 0.06231578053849410\,\mathrm{i},$$
$$\kappa^{(3)} = -0.140044348129 - 0.169224463926\,\mathrm{i}.$$

These agree with the fits (18) from the side $b < \eta$ to all of their eleven digits. At $s = 0$ the same computation reproduces $\tau_n(0) = B_n \mathrm{e}^{2\pi \mathrm{i} n a}$ to the precision of the stored $B_n$. The second coefficients $\kappa_2^{(n)}$ of Theorem 5.8 are computed by a five-point stencil ($s = \pm 10^{-12}, \pm 2 \cdot 10^{-12}$, 90 digits) applied to the same finite truncations (`mrr_second.py`), with $N = 7$ ($\operatorname{Im} z = 1$, orbits of length 2000) and $N = 10$ ($\operatorname{Im} z = 1.5$, length 3000). The series converge for $N \geq 3$; the condition $N \geq m^2 + m + 1 = 7$ is needed only for the proof. The two computations agree to 18, 16 and 12 digits respectively:

$$\kappa_2^{(1)} = -7.88992074991511334 \cdot 10^{-5} + 3.6554727325799459 \cdot 10^{-4}\,\mathrm{i},$$
$$\kappa_2^{(2)} = -1.895481905818775 \cdot 10^{-4} + 1.106655243691997 \cdot 10^{-4}\,\mathrm{i},$$
$$\kappa_2^{(3)} = 4.07137984831 \cdot 10^{-4} - 8.49506527544 \cdot 10^{-3}\,\mathrm{i}.$$

The fits from the side $b < \eta$ (`deficit_modes.py`) give $-7.88991 \cdot 10^{-5} + 3.65547 \cdot 10^{-4}$i, $-1.89548 \cdot 10^{-4} + 1.10666 \cdot 10^{-4}$i and $4.07137 \cdot 10^{-4} - 8.49507 \cdot 10^{-3}$i, in agreement to within one unit in the sixth digit.

## 6. The sewing solution: decay estimate

Throughout, $b < \eta$ is real and close to $\eta$, $\varepsilon = |\log \lambda|$, and $S$ has the real normalisation of Section 3, so $T(z) - z = \sum t_n \mathrm{e}^{2\pi \mathrm{i} n z}$ with $\operatorname{Im} t_0 = h/2$ and $t_{-n} = \overline{t_n}$ (Lemma 3.1).

**Lemma 6.1** (The band)**.** *Let $Z_b(u) = A_1 \log \frac{u-u_1}{u-u_2}$ with the branch $\operatorname{Im} Z_b \in (0, h)$ on $\mathbb{C} \setminus ((-\infty, u_1] \cup [u_2, \infty))$. This branch has $\operatorname{Im} Z_b = h/2$ on $(u_1, u_2)$ and $\operatorname{Im} Z_b < h/2$ on $\operatorname{Im} u > 0$. For $Y > 0$ let*

$$\mathcal{B}_b(Y) = \{u : \ Y < \operatorname{Im} Z_b(u) < h - Y\},$$

*a lens bounded by two circular arcs through $u_1, u_2$. There is an absolute $Y_B$ such that for $Y \geq Y_B$ and all $b$ near $\eta$:*

(i) *every $u \in \mathcal{B}_b(Y)$ has its forward orbit in $D_{r'}$ tending to $u_1$ and its principal backward orbit in $D_{r'}$ tending to $u_2$, with $\sum_{k \in \mathbb{Z}} |F_b(f_b^k u)| \leq C/Y + C\varepsilon$;*

(ii) *the real-normalised times* $\mathcal{R}_b = \Psi_b + \sum_{k\ge 0} F_b \circ f_b^k + c_R$ *and* $\mathcal{S}_b = \Psi_b - \sum_{k\ge 1} F_b \circ f_b^{-k} + c_S$ *are holomorphic and injective on* $\mathcal{B}_b(Y)$. *They are branches of* $R_b^{-1}$ *and* $S_b^{-1}$, *with* $\mathcal{S}_b$ *real on* $(u_1, u_2)$ *and* $\mathcal{R}_b$ *as in item (c) above, and*

$$\bigl|\operatorname{Im}\mathcal{S}_b(u) - (\operatorname{Im} Z_b(u) - h/2)\bigr| \le \tfrac{\pi}{3} + C/Y + C\varepsilon;$$

(iii) $S_b$ *maps the strip* $\Sigma = \{|\operatorname{Im} w'| < h/2 - Y - 2\}$ *into* $\mathcal{B}_b(Y)$ *with* $\mathcal{S}_b \circ S_b = \mathrm{id}$. *Consequently* $T = \mathcal{R}_b \circ S_b$ *is holomorphic on* $\Sigma$, *and*

$$T(w') - w' = (c_R - c_S) + \sum_{k\in\mathbb{Z}} F_b(f_b^k(S_b(w'))).$$

*Proof.* (i) Put $q = \mathrm{e}^{Z/A_1} = (u-u_1)/(u-u_2)$ and $\zeta = Z/A_1 = -\varepsilon Z$, so $\operatorname{Im}\zeta \in (-2\pi + \varepsilon Y, -\varepsilon Y)$ on the band. On $-2\pi \le \operatorname{Im}\zeta \le 0$ one has $|\mathrm{e}^\zeta - 1| \ge c\min(1, \delta(\zeta))\max(1, |\mathrm{e}^\zeta|)$ with $\delta(\zeta) = \min(|\zeta|, |\zeta + 2\pi\mathrm{i}|)$ and an absolute $c > 0$; the only zeros are 0 and $-2\pi\mathrm{i}$. On the band $\delta(\zeta) = \varepsilon\min(|Z|, |Z - \mathrm{i}h|) > \varepsilon Y$. Since $u - u_2 = (u_2 - u_1)/(q-1)$ and $|u_2 - u_1|/\varepsilon \to 2$ (Lemma 4.1), this gives

(15)

$$|u-u_2| \le C(\varepsilon + Y^{-1}), \qquad |u-u_1||u-u_2| = \frac{|q|\,|u_2-u_1|^2}{|q-1|^2} \le \begin{cases} C\,\min(|Z|, |Z-\mathrm{i}h|)^{-2}, & \delta(\zeta) \le 1, \\ C\varepsilon^2 \mathrm{e}^{-\varepsilon|\operatorname{Re} Z|}, & \delta(\zeta) > 1, \end{cases}$$

using $|q|/\max(1,|q|)^2 = \mathrm{e}^{-|\operatorname{Re}\zeta|}$. So $\mathcal{B}_b(Y) \subset D_{C/Y + |u_1| + |u_2|} \subset D_{r'/2}$ for $Y$ large. By the proof of Lemma 4.3, $Z \circ f - Z = 1 + O(|u| + \varepsilon)$, so $\operatorname{Re} Z$ increases by at least $\frac{3}{4}$ per step while the orbit is in $D_{r'/2}$.

The imaginary part is controlled through $\Psi_b = Z_b + (A_1 + A_2)\log(u - u_2)$, with step error $|F_b| \le M|u-u_1||u-u_2|$ (Lemma 4.2). Summing (15) along $\operatorname{Re} Z_k \ge \operatorname{Re} Z_0 + \frac{3}{4}k$ bounds $\sum_k |F_b(u_k)|$ by $C'/Y + C'\varepsilon$. Points with $\delta(\zeta_k) \le 1$ contribute $\sum_k (x_k^2 + Y^2)^{-1} \le C/Y$, and the others $\varepsilon^2 \sum \mathrm{e}^{-\varepsilon|x_k|} \le C\varepsilon$. Now $\operatorname{Im} Z - \operatorname{Im}\Psi = -(A_1 + A_2)\arg(u - u_2)$ with $\arg(u-u_2) \in (0, 2\pi)$ on the lens. So $\operatorname{Im} Z$ drifts by at most $C'/Y + C'\varepsilon + \frac{2\pi}{3} + o(1)$ along the orbit, which is $< Y/2$ for $Y \ge Y_B \ge 8$. By induction the orbit stays in $\mathcal{B}_b(Y/2) \subset D_{r'/2}$, and $\operatorname{Re} Z_k \to +\infty$, i.e. $u_k \to u_1$. The continued values of $Z$ along the orbit have $\operatorname{Im} \in (Y/2, h - Y/2) \subset (0, h)$, so they agree with the fixed slit-plane branch. The backward orbit is the forward orbit of the reflected inverse family $G_b(v) = -f_b^{-1}(-v)$, to which the same argument applies (its prepared form is checked in the proof of Theorem 4.4(2)). Its period is $h_2 = h + 2\pi/3 + o(1)$, so $\mathcal{B}_b(Y)$ is contained in the corresponding band of $G_b$ at height $Y(1 + O(\varepsilon)) - 1$. This is harmless for $Y \ge Y_B$.

(ii) As in the proof of Theorem 4.4(1), $\Psi_b(f^n u) - n \to A_1 \log\sigma_b(u) + \text{const}$ along forward orbits. So $\mathcal{R}_b$ is a branch of the attracting Abel coordinate, and $\mathcal{S}_b$ likewise of the repelling one. Choose the constants $c_R$ (by item (c) above) and $c_S$ so that $\mathcal{S}_b$ is real on the segment. Both $F_b$ and $f_b$ are real, so the sums are real there. On the segment $\operatorname{Im} Z = h/2$ and $\arg(u - u_2) = \pi$, which fixes $\operatorname{Im} c_S$. The displayed estimate then follows from $\operatorname{Im}\Psi = \operatorname{Im} Z + (A_1 + A_2)\arg(u - u_2)$, $A_1 + A_2 \to \frac{1}{3}$ and (i). Injectivity: if $\mathcal{S}_b(u) = \mathcal{S}_b(u')$ then the principal backward orbits satisfy $\tau_b(u_{-n}) = \tau_b(u'_{-n})$ near $u_2$, where $\tau_b$ is injective, so $u_{-n} = u'_{-n}$. Moreover $f_b$ is injective on $D_{r'}$ (its non-injectivity has period $2\pi\mathrm{i}/\log b$ in $w$, far larger than the disc), so $u = u'$. The same argument with $\sigma_b$ and forward orbits proves injectivity of $\mathcal{R}_b$.

(iii) Fix $x \in \mathbb{R}$. $S_b(x) \in (u_1, u_2) \subset \mathcal{B}_b(Y)$ and $\mathcal{S}_b(S_b(x)) = x$, since both sides are the real-normalised repelling time. Move $y'$ from 0 towards $\pm(h/2 - Y - 2)$. As long as $S_b(x + \mathrm{i}y')$ stays in the lens, continuation gives $\mathcal{S}_b(S_b(x + \mathrm{i}y')) = x + \mathrm{i}y'$. It cannot leave through the corners $u_1, u_2$, where $|\operatorname{Re}\mathcal{S}_b| \to \infty$. It cannot leave through the arcs either: there $\operatorname{Im} Z \in \{Y, h - Y\}$, so by (ii) $|\operatorname{Im}\mathcal{S}_b| \ge h/2 - Y - \pi/3 - o(1) > |y'|$. Hence $S_b(\Sigma) \subset \mathcal{B}_b(Y)$, $T = \mathcal{R}_b \circ S_b$ on $\Sigma$, and $T(w') - w' = \mathcal{R}_b(u) - \mathcal{S}_b(u)$ with $u = S_b(w')$ is the displayed bi-infinite sum. □

**Lemma 6.2** (Uniform mode bound)**.** *Put $Y_0 := Y_B + 3$. There is $M$, which can be taken $\le 1$, and $b_1 < \eta$ such that for $b \in (b_1, \eta)$ and all $n \neq 0$,*

$$|t_n| \le M\mathrm{e}^{-2\pi|n|d}, \qquad d := \tfrac{h}{2} - Y_0 - 1.$$

*Proof.* Take $Y = Y_B + 1$ in Lemma 6.1(iii), so that the estimate (ii) is available on the closure of the lens. $\Sigma$ has half-width $h/2 - Y_B - 3 = d + 1$, so the lines $\operatorname{Im} w' = \pm d$ lie in $\Sigma$. On them, by (iii) and (i), $T - \mathrm{id} - t_0 = \sum_k F_b(f_b^k S_b)$ − mean has modulus at most $2(C/Y_B + C\varepsilon)$. Let $M$ be the supremum of this over $b \in (b_1, \eta)$. Cauchy's estimate on these lines gives the bound for $n > 0$ and $n < 0$. This does not use Theorem 4.4. □

For $H > 0$ let $\mathfrak{A}_H$ be the space of 1-periodic $g = \sum g_n \mathrm{e}^{2\pi\mathrm{i}nz}$ with $\|g\|_H = \sum |g_n| \mathrm{e}^{2\pi|n|H} < \infty$. It is a Banach algebra (the weight is submultiplicative), its elements are holomorphic and bounded by $\|g\|_H$ on $|\operatorname{Im} z| \le H$, and the projections $\Pi_{<0}, \Pi_{\ge 0}$ onto negative and non-negative modes have norm 1. Also $\|\mathrm{e}^g\|_H \le \mathrm{e}^{\|g\|_H}$ and $\|\mathrm{e}^{g_1} - \mathrm{e}^{g_2}\|_H \le \|g_1 - g_2\|_H \, \mathrm{e}^{\max \|g_j\|_H}$.

**Theorem 6.3** (W2)**.** *Let $M, d$ be as in Lemma 6.2, $\rho = \frac14$, and choose $D \ge 1$ with $4\pi M\mathrm{e}^{-2\pi D} \le \rho(1 - \mathrm{e}^{-2\pi D})^2$. Put $H = d - D - \rho$ and $\tau = T - \mathrm{id} - t_0$. Assume $H > 1$, which holds for $b$ near $\eta$ since $d \sim \pi/\varepsilon$, and that $b$ is so close to $\eta$ that $C_1 \Lambda \mathrm{e}^{4\pi(Y_0+D+1)} < \frac12$, where $C_1 = 4M + 2\rho$. Then:*

1. *The map $\mathcal{T}(Q) = -\Pi_{<0}[\tau \circ (\mathrm{id} + Q)]$ is a contraction, with constant $\le \frac12$, of the ball $\{Q \in \Pi_{<0}\mathfrak{A}_H : \|Q\|_H \le \rho\}$ into itself. Its fixed point $Q^*$ and $P' := \Pi_{\ge 0}[\tau \circ (\mathrm{id} + Q^*)]$ satisfy*
$$T \circ (\mathrm{id} + Q^*) = \mathrm{id} + t_0 + P' \qquad \text{on } |\operatorname{Im} z| < H.$$
2. *$\varphi_- = \mathrm{id} + Q^*$ is univalent on $\{\operatorname{Im} z < H - 1\}$ and $\varphi_+ = \mathrm{id} + t_0 + P'$ on $\{\operatorname{Im} z > -(H-1)\}$. The function equal to $R \circ (\varphi_+ - \mathrm{i}h)$ above and $S \circ \varphi_-$ below is a solution of (1); it is the sewing solution $K_b^W$ of Proposition 3.4.*
3. *The equation $\varphi_+(z_0) = \mathrm{i}h$ has a unique solution with $|z_0 - (\mathrm{i}h - t_0)| < 1$, and $|z_0 - (\mathrm{i}h - t_0)| \le C_1 \Lambda \mathrm{e}^{4\pi(Y_0+D+1)}$. After the translation $z \mapsto z + z_0$ one has $K_b^W(0) = 1$ and $P = R^{-1} \circ K_b^W - \mathrm{id}$ with $P(0) = 0$.*
4. *For every $n \ge 1$ the Fourier coefficients of this $P$ satisfy*
$$\left| \frac{c_n}{\Lambda^n} - \tau_n(b) \right| \le C_n \Lambda, \qquad C_n = C\, M(M+1)\, \mathrm{e}^{2\pi(n+2)(Y_0+D+2)},$$
*with an absolute constant $C$.*

*Proof.* (1) Write $\tau \circ (\mathrm{id} + Q) = \sum_{m \neq 0} t_m \mathrm{e}^{2\pi\mathrm{i}mz} \mathrm{e}^{2\pi\mathrm{i}mQ}$. Then

$$\|\tau \circ (\mathrm{id} + Q)\|_H \le \sum_{m \neq 0} |t_m| \mathrm{e}^{2\pi|m|(H+\rho)} \le 2M \sum_{m \ge 1} \mathrm{e}^{-2\pi m D} \le \rho$$

by Lemma 6.2 and the choice of $D$. Similarly

$$\|\tau \circ (\mathrm{id} + Q_1) - \tau \circ (\mathrm{id} + Q_2)\|_H \le \sum_{m \neq 0} |t_m| \mathrm{e}^{2\pi|m|(H+\rho)} 2\pi|m| \, \|Q_1 - Q_2\|_H \le \frac{4\pi M \mathrm{e}^{-2\pi D}}{(1 - \mathrm{e}^{-2\pi D})^2} \|Q_1 - Q_2\|_H,$$

which is at most $\frac12 \|Q_1 - Q_2\|_H$. Since $|\operatorname{Im}(z + Q)| \le H + \rho < d$ on the strip, the Fourier series of $T$ converges at $z + Q(z)$ and equals $T$ there (Lemma 6.1). The identity follows by splitting $\tau \circ (\mathrm{id} + Q^*) = P' + \Pi_{<0}[\ldots] = P' - Q^*$.

(2) The series of $Q^*$ has only negative modes and converges on $\operatorname{Im} z \le H$, so it converges on the whole lower half-plane $\operatorname{Im} z < H$. There $|Q^{*\prime}| \le 2\pi \mathrm{e}^{-2\pi}\rho/(1 - \mathrm{e}^{-2\pi})^2 < 1$ on $\operatorname{Im} z \le H - 1$, so $\operatorname{Re} \varphi_-' > 0$ and $\varphi_-$ is univalent (Noshiro–Warschawski). The same holds for $\varphi_+$ above. On the strip, $R \circ (\varphi_+ - \mathrm{i}h) = R \circ (T \circ \varphi_- - \mathrm{i}h) = R \circ T \circ \varphi_- = S \circ \varphi_-$, because $\mathrm{i}h$ is a period of $R$ and $R \circ T = S$. The functional equation holds because $\varphi_\pm$ commute with $z \mapsto z + 1$. The resulting

function realises the sewing of Proposition 3.4: the images of $\varphi_\pm$ are the two half-cylinders and the seam in the $z$-plane is $\Gamma = \varphi_-^{-1}(\mathbb{R})$, a curve in $|\operatorname{Im} z| \le \rho$. Since $\varphi_\pm - \mathrm{id}$ is bounded, the argument principle shows that the restrictions of $\varphi_\pm$ to the two sides of $\Gamma$ map onto the two half-cylinders. By uniqueness of the uniformisation it is $K_b^W$ up to translation. (The identity $R \circ T = S$ is used on the band, where orbits stay in $D_{r'}$ and $\mathcal{R}_b$ inverts $R$; Lemma 6.1.)

(3) For $\operatorname{Im} z \ge h/2 - 1$ we have $|P'(z) - p_0'| \le \sum_{n\ge1} |p_n'| e^{-2\pi n(h/2-1)}$ and $|p_n'| \le \rho\, e^{-2\pi n H}$. Moreover $p_0' = \sum_{m\ge1} t_m [e^{2\pi i m Q^*}]_{-m}$, and $|[e^{2\pi i m Q^*}]_{-m}| \le e^{2\pi m\rho} e^{-2\pi m H}$, so $|p_0'| \le 2M e^{-2\pi(d+H-\rho)} \le 2M\Lambda e^{2\pi(2Y_0+D+2+2\rho)}$, and $\sum_{n\ge1} |p_n'| e^{-2\pi n(h/2-1)} \le 2\rho\Lambda e^{2\pi(Y_0+D+2+\rho)}$. Both are bounded by

$$\tfrac{1}{2} C_1 \Lambda e^{4\pi(Y_0+D+1)}.$$

Since $\operatorname{Im}(ih - t_0) = h/2$ and $|P''| \le \rho$ there, Rouché's theorem on the disc of radius 1 gives the unique root and the bound, using the smallness hypothesis. Then $K_b^W(z_0) = R(\varphi_+(z_0) - ih) = R(0) = 1$, and after translation $P(z) = \varphi_+(z + z_0) - ih - z$, so $P(0) = 0$.

(4) For $n \ge 1$ the $n$-th mode of $P$ is $c_n = p_n' e^{2\pi i n z_0}$. Because $Q^*$ has no constant or positive modes, $[e^{2\pi i m Q^*}]_0 = 1$ and $[e^{2\pi i m Q^*}]_k = 0$ for $k > 0$. Hence $p_n' = t_n + \sum_{m>n} t_m [e^{2\pi i m Q^*}]_{n-m}$, and

$$|p_n' - t_n| \le \sum_{m>n} M e^{-2\pi m d} e^{2\pi m\rho} e^{-2\pi(m-n)H} \le 2M e^{-2\pi n(d-\rho)} e^{-2\pi(d+H-\rho)}.$$

With $z_0 = ih - t_0 + \zeta$, $|\zeta| \le C_1\Lambda e^{4\pi(Y_0+D+1)}$ by (3), we have $e^{2\pi i n z_0} = \Lambda^n e^{-2\pi i n t_0} e^{2\pi i n\zeta}$ and $|t_n e^{-2\pi i n t_0}| = |\tau_n|$, so $|t_n| = |\tau_n| e^{-\pi n h}$. Dividing,

$$\frac{c_n}{\Lambda^n} - \tau_n = \tau_n (e^{2\pi i n\zeta} - 1) + (p_n' - t_n) e^{-2\pi i n t_0} e^{2\pi i n\zeta}.$$

In the first term $|\tau_n| = |t_n| e^{\pi n h} \le M e^{2\pi n(Y_0+1)}$ and $|e^{2\pi i n\zeta} - 1| \le 4\pi n|\zeta|$ when $2\pi n|\zeta| \le 1$; in the second, $|e^{2\pi i n\zeta}| \le e$ in that case. If instead $2\pi n|\zeta| > 1$, then $\Lambda \ge (2\pi n C_1)^{-1} e^{-4\pi(Y_0+D+1)}$, and the trivial bound $|c_n|\Lambda^{-n} + |\tau_n| \le C' M e^{2\pi n(Y_0+D+2)}$ is already $\le C_n\Lambda$ (this uses $D \ge 1$). In the second term, $|(p_n' - t_n) e^{-2\pi i n t_0}| \le 2M e^{2\pi n(Y_0+1+\rho)} \Lambda e^{2\pi(2Y_0+2+D+2\rho)}$, by $d = h/2 - Y_0 - 1$ and $H = d - D - \rho$. Adding the two gives the stated bound, since $n(Y_0 + 1 + \rho) + 2Y_0 + 2D + 2.5 \le (n+2)(Y_0 + D + 2)$. □

**Corollary 6.4.** *For the sewing solution, $\hat{c}_n(K_b^W) \to B_n e^{2\pi i n a}$ for every $n \ge 1$ as $b \to \eta^-$. Moreover, $\kappa_n(K_b^W) \to B_n / B_1^n$ and $\lim \arg \hat{c}_1 = \arg B_1 + 2\pi a$.*

*Proof.* Theorem 6.3(4) and Corollary 4.5: $C_n$ does not depend on $b$ and $\Lambda \to 0$. Proposition 2.4 makes $c_1$ nonzero for $b$ near $\eta$, justifying division and the limiting phase. □

**Corollary 6.5** (Local parabolic limit of the sewn solution)**.** *Put $u^* = 1/e - 1$ and $a = \Phi_{\mathrm{att}}(u^*)$. On some fixed disk about $z = 0$, the function*

$$F_\eta^{\mathrm{att}}(z) = e(1 + \Phi_{\mathrm{att}}^{-1}(z + a))$$

*is holomorphic, satisfies $F_\eta^{\mathrm{att}}(0) = 1$, and has $(F_\eta^{\mathrm{att}})'(0) \ne 0$. As $b \to \eta^-$ through real bases, $K_b^W \to F_\eta^{\mathrm{att}}$ locally uniformly on that disk, with all $z$-derivatives. Consequently there are $r > 0$ and $b_2 < \eta$ such that, for $b_2 < b < \eta$, the only preimage of 1 under $K_b^W$ in $B(0, r)$ is 0; both $K_b^W$ and $1/K_b^W$ are uniformly bounded there.*

*Proof.* Theorem 4.4(1) gives locally uniform convergence of $A_b(u) := R_b^{-1}(e(1+u))$ to $A_\eta(u) := \Phi_{\mathrm{att}}(u) - a$ near $u^*$, with $A_b(u^*) = 0$. The derivative $A_\eta'(u^*)$ is nonzero: the series in the proof of Theorem 4.4, Lemmas 4.2 and 4.3, and Cauchy's estimate give $\Phi_{\mathrm{att}}'(u) = 2/u^2 + O(1/u)$ on a narrower attracting petal. The Abel identity and $f_\eta'(u) \ne 0$ propagate nonvanishing back along the orbit of $u^*$. Uniform convergence of these local inverses therefore gives $R_b \to F_\eta^{\mathrm{att}}$ on a fixed disk about 0.

In Theorem 6.3, write $P_b(z) = \zeta_b + P_b'(z + z_0(b))$, where $\zeta_b = z_0(b) + t_0(b) - \mathrm{i}h(b) \to 0$. The proof of that theorem gives $p_0'(b) \to 0$ and $|p_n'(b)| \le \rho \mathrm{e}^{-2\pi n H(b)}$ for $n \ge 1$, while $H(b) \to \infty$ and $\operatorname{Im} z_0(b) = h(b)/2 + o(1)$. Hence $P_b \to 0$ uniformly on every fixed bounded $z$-disk on which the upper representation is defined. That representation contains a fixed disk about 0 for all sufficiently large $b$, and $K_b^W(z) = R_b(z + P_b(z))$ there. This proves the asserted convergence; Cauchy's formula gives convergence of derivatives. Choose a small closed disk on which $F_\eta^{\mathrm{att}}$ is nonzero and $F_\eta^{\mathrm{att}} - 1$ has only its simple zero at 0. Rouché's theorem and $K_b^W(0) = 1$ give the final assertions. □

The construction is directly computable (`weld_solve.py`). At $b = 1.10$ the iteration contracts by $\sim 10^{-4}$ per step and gives $\hat{c}_1 = 0.0864272029\,\mathrm{e}^{1.5510921403\mathrm{i}}$, against $\tau_1 = 0.0864272022\,\mathrm{e}^{1.5510921408\mathrm{i}}$. The relative difference is $8.8 \cdot 10^{-9} = 0.38\Lambda$, and at $b = 1.30$ it is $3.9 \cdot 10^{-19} = 0.40\Lambda$. At these bases $h$ is too small for the hypotheses of Theorem 6.3, so the agreement is not an instance of the theorem. It does match the first-order expansion of the correction, $c_1/(\Lambda\tau_1) - 1 \approx \Lambda\big[-4\pi^2|\tau_1|^2 - 2\pi\mathrm{i}\tau_1 - 4\pi\mathrm{i}\tau_2\bar{\tau}_1/\tau_1\big]$, whose modulus is $0.3834\Lambda$ and $0.3984\Lambda$ respectively .

## 7. Dependence on the base

This section proves that $K_b^W$ depends holomorphically on the base near $(b_1, \eta)$, with quantitative bounds; the identification with the boundary value $K_b$ of (2) is completed in Section 9.

**Theorem 7.1** (W1, local)**.** *Let $b_0 \in (b_1, \eta)$. There is a disc $\Delta \ni b_0$ in the base plane such that:*

1. *For $b' \in \Delta$, with $L(b'), L_2(b')$ the continued fixed points, $R_{b'}, S_{b'}$ the regular superfunctions, $\varpi(b') = 2\pi\mathrm{i}/\log\lambda(b')$ the period of $R_{b'}$, and $\hat{T}_{b'} = R_{b'}^{-1} \circ S_{b'} + \varpi/2$, the contraction of Theorem 6.3 works with $\rho$ and $D$ unchanged, $2M$ in place of $M$, and $H - \frac{1}{2}$ in place of $H$. The solution is $K_{b'}^W = R_{b'} \circ (\varphi_+ + \varpi)$ above, where $\varphi_+ = T_{b'} \circ \varphi_-$, and $S_{b'} \circ \varphi_-$ below. It is normalised by the root of $\varphi_+(z_0) = -\varpi$ in the disc of radius 1 about $-\varpi - t_0$. (For real $b'$, $\varpi = -\mathrm{i}h$.) Its fixed point, the resulting solution $K_{b'}^W$, and the normalising root $z_0(b')$ are holomorphic in $b'$.*
2. *For $b' \in \Delta$ with $\operatorname{Im} b' > 0$ one has $\arg\lambda > 0$, $\arg\lambda_2 < 0$, and $K_{b'}^W(z) \to L(b')$ as $\operatorname{Im} z \to +\infty$, $K_{b'}^W(z) \to L_2(b')$ as $\operatorname{Im} z \to -\infty$, uniformly for $\operatorname{Re} z$ in compacts. In the upper and lower half-planes (of the translated coordinate) it has the representations $R_{b'} \circ (\mathrm{id} + \theta_{\mathrm{up}})$ and $S_{b'} \circ (\mathrm{id} + \theta_{\mathrm{dn}})$. These are the defining representations of the complex-base construction in `_cbuild.py`.*
3. *(Local uniqueness.) If a solution $F$ with $F(0) = 1$ admits such representations and, after the translation that puts its seam representation in the form $T \circ (\mathrm{id} + Q) = \mathrm{id} + t_0 + P'$, has $\|Q\|_H \le \rho$ and its normalising root in that disc, then $F = K_{b'}^W$. (Without the root condition the statement is false: the translates of $K_{b'}^W$ by the other roots $c_j$, $j \ge 1$, have the same $Q$.)*

*Proof.* (1) By Lemma 6.1(iii), $T$ is holomorphic on the strip of half-width $d + 1$, so there is slack 1 in $d$; we use half of it. The Koenigs maps at $L(b')$, $L_2(b')$ and the principal logarithms defining $R^{-1}$ depend holomorphically on $b'$. To choose one parameter disc for the unbounded strip, first work on the compact fundamental rectangle $0 \le \operatorname{Re} z \le 1$, $|\operatorname{Im} z| \le d(b_0) + \frac{1}{2}$. Its image under $S_{b_0}$ is compactly contained in the common domain of the two chosen Abel charts; their local parameter continuations therefore give a single disc $\Delta$ on this rectangle. The identity $T_{b'}(z+1) = T_{b'}(z) + 1$ extends the result to the whole strip. Thus $\hat{T}_{b'} - \mathrm{id}$ is jointly holomorphic on $\Delta \times \{|\operatorname{Im} z| < d(b_0) + \frac{1}{2}\}$. Shrinking $\Delta$ again, its modes satisfy $|t_n(b')| \le 2M\mathrm{e}^{-2\pi|n|(d(b_0)-1/2)}$ (Cauchy), and the estimates of Theorem 6.3(1) hold with $2M$ in place of $M$: the original choice of $D$ gives a Lipschitz constant at most $2\rho = \frac{1}{2}$ and a ball-image norm at most $\rho(1 - \mathrm{e}^{-2\pi D})/\pi < \rho$.

The iterates $\mathcal{T}^k(0)$ are holomorphic in $b'$ and converge uniformly, so $Q^*$ is holomorphic in $b'$. So are $P'$ and, by the implicit function theorem (Rouché), $z_0$.

(2) From $\lambda e^{-\lambda} = \log b'$, $\partial\lambda/\partial b = e^\lambda/(b(1-\lambda))$ is positive for $\lambda < 1$ and negative for $\lambda_2 > 1$. Hence $\operatorname{Im} b' > 0$ gives $\operatorname{Im}\lambda > 0 > \operatorname{Im}\lambda_2$ for $\Delta$ small. Above, $K^W = R(z + c + o(1))$, and $R(w) = \sigma^{-1}(c'\lambda^w) \to L$ when $\operatorname{Re}(w\log\lambda) \to -\infty$. This is the case as $\operatorname{Im} w \to +\infty$, since $\operatorname{Re}(w\log\lambda) = \operatorname{Re} w \log|\lambda| - \operatorname{Im} w \arg\lambda$. Below, the same argument with $\arg\lambda_2 < 0$ applies. The representations are those of Theorem 6.3(2), transported by $z \mapsto z + z_0$.

(3) The fixed point of a contraction is unique in its ball. $P'$ and $z_0$ are then determined, the latter by $F(0) = 1$ together with the root condition. □

**Corollary 7.2.** *Suppose that, for all $b'$ in a nonempty open subset of $\Delta \cap \{\operatorname{Im} b' > 0\}$, the Kneser-type solution $\kappa_{b'}$ continued from $b > \eta$ satisfies the hypothesis of Theorem 7.1(3), and that $\kappa$ is jointly holomorphic in $(b', z)$ off a closed singular set whose complement is connected in $z$. Then $\kappa_{b'} = K^W_{b'}$ on $\Delta \cap \{\operatorname{Im} b' > 0\}$. The boundary value $K_b$ exists on $\Delta \cap \mathbb{R}$ and equals $K^W_b$. The upper family continues analytically across $\Delta \cap \mathbb{R}$. The conclusions of Corollary 6.4 then hold for $K_b$, and in particular so do the limits of Corollary 1.1.*

*Proof.* On the open set, Theorem 7.1(3) gives $\kappa = K^W$. Both families are holomorphic in $(b', z)$ for $z$ in a lower half-plane, where $K^W = S \circ \varphi_-$ has no cuts. So they agree on $\Delta \cap \{\operatorname{Im} b' > 0\}$ by the identity theorem, and then everywhere by continuation in $z$. The rest is Theorem 7.1(1) and Corollary 6.4. □

The hypothesis of Corollary 7.2 is an inequality on an open set of bases. It concerns only the lower correction of $\kappa$ near the seam. The ladders are consistent with it: they reproduce $\tau_1$ and $t_2/t_1^2$ to their precision (Table 1), and the explicit sewing solution matches the ladder value of $\hat{c}_1$ at $b = 1.10$ to eight digits. This check is numerical and not certified. A proof free of numerics would need a global uniqueness theorem for Kneser-type solutions at complex bases inside the Shell–Thron region, one that selects the no-winding solution: the class defined only by the limits at $\pm i\infty$ also contains solutions of the form $R \circ (z + 2\pi i m z/\log\lambda + \dots)$. The base continuation from $b > \eta$ passes the Shell–Thron boundary, where $|\Lambda| = 1$ and the present contraction does not apply. Section 9 supplies this input by a different route, a tilted initial region that is insensitive to $|\Lambda|$, and identifies the continued family with $K^W$ without numerics.

## 8. Characterisation and boundary values

Corollary 7.2 needs a quantitative input about $\kappa$ at one base. Here we replace it with a qualitative characterisation. From it follows a boundary-value theorem whose only hypothesis is a uniform bound on the two $\theta$-representations along a ladder $b + iy$, $y \downarrow 0$.

**Definition 8.1.** For real $b \in (1, \eta)$ and $Y > 0$ let $\mathcal{C}_b(Y)$ be the set of solutions $F$ of (1) with $F(0) = 1$ such that ("solution" means holomorphic off a closed singular set and satisfying the equation where defined)

(a) $F = R \circ (\mathrm{id} + P)$ on $\{\operatorname{Im} z > -2Y\}$, with $P$ holomorphic, 1-periodic and bounded there, and $P(0) = 0$ (the branch $R^{-1}(1) = 0$);
(b) $F = S \circ (\mathrm{id} + Q)$ on $\{\operatorname{Im} z < -Y\}$, with $Q$ holomorphic, 1-periodic and bounded there, and $|\operatorname{Im} Q - h/2| \le 1$ on the strip $O = \{-2Y < \operatorname{Im} z < -Y\}$.

Condition (b) says that the real line of repelling time, i.e. the segment $(L, L_2)$, sits half a period of attracting time below the real axis of $F$. Heuristically, some such condition is necessary. For the sewing solution the roots of $\varphi^W_+(c) \in ih\mathbb{Z}$ near the axis are $c_j = c_0 + ijh + O(\Lambda)$. The translates $K^W(\cdot + c_j)$ keep an $R$-representation above and an $S$-representation on some lower half-plane, and give $|c_1| \asymp \Lambda^{1+j}$. Condition (b) excludes $j \neq 0$, as the proof shows.

**Theorem 8.2** (Characterisation)**.** *There are $b_1 < \eta$ and $Y_B$ such that for $b \in (b_1, \eta)$ and $Y_0 + D + 3 < Y < (h - Y_0 - D - 3)/2$ (with $Y_0, D$ as in Theorem 6.3),*

$$\mathcal{C}_b(Y) = \{K_b^W\},$$

*with $K_b^W$ normalised as in Theorem 6.3(3). In particular $\hat{c}_n \to B_n \mathrm{e}^{2\pi \mathrm{i} n a}$ for every $n$ holds for the unique element of $\mathcal{C}_b(Y)$.*

*Proof.* $K_b^W \in \mathcal{C}_b(Y)$. In the normalised coordinate, $\varphi_-(z) = z + z_0 + Q^*(z + z_0)$ is defined for $\mathrm{Im}(z + z_0) < H$, i.e. for $\mathrm{Im}\, z < -(Y_0 + 1 + D + \rho) + O(\Lambda)$, which contains $\{\mathrm{Im}\, z < -Y\}$. There $|Q - z_0| \le \rho$, so $|\mathrm{Im}\, Q - h/2| \le \rho + O(\Lambda) < 1$. The $R$-representation holds for $\mathrm{Im}(z + z_0) > -(H - 1)$, i.e. above $-(h - Y_0 - D - 2 - \rho)$, which contains $\{\mathrm{Im}\, z > -2Y\}$. $P$ and $Q$ are bounded there.

*Uniqueness.* Let $F \in \mathcal{C}_b(Y)$, $\varphi_+ = \mathrm{id} + P$ and $\varphi_- = \mathrm{id} + Q$. Work with the strip $\Sigma_q = \{|\mathrm{Im}\, w'| < d - D\}$ of $S$-time. By Lemma 6.2 and the choice of $D$, on $\overline{\Sigma_q}$

$$|T - \mathrm{id} - t_0| \le \frac{2M\mathrm{e}^{-2\pi D}}{1 - \mathrm{e}^{-2\pi D}} \le \frac{\rho}{2\pi}, \qquad |T' - 1| \le \rho < 1,$$

so $T$ is univalent on the convex set $\Sigma_q$ (Noshiro–Warschawski), continuous up to the boundary, and within $\rho/2\pi$ of a translation.

(i) *The overlap.* Let $O = \{-2Y < \mathrm{Im}\, z < -Y\}$. For $z \in O$, $\mathrm{Im}\, \varphi_-(z) \in (h/2 - 2Y - 1,\ h/2 - Y + 1) \subset (-(d - D), d - D)$, because $2Y < h - Y_0 - D - 2$ and $Y > Y_0 + D + 2$. Since $\sigma \circ R(w) = \sigma(1)\lambda^w$ wherever $R$ is defined, $R \circ \varphi_+ = S \circ \varphi_- = R \circ T \circ \varphi_-$ gives $\lambda^{\varphi_+} = \lambda^{T \circ \varphi_-}$. Hence $\varphi_+ = T \circ \varphi_- - \mathrm{i}h + k\mathrm{i}h$ on $O$ for a constant $k \in \mathbb{Z}$. No injectivity is needed for this step.

(ii) *The lifted sewn surface.* Let $V_- = \{\mathrm{Im}\, w' < d - D\}$ (repelling time) and let $V_+$ be the region above the curve $T(\{\mathrm{Im}\, w' = -(d - D)\}) - \mathrm{i}h$ (attracting time). Let $\tilde{X}$ be $V_- \sqcup V_+$ modulo $w = T(w') - \mathrm{i}h$ for $w' \in \Sigma_q$. Since $T$ is a homeomorphism of $\overline{\Sigma_q}$ onto its image, the identification graph is closed and $\tilde{X}$ is a Hausdorff Riemann surface. Translation by 1 acts freely on it, and $X = \tilde{X}/\mathbb{Z} \cong \mathbb{C}^*$, both ends being punctured discs. The function $\tilde{K}$, equal to $R$ on $V_+$ and $S$ on $V_-$, is well defined, because $R(T(w') - \mathrm{i}h) = S(w')$ on the band. The sewing solution is $\tilde{K} \circ \tilde{\Phi}^W$, where $\tilde{\Phi}^W$ is the lift of its uniformisation; sewing along the real line or along any other essential curve in $\Sigma_q$ gives the same $\tilde{X}$.

Define $\tilde{\Phi} : \mathbb{C} \to \tilde{X}$ by $\varphi_+ - k\mathrm{i}h$ on $\{\mathrm{Im}\, z > -2Y\}$ and $\varphi_-$ on $\{\mathrm{Im}\, z < -Y\}$. These agree on $O$ by (i). They land in $V_\mp$ as follows. $\mathrm{Im}\, Q$ is bounded and harmonic on a half-cylinder, hence extends harmonically over the puncture. So on $\{\mathrm{Im}\, z \le -Y - \delta\}$ it is bounded by its values on the line $\mathrm{Im}\, z = -Y - \delta \subset O$. This gives $\mathrm{Im}\, \varphi_- \le -Y + h/2 + 1 < d - D$. Likewise, on a line in $O$ near $\mathrm{Im}\, z = -2Y$, (i) gives $\mathrm{Im}(\varphi_+ - k\mathrm{i}h) \ge \mathrm{Im}\, z - 1 - \rho/2\pi$. The minimum principle then gives $\mathrm{Im}(\varphi_+ - k\mathrm{i}h) \ge -2Y - 1 - \rho/2\pi$ above that line. This exceeds the height $-h/2 - d + D + \rho/2\pi$ of the lower boundary of $V_+$, because $2Y < h - Y_0 - D - 3$.

(iii) $\tilde{\Phi}$ *is a translate of* $\tilde{\Phi}^W$. $\tilde{\Phi}$ commutes with $z \mapsto z + 1$ and descends to a holomorphic $\Phi : \mathbb{C}/\mathbb{Z} \to X \cong \mathbb{C}^*$. $\Phi$ is proper, since $\varphi_\pm - \mathrm{id}$ are bounded, and has local degree one at each end, since $P \to p_0$ and $Q \to q_0$. A proper holomorphic map of $\mathbb{C}^*$ that fixes the ends with local degree one is linear. Hence $\tilde{\Phi} = \tilde{\Phi}^W(\cdot + c)$ for some $c \in \mathbb{C}$, and $F = \tilde{K} \circ \tilde{\Phi} = K_b^W(\cdot + c)$ in the coordinate of Theorem 6.3(1)–(2). Univalence of $\varphi_\pm$ is a conclusion, not an assumption.

(iv) *The root.* Take a line $\ell \subset O$ with $\mathrm{Im}\, z = -Y''$ and $Y''$ close to $2Y$. There $\mathrm{Im}\, \varphi_- \le -2Y + h/2 + 1 + o(1) < H - \rho$, so $\varphi_-(\ell)$ lies in $\varphi_-^W(\{\mathrm{Im} < H\})$. The inclusion $\varphi_-^W(\{\mathrm{Im} < H\}) \supset \{\mathrm{Im} < H - \rho\}$ holds because $|Q^*| \le \rho$: for $w$ in the smaller set, $\varphi_-^W - w$ has winding number 1 around 0 on the boundary of a large fundamental rectangle. By injectivity of $\tilde{\Phi}^W$, the points $z + c$ with $z \in \ell$ lie in $\{\mathrm{Im} < H\}$. There $Q(z) = c + Q^*(z + c)$, so $|\mathrm{Im}\, c - h/2| \le 1 + \rho$. On the upper chart, $\varphi_+(z) - k\mathrm{i}h = \varphi_+^W(z + c) - \mathrm{i}h$, and $P(0) = 0$ gives $\varphi_+^W(c) = \mathrm{i}h(1 - k)$. Since

$|P'| \le \rho$ where $\varphi_+^W$ is defined, $|c - (ih(1-k) - t_0)| \le \rho$. Comparing imaginary parts with the bound on $\operatorname{Im} c$ gives $|k|h \le 1 + 2\rho$, so $k = 0$. Then $c$ lies in the Rouché disc of Theorem 6.3(3), so $c = z_0$ and $F = K_b^W$. The last sentence of the theorem is Corollary 6.4. □

**Theorem 8.3** (Boundary values)**.** *Let $b \in (b_1, \eta)$ and $Y$ be as in Theorem 8.2. Suppose that for every sequence $y_k \downarrow 0$ the Kneser-type solutions $\kappa_k = \kappa_{b+iy_k}$ admit representations (a) and (b) with $R_{b+iy_k}$, $S_{b+iy_k}$ in place of $R_b, S_b$. Suppose also that $P_k, Q_k$ are bounded uniformly in $k$, and that $|\operatorname{Im} Q_k - h(b_k)/2| \le \frac{1}{2}$ on $O$, with $b_k = b + iy_k$. Then $\kappa_k \to K_b^W$ locally uniformly off the (moving) singular sets, the boundary value (2) exists, and its separation coefficients satisfy $|\hat{c}_n(b) - \tau_n(b)| \le C_n \Lambda$. Consequently, if the hypothesis holds for all $b$ near $\eta$, the limits of Corollary 1.1 hold for $K_b$, together with $\hat{c}_n \to B_n e^{2\pi i n a}$ for all $n$.*

*Proof.* $R_{b'} \to R_b$ and $S_{b'} \to S_b$ locally uniformly (Koenigs maps depend holomorphically on the base), and so do the constants $h, \lambda$. By Montel's theorem a subsequence of $(P_k, Q_k)$ converges locally uniformly to $(P, Q)$, bounded and periodic, with $P(0) = 0$ and $|\operatorname{Im} Q - h/2| \le \frac{1}{2}$. The limit $F = R_b \circ (\mathrm{id} + P) = S_b \circ (\mathrm{id} + Q)$ solves (1) with $F(0) = 1$, so $F \in \mathcal{C}_b(Y)$ and $F = K_b^W$ by Theorem 8.2. Every subsequence has the same limit. The Fourier coefficients of $P_k$ converge to those of $P$, and Theorem 6.3(4) applies. □

The hypothesis of Theorem 8.3 is the form in which the complex-base construction computes $\kappa$: `_cbuild.py` solves precisely for bounded $\theta_{\mathrm{up}}, \theta_{\mathrm{dn}}$ with these two representations. Numerically it holds along every ladder we ran. The ladder values of $c_1$ are of size $\Lambda$, not $O(1)$ or $\Lambda^2$. This is consistent with the height condition, but it pins the offset only in combination with the value of $|\tau_1|$. For the family continued from $b > \eta$ around the cusp, the conclusion of the theorem is obtained directly in Section 9, without verifying the hypothesis along ladders.

8.1. **Uniform bounds.** The hypothesis of Theorem 8.3 asks for bounds that are uniform along a ladder. We show that uniformity is automatic. It suffices that the continued family lie in the characterisation class on some open set of bases, with no quantitative condition.

Let $N = \bigcup_{b_0 \in (b_1, \eta)} \Delta(b_0)$ be the union of the discs of Theorem 7.1, and $N^+ = N \cap \{\operatorname{Im} b' > 0\}$. Both are connected: the discs are centred on an interval, and consecutive upper half-discs overlap. For complex $b'$, Definition 8.1 is read with $R_{b'}, S_{b'}$, the period $\varpi(b') = 2\pi i / \log \lambda(b')$ in place of $-ih$, and $h(b') = |\operatorname{Im} \varpi(b')|$.

**Theorem 8.4** (Uniformity)**.** *Shrinking the discs if necessary:*

1. *The sewing solutions $K_{b'}^W$ of the discs agree on overlaps and form one family, holomorphic in $(b', z)$ on $N \times \mathbb{C}$ minus the singular set. Its representation functions $P_{b'}, Q_{b'}$ are bounded uniformly on $\{\operatorname{Im} z > -2Y\}$ and $\{\operatorname{Im} z < -Y\}$ respectively, for $b'$ in any compact subset of $N$, including compact pieces of the real segment.*
2. *For every $b' \in N$, $\mathcal{C}_{b'}(Y) = \{K_{b'}^W\}$.*
3. *Let $\kappa$ be any family of solutions, jointly holomorphic in $(b', z)$ off a closed singular set, whose regular set in $z$ is connected for each $b'$ and which contains $N^+ \times W$ for some open $W \subset \{\operatorname{Im} z < -Y\}$, with $\kappa_{b'} \in \mathcal{C}_{b'}(Y)$ for all $b'$ in a nonempty open subset of $N^+$. Then $\kappa = K^W$ on $N^+$. Its $\theta$-representations are bounded uniformly along every ladder $b + iy$, $y \downarrow 0$, $b \in (b_1, \eta)$. So the hypothesis of Theorem 8.3 holds, and with it the conclusions there, including the limits of Corollary 1.1, for $\kappa$.*

*Proof.* (1) On $\Delta(b_0) \cap \Delta(b_0')$ let $H \le H'$ be the two strip widths. Since $\|\cdot\|_H \le \|\cdot\|_{H'}$, the fixed point of the $H'$-contraction lies in the $H$-ball. It is a fixed point there, so it equals the $H$-fixed point by uniqueness. The normalising roots coincide by Rouché. The singular set $\{(b', z) : \varphi_+(z) + \varpi(b') \in \text{cuts of } R_{b'}\}$ is closed in $N \times \mathbb{C}$, because the cuts move continuously with $b'$. Uniform bounds on compacta follow from continuity of the fixed point in the base

and from the explicit bounds $\|Q^*\|_H \le \rho$, $\|P'\|_H \le \rho$ of Theorem 6.3, which hold throughout each disc by Theorem 7.1(1). The continuation of the representations to the half-planes of Definition 8.1 is as in the proof of Theorem 8.2: through the band, which persists for $b'$ near the real axis because $T_{b'}$ depends holomorphically on $b'$.

(2) The proof of Theorem 8.2 runs verbatim with the following per-base data. The admissible range is $Y_0 + D + 3 < Y < (h(b') - Y_0 - D - 3)/2$ with $H - \frac{1}{2}$ in place of $H$; it contains a fixed $Y$ for all $b'$ in each small disc. The strip $\Sigma_q$, on which $|\hat{T}_{b'} - \mathrm{id} - \hat{t}_0| \le \rho/2\pi$ and $|\hat{T}'_{b'} - 1| \le \rho$, persists by continuity in $b'$. So does the identity $R_{b'} \circ T_{b'} = S_{b'}$ on it, by analytic continuation in $b'$ of an identity valid at real $b'$. The identity $\sigma \circ R(w) = \sigma(1)\lambda^w$ holds for all $b'$. The sewn surface is again $\mathbb{C}^*$. Proper maps of $\mathbb{C}^*$ fixing the ends with local degree one are linear. Finally, the normalising roots $c_j = c_0 - j\varpi + O(|\Lambda|)$ have imaginary parts separated by $h(b')$. None of these steps uses reality of the base. The one place it entered, $\operatorname{Im} t_0 = h/2$, is replaced by $\operatorname{Im} \hat{t}_0 \approx 0$ for the shifted map $\hat{T}_{b'}$, which holds after shrinking the discs.

(3) By (2), $\kappa_{b'} = K^W_{b'}$ on a nonempty open subset of $N^+$. For $z \in W$, where $K^W_{b'} = S_{b'} \circ \varphi_-$ has no cuts, both sides are holomorphic in $b'$ on the connected set $N^+$. So they agree on $N^+ \times W$, and then for all $z$ off the singular set by continuation in $z$ (the complement of the cuts is connected). Uniform boundedness is then (1). □

**Corollary 8.5** (A bounded complex cusp corridor)**.** *There are $b_3 < \eta$, $r > 0$, and a connected open set $N_\eta \subset \mathbb{C}$ containing $(b_3, \eta)$, with $\eta \in \partial N_\eta$, such that $K^W$ is jointly holomorphic on $N_\eta \times B(0,r)$ and $K^W_b \to F^{\mathrm{att}}_\eta$ locally uniformly there as $b \to \eta$ within $N_\eta$, together with all $z$-derivatives. Throughout $N_\eta$, the only preimage of 1 in $B(0,r)$ is 0, and $K^W_b$ and $1/K^W_b$ are uniformly bounded on that disk. No lower bound on the width of $N_\eta$ is asserted.*

*Proof.* Choose $r$ inside the disk of Corollary 6.5 so that $F^{\mathrm{att}}_\eta$ has no zero on $\overline{B(0,r)}$ and $F^{\mathrm{att}}_\eta - 1$ has just its simple zero at 0. The minimum of $|F^{\mathrm{att}}_\eta|$ on the closed disk and of $|F^{\mathrm{att}}_\eta - 1|$ on its boundary is positive; let one third of the smaller minimum be $m$. By that corollary, for every real $x \in (b_3, \eta)$, with $b_3$ sufficiently close to $\eta$, $\|K^W_x - F^{\mathrm{att}}_\eta\|_{\overline{B(0,r)}} < m$. Theorem 7.1 gives a base disk $U_x$ centred at $x$ on which $K^W_b$ is jointly holomorphic near the closed $z$-disk. Shrink it so that $|b - x| < (\eta - x)/4$ and

$$\|K^W_b - K^W_x\|_{\overline{B(0,r)}} < \min(m, \eta - x) \qquad (b \in U_x).$$

Two overlapping disks centred on the real line have an overlapping real interval. Both families equal the uniquely normalised real solution of Theorem 6.3 there, so the identity theorem patches them. Their union $N_\eta = \bigcup_x U_x$ is connected and excludes $\eta$. The preceding bounds give zero-freeness and, by Rouché's theorem, exactly one preimage of 1 in $B(0,r)$, necessarily 0. They also bound the functions and their reciprocals uniformly. If $b \to \eta$ inside $N_\eta$ and $b \in U_x$, then $|b - x| < (\eta - x)/4$ forces $x \to \eta$; the last displayed bound and Corollary 6.5 give the stated convergence. Cauchy's formula gives convergence of derivatives on smaller disks. □

**Remark 8.6.** The open-set hypothesis in (3) is implied by membership at a single base $b'_*$ with strict inequality $|\operatorname{Im} Q - h/2| < 1$, provided the limits $\kappa_{b'}(z) \to L(b')$, $L_2(b')$ as $\operatorname{Im} z \to \pm\infty$ are locally uniform in $b'$. Indeed, at a complex base these limits give $R$- and $S$-representations near $\pm i\infty$ with bounded periodic corrections, provided there is no winding. In detail, $H(q) = \lambda^P$, with $q = \mathrm{e}^{2\pi i z}$, satisfies $|H| = o(|q|^{-\arg\lambda/2\pi})$ with $\arg\lambda < 2\pi$, so it extends holomorphically to $q = 0$. A zero of $H$ at $q = 0$ is exactly winding. We take $\operatorname{Im} z$ large enough to avoid the other zeros of $H$ (points where $\kappa = L$). The winding number is an integer that varies continuously, so it persists. The $S$-side is the same; $\tau^{-1}$ has no critical points. On the remaining strip, which is compact modulo 1, the representations of $\kappa_{b'_*}$ continue to nearby $b'$ because $\kappa_{b'} \to \kappa_{b'_*}$ uniformly

there, and the height condition is open. Local uniformity of the defining limits in the base would suffice here; it is not established by the parameter argument in [24, Prop. 3 and §5].

Consequently, no quantitative estimate on $\kappa$ is needed once a jointly holomorphic base continuation and membership in the class of Definition 8.1 on an open set are known; Section 9 proves the identification with $K^W$ directly for the continuation around the cusp. The pointwise construction in `_cbuild.py` and the ladders are consistent with membership, but do not establish it. Proposition 2 of [24] cannot provide the required uniqueness (Remark 1.2), and its parameter-continuation argument lacks the uniform estimates noted in Section 10.6.

## 9. Continuation of Kneser's solution around the cusp

This section makes the hypothesis of Theorem D unnecessary for the continuation that passes near $\eta$. The continuation leaves the real exterior ray, crosses the upper Shell–Thron boundary arc close to the cusp, and returns to the real axis below $\eta$ from the upper half-plane. Near the cusp the two fixed points are close, and Kneser's initial region (the chord between the fixed points together with its image) can be tilted continuously into an initial region for the attracting–repelling configuration below $\eta$. The multiplier of the upper fixed point crosses the unit circle on the way, but this plays no role. In the logarithmic coordinate of the pair of fixed points the tilted region is a thin strip on which the map is a translation up to a relative error $O(|\varepsilon|)$, so its quotient is quasiconformally a flat cylinder, whether or not the fixed points are linearisable. At the end of the path the tilted region is a fundamental domain for the sewn surface of Proposition 3.4, and a proper-map argument identifies the continued solution with $K_b^W$.

*Conventions for this section.* Here $\varepsilon$ is the complex parameter $1-\lambda_1$ defined below; for real $b<\eta$ it differs from the quantity $|\log\lambda|$ called $\varepsilon$ in Sections 6–8 by $O(\varepsilon^2)$, and we use results of those sections only in the form stated there. The letters $\alpha$ (a tilt angle), $\varsigma$ (a normal coordinate), $s\in[0,1]$ (an interpolation parameter), $H$, $X$ and $M$ have local meanings. $L_1$ denotes the fixed point with multiplier $1-\varepsilon$: it is the upper fixed point $L_+$ for real $b>\eta$ and the attracting fixed point $L$ for real $b<\eta$. Constants $C, c$ do not depend on $\varepsilon$; $r_0$ is decreased finitely many times.

### 9.1. The cusp parameter.

For small complex $\varepsilon$ put

$$\lambda_1 = 1-\varepsilon, \qquad \log b(\varepsilon) = \lambda_1 e^{-\lambda_1} = (1-\varepsilon)e^{\varepsilon-1},$$

and let $\lambda_2(\varepsilon)$ be the other root of $\lambda e^{-\lambda} = \log b(\varepsilon)$ near 1. It is holomorphic, $\lambda_2 = 1+\varepsilon+\frac{2}{3}\varepsilon^2+O(\varepsilon^3)$. By Lemma 2.1, $L_j = e^{\lambda_j}$ are fixed points of $E_b$ with multipliers $\lambda_j$. In the coordinate $w = e(1+u)$,

$$f_\varepsilon(u) = e^{\mu(1+u)-1}-1, \qquad \mu = e\log b = (1-\varepsilon)e^{\varepsilon},$$

with fixed points $u_j = e^{\lambda_j-1}-1$, so that $u_1-u_2 = -2\varepsilon(1+O(\varepsilon))$. For $0<r\le\frac{1}{10}$ let

$$\mathcal{S}_r^{\mathrm{c}} = \Big\{\varepsilon:\ 0<|\varepsilon|<r,\ -\tfrac{7\pi}{12}<\arg\varepsilon<\tfrac{\pi}{12}\Big\}, \qquad \mathcal{S}_r^{\mathrm{c}\prime} = \Big\{\varepsilon\in\mathcal{S}_r^{\mathrm{c}}:\ -\tfrac{7\pi}{12}+\tfrac{1}{10}<\arg\varepsilon<\tfrac{\pi}{12}-\tfrac{1}{10}\Big\}.$$

The lemmas are stated on $\mathcal{S}_r^{\mathrm{c}}$; the family is built on $\mathcal{S}_r^{\mathrm{c}\prime}$, so that the parameter discs of Proposition 9.9 stay in $\mathcal{S}_r^{\mathrm{c}}$. Put $u^* = e^{-1}-1$, the base point $w=1$, and $u_n^* = f_\varepsilon^n(u^*)$.

**Lemma 9.1** (Cusp parameter)**.** *There are $r_0, r_1>0$ with the following properties.*

(1) *$\varepsilon\mapsto b(\varepsilon)$ is injective on $\mathcal{S}_{r_0}^{\mathrm{c}}$. For every $r\le r_0$ there is $r'>0$ such that $b(\mathcal{S}_r^{\mathrm{c}\prime})$ contains $\{b:|b-\eta|<r',\ \operatorname{Im} b>0\}$ and the segments $(\eta-r',\eta)$ and $(\eta,\eta+r')$. Real $b\in(\eta-r_1,\eta)$ corresponds to real $\varepsilon\in(0,r_0)$, and then $L_1=L$ is the attracting and $L_2$ the repelling real fixed point. Real $b\in(\eta,\eta+r_1)$ corresponds to $\varepsilon(\theta) = 1-\theta\cot\theta-\mathrm{i}\theta = \frac{1}{3}\theta^2-\mathrm{i}\theta+O(\theta^4)$, $0<\theta<\theta_1$; then $L_1=L_+$ is the fixed point in the upper half-plane used in Proposition 10.1, and $\lambda_2(\varepsilon(\theta)) = \overline{\lambda_1}$, $u_2=\overline{u_1}$.*

(2) *On* $D_{\rho_0}$, $\rho_0=\frac12$, $f_\varepsilon(u)-u=(u-u_1)(u-u_2)k_\varepsilon(u)$ *with* $k_\varepsilon$ *jointly holomorphic in* $(\varepsilon,u)$ *and* $|k_\varepsilon-\frac12|\le\frac14$. *Moreover* $f_\varepsilon'(u_j)=\lambda_j$.
(3) $g_\varepsilon(v)=(\mathrm{Log}(1+v)+1)/\mu-1$ *is jointly holomorphic on* $\{\mathrm{Re}(1+v)>0\}$, *and* $g_\varepsilon(f_\varepsilon(u))=u$ *whenever* $|\operatorname{Im}(\mu(1+u))|<\pi$, *in particular on* $D_{\rho_0}$ *and near the real segment* $[u^*,0]$.

*Proof.* (1) $(1-\varepsilon)\mathrm{e}^\varepsilon-1=-\frac12\varepsilon^2(1+O(\varepsilon))$, so $\log b-\log\eta=-(2\mathrm{e})^{-1}\psi_0(\varepsilon)^2$ with $\psi_0$ holomorphic and $\psi_0(\varepsilon)=\varepsilon(1+O(\varepsilon))$. For small $r_0$, $\psi_0$ is injective and maps $\mathcal{S}^{\mathrm{c}}_{r_0}$ into a sector of opening $2\pi/3+o(1)<\pi$, on which squaring is injective. The image of $\mathcal{S}^{\mathrm{c}\prime}_r$ under $-\psi_0^2$ contains, near 0, all directions in $(-\frac{\pi}{6}+0.2+o(1),\frac{7\pi}{6}-0.2-o(1))\supset[0,\pi]$. Since $b\mapsto\log b$ is biholomorphic near $\eta$, (1) follows except for the identification of the real segments. For real $\varepsilon\in(0,1)$, $\lambda_1\in(0,1)$ and Lemma 2.1 applies. For real $b>\eta$ the upper fixed point has multiplier $L_+\log b=\theta\cot\theta+\mathrm{i}\theta$ with $\theta=\operatorname{Im}(L_+\log b)\in(0,\pi)$ (the computation in the proof of Proposition 10.1), and $\lambda_1=1-\varepsilon$ gives $\varepsilon(\theta)$, whose argument lies in $(-\pi/2,-\pi/2+\theta)$. The conjugate fixed point $L_-=\overline{L_+}$ is the other one near e, so $\lambda_2=\overline{\lambda_1}$.

(2) $k_0(u)=(\mathrm{e}^u-1-u)/u^2=\frac12+\sum_{n\ge1}u^n/(n+2)!$ differs from $\frac12$ by less than 0.1 on $D_{1/2}$. Division of $f_\varepsilon(u)-u$ by $(u-u_1)(u-u_2)$ (Cauchy's formula on $|u|=1$) gives joint holomorphy and continuity at $\varepsilon=0$. Differentiating the product at $u_j$ gives the multipliers.

(3) $\mathrm{Log}(\mathrm{e}^{\mu(1+u)-1})=\mu(1+u)-1$ when $|\operatorname{Im}(\mu(1+u))|<\pi$. □

9.2. **The logarithmic coordinate.** Put $q=(u-u_1)/(u-u_2)$ and $\chi=\log q$, so that

$$u=u(\chi)=u_2+\frac{u_1-u_2}{1-\mathrm{e}^\chi}.$$

The points $\chi\in2\pi\mathrm{i}\mathbb{Z}$ (the *holes*) correspond to $u=\infty$. Let $\mathcal{G}^{\mathrm{c}}=\{\chi:|\chi-2\pi\mathrm{i}k|\ge\frac12\text{ for all }k\in\mathbb{Z}\}$, and put $A_1=1/\operatorname{Log}\lambda_1=-\varepsilon^{-1}(1+O(\varepsilon))$ and $Z=A_1\chi$.

**Lemma 9.2** (The step in the logarithmic coordinate)**.** *For* $0<|\varepsilon|<r_0$ *the following hold (the proofs of (1)–(5) do not use the sector).*

(1) *On* $\mathcal{G}^{\mathrm{c}}$, $|1-\mathrm{e}^\chi|\ge\frac16\max(1,|\mathrm{e}^\chi|)$; *hence* $|u(\chi)|\le C|\varepsilon|$ *and* $|u-u_1||u-u_2|\le C|\varepsilon|^2\min(|\mathrm{e}^\chi|,|\mathrm{e}^{-\chi}|)$. *On* $\{|\chi|\le1\}$, $|1-\mathrm{e}^\chi|\ge0.63|\chi|$.
(2) *For* $u\in D_{\rho_0}\setminus\{u_1,u_2\}$,

$$\frac{q(f(u))}{q(u)}=1+\beta(u),\qquad\beta=\frac{(u_1-u_2)k(u)}{1+(u-u_1)k(u)},$$

*and* $\Delta(u):=\mathrm{Log}(1+\beta(u))=-\varepsilon(1+\vartheta(u))$ *with* $|\vartheta|\le C(|u|+|\varepsilon|)$ *and* $|\Delta'(u)|\le C|\varepsilon|$ *on* $D_{\rho_0/2}$. *Thus* $\chi(f(u))=\chi(u)+\Delta(u)$ *on the branch continued along the short segment from* $\chi(u)$, *and* $A_1\Delta=1+O(|u|+|\varepsilon|)$.
(3) *On* $\mathcal{G}^{\mathrm{c}}$, *regarded as functions of* $\chi$, $|\vartheta|\le C|\varepsilon|$ *and* $|\partial_\chi\Delta|\le C|\varepsilon|^2$. *On* $\{|\chi|\le1\}\cap\{|u(\chi)|\le\rho_0/2\}$, $|\partial_\chi\Delta|\le C|\varepsilon|^2/|\chi|^2$.
(4) $\Delta(u_1)=\operatorname{Log}\lambda_1$ *and* $\Delta(u_2)=-\operatorname{Log}\lambda_2$. *For* $\chi\in\mathcal{G}^{\mathrm{c}}$, $|\Delta-\operatorname{Log}\lambda_1|+|\partial_\chi\Delta|\le C|\varepsilon|^2|\mathrm{e}^\chi|$ *if* $\operatorname{Re}\chi\le0$, *and* $|\Delta+\operatorname{Log}\lambda_2|+|\partial_\chi\Delta|\le C|\varepsilon|^2|\mathrm{e}^{-\chi}|$ *if* $\operatorname{Re}\chi\ge0$.
(5) $|\partial_\varepsilon\Delta|\le1+C|\varepsilon|$ *at fixed* $\chi\in\mathcal{G}^{\mathrm{c}}$.

*All functions here are jointly holomorphic in* $\varepsilon$ *and* $u$ *(or* $\chi$*).*

*Proof.* (1) On the circle $|\chi|=\frac12$ the minimum of $|\mathrm{e}^\chi-1|$ is $1-\mathrm{e}^{-1/2}>0.39$, attained at $\chi=-\frac12$. The function $1/(\mathrm{e}^\chi-1)$ is holomorphic on the part of the strip $|\operatorname{Im}\chi|\le\pi$ outside the disc, is bounded by 1 on its horizontal edges (where $|\mathrm{e}^\chi-1|=1+\mathrm{e}^{\operatorname{Re}\chi}$), and tends to $-1$ and 0 at the two ends; the maximum principle and periodicity give $|\mathrm{e}^\chi-1|\ge0.39$ on $\mathcal{G}^{\mathrm{c}}$. For $\operatorname{Re}\chi\ge\log2$, $|\mathrm{e}^\chi-1|\ge\frac12|\mathrm{e}^\chi|$, and for $0\le\operatorname{Re}\chi\le\log2$, $0.39\ge\frac16|\mathrm{e}^\chi|$. The bounds on $u$ follow from $u-u_2=(u_1-u_2)/(1-\mathrm{e}^\chi)$ and $u-u_1=\mathrm{e}^\chi(u-u_2)$. The last bound is the minimum of $|(\mathrm{e}^\chi-1)/\chi|$ on the closed unit disc, $1-\mathrm{e}^{-1}$.

(2) The identity follows from $(f(u)-u_j)/(u-u_j)=1+(u-u_{3-j})k(u)$. By Lemma 9.1(2) and $u_1-u_2=-2\varepsilon(1+O(\varepsilon))$, $\beta=-\varepsilon(1+O(|u|+|\varepsilon|))$, and $\operatorname{Log}(1+\beta)=\beta(1+O(|\beta|))$. Cauchy's estimate on $D_{\rho_0}$, where $\Delta=O(|\varepsilon|)$, bounds $\Delta'$.

(3) The first bound is (2) with (1). For the derivatives, $\partial_\chi\Delta=\Delta'(u)\,du/d\chi$ with $du/d\chi=(u_1-u_2)\mathrm{e}^\chi/(1-\mathrm{e}^\chi)^2$, which is $O(|\varepsilon|)$ on $\mathcal{G}^{\mathrm{c}}$ by (1) and $O(|\varepsilon|/|\chi|^2)$ on $|\chi|\le 1$.

(4) $1+\beta(u_1)=1+(u_1-u_2)k(u_1)=f'(u_1)=\lambda_1$, and $1+\beta(u_2)=1/(1+(u_2-u_1)k(u_2))=1/\lambda_2$; all logarithms are principal and close to 0. Since $\Delta'(u)=O(|\varepsilon|)$, $|\Delta(u)-\Delta(u_1)|\le C|\varepsilon||u-u_1|\le C|\varepsilon|^2|\mathrm{e}^\chi|$ by (1) when $\operatorname{Re}\chi\le 0$, and $|\partial_\chi\Delta|=|\Delta'(u)|\,|u_1-u_2|\,|\mathrm{e}^\chi|/|1-\mathrm{e}^\chi|^2\le C|\varepsilon|^2|\mathrm{e}^\chi|$. The case of $u_2$ is symmetric, with $\mathrm{e}^{-\chi}$ in place of $\mathrm{e}^\chi$.

(5) At fixed $\chi$, $\Delta=-\varepsilon(1+\vartheta)$ with $\vartheta$ holomorphic in $\varepsilon$ and $O(|\varepsilon|)$; Cauchy's estimate on a disc of radius $|\varepsilon|/2$ gives $\varepsilon\,\partial_\varepsilon\vartheta=O(|\varepsilon|)$. □

9.3. **Tilted initial regions.** Write $\omega=\mathrm{e}^{\mathrm{i}\alpha}$. For $x_0\in[-\frac12,\frac12]$ let

$$\ell=\ell_{\alpha,x_0}=\{\mathrm{i}\pi+x_0+t\omega:\ t\in\mathbb{R}\},$$

and use the coordinates $\chi=\mathrm{i}\pi+x_0+(t+\mathrm{i}\varsigma)\omega$, $t,\varsigma\in\mathbb{R}$. We call $\ell$ *admissible for* $\varepsilon$ if

$$\cos\alpha\ge\tfrac35,\qquad -\operatorname{Im}(\varepsilon\bar\omega)\ge\tfrac12|\varepsilon|. \tag{16}$$

For $\varepsilon\in\mathcal{S}_r^{\mathrm{c}}$ put $\alpha(\varepsilon)=\frac12(\arg\varepsilon+\frac\pi2)\in(-\frac{\pi}{24},\frac{7\pi}{24})$. Then $\cos\alpha(\varepsilon)>\cos\frac{7\pi}{24}>0.6$ and $-\operatorname{Im}(\varepsilon\bar\omega)/|\varepsilon|=\sin(\frac\pi4-\frac12\arg\varepsilon)>\sin\frac{5\pi}{24}>0.6$. Hence, for $\varepsilon_0\in\mathcal{S}_r^{\mathrm{c}\prime}$, the line $\ell_{\alpha,x_0}$ with $|\alpha-\alpha(\varepsilon_0)|\le\frac{1}{20}$ is admissible for every $\varepsilon\in\mathcal{S}_r^{\mathrm{c}}$ with $|\arg\varepsilon-\arg\varepsilon_0|\le\frac{1}{20}$: then $\alpha\in(-\frac{\pi}{24},\frac{7\pi}{24})$, so $\cos\alpha>\cos\frac{7\pi}{24}>\frac35$, and $-\operatorname{Im}(\varepsilon\bar\omega)/|\varepsilon|\ge\sin(\frac{5\pi}{24}-\frac{1}{10})>\frac12$. At $\varepsilon(\theta)$ (real $b>\eta$), $\alpha=0$ is admissible; at real $\varepsilon>0$, $\alpha(\varepsilon)=\frac\pi4$. For fixed $\varepsilon$ the admissible pairs $(\alpha,x_0)$ form a rectangle. The distance from $2\pi\mathrm{i}k$ to $\ell$ is $|(2k-1)\pi\cos\alpha+x_0\sin\alpha|\ge\frac35\pi-\frac25>1.48$, so the strip $\{|\varsigma|\le\frac14\}$ lies in $\mathcal{G}^{\mathrm{c}}$.

**Lemma 9.3** (Tilted initial region)**.** *Let $\varepsilon\in\mathcal{S}_{r_0}^{\mathrm{c}}$, let $\ell$ be admissible for $\varepsilon$, and let $f_\chi(\chi)=\chi+\Delta(\chi)$ be $f_\varepsilon$ in the coordinate $\chi$.*

1. *On $\mathcal{G}^{\mathrm{c}}$ the normal component $\operatorname{Im}(\Delta\bar\omega)$ lies in $[\frac25|\varepsilon|,\frac65|\varepsilon|]$, and $f_\chi$ is injective on every convex subset of $\mathcal{G}^{\mathrm{c}}$.*
2. *$f_\chi(\ell)$ is the graph $\varsigma=g(t)$ of a real-analytic function with $\frac25|\varepsilon|\le g\le\frac65|\varepsilon|$ and $|g'|\le C|\varepsilon|^2$.*
3. *$M(t,s)=\mathrm{i}\pi+x_0+t\omega+s\Delta(\mathrm{i}\pi+x_0+t\omega)$ is an orientation-preserving real-analytic diffeomorphism of $\mathbb{R}\times[0,1]$ onto the closed strip $\hat H=\{0\le\varsigma\le g(t)\}$ between $\ell$ and $f_\chi(\ell)$.*
4. *$\chi\mapsto u(\chi)$ is injective on $\hat H$. Its image $H=H_\varepsilon^\ell$ together with $u_1,u_2$ is a closed Jordan region in $D_{C|\varepsilon|}$, bounded by the arcs $\gamma=u(\ell)$ and $f(\gamma)=u(f_\chi(\ell))$, which join $u_2$ ($t\to+\infty$) to $u_1$ ($t\to-\infty$) and meet only there.*
5. *$f_\varepsilon$ maps a neighbourhood of $\gamma$ biholomorphically onto a neighbourhood of $f(\gamma)$; in the coordinate $\chi$ it is $f_\chi$, which is injective on the convex strip $\{|\varsigma|\le\frac14\}$.*

*Proof.* (1) By Lemma 9.2(3), $\operatorname{Im}(\Delta\bar\omega)=-\operatorname{Im}(\varepsilon\bar\omega)-\operatorname{Im}(\varepsilon\vartheta\bar\omega)\in[\frac12|\varepsilon|-C|\varepsilon|^2,|\varepsilon|+C|\varepsilon|^2]$. If $\chi,\chi'$ lie in a convex subset of $\mathcal{G}^{\mathrm{c}}$, then $|f_\chi(\chi)-f_\chi(\chi')-(\chi-\chi')|\le C|\varepsilon|^2|\chi-\chi'|<|\chi-\chi'|$.

(2) The $t$-coordinate of $f_\chi(\mathrm{i}\pi+x_0+t\omega)$ is $t+\operatorname{Re}(\Delta\bar\omega)$, whose derivative is $1+O(|\varepsilon|^2)>0$; it is therefore a real-analytic bijection of $\mathbb{R}$, and (1) and $|\partial_\chi\Delta|\le C|\varepsilon|^2$ bound $g$ and $g'$.

(3) $\partial_tM=\omega(1+O(|\varepsilon|^2))$ and $\partial_sM=\Delta$, so the Jacobian is $\operatorname{Im}(\Delta\bar\omega)(1+O(|\varepsilon|^2))\ge\frac13|\varepsilon|>0$. $M$ maps $s=0$ onto $\ell$ and $s=1$ onto $f_\chi(\ell)$ bijectively, each segment $t=\text{const}$ to a segment of length $\le2|\varepsilon|$ joining them, and is proper. A proper local diffeomorphism that is a bijection on the boundary lines is a diffeomorphism onto the region between them.

(4) Points of the strip $\{|\varsigma|\le\frac14\}$ that differ by $2\pi\mathrm{i}k$, $k\ne0$, have $\varsigma$-coordinates differing by $2\pi|k|\cos\alpha>1$; so exp is injective on $\hat H$, and the Möbius map $q\mapsto u$ is injective. Along

$\ell$, $\operatorname{Re}\chi = x_0 + t\cos\alpha - \varsigma\sin\alpha$ tends to $\mp\infty$ as $t \to \mp\infty$, so $q \to 0$ ($u \to u_1$) and $q \to \infty$ ($u \to u_2$). The two boundary curves of $\hat H$ are disjoint by (2), and their images are disjoint arcs with common endpoints $u_1, u_2$; the image of $\hat H$ is the closure of the complementary component they bound that contains $u(\operatorname{int}\hat H)$. The inclusion in $D_{C|\varepsilon|}$ is Lemma 9.2(1).

(5) By Lemma 9.2(2), $f_\chi$ represents $f_\varepsilon$ in $\chi$, and (1) applies to the convex strip. □

Let $X = X_\varepsilon^\ell$ be the Riemann surface obtained from $H$ by identifying $u \in \gamma$ with $f_\varepsilon(u) \in f(\gamma)$, with charts near the seam given by Lemma 9.3(5). Since $\chi$ is injective on $\hat H$ we use it (or $Z = A_1\chi$) as the coordinate on $H$. We call the branch of $\chi$ obtained by applying $f_\chi^{\pm1}$ to points of $\hat H$ the *continued branch*.

**Lemma 9.4** (Flat model)**.** *Let $\ell$ be admissible for $\varepsilon \in \mathcal{S}^{\mathrm{c}}_{r_0}$. In the coordinate $Z$ let $\hat H_0$ be the straight strip between the line $A_1\ell$ and $A_1\ell + 1$, and let $C_0 = \hat H_0/(\zeta \sim \zeta + 1) = \mathbb{C}/\mathbb{Z}$. The map*

$$\mathcal{H}(A_1(\mathrm{i}\pi + x_0 + t\omega) + s) = A_1 M(t, s), \qquad t \in \mathbb{R},\ 0 \le s \le 1,$$

*induces a quasiconformal homeomorphism $\mathcal{H} : C_0 \to X$ with $|\mathcal{H}(\zeta) - \zeta| \le C|\varepsilon|$ on $\hat H_0$ and Beltrami coefficient $|\upsilon_{\mathcal{H}}| \le C|\varepsilon|$. Consequently:*

1. *$X$ is biholomorphic to $\mathbb{C}^*$, the ends at $u_1$ and $u_2$ corresponding to the punctures. Let $Z_X : X \to \mathbb{C}^*$ be a uniformisation with the $u_1$-end at $0$, and $z = (2\pi\mathrm{i})^{-1}\log Z_X$. Crossing the seam from $\gamma$ to $f(\gamma)$ increases $z$ by $1$. The inverse $z \mapsto u$, extended by $\tilde F(z+1) = f_\varepsilon(\tilde F(z))$ and by one application of $f_\chi^{-1}$ on the strip $\{|\varsigma| \le \frac14\}$, is a holomorphic solution of this equation on the region to the right of the lift of $f_\chi^{-1}(\ell)$.*
2. *If $\mathsf{p} \in X$ and the lift $z$ is normalised at $\mathsf{p}$, then for $|\operatorname{Im}(z - z(\mathsf{p}))| \le 1$ and $|\operatorname{Re}(z - z(\mathsf{p}))| \le 2$ the $Z$-coordinate $Z(z)$ of $\tilde F(z)$ on the continued branch satisfies $|Z(z) - Z(\mathsf{p}) - (z - z(\mathsf{p}))| \le C|\varepsilon|$.*

*Proof.* $A_1\Delta = 1 + \vartheta_1$ with $|\vartheta_1| \le C|\varepsilon|$ on $\mathcal{G}^{\mathrm{c}}$ (Lemma 9.2(2),(3)). In the source coordinate $\zeta = A_1(\mathrm{i}\pi + x_0 + t\omega) + s$ one has $\mathcal{H}(\zeta) = \zeta + s\vartheta_1$, so $|\mathcal{H}(\zeta) - \zeta| \le C|\varepsilon|$. The partial derivatives are $\partial_t\mathcal{H} = A_1\omega(1 + O(|\varepsilon|^2))$ and $\partial_s\mathcal{H} = 1 + \vartheta_1$, against $A_1\omega$ and $1$ for the identity. The unit vectors $A_1\omega/|A_1\omega|$ and $1$ make an angle whose sine is $|\operatorname{Im}(\omega\overline{\operatorname{Log}\lambda_1})|/|\operatorname{Log}\lambda_1| \ge \frac12 - C|\varepsilon|$, by $\operatorname{Log}\lambda_1 = -\varepsilon(1 + O(\varepsilon))$ and (16). So, in this basis, $D\mathcal{H} = I + O(|\varepsilon|)$ and $|\upsilon_{\mathcal{H}}| \le C|\varepsilon|$. $\mathcal{H}$ is a homeomorphism of the strips by Lemma 9.3(3); it maps the edge $s = 0$ to $A_1\ell$ and $s = 1$ to $A_1 f_\chi(\ell)$ with the same $t$, and hence respects the identifications $\zeta \sim \zeta + 1$ and $u \sim f_\varepsilon(u)$. Across $k$ seams, $\mathcal{H}(\zeta + k)$ is represented on the continued branch by $A_1 f_\chi^k(\mathcal{H}(\zeta)/A_1)$, and $A_1 f_\chi^k(\chi) - A_1\chi - k = O(|k||\varepsilon|)$ for $|k| \le 3$ by Lemma 9.2(2).

(1) $C_0 \cong \mathbb{C}^*$. Each end of $X$ is the image of an end of $C_0$, a punctured disc; quasiconformal maps preserve the infinite modulus of an annular end [19], so each end of $X$ is a punctured disc and $X \cong \mathbb{C}^*$. In $C_0$ the passage from the edge $s = 0$ to $s = 1$ adds 1. The $u_1$-end is $t \to -\infty$, where $\operatorname{Im}\zeta = t\operatorname{Im}(A_1\omega) + O(1) \to +\infty$ because $\operatorname{Im}(A_1\omega) = -\operatorname{Im}(\operatorname{Log}\lambda_1\bar\omega)/|\operatorname{Log}\lambda_1|^2 < 0$. Since $\mathcal{H}$ preserves orientation and ends, the uniformisation with the $u_1$-end at $0$ has the same properties. The last assertion is the usual lift of the uniformisation; the extra step $f_\chi^{-1}$ is allowed by Lemma 9.3(5).

(2) Put $\xi = \exp(2\pi\mathrm{i}(\zeta - \zeta_{\mathsf{p}}))$ on $C_0$, $\zeta_{\mathsf{p}} = \mathcal{H}^{-1}(\mathsf{p})$, and let $\upsilon$ be the Beltrami coefficient of $\mathcal{H}$ transported to $\hat{\mathbb{C}}$. For complex $\mathsf{t}$ with $|\mathsf{t}| < R := 1/(2\|\upsilon\|_\infty)$ let $\mathcal{W}_{\mathsf{t}}$ be the solution with coefficient $\mathsf{t}\upsilon$ fixing $0, 1, \infty$. By [20, Thm. 11], $\mathsf{t} \mapsto \mathcal{W}_{\mathsf{t}}(\xi)$ is holomorphic; $\mathcal{W}_0 = \mathrm{id}$; and the $\mathcal{W}_{\mathsf{t}}$ are 3-quasiconformal and normalised, hence uniformly bounded, by some $M_0$, on the annulus $A = \{\mathrm{e}^{-3\pi} \le |\xi| \le \mathrm{e}^{3\pi}\}$ [19, Ch. II]. Schwarz's lemma applied to $\mathsf{t} \mapsto \mathcal{W}_{\mathsf{t}}(\xi) - \xi$ on $|\mathsf{t}| < R$ gives $|\mathcal{W}_1(\xi) - \xi| \le 2(M_0 + \mathrm{e}^{3\pi})\|\upsilon\|_\infty \le C|\varepsilon|$ on $A$, since $R \ge 1$ for small $|\varepsilon|$. By a degree argument on the boundary circles of $A$, $\mathcal{W}_1^{-1}$ also moves points of $\{\mathrm{e}^{-2\pi} \le |\xi| \le \mathrm{e}^{2\pi}\}$ by $O(|\varepsilon|)$.

$\mathcal{W}_1 \circ \exp(2\pi i(\mathcal{H}^{-1} - \zeta_{\mathsf{p}}))$ is a uniformisation of $X$ normalised at $\mathsf{p}$, so it equals $\exp(2\pi i(z - z(\mathsf{p})))$. Taking logarithms on the annulus, and combining with the bounds on $\mathcal{H}$ across at most three seams, gives (2). □

**Lemma 9.5** (The base orbit and the marked point)**.** *There are $r_2 \le r_0$, a small $\rho_1 > 0$ and $N_1$ such that the following holds for $\varepsilon \in \mathcal{S}^{\mathrm{c}}_{r_2}$ and every line $\ell$ admissible for $\varepsilon$. Let $\chi^*_n = \chi(u^*_n)$, $n \ge N_1$, be the branch continued from the principal value at $n = N_1$ (so $\chi^*_{n+1} = f_\chi(\chi^*_n)$), and $Z^*_n = A_1\chi^*_n$.*

1. *For $N_1 \le n \le N_1 + 10/|\varepsilon|$, $Z^*_n = Z^*_{N_1} + (n - N_1) + \mathcal{E}_n$ with $|\mathcal{E}_n| \le C\log n + C|\varepsilon|n$, and $Z^*_{N_1}$ is within relative distance $\frac{1}{5}$ of a real number $\ge 4/\rho_1$.*
2. *On that range the normal coordinate $\varsigma(\chi^*_n)$ is strictly increasing. Let $n_0$ be the first index with $\varsigma(\chi^*_{n_0}) \ge 0$. Then $n_0 \le N_1 + 8/|\varepsilon|$ and $\chi^*_{n_0} \in \hat{H} \setminus f_\chi(\ell)$.*
3. *For $N_1 \le n \le n_0 + 1$, the $3|\varepsilon|$-neighbourhood of $\chi^*_n$ (equivalently, approximately, the 3-neighbourhood of $Z^*_n$) lies in $\mathcal{G}^{\mathrm{c}} \cup \{|\chi| < 1\}$ and in $\{|u(\chi)| \le 2\rho_1\}$.*

*The* marked point *of $X^\ell_\varepsilon$ is the point represented by $\chi^*_{n_0} \in \hat{H}$.*

*Proof.* At $\varepsilon = 0$ the orbit $f_0^n(u^*)$ increases to 0 along the negative axis, with $f_0^n(u^*) = -2/n + O(\log n/n^2)$. Choose $N_1$ with $|f_0^{N_1}(u^*)| \le \rho_1/2$, and then $r_2$ so small that $|u^*_{N_1} - f_0^{N_1}(u^*)| \le \frac{1}{10}|f_0^{N_1}(u^*)|$ for $|\varepsilon| < r_2$. Since $q(u) = 1 + (u_2 - u_1)/(u - u_2)$, $Z^*_{N_1} = -2/u^*_{N_1}\,(1 + O(|\varepsilon|/\rho_1) + O(\varepsilon))$, which gives the statement about $Z^*_{N_1}$. By Lemma 9.2(2), $Z^*_{n+1} - Z^*_n = A_1\Delta(u^*_n) = 1 + e_n$ with $|e_n| \le C(|u^*_n| + |\varepsilon|)$ while $u^*_n \in D_{\rho_0}$. While $|\chi| \le 1$, Lemma 9.2(1) gives $|u| \le |u_2| + |u_1 - u_2|/(0.63|\chi|) \le 3.2/|Z| + C|\varepsilon|$; on $\mathcal{G}^{\mathrm{c}}$, $|u| \le C|\varepsilon|$. By induction $\operatorname{Re} Z^*_n$ increases by at least $\frac{1}{2}$ per step, and $|Z| \ge |Z^*_{N_1}| - 3 \ge 3/\rho_1$ on the 3-neighbourhoods, so $|u| \le 1.1\rho_1 + C|\varepsilon|$ there; this gives (1) and the last part of (3). Multiplying by $1/A_1 = -\varepsilon(1 + O(\varepsilon))$, $\chi^*_n$ lies within $C|\varepsilon|\log(1/|\varepsilon|) + C|\varepsilon|/\rho_1$ of the ray $\{-\varepsilon\tau : \tau \ge 0\}$ in the range of (1).

The origin has $\varsigma(0) = -\pi\cos\alpha + x_0\sin\alpha \le -1.48$ and $|\varsigma(0)| \le \sqrt{\pi^2 + 1/4}$, and along the ray $\varsigma$ increases at rate $-\operatorname{Im}(\varepsilon\bar{\omega}) \ge |\varepsilon|/2$; so the ray meets $\ell$ at $\tau \le 6.4/|\varepsilon|$. Before that crossing the ray stays at distance $\ge 2\pi\sin\frac{5\pi}{12} > 6$ from $2\pi i k$, $k \le -1$, and in $\{\varsigma < 0\}$, at distance $\ge 1.48$ from $2\pi i k$, $k \ge 1$. With the smallness of the deviation this gives (3). The normal component of each step is at least $|\varepsilon|(\frac{1}{2} - C\rho_1 - C|\varepsilon|) > 0$, so $\varsigma(\chi^*_n)$ is strictly increasing, and $n_0$ exists with the stated bound. Then $-\frac{6}{5}|\varepsilon| \le \varsigma(\chi^*_{n_0-1}) < 0$, so $\chi^*_{n_0-1}$ lies in the convex strip $\{|\varsigma| \le \frac{1}{4}\}$, on which each $\mathrm{id} + s\Delta$, $0 \le s \le 1$, is injective. This isotopy carries $\ell$ to $f_\chi(\ell)$ and the side $\{\varsigma < 0\}$ of $\ell$ into the corresponding side of $f_\chi(\ell)$; hence $\chi^*_{n_0} \in \hat{H} \setminus f_\chi(\ell)$. □

The physical point $u^*_n$ may lie in $H$ for several $n$, because $H$ spirals around $u_1$ and $u_2$ when $\alpha \ne 0$. These other visits are points of $X$ different from the marked point, and correspond to the translated roots $c_j$ discussed after Definition 8.1. Only the first crossing on the continued branch is used.

**Lemma 9.6** (Shadowing the base orbit)**.** *Let $W_0 = \{z : \operatorname{dist}(z, [0,1]) < \frac{1}{4}\}$. Under the hypotheses of Lemma 9.5, let $z$ be the lift of Lemma 9.4 normalised by $z = n_0$ at the marked point. Then for $0 \le k \le n_0 - N_1$ the branch $g^k_\varepsilon$ is holomorphic on $\tilde{F}(n_0 + W_0)$, and the $Z$-coordinates of $g^k_\varepsilon(\tilde{F}(n_0 + w))$ lie within $\frac{1}{10}$ of $Z^*_{n_0-k} + w$ ($w \in W_0$). For $|\varepsilon|$ small the $N_1$ further applications of the same branch $g_\varepsilon$ (principal logarithm) are defined and jointly holomorphic on $g_\varepsilon^{n_0-N_1}(\tilde{F}(n_0 + W_0))$ and its images, which stay in $\{\operatorname{Re}(1+v) > 0\}$ near the real segment $[u^*, 0]$.*

*Proof.* By Lemma 9.4(2) with $\mathsf{p}$ the marked point, the $Z$-coordinates of $\tilde{F}(n_0 + w)$ lie within $C|\varepsilon|$ of $Z^*_{n_0} + w$. In the coordinate $Z$, $f_\varepsilon$ is $Z \mapsto Z + A_1\Delta$, whose derivative is $1 + \partial_\chi\Delta$, and $g_\varepsilon$ is its inverse $g_Z$, with $|g'_Z - 1| \le 2|\partial_\chi\Delta|$ evaluated at $g_Z(Z)$. By Lemma 9.2(3) and $|\chi| = |Z|/|A_1|$, $|g'_Z - 1| \le \delta(Z) := C|\varepsilon|^2 + C/|Z|^2$ on the 3-neighbourhood of the orbit segment,

where Lemma 9.5(3) applies and $|Z| \geq |Z^*_{N_1}| - 3$. Since $g^k_Z(Z^*_{n_0}) = Z^*_{n_0-k}$, the mean-value inequality gives, as long as the images stay in that neighbourhood,

$$|g^k_Z(Z) - Z^*_{n_0-k} - (Z - Z^*_{n_0})| \leq \Big(\prod_{j<k}(1 + 2\delta(Z^*_{n_0-j})) - 1\Big)|Z - Z^*_{n_0}| \leq (\mathrm{e}^{C|\varepsilon|+C\rho_1} - 1) \cdot 2,$$

because on the 3-neighbourhood of $Z^*_{n_0-j}$ one has $\delta \leq 2\delta(Z^*_{n_0-j})$, using $\sum_j |\varepsilon|^2 \leq C|\varepsilon|$ over $n_0 \leq N_1 + 8/|\varepsilon|$ steps and $\sum_j |Z^*_{n_0-j}|^{-2} \leq C/|Z^*_{N_1}| \leq C\rho_1$ (the real parts grow by at least $\frac{1}{2}$ per step). For small $\rho_1$ and $|\varepsilon|$ the right side is below $\frac{1}{20}$, so by induction the images stay in the neighbourhood and the claim holds. These sets lie in $\{|u| \leq 2\rho_1\} \subset D_{\rho_0}$, where $g_\varepsilon \circ f_\varepsilon = \mathrm{id}$ (Lemma 9.1(3)).

At $k = n_0 - N_1$ the set lies within $\frac{1}{10}$, in $Z$, of $Z^*_{N_1} + W_0$. As $\varepsilon \to 0$, $Z = A_1\chi$ converges on this region to $-2/u$ locally uniformly, so the set lies, for small $|\varepsilon|$, in a neighbourhood of the compact set $\overline{K_0}$, $K_0 = \{u : -2/u \in Z_0 + W_0^{1/5}\}$, where $Z_0 = -2/f_0^{N_1}(u^*)$ and $W_0^{1/5}$ is the $\frac{1}{5}$-neighbourhood of $W_0$. At $\varepsilon = 0$ the attracting Fatou coordinate is $\Phi_{\mathrm{att}}(u) = -2/u + \frac{1}{3}\log(-u) + O(u)$; on $\overline{K_0}$ the term $\frac{1}{3}\log(-u)$ varies by $O(\rho_1)$, so $\Phi_{\mathrm{att}}(\overline{K_0}) \subset c_0 + N_1 + W_0^{1/5+C\rho_1}$ for a real constant $c_0$. The Fatou parametrisation $P_0 = \Phi_{\mathrm{att}}^{-1}$ extends to an entire function with $P_0(z+1) = f_0(P_0(z))$, and $g_0(P_0(z)) = P_0(z-1)$ near the real axis, where $1 + P_0(z-1)$ is close to $(0, 1]$. Hence $g_0^{N_1}$ is holomorphic on a neighbourhood of $\overline{K_0}$, and all intermediate images lie in a compact subset of $\{\mathrm{Re}(1+v) > 0\}$ near $[u^*, 0]$. This is an open condition, and $g_\varepsilon \to g_0$ uniformly on compact subsets of $\{\mathrm{Re}(1+v) > 0\}$; by compactness it persists for small $|\varepsilon|$, jointly holomorphically in $\varepsilon$. □

**Definition 9.7.** For $\varepsilon \in \mathcal{S}^{\mathrm{c}}_{r_2}$ and an admissible line $\ell$, the *normalised tilted solution* is

$$F^\ell_\varepsilon(z) = g^{n_0}_\varepsilon(\tilde{F}(z + n_0)), \qquad z \in W_0,$$

with $\tilde{F}$ lifted so that the marked point has $z = n_0$ and $g_\varepsilon$ the principal branch of Lemma 9.1(3), which is holomorphic along the way by Lemma 9.6. It is holomorphic on $W_0$, $F^\ell_\varepsilon(0) = u^*$, and $F^\ell_\varepsilon(z+1) = f_\varepsilon(F^\ell_\varepsilon(z))$ where both sides are defined; in $w = \mathrm{e}(1+u)$ it satisfies $F(z+1) = b^{F(z)}$ and $F(0) = 1$.

**Lemma 9.8** (Change of line)**.** *Let $\ell, \ell'$ be admissible for the same $\varepsilon \in \mathcal{S}^{\mathrm{c}}_{r_2}$. Then $F^\ell_\varepsilon = F^{\ell'}_\varepsilon$.*

*Proof.* Since the admissible pairs form a rectangle, it suffices to treat $|\alpha - \alpha'| \leq \frac{1}{100}$ and $|x_0 - x_0'| \leq \frac{1}{100}$. Let $\chi = \mathrm{i}\pi + x_0 + (t + \mathrm{i}\varsigma)\omega \in \hat{H}$ and let $\varsigma'$ be its normal coordinate for $\ell'$. Then $|\varsigma'| \leq \frac{1}{100}(1+|t|) + O(|\varepsilon|)$. Each application of $f^{\pm 1}_\chi$ changes $\varsigma'$ by at least $\frac{2}{5}|\varepsilon|$ in the same direction (Lemma 9.3(1)) and moves the point by at most $2|\varepsilon|$. So after at most $(1 + |t|)/(40|\varepsilon|)$ steps, forward or backward, the orbit of $\chi$ on the continued branch reaches the strip $\hat{H}'$ of $\ell'$, within distance $(1 + |t|)/20$ of $\chi$. The distance from $\chi$ to the holes is at least $\max(1.48, 0.6|t| - 0.5)$, which exceeds $(1 + |t|)/20 + \frac{1}{2}$; so this part of the orbit stays in a disc about $\chi$ contained in $\mathcal{G}^{\mathrm{c}}$, on which $f_\chi$ is injective. As in the proof of Lemma 9.5(2), exactly one iterate $f^{n(\chi)}_\chi(\chi)$ lies in $\hat{H}' \setminus f_\chi(\ell')$. The map $\chi \mapsto f^{n(\chi)}_\chi(\chi)$ is locally a fixed iterate of $f_\chi$, it jumps by one iterate exactly across $\ell'$, where the two values are identified in $X^{\ell'}$, and it satisfies $n(f_\chi(\chi)) = n(\chi) - 1$ on $\ell$. It therefore induces a holomorphic map $X^\ell \to X^{\ell'}$, and the same construction from $\ell'$ gives its inverse. It preserves the two ends. The continued base orbit $\chi^*_n$ is common to both lines, and both marked points are its first crossings; the orbit segment between them lies in the region just described, so the map sends the marked point $\chi^*_{n_0}$ of $\ell$ to $f^{n_0'-n_0}_\chi(\chi^*_{n_0}) = \chi^*_{n_0'}$, the marked point of $\ell'$. In the lifts normalised at the marked points ($z = n_0$ and $z = n_0'$), and since

$f_\chi$ corresponds to $z \mapsto z+1$, one gets $\tilde F'(z+n_0') = f_\varepsilon^{n_0'-n_0}(\tilde F(z+n_0))$ near $z=0$. Applying the inverse branches of Definition 9.7 gives $F_\varepsilon^{\ell'} = F_\varepsilon^{\ell}$ on $W_0$. □

We write $F_\varepsilon$ for the common normalised tilted solution.

### 9.4. **Holomorphic dependence.**

**Proposition 9.9** (The tilted family is holomorphic)**.** *There is $r_3 \le r_2$ such that $(\varepsilon, z) \mapsto F_\varepsilon(z)$ is jointly holomorphic on $\mathcal{S}^{c\prime}_{r_3} \times W_0$.*

*Proof.* Fix $\varepsilon_0 \in \mathcal{S}^{c\prime}_{r_3}$ and the disc $U_0 = \{|\varepsilon - \varepsilon_0| < |\varepsilon_0|/40\}$; it lies in $\mathcal{S}^{c}_{r_2}$ and $|\arg\varepsilon - \arg\varepsilon_0| < \frac{1}{20}$ on it. Choose $\alpha \neq 0$ with $|\alpha - \alpha(\varepsilon_0)| \le \frac{1}{20}$, and then $x_0 \in [-\frac12, \frac12]$, such that $\chi^*_{n_0}(\varepsilon_0)$ is not on $\ell$ or $f_\chi(\ell)$; for $\alpha \neq 0$ a change of $x_0$ moves the line in its normal direction by $x_0 \sin\alpha$, so all but finitely many small values of $x_0$ will do. By the admissibility remark above $\ell = \ell_{\alpha,x_0}$ is admissible for all $\varepsilon \in U_0$. Shrink $U_0$ so that $n_0$ is constant and $\chi^*_{n_0}(\varepsilon)$ stays in the interior of $\hat H_\varepsilon$. By Lemma 9.8 this choice does not affect $F_\varepsilon$.

Let $M_\varepsilon$ be the diffeomorphism of Lemma 9.3(3) for $\ell$. The map $\mathcal{J}_\varepsilon = M_\varepsilon \circ M_{\varepsilon_0}^{-1} : \hat H_{\varepsilon_0} \to \hat H_\varepsilon$ respects the seam identifications and induces a homeomorphism $X_{\varepsilon_0} \to X_\varepsilon$. At each point its derivative sends $v_1 = \partial_t M_{\varepsilon_0} = \omega(1 + O(|\varepsilon_0|^2))$ to $w_1 = \partial_t M_\varepsilon = \omega(1 + O(|\varepsilon_0|^2))$ and $v_2 = \Delta_{\varepsilon_0}$ to $w_2 = \Delta_\varepsilon$, with $|w_2 - v_2| \le 1.1|\varepsilon - \varepsilon_0| \le 0.03|\varepsilon_0|$ by Lemma 9.2(5). Writing it as $\zeta \mapsto \mathsf{a}\zeta + \mathsf{b}\bar\zeta$,

$$\mathsf{b} = \frac{(w_1 - v_1)v_2 - (w_2 - v_2)v_1}{2\mathrm{i}\,\mathrm{Im}(v_2\bar v_1)}, \qquad |\,\mathrm{Im}(v_2\bar v_1)| \ge \tfrac25|\varepsilon_0|(1 - C|\varepsilon_0|),$$

so $|\mathsf{b}| \le 0.05$, $|\mathsf{a} - 1| \le 0.05$, and the Beltrami coefficient $\upsilon_\varepsilon = \mathsf{b}/\mathsf{a}$ of $\mathcal{J}_\varepsilon$ satisfies $|\upsilon_\varepsilon| \le \frac{1}{10}$. It is holomorphic in $\varepsilon$ at each point, since $\mathsf{a}, \mathsf{b}$ are affine in $w_1, w_2$; being uniformly bounded, it is holomorphic as an $L^\infty$-valued function. Transport $\upsilon_\varepsilon$ to $\hat{\mathbb{C}}$ by a fixed uniformisation $Z_{\varepsilon_0}$ of $X_{\varepsilon_0}$, and let $\mathcal{W}_\varepsilon$ be the normalised solution fixing $0, \infty$ (the ends) and 1. By [20, Thm. 11], $\mathcal{W}_\varepsilon(\xi)$ is holomorphic in $\varepsilon$ for each $\xi$, and $Z_\varepsilon = \mathcal{W}_\varepsilon \circ Z_{\varepsilon_0} \circ \mathcal{J}_\varepsilon^{-1}$ is a uniformisation of $X_\varepsilon$ with the ends at $0, \infty$.

In the interior of $\hat H_{\varepsilon_0}$, $\mathcal{J}_\varepsilon$ is a real-analytic diffeomorphism, holomorphic in $\varepsilon$ at fixed source points, and $\upsilon_\varepsilon$ is real-analytic in the source point and holomorphic in $\varepsilon$. By the interior Schauder estimates for the Beltrami equation [18, Ch. V], applied to difference quotients in $\varepsilon$, $(\varepsilon, \xi) \mapsto \mathcal{W}_\varepsilon(\xi)$ is $C^1$ there. The maps $\mathcal{W}_\varepsilon$ and $\mathcal{J}_\varepsilon \circ Z_{\varepsilon_0}^{-1}$ have the same Beltrami coefficient $\upsilon$ (transported), i.e. $Z_\varepsilon$ is conformal. Put $\xi_\varepsilon(\chi) = Z_{\varepsilon_0}(\mathcal{J}_\varepsilon^{-1}(\chi))$. Differentiating $\mathcal{J}_\varepsilon(Z_{\varepsilon_0}^{-1}(\xi_\varepsilon(\chi))) = \chi$ in $\bar\varepsilon$ at fixed $\chi$, and using that $\mathcal{J}_\varepsilon$ is holomorphic in $\varepsilon$ at fixed source points, gives $\partial_{\bar\varepsilon}\xi_\varepsilon + \upsilon\,\partial_{\bar\varepsilon}\overline{\xi_\varepsilon} = 0$. With $\partial_{\bar\xi}\mathcal{W}_\varepsilon = \upsilon\,\partial_\xi\mathcal{W}_\varepsilon$ and $\partial_{\bar\varepsilon}\mathcal{W}_\varepsilon = 0$ this gives $\partial_{\bar\varepsilon}[\mathcal{W}_\varepsilon(\xi_\varepsilon(\chi))] = \partial_\xi\mathcal{W}_\varepsilon\,(\partial_{\bar\varepsilon}\xi_\varepsilon + \upsilon\,\partial_{\bar\varepsilon}\overline{\xi_\varepsilon}) = 0$. So $(\varepsilon, \chi) \mapsto Z_\varepsilon(\chi)$ is jointly holomorphic in the interior of the strips.

With $\chi^*_{n_0}(\varepsilon)$ in the interior, $\zeta_0(\varepsilon) = (2\pi\mathrm{i})^{-1}\log Z_\varepsilon(\chi^*_{n_0}(\varepsilon))$ is holomorphic, and $\tilde F_\varepsilon(z + n_0) = u_\varepsilon(Z_\varepsilon^{-1}(\mathrm{e}^{2\pi\mathrm{i}(z+\zeta_0(\varepsilon))}))$ is jointly holomorphic wherever the point lies in the interior of $\hat H_\varepsilon$, by the holomorphic inverse function theorem. Points of $n_0 + W_0$ whose images lie on the seam, or in the extra step of Lemma 9.4(1), are handled in the same way with finitely many further admissible angles $\alpha$ (by compactness of $\overline{W_0}$), for which they are interior; Lemma 9.8 shows that the two descriptions define the same function. Composing with the inverse branches of Lemma 9.6, which are jointly holomorphic, gives joint holomorphy on $U_0 \times W_0$. On overlapping discs the local families coincide by Lemma 9.8. □

### 9.5. **Abel coordinates on the tilted region.**

**Lemma 9.10** (Koenigs times along orbits)**.** *Let $b < \eta$ be real, so $\varepsilon > 0$, and put $\Psi = A_1\log(u - u_1) + A_2\log(u - u_2)$, $A_2 = 1/\log\lambda_2$, and $F_b = \Psi \circ f_b - \Psi - 1$ as in Lemma 4.2. Let $P$ be a*

*path in the $\chi$-plane such that for each $\chi \in P$ the forward orbit $f_\chi^k(\chi)$ stays in $\mathcal{G}^c$ and $\operatorname{Re} f_\chi^k(\chi)$ decreases by at least $\varepsilon/2$ per step. Then, along $P$, the branch of $R^{-1}$ continued along $P$ satisfies*

$$R^{-1} = \Psi + \sum_{k\geq 0} F_b \circ f^k + c_+, \qquad \sum_{k\geq 0} |F_b(f^k(u))| \leq C\varepsilon,$$

*with a constant $c_+$. The same holds for $S^{-1}$ with backward orbits ($f_\chi^{-k}$, Re increasing by at least $\varepsilon/2$ per step) and $\Psi - \sum_{k\geq 1} F_b \circ f^{-k} + c_-$.*

*Proof.* By Lemma 4.2 and Lemma 9.2(1), $|F_b(f^k(u))| \leq MC\varepsilon^2 \min(|e^{\chi_k}|, |e^{-\chi_k}|) \leq C\varepsilon^2 e^{-|\operatorname{Re}\chi_k|}$; since $\operatorname{Re}\chi_k$ changes by at least $\varepsilon/2$ per step, the sum is at most $C\varepsilon^2 \cdot \sum_k e^{-|\operatorname{Re}\chi_0 - k\varepsilon/2|} \leq C\varepsilon$. The orbit converges to $u_1$ because $\operatorname{Re}\chi_k \to -\infty$. As in the proof of Theorem 4.4(1), $\lim_n [\Psi(f^n u) - n] = \Psi(u) + \sum_{k\geq 0} F_b(f^k u)$ exists and equals $A_1 \log \sigma_1(u)$ plus a constant, because $\sigma_1(f^n u) = \lambda_1^n \sigma_1(u)$ and $A_1 \log \lambda_1 = 1$; the constant depends only on the branches, which are continued along $P$. The backward case is identical with the reflected inverse family of Theorem 4.4(2). □

9.6. **The two ends of the corridor.**

**Proposition 9.11** (Exterior end)**.** *For $0 < \theta < \theta_1$ (real $b \in (\eta, \eta + r_1)$), $F_{\varepsilon(\theta)}$ is Kneser's real-normalised solution on $W_0$.*

*Proof.* The line $\ell = i\pi + \mathbb{R}$ ($\alpha = 0$, $x_0 = 0$) is admissible for $\varepsilon(\theta)$, because $-\operatorname{Im}\varepsilon(\theta) = \theta$. Since $\chi \in i\pi + \mathbb{R}$ means $q < 0$ and $u_2 = \overline{u_1}$, $\gamma = u(\ell)$ is the open segment between the conjugate fixed points, i.e. in $w = e(1+u)$ the chord $\gamma_b$ of Proposition 10.1, and $f(\gamma)$ is its image $E_b(\gamma_b)$. Hence $H$ is the initial region $H_b$ and $X$ is the cylinder of that proposition, with the upper end at $L_1 = L_+$. There the uniformisation takes the upper end to 0 and is normalised so that $z(1) = 0$, with $z$ real and increasing along the positive axis. For conjugate $u_1, u_2$ the real axis is $\{|q| = 1\}$, i.e. $\operatorname{Re}\chi = 0$, and the continued branch of the real base orbit moves up the imaginary $\chi$-axis from 0; it crosses the chord between $u^*_{n_0-1}$ and $u^*_{n_0}$, and $u^*_{n_0} = E_b^{n_0}(1)$ has Kneser coordinate $n_0$. So our lift is Kneser's, and our inverse branches are the real logarithms along the orbit, as in Kneser's normalisation. By Lemma 9.8 the admissible line used in the family may be replaced by this one. □

**Proposition 9.12** (Interior end)**.** *There is $r_4 > 0$ such that $F_\varepsilon = K^W_{b(\varepsilon)}$ on $W_0$ for real $\varepsilon \in (0, r_4)$. Here $r_4$ is chosen so that $b(\varepsilon) \in (b_1, \eta)$ with the closeness conditions of Theorem 6.3, and so that $0.39h + C_{\mathrm{im}} < d - D$, where $C_{\mathrm{im}}$ is the constant in (17) below and $d, D$ are as in Lemma 6.2 and Theorem 6.3.*

*Proof.* Let $\varepsilon \in (0, r_4)$ be real, so $b < \eta$ is real, $\lambda_1 \in (0,1)$ and $u_1 < 0 < u_2$. Take $\alpha = \frac{\pi}{4}$, $x_0 = 0$, so that $\chi = i\pi + (t + i\varsigma)e^{i\pi/4}$ has $\operatorname{Re}\chi = (t - \varsigma)/\sqrt{2}$ and $\operatorname{Im}\chi = \pi + (t+\varsigma)/\sqrt{2}$. Put $h = 2\pi/|\log\lambda_1|$. The real line of the $u$-plane appears as follows: $u < u_1$ is $\{\operatorname{Im}\chi \in 2\pi\mathbb{Z},\ \operatorname{Re}\chi < 0\}$, the segment $(u_1, u_2)$ is $\{\operatorname{Im}\chi \in \pi + 2\pi\mathbb{Z}\}$, and $u > u_2$ is $\{\operatorname{Im}\chi \in 2\pi\mathbb{Z},\ \operatorname{Re}\chi > 0\}$; the lower half-plane is $\{0 < \operatorname{Im}\chi < \pi\}$ modulo $2\pi$, because $q \mapsto u$ has negative determinant $u_1 - u_2$. Since $f_\varepsilon$ is real and $\Delta$ is real at $u_1, u_2$, Lemma 9.2 gives $\operatorname{Im}\Delta = O(\varepsilon^2)$ on $\mathcal{G}^c$ and $\operatorname{Im}\Delta = O(\varepsilon^2 |e^{\mp\chi}|)$ near the two ends.

*Marked point.* The base orbit is real, $u^*_n < u_1$, so $\chi^*_n$ is real and moves to the left. On the real axis $\varsigma = -(\operatorname{Re}\chi + \pi)/\sqrt{2}$, so the marked point $p_c := \chi^*_{n_0}$ lies in $[-\pi - 2\varepsilon, -\pi]$, with $t \approx -\pi\sqrt{2}$.

*The two charts.* Let $U \subset X$ and $V \subset X$ be the images of $M(\{t < \pi\sqrt{2} - 1\})$ and $M(\{t > -\pi\sqrt{2} + 1\})$. They are open, and saturated for the seam identification, since $M(t, 0) \sim M(t, 1)$ preserves $t$. For $\chi \in M(\{t < \pi\sqrt{2} - 1\})$ the forward orbit $f_\chi^k(\chi)$ moves left by at least $\varepsilon/2$ per step, and its imaginary part drifts by $O(\varepsilon)$ in total (the sum of $O(\varepsilon^2)$ over $O(1/\varepsilon)$ steps in $|\operatorname{Re}\chi| \leq 4$

plus the geometric tail at the end). Such points with $\operatorname{Re}\chi \geq -\frac{1}{2}$ have $\operatorname{Im}\chi \in (\pi-1, 2\pi-0.7)$, so these orbits stay in $\mathcal{G}^c$. Symmetrically, backward orbits of points of $M(\{t > -\pi\sqrt{2}+1\})$ move right, stay in $\mathcal{G}^c$ and converge to $u_2$. So Lemma 9.10 applies on both sets, and the attracting and repelling Koenigs coordinates $\sigma_1$ and $\tau$ of Section 3 are defined and nonzero along these orbits. Let $r_+$ be the branch of $R^{-1} = \log(\sigma_1/\sigma_1(u^*))/\log\lambda_1$ on $M(\{t < \pi\sqrt{2}-1\})$ continued from $p_c$, where $r_+(p_c) = n_0$ is real ($\sigma_1(u^*_{n_0})/\sigma_1(u^*) = \lambda_1^{n_0} > 0$). By Lemma 9.10, $\partial_\chi r_+ = A_1 + O(1)$ on convex subsets at distance $\geq \frac{1}{2} + \frac{1}{10}$ from the holes. For $\chi \in \ell$, $\sigma_1 \circ f = \lambda_1\sigma_1$ gives $r_+(f_\chi(\chi)) - r_+(\chi) \in 1 + \mathrm{i}h\mathbb{Z}$ for the continuation along $s \mapsto M(t,s)$, a segment of length $O(\varepsilon)$ on which $r_+$ changes by $O(1)$, far less than $h$. Hence $r_+ \circ f_\chi = r_+ + 1$ on $\ell$, and $r_+$ descends to a holomorphic map $U \to \mathbb{C}/\mathbb{Z}$.

Let $(t_c, \varsigma_c)$ be the coordinates of $p_c$, so $0 \leq \varsigma_c < g(t_c)$, and follow the curve $\varsigma = \varsigma_c\, g(t)/g(t_c)$, which lies in $\hat{H}$, in the direction of increasing $t$. Since $|g'| \leq C\varepsilon^2$ and $g \geq \frac{2}{5}\varepsilon$, $\operatorname{Im}\chi = \pi + (t+\varsigma)/\sqrt{2}$ increases strictly along it, from 0 at $p_c$. Let $u_m \in (u_1, u_2)$ be the point where it equals $\pi$. Let $r_-$ be the branch of $S^{-1}$ on $M(\{t > -\pi\sqrt{2}+1\})$ that is real at $u_m$; it descends to $V$ in the same way.

*Imaginary parts.* In Lemma 9.10, $\Psi = A_1\chi + (A_1 + A_2)\log(u - u_2)$, with $A_1 = -h/2\pi$, $A_2 = \frac{h}{2\pi}(1 + O(\varepsilon))$ and $A_1 + A_2 \to \frac{1}{3}$ (Lemma 4.1), and $\arg(u - u_2) = \arg(u_1 - u_2) - \arg(1 - e^\chi)$. On $U$ the continued $\arg(1 - e^\chi)$ is bounded, because $\operatorname{Im}\chi < 2\pi - 0.7$ wherever $\operatorname{Re}\chi \geq -\frac{1}{2}$ and $|e^\chi| \leq e^{-1/2}$ elsewhere; on $U \cap V$ the range of $\operatorname{Im}\chi$ is bounded. Hence, with the branches fixed above,

$$\left|\operatorname{Im} r_+ + \tfrac{h}{2\pi}\operatorname{Im}\chi\right| \leq C_{\mathrm{im}} \text{ on } U, \qquad \left|\operatorname{Im} r_- + \tfrac{h}{2\pi}(\operatorname{Im}\chi - \pi)\right| \leq C_{\mathrm{im}} \text{ on } U \cap V. \tag{17}$$

On $V \setminus U$, $\operatorname{Im}\chi > 2\pi - 1/\sqrt{2}$. There $\arg(1 - e^\chi) = \operatorname{Im}\chi - \pi + O(1)$ where $\operatorname{Re}\chi \geq 0$, and $A_1 - (A_1 + A_2) = -A_2$, so $\operatorname{Im} r_- \leq -\min(|A_1|, A_2)(\operatorname{Im}\chi - \pi) + C$. Along the curve from $p_c$ to $u_m$, $\operatorname{Im}\chi$ increases from 0 to $\pi$, so the path runs through the lower half-plane: $\arg(u - u_1)$ increases by $\pi$ and $\arg(u - u_2)$ returns to its initial value. Hence $\operatorname{Im} r_+(u_m) = A_1\pi + O(\varepsilon) = -h/2 + O(\varepsilon)$. By Lemma 3.1, $\sigma_1(u_m)/\sigma_1(u^*) < 0$, so $\operatorname{Im} r_+(u_m) \in -h/2 + h\mathbb{Z}$; therefore $\operatorname{Im} r_+(u_m) = -h/2$ exactly. This is the seam branch of Proposition 3.4 and Definition 8.1: by item (c) in Section 4, the path from the base point to the segment passing above $u_1$ gives $+h/2$, and the one passing below gives $-h/2$. (The conjugate branch $K^{W,\#}$ would require the path above $u_1$, i.e. a line of negative slope at real $\varepsilon$, which is not admissible.)

*Transition.* On $U \cap V$, $R(r_+) = u = S(r_-)$, so $r_+ - T(r_-) \in \mathrm{i}h\mathbb{Z}$ is constant, with $T$ the map of Section 3 (branch $\operatorname{Im} t_0 = h/2$). At $u_m$, $r_-$ is real and $\operatorname{Im} T(r_-) = h/2$ (Lemma 3.1), while $\operatorname{Im} r_+ = -h/2$; hence $r_+ = T \circ r_- - \mathrm{i}h$ on $U \cap V$. By (17) and $|\operatorname{Im}\chi - \pi| \leq \pi - 1/\sqrt{2}$ on $U \cap V$, $r_-(U \cap V)$ lies in $\{|\operatorname{Im} w'| < 0.39h + C_{\mathrm{im}}\}$, which is inside the strip $\Sigma_q = \{|\operatorname{Im} w'| < d - D\}$ of the proof of Theorem 8.2 by the choice of $r_4$. There $T$ is univalent, $T(\mathbb{R}) = \{\operatorname{Im} w = h/2\}$, and $T$ maps $\{\operatorname{Im} w' \lesseqgtr 0\} \cap \Sigma_q$ below and above that line.

*The map to the sewn surface.* Let $X^W$ be the surface of Proposition 3.4: $C_+ = \{\operatorname{Im} w \geq -h/2\}/\mathbb{Z}$ and $C_- = \{\operatorname{Im} w' \leq 0\}/\mathbb{Z}$ glued by $w = T(w') - \mathrm{i}h$. Define $\Theta : X \to X^W$ by $\Theta = r_+$ where $\operatorname{Im} r_+ \geq -h/2$ and $\Theta = r_-$ where $\operatorname{Im} r_- \leq 0$. By (17) and the estimate on $V \setminus U$, points of $U \setminus V$ have $\operatorname{Im}\chi < 1/\sqrt{2}$, hence $\operatorname{Im} r_+ > -0.12h - C > -h/2$, and points of $V \setminus U$ have $\operatorname{Im} r_- < -0.38h + C < 0$. On $U \cap V$, $\operatorname{Im} r_+ \geq -h/2$ if and only if $\operatorname{Im} r_- \geq 0$, by the previous paragraph. So every point of $X$ is covered, and on the curve $\operatorname{Im} r_- = 0$ both definitions give the same point of $X^W$. Thus $\Theta$ is continuous, holomorphic off an analytic curve, hence holomorphic. As a point of $X$ tends to the $u_1$-end, $t \to -\infty$ and $\operatorname{Im} r_+ \to +\infty$; as it tends to the $u_2$-end, $\operatorname{Im} r_- \to -\infty$. So $\Theta$ extends to a holomorphic map of the compactified spheres with $\Theta^{-1}(\mathrm{end}) = \mathrm{end}$.

It has local degree one at the $u_1$-end; in fact $r_+$ is injective on $U$. Let $x, x' \in U$ be represented by $\chi, \chi' \in \hat{H}$ with $r_+(\chi') - r_+(\chi) = m \in \mathbb{Z}$. By Lemma 9.10, $r_+ = A_1\chi + \phi_+(\chi)$ where $\phi_+$ is bounded with bounded derivative on the $\frac{1}{10}$-neighbourhood of $\hat{H} \cap U$, which lies in $\mathcal{G}^{\mathrm{c}}$. Equal imaginary parts give $|\operatorname{Im}\chi - \operatorname{Im}\chi'| \le C\varepsilon$; since $\hat{H}$ is a strip of normal width $\le \frac{6}{5}\varepsilon$ along a line of slope 1, this forces $|\chi - \chi'| \le C_1\varepsilon$, and then $|m| \le |A_1||\chi - \chi'|(1 + C\varepsilon) + C \le C_2$. On the convex disc $N$ of radius $(C_1 + 3C_2)\varepsilon$ about $\chi$, $\partial_\chi r_+ = A_1(1 + O(\varepsilon))$, so $r_+$ is injective on $N$; the points $\chi'$ and $f_\chi^m(\chi)$ lie in $N$, and $r_+ \circ f_\chi^m = r_+ + m$ gives $r_+(f_\chi^m(\chi)) = r_+(\chi')$, hence $\chi' = f_\chi^m(\chi)$. Since $\hat{H}$ is a fundamental strip for $f_\chi$, $x = x'$ in $X$. Hence $\Theta$ has degree one and is biholomorphic.

*Normalisation.* $\Theta$ maps the marked point $p_c$ to the point of $C_+$ with attracting time $r_+(p_c) = n_0$. In Proposition 3.4 the lift is fixed by $K^W(0) = R(0) = 1$ on the branch $w(0) = 0$, i.e. attracting time 0 at $z = 0$, hence time $n_0$ at $z = n_0$. So the lifts agree, and $\tilde{F}(z + n_0) = R(r_+) = K_b^W(z + n_0)$ near $z = 0$. Near $z = 0$ the points $K_b^W(z + k)$, $0 \le k \le n_0$, stay near $u_k^*$, where the inverse branches of Definition 9.7 invert $f_\varepsilon$ (Lemma 9.1(3)); hence $F_\varepsilon = K_b^W$ near $z = 0$, and by the identity theorem on $W_0$. □

## 9.7. **The continuation theorem.**

**Theorem 9.13** (Continuation around the cusp)**.** *There are $r_5 > 0$ and an open set $\Omega \supset D^+ \cup (\eta - r_5, \eta) \cup (\eta, \eta + r_5)$, where $D^+ = \{b : |b - \eta| < r_5,\ \operatorname{Im} b > 0\}$, with the following properties.*

1. *Kneser's real-normalised solution $\kappa_b$, $b \in (\eta, \eta + r_5)$, continues analytically in the base to a family on $\Omega$ that is jointly holomorphic on $\Omega \times W_0$, $W_0 = \{z : \operatorname{dist}(z, [0, 1]) < \frac{1}{4}\}$. In particular it continues across the arc of the Shell–Thron boundary contained in $D^+$, whatever the arithmetic of the neutral multiplier there.*
2. *On $(\eta - r_5, \eta)$ the continued family equals $K_b^W$ on $W_0$.*
3. *Let $N_W$ be the neighbourhood of $(1, \eta)$ of Proposition 3.5. Continuing the family from a neighbourhood of $(\eta - r_5, \eta)$ inside $N_W$, on a $b$-dependent neighbourhood of $z = 0$ contained in the regular domain of $K_b^W$, one obtains $K^W$. Hence for every $b \in (1, \eta)$, if $\kappa_{b+\mathrm{i}y}$ is continued first through $D^+$ and then inside $N_W \cap \{\operatorname{Im} b > 0\}$, the limit $\lim_{y \downarrow 0} \kappa_{b+\mathrm{i}y}$ exists near $z = 0$ and equals $K_b^W$; by the functional equation and continuation in $z$ it equals $K_b^W$ on its whole regular domain.*

*Proof.* (1) By Lemma 9.1, $\varepsilon \mapsto b(\varepsilon)$ maps $\mathcal{S}_{r_3}^{\mathrm{c}\prime}$ biholomorphically onto an open set $\Omega$ that contains $D^+$ and the two real segments, for $r_5$ small. By Proposition 9.9, $(b, z) \mapsto F_{\varepsilon(b)}(z)$ is jointly holomorphic on $\Omega \times W_0$. By Proposition 9.11 it equals $\kappa_b$ on $(\eta, \eta + r_5)$. Since $\Omega$ is connected, it is the analytic continuation of $\kappa$ along every path in $\Omega$. The Shell–Thron boundary near $\eta$ consists of $b = \exp(\lambda \mathrm{e}^{-\lambda})$ with $|\lambda| = 1$. Its arc in the upper half-plane is $\varepsilon = 1 - \mathrm{e}^{\mathrm{i}\phi} = -\mathrm{i}\phi + O(\phi^2)$ with small $\phi > 0$, since then $b - \eta$ is a positive multiple of $-\varepsilon^2 = \phi^2 + \mathrm{i}\phi^3 + O(\phi^4)$; it lies in $\mathcal{S}_{r_3}^{\mathrm{c}\prime}$.

(2) is Proposition 9.12.

(3) By Proposition 3.5, $K^W$ is jointly holomorphic on the set of $(b, z)$ with $b \in N_W$ and $z$ in a neighbourhood of 0 that depends continuously on $b$; the upper chart contains $w = 0$, where $R$ is regular. Shrinking the discs $\Delta_{b_0}$ of that proposition, we may assume $\Delta_{b_0} \subset \{|b - b_0| < (\eta - b_0)/2\}$; then $N_W \cap \{\operatorname{Im} b > 0\}$ is a union of upper half-discs centred on $(1, \eta)$, which is connected since consecutive ones overlap. Each $\Delta_{b_0} \cap D^+$ is convex, and if it is nonempty its closure meets $(\eta - r_5, \eta)$. On a neighbourhood of $(\eta - r_5, \eta)$ times a small disc about $z = 0$, the two families agree by (2) and the identity theorem; hence they agree on every $\Delta_{b_0} \cap D^+$, and the continuation from $D^+$ into $N_W \cap \{\operatorname{Im} b > 0\}$ is unambiguous and equals $K^W$. At every $b \in (1, \eta)$ the boundary

value is therefore $K_b^W$ near $z = 0$. Both sides satisfy $F(z+1) = b^{F(z)}$, so they agree on the whole regular domain. □

**Corollary 9.14** (Separation limits for the continued Kneser family)**.** *For the continuation of Kneser's solution in Theorem 9.13, $K_b = \lim_{y\downarrow 0} \kappa_{b+\mathrm{i}y}$ exists for $b \in (1,\eta)$ and equals $K_b^W$. Its separation coefficients satisfy $c_n/\Lambda^n \to B_n \mathrm{e}^{2\pi \mathrm{i} n a}$ for every $n \geq 1$ as $b \to \eta^-$. In particular $|c_1|/\Lambda \to |B_1| = 0.0890584\ldots$, with $B_1 \neq 0$. This confirms the limit suggested by the numerics of Section 10.1: the first separation mode of the continued Kneser family, scaled by $\Lambda$, tends to the first coefficient of the inverse horn map of $\mathrm{e}^u - 1$.*

*Proof.* Theorem 9.13 gives $K_b = K_b^W$; apply Corollary 1.1 and Proposition 2.4. □

**Remark 9.15** (Lower half-plane, and what is not claimed)**.** (1) Complex conjugation of the whole construction uses the sector $\{\bar{\varepsilon} : \varepsilon \in \mathcal{S}_r^{\mathrm{c}}\}$, the lines $\overline{\ell_{\alpha,x_0}}$ through $-\mathrm{i}\pi \equiv \mathrm{i}\pi \pmod{2\pi\mathrm{i}}$ with slope $-\alpha$, and the conjugate admissibility condition $+\operatorname{Im}(\varepsilon\bar{\omega}) \geq \frac{1}{2}|\varepsilon|$. It shows that continuation of Kneser's solution through the lower half-disc yields the reflected branch $K_b^{W,\#}$ of Corollary 3.9 on $(\eta - r_5, \eta)$. The two continuations differ by at least $2c\Lambda$ at the height $x_*$; this is the monodromy of Kneser's family around $\eta$, consistent with Corollary 3.8.

(2) The proof does not use any arithmetic property of the multiplier on the Shell–Thron boundary. The ends of the tilted region are punctures for Siegel, Cremer and parabolic multipliers alike, because the only inputs are the step estimate $A_1\Delta = 1 + O(\varepsilon)$ and the transversality $\operatorname{Im}(\operatorname{Log}\lambda_1\bar{\omega}) > 0$. In particular no Brjuno condition, Siegel-disc estimate or boundary-value (Painlevé-type) argument is needed near $\eta$.

(3) The theorem asserts holomorphy in $(b,z)$ for heights near $[0,1]$, and by the functional equation on the regular domain reached from there. No claim is made about the behaviour of the continued solution as $\operatorname{Im} z \to \pm\infty$ at parameters with a neutral multiplier; the construction only uses that each end of the tilted region is a puncture of the quotient.

(4) The theorem concerns continuation along the stated paths. The width of $N_W$ is not quantified, so for $b$ far from $\eta$ the path runs close to the real axis. The family of [24] is defined by continuation along paths in the upper half of the Shell–Thron region; it coincides with the family here if continuation along those paths is path-independent, which is Open problem 1; the nine-digit agreement at $b = \sqrt{2}$ (Section 10.1) supports it.

## 10. Numerical results

10.1. **Ladder computations of a two-sided numerical solution.** In this numerical section, $K_b$ denotes the extrapolated values from a pointwise two-fixed-point construction; no base continuation is assumed. The solution is computed at complex bases $b + \mathrm{i}y$ by a two-fixed-point $\theta$-mapping construction. The program represents it as $R_{b+\mathrm{i}y} \circ (\mathrm{id} + \theta_{\mathrm{up}})$ in the upper half-plane and $S_{b+\mathrm{i}y} \circ (\mathrm{id} + \theta_{\mathrm{dn}})$ in the lower one. The computation runs along ladders $y \downarrow 0$ at working precisions up to 52 digits and is extrapolated to $y = 0$. Then $c_1$ is read off from $D = K - R$. First-mode dominance is confirmed by the constancy of $q(z) = D(z)/(R'(z)(\mathrm{e}^{2\pi\mathrm{i}z} - 1))$ over four points, including $z = \frac{1}{2} - \frac{\mathrm{i}}{2}$, where the first mode is amplified by $\mathrm{e}^{\pi}$ (Table 2). Omitting the constant mode $c_0 = -\sum_{n\geq 1} c_n$ produces a spread of 1.917 instead.

Table 2. The ratio $|q|$ at four points. The last column is $\max|q|/\min|q|$.

| base | $z = \frac{1}{2}$ | $z = -\frac{1}{2}$ | $z = \frac{1}{4}$ | $z = \frac{1}{2} - \frac{\mathrm{i}}{2}$ | spread |
|---|---|---|---|---|---|
| $1.1 + 0.0005\mathrm{i}$ | $1.987\,636 \times 10^{-9}$ | $1.987\,636 \times 10^{-9}$ | $1.987\,636 \times 10^{-9}$ | $1.987\,637 \times 10^{-9}$ | 1.0000 |
| $1.05 + 0.003\mathrm{i}$ | $1.422\,053 \times 10^{-7}$ | $1.422\,053 \times 10^{-7}$ | $1.422\,053 \times 10^{-7}$ | $1.422\,061 \times 10^{-7}$ | 1.0000 |

Table 3 lists $|\hat{c}_1| = |c_1|/\Lambda$. Over the range shown, $|D|$ varies by forty-five orders of magnitude while $|\hat{c}_1|$ varies by less than 5% and increases towards $|B_1|$. The row $b = \sqrt{2}$ uses Paulsen's value $\operatorname{Im} K_{\sqrt{2}}(\frac{1}{2}) \approx -1.18899697 \cdot 10^{-48}$, from a 120-digit contour computation [24]. It shares neither code nor method with ours. Solving the sewing equation directly at $b = \sqrt{2}$ (80 digits, 32 Fourier modes, residual $< 10^{-75}$; `paulsen_sqrt2_check.py`) gives $c_1 = -2.0924123 \cdot 10^{-50} + 1.4780488648 \cdot 10^{-48}\,\mathrm{i}$, $R'_{\sqrt{2}}(\frac{1}{2}) = 0.4022184246$, and $\operatorname{Im} K^{W}_{\sqrt{2}}(\frac{1}{2}) = \operatorname{Im}[R'(\frac{1}{2})P(\frac{1}{2})] + O(\Lambda^2) = -1.18899697185 \cdot 10^{-48}$. This matches Paulsen's value to relative accuracy $1.6 \cdot 10^{-9}$, the rounding of his quoted digits.

Table 3. Ladder values of $|\hat{c}_1|$ and their ratio to $|B_1|$.

| $b$ | $\lambda$ | $\Lambda$ | $\lvert\hat{c}_1\rvert$ | $\lvert\hat{c}_1\rvert/\lvert B_1\rvert$ | source |
|---|---|---|---|---|---|
| 1.05 | 0.051362 | $1.68 \times 10^{-6}$ | 0.085017 | 0.95462 | ladder, 20 digits |
| 1.10 | 0.105964 | $2.30 \times 10^{-8}$ | 0.086427 | 0.97045 | ladder, 20 digits |
| 1.15 | 0.164802 | $3.10 \times 10^{-10}$ | 0.087205 | 0.97919 | ladder, 20 digits |
| 1.20 | 0.229312 | $2.28 \times 10^{-12}$ | 0.087732 | 0.98511 | ladder, 20 digits |
| 1.25 | 0.301735 | $4.91 \times 10^{-15}$ | 0.088125 | 0.98952 | ladder, 24 digits |
| 1.30 | 0.385935 | $9.82 \times 10^{-19}$ | 0.088434 | 0.99299 | ladder, 32 digits |
| $\sqrt{2}$ | 0.693147 | $1.66 \times 10^{-47}$ | 0.088941 | 0.99868 | [24] |

The ladder values agree with the Kneser-free prediction $\tau_1$ and $t_2/t_1^2$, moduli and arguments, to the precision of the extrapolation (Table 1). Solving the sewing equation of Theorem 6.3 numerically gives $\hat{c}_1 = \tau_1(1 + \vartheta)$ with $|\vartheta| = 0.38\Lambda$ at $b = 1.10$ and $0.40\Lambda$ at $b = 1.30$. This matches the first-order formula $\vartheta \approx \Lambda[-4\pi^2|\tau_1|^2 - 2\pi\mathrm{i}\tau_1 - 4\pi\mathrm{i}\tau_2\bar{\tau}_1/\tau_1]$.

10.2. **Approach to the parabolic limit.** The transition map needs no ladder. Its modes are measured on the line $\operatorname{Im} z = -h_2/2 + Y$ of repelling time, near the gate carrying the positive modes, so the working precision needed is independent of $h$. Results for $Y = 1, 1.5, 2, 2.5, 3$ agree to all printed digits. Table 4 follows the invariants to $\eta - b = 2.7 \cdot 10^{-5}$.

Table 4. Kneser-free invariants near $\eta$. Here $\delta_1 = 1 - |\tau_1|/|B_1|$ and $p = |\log\lambda| \log\lambda_2$.

| $\lambda$ | $\eta - b$ | $\lvert\tau_1\rvert/\lvert B_1\rvert$ | $\delta_1/p$ | $\arg\tau_1$ | $\lvert t_2/t_1^2\rvert$ | $\lvert t_3/t_1^3 - B_3/B_1^3\rvert$ |
|---|---|---|---|---|---|---|
| 0.5 | $9.04 \times 10^{-2}$ | 0.99601457 | 0.0102076 | 1.581415 | 1.555683 | 0.148 |
| 0.8 | $1.21 \times 10^{-2}$ | 0.99952725 | 0.0102002 | 1.585920 | 1.572187 | 0.0177 |
| 0.9 | $2.84 \times 10^{-3}$ | 0.99989062 | 0.0101993 | 1.586391 | 1.573905 | 0.00410 |
| 0.95 | $6.87 \times 10^{-4}$ | 0.99997362 | 0.0101991 | 1.586499 | 1.574298 | 0.000989 |
| 0.98 | $1.08 \times 10^{-4}$ | 0.99999587 | 0.0101990 | 1.586528 | 1.574403 | 0.000155 |
| 0.99 | $2.67 \times 10^{-5}$ | 0.99999897 | 0.0101990 | 1.586532 | 1.574418 | 0.0000385 |
| limit | 0 | 1 | 0.0101990063 | 1.5865331 | 1.5744227 | 0 |

10.3. **The deficit law.** The natural parameter is the product of the logarithmic multipliers, $p = |\log\lambda| \log\lambda_2 = \varepsilon^2(1 - \varepsilon/3 + O(\varepsilon^2))$. Up to an analytic factor, this is the canonical parameter of the Mardešić–Roussarie–Rousseau normal form [21]. Fits of degree 2, 3 and 4 in $p$ over twelve

bases agree to eleven digits:

$$\log \frac{\tau_n}{B_n e^{2\pi i n a}} = \kappa^{(n)} p + O(p^2), \qquad \begin{aligned} \kappa^{(1)} &= -0.0101990063459 - 0.0132509561825\,\mathrm{i}, \\ \kappa^{(2)} &= -0.0510565562 - 0.0623157805\,\mathrm{i}, \\ \kappa^{(3)} &= -0.1400443481 - 0.1692244639\,\mathrm{i}, \end{aligned} \tag{18}$$

with $p^2$-coefficients of order $10^{-4}$ and constant terms below $10^{-16}$. Hence $\delta_1 = Cp + O(p^2)$ with $C = 0.0101990063$. Theorem 5.6 proves the first-order statement and computes $\kappa^{(n)}$ independently of any fit (Section 5). By Theorem 6.3, the Kneser-side $\hat{c}_n$ inherit (18) up to $O(\Lambda)$, which is beyond all orders in $p$. The $\kappa^{(n)}$ are not linear in $n$, so the first-order effect of the unfolding is not a translation of Fatou coordinates.

10.4. **Identification with an independent Kneser-type implementation.** The sewing solution can also be compared with a completely independent two-fixed-point computation at individual bases. Sheldon Levenstein's PARI/GP program `fatou.gp` builds the "merged" two-fixed-point solution for complex bases. At each base it constructs, independently, the solution with a $\theta$-mapping at each of the two fixed points: the two regular representations of [24, Prop. 1 and Lemma 1]. Its author regards this as the continuation of Kneser's solution in the base. We use the unmodified program (sha256 `70559daf...`): published reference values with 100 verified digits at $b = 0.8 + 0.4\mathrm{i}$ and $b = 1 + \mathrm{i}$, and our own runs at `\p 40` at $b = 1.2 + 0.1\mathrm{i}$ and $b = 1.3 + 0.2\mathrm{i}$, near the interval $(1, \eta)$. In those runs $\mathrm{sexp}(0) = 1$ to $10^{-53}$, and the functional equation holds to $10^{-54}$.

We solved the sewing equation of Theorem 6.3 directly at these complex bases (script `weld_complex.py`). $R$ is the regular solution at the attracting fixed point and $S$ the one at the repelling fixed point; the seam lies in the band of holomorphy of $T = R^{-1} \circ S$, and the root is selected as in Definition 8.1. Table 5 shows the result. At the three bases where the band is not too narrow, the sewing solution agrees with the independent value *to the full working precision of* 40 *digits*. That is 27 to 35 orders of magnitude below the size of the separation $|\kappa - R|$ itself. At $1 + \mathrm{i}$ the band is narrowest and the agreement $(2 \cdot 10^{-25})$ is limited by the number of samples. The neighbouring roots $c_{\pm 1}$ reproduce $R$ instead, and miss by the full separation. The $\theta$-mapping builder used for the ladders does not converge at $0.8 + 0.4\mathrm{i}$, whereas the sewing iteration converges in a few dozen steps.

TABLE 5. Sewing solution versus the independent `fatou.gp` values of $\kappa_b(\frac{1}{2})$ (working precision 40 digits).

| $b$ | $\lvert\Lambda\rvert$ | $\lvert\kappa_b(\frac{1}{2}) - R_b(\frac{1}{2})\rvert$ | $\lvert K_b^W(\frac{1}{2}) - \kappa_b(\frac{1}{2})\rvert$ | $\lvert\hat{c}_1\rvert$ |
|---|---|---|---|---|
| $1.2 + 0.1\mathrm{i}$ | $1.91 \times 10^{-11}$ | $6.91 \times 10^{-13}$ | $4.6 \times 10^{-40}$ | 0.0869956 |
| $1.3 + 0.2\mathrm{i}$ | $2.92 \times 10^{-11}$ | $1.81 \times 10^{-12}$ | $8.3 \times 10^{-41}$ | 0.0876095 |
| $0.8 + 0.4\mathrm{i}$ | $1.05 \times 10^{-3}$ | $9.86 \times 10^{-5}$ | $5.5 \times 10^{-39}$ | 0.0856072 |
| $1 + \mathrm{i}$ | $2.13 \times 10^{-2}$ | $4.41 \times 10^{-3}$ | $2.0 \times 10^{-25}$ | 0.0873850 |

The agreement identifies the sewing solution with an independent implementation of the two-sided regular representation, at four bases, two of them near $(1, \eta)$. It does *not* test the analytic continuation across the Shell–Thron boundary: both programs work at one base at a time. The continuation itself is proved near the cusp in Section 9.

10.5. **The jump data of [?].** For $1 < b < \eta$ the base-continued values from above and below the real axis are complex conjugate. Paulsen [**?**, (29)] lists $G(b^+) - G(b^-) = 2\mathrm{i}\,\mathrm{Im}\,\mathcal{K}_{b^+}(\frac{1}{2})$ at seven bases in $[1.175, \sqrt{2}]$ and fits them by $K(\eta - b)^m \exp(C/(b-\eta))$, with $m \approx 20.7$. By Corollary **??** these are values of $K_b^W$, and by Corollary 1.1 their scale is $\Lambda = \exp(4\pi^2/\log\lambda)$, which behaves like $\exp(-C_\eta(\eta - b)^{-1/2})$, not like $\exp(C/(b-\eta))$. The quotient by $\Lambda$ ($\lambda = -W_0(-\log b)$) is

| $b$ | 1.175 | 1.2 | 1.25 | 1.3 | 1.35 | 1.4 | $\sqrt{2}$ |
|---|---|---|---|---|---|---|---|
| $\log\Lambda$ | $-24.24$ | $-26.81$ | $-32.95$ | $-41.47$ | $-55.30$ | $-86.87$ | $-107.71$ |
| $\lvert\mathrm{Im}\,\mathcal{K}_b(\frac{1}{2})\rvert/\Lambda$ | 0.0273 | 0.0318 | 0.0410 | 0.0503 | 0.0596 | 0.0689 | 0.0716 |

Over this range $|\mathrm{Im}\,\mathcal{K}_b(\frac{1}{2})|$ changes by 36 orders of magnitude and the quotient by a factor 2.6. The exponent $m$ of the fit absorbs the difference between the two laws.

10.6. **A pointwise test near the Shell–Thron boundary.** The analytic continuation of Kneser's solution across the Shell–Thron boundary is argued in [24, §5] only through the continuity of a numerical scheme. In particular, continuity of each finite quadrature approximant does not imply continuity, let alone holomorphy, of its limit without locally uniform convergence in the base parameter. The transfer in [24, Prop. 3] of fixed-point limits from real to complex bases likewise requires such uniformity. We tested whether the pointwise two-fixed-point constructions on the two sides are numerically compatible near the irrational boundary point $\lambda_c = \mathrm{e}^{\mathrm{i}\phi}$, $\phi = 2\pi(\sqrt{2}-1)/5$, where $b_c = \exp(\lambda_c \mathrm{e}^{-\lambda_c}) \approx 1.5217 + 0.0148\mathrm{i}$. Along the normal $\lambda = r\mathrm{e}^{\mathrm{i}\phi}$ we computed the pointwise output $F_b^{\mathrm{GP}}(\frac{1}{2})$ of `fatou.gp` (`\p 24`). We used three bases inside ($r = 0.98, 0.99, 0.995$, one attracting fixed point) and three outside ($r = 1.005, 1.01, 1.02$, both fixed points repelling). So the two sides are computed with structurally different representations. A degree-4 polynomial in $b$ fitted to five of the six values predicts the sixth to $10^{-12}$–$10^{-13}$, uniformly on both sides. The quadratic through the three inside values extrapolates to the outside values with errors $1.3 \cdot 10^{-8}$, $3.2 \cdot 10^{-8}$ and $1.1 \cdot 10^{-7}$, growing with the distance as the cubic truncation term of a smooth function should. The reverse extrapolation gives the mirror-image errors. The errors are consistent with a smooth join of these pointwise constructions at the tested scales. Finite sampling cannot exclude a smaller jump, a higher-order nonanalytic term, or failure of continuation elsewhere on the boundary. Nor does the pointwise algorithm identify its outputs with the base-continued family $\kappa$. This is numerical evidence compatible with the continuation claimed in [24], not a proof.

We repeated the six-base calculation at five noninteger heights, using the unmodified `fatou.gp` at 50-digit precision (`boundary_multiheight.py`; raw values in `data/boundary-multiheight.json`). Table 6 reports the largest leave-one-out error when a degree-4 polynomial in $b$ through five bases predicts the sixth. It also reports the difference at $b_c$ between quadratics fitted separately to the three inside and three outside values. Recomputing the two closest bases at 70-digit precision changed all ten values by less than $5 \cdot 10^{-50}$. The five tested heights form only a finite set, and sampling at one boundary point cannot verify condition (P2).

TABLE 6. Finite two-sided boundary comparison at five noninteger heights. The quadratic gap is an extrapolation diagnostic, not a bound on a jump.

| $z$ | max. degree-4 leave-one-out error | two-sided quadratic gap at $b_c$ |
|---|---|---|
| 1/6 | $1.21 \cdot 10^{-12}$ | $4.25 \cdot 10^{-9}$ |
| 1/3 | $1.62 \cdot 10^{-12}$ | $6.64 \cdot 10^{-9}$ |
| 1/2 | $1.45 \cdot 10^{-12}$ | $7.02 \cdot 10^{-9}$ |
| 2/3 | $9.68 \cdot 10^{-13}$ | $5.57 \cdot 10^{-9}$ |
| 5/6 | $4.28 \cdot 10^{-13}$ | $2.89 \cdot 10^{-9}$ |

**Proposition 10.1** (Local base analyticity on the real exterior ray)**.** *For every real $b_0 > \eta$, Kneser's real-normalised solution admits a jointly holomorphic extension in $(b, z)$ to $\Delta \times W$, where $\Delta$ is a complex disc about $b_0$ and $W$ is a complex neighbourhood of $[0, 1]$. These local extensions agree on overlap, so the classical exterior branch has a jointly holomorphic germ along the entire real ray $(\eta, \infty)$. In particular, every compact subinterval of that ray has a common complex base neighbourhood and a common complex height neighbourhood of $[0, 1]$ on which this family is jointly holomorphic. The assertion supplies no corridor reaching a non-cusp Shell–Thron boundary arc.*

*Proof.* Write $a = \log b$ and choose the principal fixed-point pair $L_+(b)$ and $L_-(b)$, conjugate when $b > \eta$ and characterised there by $0 < a \operatorname{Im} L_+ < \pi$. They are simple fixed points and hence holomorphic in $b$ near $b_0$. Set $u_b = (L_+ + L_-)/2$, $v_b = (L_+ - L_-)/(2\mathrm{i})$, and $\gamma_b(t) = u_b + \mathrm{i}v_b t$ for $-1 \le t \le 1$. At a real base, $u_b, v_b \in \mathbb{R}$, $v_b > 0$, and

$$E_b(\gamma_b(t)) = r_b e^{\mathrm{i}\theta_b t}, \qquad r_b = e^{au_b} = |L_+|, \quad \theta_b = av_b \in (0, \pi).$$

Since $v_b = r_b \sin\theta_b$, the multipliers satisfy $|E_b'(L_\pm)| = ar_b = \theta_b / \sin\theta_b > 1$. The straight chord $\gamma_b$ and this circular arc meet only at $L_\pm$ and bound the initial region $H_b = \{\operatorname{Re} w \ge u_b,\ |w| \le r_b\} \setminus \{L_+, L_-\}$. This is Kneser's initial region as described in [30, §4, Lem. 4]. The arcs depend holomorphically on $b$ through their real-analytic parameter $t$. They remain disjoint away from their endpoints after shrinking a disc $\Delta$ about $b_0$; at the endpoints their tangents remain distinct because the fixed-point multipliers are nonreal. The bounded Jordan interiors vary continuously, so their union in $\Delta \times \mathbb{C}$ is open; the ambient coordinates $(b, w)$ make its projection a holomorphic family. Identify thin analytic collars of the two arcs by $(b, w) \mapsto (b, E_b(w))$. The map is injective on the collar after shrinking $\Delta$, since it is injective on the compact chord at $b_0$ and its derivative never vanishes. This gives a Hausdorff holomorphic family of cylinders with one end at each $L_\pm(b)$.

The ends can be filled holomorphically. Indeed, parameter-dependent Koenigs coordinates $\sigma_{\pm,b}$, normalised by $\sigma_{\pm,b}(L_\pm) = 0$ and $\sigma'_{\pm,b}(L_\pm) = 1$, linearise $E_b$ near the two repelling fixed points. If $\ell_{\pm,b} = \log E_b'(L_\pm(b))$ is chosen continuously, then $\exp(2\pi\mathrm{i} \log \sigma_{\pm,b} / \ell_{\pm,b})$ (or its reciprocal) is a puncture coordinate on the corresponding glued end. Filling both punctures yields a proper holomorphic submersion of compact genus-zero surfaces over $\Delta$, just as in the collar construction of Proposition 3.5.

We also have a holomorphic marked section represented by $w = 1$. For real $b > \eta$, $E_b(x) > x$ for every real $x$: the difference is positive at $x \to \pm\infty$ and has no real zero. The adjacent real intervals between the chord and its image are $(u_b, r_b)$, with $r_b = E_b(u_b) > u_b$. The forward endpoints $E_b^n(u_b)$ increase to $+\infty$, since a finite limit would be a real fixed point. For any $x > 0$ below $u_b$, forward iteration eventually enters $(u_b, r_b)$ or one of its boundary seams; for $x > r_b$, repeated real logarithms eventually do the same. Indeed, a monotone orbit staying on either side would have a finite real fixed-point limit. Thus these dynamical translates and their seam

collars cover the positive real axis. In particular, 1 is in the interior of one such interval or on a seam. A finite chain of holomorphic dynamical charts represents it near $b_0$; at a seam the analytic collar gives the same section. The entire compact real segment from 1 to $b = E_b(1)$ is covered by finitely many of these charts.

The two filled ends and this marked point are three disjoint holomorphic sections. Fischer–Grauert local triviality [9] and the fibrewise three-point Möbius normalisation give a jointly holomorphic coordinate $q_b$ on the compactified cylinder, taking the upper end to 0, the lower end to $\infty$, and $w = 1$ to 1. The upper puncture coordinate $\exp(2\pi\mathrm{i}\log\sigma_{+,b}/\ell_{+,b})$ shows that the logarithmic lift $z = (2\pi\mathrm{i})^{-1}\log q_b$, with $z(1) = 0$, gains $+1$ under the edge identification by $E_b$. On the real slice, reflection exchanges the two ends and fixes the marked point, so $q_b(\bar{w}) = 1/\overline{q_b(w)}$ in the real dynamical charts. Thus $z$ is real there; its local injectivity and $z(E_b(w)) = z(w) + 1$ make it strictly increasing along the positive axis. Inverting on the finite chart chain covering the real segment from 1 to $b$ gives a jointly holomorphic superfunction on $\Delta \times W$, after shrinking both.

For real $b$, the lift restricted to $H_b$ is injective and its integer translates cover the universal-cover plane. Choosing any $d \in H_b$ and subtracting its value gives the initial-region criterion of [30, Thm. 1]; the injectivity and tiling assertions for the Kneser generalisation are proved in [30, Thm. 5]. We normalise at $d$ here, since 1 need not itself belong to $H_b$. Their Abel coordinates differ only by a constant on $H_b$, and continuation along the real dynamical charts to $w = 1$ makes that constant zero. Thus the inverse just constructed is the classical real-normalised Kneser branch. The marking fixes the local family uniquely, and the identity theorem patches the parameter discs on overlaps. A finite subcover of any compact real base interval, followed by intersection of its finitely many height neighbourhoods, gives the final uniformity assertion. □

**Proposition 10.2** (Persistence of the exterior initial curve near the cusp)**.** *Put, for small $\theta \neq 0$,*

$$r(\theta) = \mathrm{e}^{\theta\cot\theta}, \qquad a(\theta) = \frac{\theta\mathrm{e}^{-\theta\cot\theta}}{\sin\theta}, \qquad b(\theta) = \mathrm{e}^{a(\theta)}, \qquad L_\pm(\theta) = r(\theta)\mathrm{e}^{\pm\mathrm{i}\theta}.$$

*For every fixed $M > 0$ there is $p_0 > 0$ such that, whenever $\theta = p + \mathrm{i}q$, $0 < p < p_0$ and $|q| \le Mp$, the chord $\gamma_\theta(t) = r(\theta)(\cos\theta + \mathrm{i}t\sin\theta)$ and its image $E_{b(\theta)}(\gamma_\theta(t)) = r(\theta)\mathrm{e}^{\mathrm{i}\theta t}$ are individually injective and meet only at their common endpoints $L_\pm(\theta)$. Thus the exterior initial Jordan curve persists geometrically in this complex parameter cone.*

*There is moreover a real-analytic function $q_c(p) = p^2/6 + O(p^3)$ such that, for $0 < p < p_0$ and $0 \le q < q_c(p)$, both fixed-point multipliers have modulus greater than one. The upper multiplier has modulus one at $q = q_c(p)$; this is a non-cusp Shell–Thron boundary arc. Therefore the unmarked two-end quotient surfaces of Proposition 10.1 exist locally throughout this exterior wedge, arbitrarily close to that boundary arc. The map $\theta \mapsto b(\theta)$ is injective on a sufficiently small such wedge, so its image is a genuine complex base corridor attached to the real exterior ray. This geometric assertion does not supply a holomorphic section represented by $w = 1$ on a common parameter corridor or a common regular height domain there.*

*Proof.* The identities $ar\sin\theta = \theta$ and $\log r = \theta\cot\theta$ show that $L_\pm$ are fixed points and give the displayed image arc. The functions $a(\theta)$ and $b(\theta)$ are even and holomorphic at 0, with $b(\theta) = \eta(1+\theta^2/(2\mathrm{e})+O(\theta^4))$; in particular $b'(\theta) \neq 0$ for small $\theta \neq 0$. More strongly, $b(\theta) = B(\theta^2)$ with $B'(0) = \eta/(2\mathrm{e}) \neq 0$. The map $B$ is locally injective, as is $\theta \mapsto \theta^2$ in $\mathrm{Re}\,\theta > 0$; hence $b$ is injective on the small wedge.

An intersection at an interior arc point would require the number

$$s_\theta(t) = \frac{\mathrm{e}^{\mathrm{i}\theta t} - \cos\theta}{\mathrm{i}\sin\theta} \qquad (-1 < t < 1)$$

to be real. Taylor expansion with its endpoint zeros factored out gives, uniformly for $-1 \le t \le 1$,

$$e^{i\theta t} - \cos\theta - it\sin\theta = (1-t^2)\theta^2\big(\tfrac{1}{2} + O(|\theta|)\big), \qquad \operatorname{Im} s_\theta(t) = -(1-t^2)\big(\tfrac{p}{2} + O(|\theta|^2)\big).$$

The error is uniform even as $t \to \pm 1$: every term of order at least two in the power series has the factor $1-t^2$, with the remaining polynomial bounded on $[-1,1]$. In the stated cone, $|\theta|^2 \le (1+M^2)p^2$, so the last imaginary part is strictly negative for $-1 < t < 1$ and small $p$. The chord is injective since $\sin\theta \ne 0$, and the arc is injective since $e^{i\theta(t_1-t_2)} = 1$ with $|t_1 - t_2| \le 2$ and small $|\theta|$ forces $t_1 = t_2$. The same expansion gives distinct endpoint tangents, hence a Jordan curve.

The multipliers are $\lambda_\pm(\theta) = a(\theta)L_\pm(\theta) = \theta\cot\theta \pm i\theta$. For $\theta = p + iq$ near zero,

$$|\lambda_+|^2 = 1 + \frac{p^2}{3} - 2q + O(p^3 + p|q| + q^2), \qquad |\lambda_-|^2 = 1 + \frac{p^2}{3} + 2q + O(p^3 + p|q| + q^2).$$

At $q = 0$ both are repelling, while $\partial_q|\lambda_+|^2 = -2 + O(p + |q|)$. The real implicit function theorem gives a unique nearby zero $q_c(p) = p^2/6 + O(p^3)$, and the signs above show that both multipliers are repelling for $0 \le q < q_c(p)$. Since $a(\theta) = \lambda_+(\theta)e^{-\lambda_+(\theta)}$, the locus $|\lambda_+| = 1$ is precisely the stated Shell–Thron boundary parameterisation. The collar gluing and repelling-end filling in Proposition 10.1 apply locally at every point of the open wedge. □

**Proposition 10.3** (A certified marked exterior path)**.** *Set $\theta(q) = \frac{1}{10} + iq$ and use $b(\theta)$, $L_\pm(\theta)$ and the initial region $H_\theta$ of Proposition 10.2. There is a unique $q_c \in (0, 1667/10^6)$ with $|\lambda_+(\theta(q_c))| = 1$. For every $0 \le q \le q_c$,*

$$E^{\circ 28}_{b(\theta(q))}(1) \in \operatorname{int} H_{\theta(q)}.$$

*For $0 \le q < q_c$ both fixed points are repelling, and the exterior Kneser germ normalised by $F(0) = 1$ analytically continues along the embedded base path $q \mapsto b(\theta(q))$. The result concerns the germ at height $z = 0$; it does not give a single regular height domain containing $[0,1]$ uniformly as $q \uparrow q_c$.*

*Proof.* The following strict enclosures are certified with outward-rounded Arb complex ball arithmetic by the reproducible script `docs/certify_exterior_mark.py`. It uses only the rational inputs $p = 1/10$, $Q = 1667/10^6$, 28 iterates, and 1000 equal subintervals of $[0, Q]$; the recorded run used `python-flint` 0.9.0 with FLINT 3.6.0. For $0 \le q \le Q$ it checks that the arc coordinate $s_\theta(t) = (e^{i\theta t} - \cos\theta)/(i\sin\theta)$ has $\partial_t \operatorname{Re} s_\theta(t) > 0.98$ for $-1 \le t \le 1$, with $\operatorname{Re} s_\theta(-1/10) < -0.08$ and $\operatorname{Re} s_\theta(1/10) > 0.08$. The estimate behind the remaining whole-arc check is elementary: writing

$$e^{i\theta t} - \cos\theta - it\sin\theta = (1-t^2)\theta^2 H(\theta, t), \qquad |H(\theta,t) - \tfrac{1}{2}| \le \frac{e^R - 1 - R}{2R}, \quad R = 0.101,$$

the termwise interpolation remainder and $|\sin\theta/\theta - 1| \le \sinh(R)/R - 1$ imply $\operatorname{Im} s_\theta(t) < -0.04(1-t^2)$ for $-1 < t < 1$. Indeed, after division by $1-t^2$ the order-$n$ interpolation polynomial is a sum of at most $n/2$ powers of $t$, each of modulus at most one; this gives the stated bound on $H$. Thus the image arc is a graph strictly below the chord in the $s$-plane. On the horizontal interval $[-0.01, 0.04]$ its height is below $-0.0396$.

Let $w_{28}(\theta) = E^{\circ 28}_{b(\theta)}(1)$ and $\zeta = (w_{28}/r - \cos\theta)/(i\sin\theta)$. The same script encloses $\zeta$ on every parameter subinterval and verifies $-0.01 < \operatorname{Re}\zeta < 0.04$ and $-0.03 < \operatorname{Im}\zeta < -0.01$. Hence $w_{28}$ lies strictly between the two arcs. It also verifies that $|\lambda_+|^2 - 1$ is positive at $q = 0$, negative at $q = Q$, and has derivative less than $-1$ throughout $[0, Q]$; that $|\lambda_-|^2 > 1.003$ there; and that $\operatorname{Im}(db(\theta(q))/dq) > 0.05$. These strict inequalities prove the root, repelling and embedded-path claims.

For $q < q_c$, the two repelling ends and the analytic collars give a holomorphic family of quotient spheres locally in $q$. The point $w_{28}(\theta)$ is a holomorphic section in the initial region, so it marks the physical point $w = 1$ through the finite iterate $E_b^{\circ 28}$; that iterate has nonzero derivative at 1. After three-point normalisation choose the logarithmic Abel lift with value 28 at $w_{28}$, and pull it back by the local inverse of $E_b^{\circ 28}$ to obtain value 0 at $w = 1$. Local inversion of this Abel coordinate gives a jointly holomorphic normalised superfunction germ at $z = 0$. At $q = 0$ it is the classical Kneser germ by Proposition 10.1; the embedded path continues that germ uniquely. □

**Proposition 10.4** (A prescribed Brjuno endpoint for the marked path)**.** *Let*

$$\alpha = \frac{\sqrt{2}-1}{26}, \qquad \lambda_c = \mathrm{e}^{2\pi \mathrm{i}\alpha}.$$

*There is a unique solution $\theta_c = p_c + \mathrm{i}q_c$ of $\lambda_+(\theta_c) = \lambda_c$ in the coordinate square of radius $10^{-8}$ centred at*

$$\theta_0 = 0.1000435842 + 0.0016682119\,\mathrm{i}.$$

*The number $\alpha$ is Brjuno. Along the embedded path $\theta(q) = p_c + \mathrm{i}q$, $0 \le q < q_c$, the two fixed points remain repelling and $E_{b(\theta(q))}^{\circ 28}(1)$ remains strictly inside $H_{\theta(q)}$, also at $q = q_c$ in the latter geometric assertion. Consequently the classical real Kneser germ has a marked analytic continuation along an exterior path ending at a specified non-cusp Brjuno boundary point. No common regular $z$-domain containing $[0, 1]$ is asserted.*

*Proof.* The number $\alpha$ is a root of $676X^2 + 52X - 1$. For a rational $m/n$ near it, that integer polynomial has nonzero value of modulus at least $1/n^2$ at $m/n$. Its derivative is bounded nearby, hence $|\alpha - m/n| \ge c/n^2$ for some $c > 0$. The denominators of its continued-fraction convergents therefore satisfy $n_{k+1} \le c^{-1} n_k$, while they grow at least as fast as Fibonacci numbers. Thus $\sum_k (\log n_{k+1})/n_k < \infty$, the Brjuno condition.

The remaining claims are certified with outward-rounded Arb complex balls by `docs/certify_brjuno_exterior_mark.py`. Put $f(\theta) = \theta \cot\theta + \mathrm{i}\theta - \lambda_c$ and $T(\theta) = \theta + \mathrm{i}f(\theta)$. On the stated closed square the script verifies

$$|f(\theta_0)| < 3 \cdot 10^{-11}, \qquad \sup |T'(\theta)| < 0.07, \qquad 3 \cdot 10^{-11} + 0.07\sqrt{2} \cdot 10^{-8} < 10^{-8}.$$

The contraction theorem gives the unique root $\theta_c$ in the square. In particular $0 < q_c < 0.00166823$.

The same script subdivides $[0, 0.00166823]$ into 1000 rational intervals and encloses the entire coordinate box $|p - 0.1000435842| \le 10^{-8}$. Uniformly there, it verifies the strict arc-graph and multiplier inequalities used in Proposition 10.3, with the whole-arc bound $\operatorname{Im} s_\theta(t) < -0.04(1 - t^2)$ obtained now using $|\theta| < 0.102$. It verifies for $\zeta = (E_{b(\theta)}^{\circ 28}(1)/r - \cos\theta)/(\mathrm{i}\sin\theta)$ that

$$-0.01 < \operatorname{Re}\zeta < 0.04, \qquad -0.03 < \operatorname{Im}\zeta < -0.01.$$

The arc is below $-0.0396$ on this horizontal range, so the marked iterate lies strictly inside $H_\theta$. Finally the script checks $\partial_q |\lambda_+|^2 < -1$, $|\lambda_-|^2 > 1.003$, and $\operatorname{Im}(db(p + \mathrm{i}q)/dq) > 0.05$ throughout the enclosing box. Since $|\lambda_+(\theta_c)| = 1$, both multipliers are repelling for $q < q_c$ and the base path is embedded. The marked quotient construction and local inversion from Proposition 10.3 now apply along this path. □

**Proposition 10.5** (A marked pullback cell along the Brjuno path)**.** *For every $0 \le q \le q_c$ on the path of Proposition 10.4, the inverse branch of $E_{b(q)}^{\circ 28}$ taking $w_{28}(q)$ to 1 extends conformally to a neighbourhood of $\overline{H_q}$. Denote it by $\Phi_q$. It depends continuously on $q$ up to $q_c$ and holomorphically in the base parameter locally along the path. Moreover*

$$\Phi_q(E_{b(q)}(w)) = E_{b(q)}(\Phi_q(w)) \qquad (w \in \gamma_q([-1,1])).$$

*Consequently $D_q = \Phi_q(H_q)$ is a Jordan cell containing $w = 1$; its two boundary arcs are $\Phi_q(\gamma_q)$ and $E_{b(q)}(\Phi_q(\gamma_q))$, and their endpoints are the original fixed points $L_\pm(q)$. This supplies a global physical pullback of the marked initial region, including at the Brjuno endpoint.*

*Proof.* The certificate `docs/certify_brjuno_exterior_mark.py` now also encloses the singular orbit $E_{b(q)}^{\circ k}(0)$ for every $0 \le k \le 28$ on all 1000 parameter slabs. In the chord coordinate $\zeta = (w/r - \cos\theta)/(\mathrm{i}\sin\theta)$ it verifies

$$-0.1 < \operatorname{Re}\zeta < 0.2, \qquad \operatorname{Im}\zeta > 0.02 \quad (w = E_{b(q)}^{\circ k}(0),\ 0 \le k \le 28).$$

The closed cell $\overline{H_q}$ lies below the chord $\operatorname{Im}\zeta = 0$, so it avoids every one of these orbit points with a uniform margin. Choose a simply connected thin neighbourhood $U_q$ of $\overline{H_q}$ which still avoids them.

Set $g_0(w) = w$. Inductively choose a logarithm on $U_q$ so that

$$E_{b(q)}(g_j(w)) = g_{j-1}(w), \qquad g_j(w_{28}(q)) = E_{b(q)}^{\circ(28-j)}(1) \quad (1 \le j \le 28).$$

For $j = 1$, $g_0$ never vanishes on $U_q$. If $g_{j-1}$ vanished at $w \in U_q$, then $w = E_{b(q)}^{\circ(j-1)}(0)$, contrary to the certificate. Thus each logarithm exists on the simply connected $U_q$. It is univalent, and the induction gives $E_{b(q)}^{\circ j} \circ g_j = \mathrm{id}$. In particular $\Phi_q = g_{28}$ is the asserted inverse branch. The uniform clearance and the analytic motion of the two boundary arcs allow the $U_q$ and logarithm branches to be chosen locally uniformly in $q$, including at $q_c$. This proves the stated parameter dependence.

It remains to check the seam identity; it does not follow merely from the equation $E_b^{\circ 28} \circ \Phi_q = \mathrm{id}$. At the real base $q = 0$, put $u = \gamma_0(0)$ and $r = E_{b(0)}(u)$. The branch $\Phi_0$ is the real inverse of the strictly increasing map $E_{b(0)}^{\circ 28}$ on the real interval through $w_{28}(0)$. Hence $\Phi_0(r) = E_{b(0)}(\Phi_0(u))$. For arbitrary $(q,t)$ in the connected rectangle $[0,q_c] \times [-1,1]$, both $\Phi_q(E_{b(q)}(\gamma_q(t)))$ and $E_{b(q)}(\Phi_q(\gamma_q(t)))$ are inverse images of $E_{b(q)}(\gamma_q(t))$ under $E_{b(q)}^{\circ 28}$. They vary continuously, and $E_b^{\circ 28}$ has nonzero derivative everywhere. Their equality set is therefore both open and closed in the rectangle. It contains $(0,0)$, so it is the whole rectangle.

The conformal image $D_q = \Phi_q(H_q)$ is Jordan and contains $\Phi_q(w_{28}(q)) = 1$. The seam identity identifies its second boundary arc with the image under $E_b$ of its first. At either endpoint $L_\pm$, it says that $\Phi_q(L_\pm)$ is fixed by $E_b$; since $E_b^{\circ 28}(\Phi_q(L_\pm)) = L_\pm$, it follows that $\Phi_q(L_\pm) = L_\pm$. These are the claimed endpoints. □

**Proposition 10.6** (A quantitative flat-cylinder model)**.** *For every parameter on the marked Brjuno path, including its boundary endpoint, the sewn two-end cylinder has a quasiconformal parameterisation from the standard cylinder $\mathbb{C}/\mathbb{Z}$ with maximal dilatation less than $11/9$. In the chord coordinate of Proposition 10.2, it is induced by*

$$z = x + 20\mathrm{i}y, \quad t = \tanh y, \quad \zeta_q(z) = t + x\big(s_{\theta(q)}(t) - t\big), \qquad 0 \le x \le 1, \quad y \in \mathbb{R},$$

*where the sides $x = 0$ and $x = 1$ are identified by $E_{b(q)}$. The bound is uniform as $q \uparrow q_c$ and controls the entire punctured cylinder, not only a compact middle part. It does not by itself assert that the image of the flat seam meets a normalised height circle only once.*

*Proof.* The chord and image arc are disjoint Jordan edges, and the latter is an increasing graph in the chord coordinate by the certified inequalities of Proposition 10.4. Write

$$s_\theta(t) - t = (1 - t^2)D_\theta(t), \qquad D_\theta(t) = \frac{\theta^2 H(\theta, t)}{\mathrm{i}\sin\theta},$$

where the entire interpolation remainder $H$ is defined by

$$e^{\mathrm{i}\theta t} - \cos\theta - \mathrm{i}t\sin\theta = (1 - t^2)\theta^2 H(\theta, t).$$

Set $A_\theta(t, x) = 1 + x(s'_\theta(t) - 1)$. Since $dt/dy = 1 - t^2$, differentiation in $z = x + 20\mathrm{i}y$ gives the Beltrami coefficient

$$\mu_\theta(t, x) = \frac{D_\theta(t) + \mathrm{i}A_\theta(t, x)/20}{D_\theta(t) - \mathrm{i}A_\theta(t, x)/20}.$$

The affine change from $\zeta$ to the physical $w$-coordinate is conformal and leaves this coefficient unchanged.

The reproducible outward-rounded Arb calculation in `certify_exterior_flat_cylinder.py` encloses this formula on the whole box $|\operatorname{Re}\theta - 0.1000435842| \le 10^{-8}$, $0 \le \operatorname{Im}\theta \le 0.00166823$, $-1 \le t \le 1$, $0 \le x \le 1$, and proves $|\mu_\theta| < 1/10$. To avoid interval division by $1 - t^2$ at the ends, it evaluates the Taylor terms of orders 2 through 11 and bounds the remaining modulus by

$$\frac{R^{10}}{2\,11!\,(1 - R/12)}, \qquad R = 0.102.$$

Thus the interpolation has positive Jacobian in its interior. Its boundary maps once around the certified Jordan curve; the degree formula then makes it a homeomorphism onto $H_q$ and a local diffeomorphism inside. The two vertical sides match under $E_b$, so it descends to a quasiconformal homeomorphism of the sewn punctured cylinder. The standard inequality $K \le (1 + \|\mu\|_\infty)/(1 - \|\mu\|_\infty)$ gives $K < 11/9$. The uniform bound also makes the two isolated ends removable for the pulled-back complex structure. □

**Proposition 10.7** (Marked exterior germ limits on a boundary subarc)**.** *Let* $I = [0.1000435842 - 10^{-8}, 0.1000435842 + 10^{-8}]$. *For every* $p \in I$ *there is a unique* $q_c(p) \in (0, 0.00166823)$ *with* $|\lambda_+(p + \mathrm{i}q_c(p))| = 1$. *On the exterior side* $0 \le q < q_c(p)$, *the real-normalised Kneser germ transported from* $q = 0$ *has a common complex height disc* $V$ *about* $z = 0$, *independent of* $(p, q)$ *in this parameter region. As* $q \uparrow q_c(p)$, *these germs converge locally uniformly on* $V$ *to nonconstant holomorphic boundary germs, uniformly in* $p \in I$. *Every limit takes the value* 1 *and has nonzero derivative at* 0. *In particular this holds at the prescribed Brjuno multiplier of Proposition 10.4; the existence of these local boundary germs does not require a Brjuno hypothesis. No common domain containing* $[0, 1]$ *or matching with an inner branch is asserted.*

*Proof.* The interval certificate of Proposition 10.4 verifies $|\lambda_+(p)| > 1$ throughout $I$ and $\partial_q |\lambda_+(p + \mathrm{i}q)|^2 < -1$ on the whole enclosing box. It also subdivides $I$ into 100 rational intervals and checks $|\lambda_+(p + 0.00166823\,\mathrm{i})| < 1$ on each one. The intermediate-value theorem and strict derivative give the unique $q_c(p)$, which depends real-analytically on $p$. The same certificate puts the marked iterate in $H_{p+\mathrm{i}q}$ with a uniform strict margin for the entire closed region $K = \{(p, q) : p \in I,\ 0 \le q \le q_c(p)\}$.

Use the standard punctured sphere coordinate $\xi = \exp(2\pi \mathrm{i} z)$ on the flat cylinder of Proposition 10.6. Pulling back the physical complex structure of the sewn cylinder gives a Beltrami coefficient $\mu_{p,q}$ on this fixed sphere, with $\|\mu_{p,q}\|_\infty < 1/10$. The explicit formula in that proposition, including its factored endpoint limits, shows that $(p, q) \mapsto \mu_{p,q}$ is continuous in the $L^\infty$ norm on $K$. The normalised solutions of the Beltrami equation therefore vary continuously on the sphere by [20, Thm. 8]. Their source-coordinate marked points also vary continuously, because $w_{28}(p, q)$ stays uniformly inside the Jordan cell and the flat-cylinder parameterisations are homeomorphisms varying continuously. Three-point normalisation at the two punctures and $w_{28}$ consequently gives conformal coordinates $Q_{p,q}$ that vary locally uniformly in the physical chart near the marked point. The boundary coordinates are defined by this same uniformisation, even when the upper multiplier is neutral. For $q < q_c(p)$ they coincide with the repelling-end quotient coordinates used to transport the classical exterior germ, by uniqueness of the three-point normalisation.

The certified interior margin supplies a fixed physical disc about $w_{28}(p, q)$ in the moving cells. On this compact family of discs, $Q_{p,q}$ and all its $w$-derivatives depend continuously on $(p, q)$ by Cauchy's formula. Their first derivatives at the marked points never vanish, since each $Q_{p,q}$ is conformal. Compactness of $K$ and the inverse function theorem give a single small disc $V$ on which the inverses of the normalised Abel charts

$$A_{p,q}(w) = \frac{1}{2\pi\mathrm{i}} \log Q_{p,q}(E^{\circ 28}_{b(p+\mathrm{i}q)}(w)), \qquad A_{p,q}(1) = 0,$$

exist and depend continuously on $(p, q)$. For $q < q_c(p)$ their inverses are precisely the transported Kneser germs. Uniform continuity on the compact set $K$ proves the asserted boundary convergence and nondegeneracy, uniformly in $p$. □

**Proposition 10.8** (An exterior analytic base corridor for the germ)**.** *There are a convex complex $\theta$-neighbourhood $U$ of the closed rectangle*

$$\widehat{K} = \{\theta : |\operatorname{Re}\theta - 0.1000435842| \le 10^{-8},\ |\operatorname{Im}\theta| \le 10^{-2}\}$$

*and a disc $V$ about $z = 0$ such that the real-normalised exterior Kneser germ extends jointly holomorphically to $U \times V$. The map $\theta \mapsto b(\theta)$ is injective on $U$, so its image is a genuine complex base corridor containing the real exterior seed segment and crossing a non-cusp Shell–Thron boundary subarc. The assertion is about the germ on $V$; it does not give a common regular neighbourhood of $[0, 1]$ or identify the continuation with the inner sewn branch.*

*Proof.* The outward-rounded certificate `certify_exterior_wide_corridor.py` checks on $\widehat{K}$ that the image arc remains an increasing graph below the chord, $E_b^{\circ 28}(1)$ remains inside the Jordan cell, and the singular orbit $E_b^{\circ k}(0)$ lies above its chord for $0 \le k \le 28$. It also gives $|\mu_\theta| < 1/10$ on the entire flat cylinder and $\operatorname{Re} b'(\theta) > 0.05$. All inequalities are strict, so they persist on a slightly larger convex open rectangle $U$. In the fixed flat-cylinder coordinate $\xi = e^{2\pi i z}$ of Proposition 10.6, the pulled-back Beltrami coefficient $\mu_\theta$ depends holomorphically on $\theta$ as an $L^\infty$-valued function: its explicit formula is holomorphic in $\theta$ pointwise, uniformly bounded away from modulus one, and has uniform Taylor bounds on a slightly larger parameter rectangle. The Ahlfors–Bers holomorphic-parameter theorem [20, Thm. 11] supplies a solution $h_\theta(\xi)$ of the Beltrami equation, normalised at the fixed source points $0, 1, \infty$, that is holomorphic in $\theta$ for each fixed $\xi$.

One must account for the fact that the inverse of the explicit quasiconformal map $\Psi_\theta$ from the flat cylinder to the physical Jordan cell is not holomorphic in the parameter at fixed physical $w$. Put $\xi_\theta(w) = \Psi_\theta^{-1}(w)$ near the marked point. The maps $h_\theta$ and $\Psi_\theta$ have the same Beltrami coefficient in $\xi$, while $\Psi_\theta(\xi)$ is holomorphic in $\theta$ for fixed $\xi$. Differentiating $\Psi_\theta(\xi_\theta(w)) = w$ at fixed $w$ gives

$$\partial_{\bar\theta}\xi_\theta(w) + \mu_\theta(\xi_\theta(w))\partial_{\bar\theta}\overline{\xi_\theta(w)} = 0.$$

The same identity and $\partial_{\bar\xi} h_\theta = \mu_\theta \partial_\xi h_\theta$ imply $\partial_{\bar\theta}[h_\theta(\xi_\theta(w))] = 0$. Differentiating instead at fixed $\theta$ with respect to $\bar w$ proves that this composition is holomorphic in $w$. These calculations take place in a smooth interior chart, where all derivatives are classical; hence the composition is jointly holomorphic. They apply also at the holomorphic marked section $w_{28}(\theta) = E_{b(\theta)}^{\circ 28}(1)$.

Divide this composition by its nonzero value at $w_{28}(\theta)$. The resulting coordinate $Q_\theta(w)$ takes the two punctures to $0, \infty$ and the marked point to 1, and is jointly holomorphic in $(\theta, w)$ near the marked section. Composing with $E_{b(\theta)}^{\circ 28}$, taking the logarithm with value zero at $w = 1$, and applying the parameter-dependent inverse function theorem yields a jointly holomorphic normalised superfunction on $U \times V$, after shrinking $U, V$ if necessary. At $q = 0$ it is the classical Kneser germ, so uniqueness of analytic continuation identifies it throughout the connected corridor.

Finally the wide certificate checks $\operatorname{Im}(db(p + iq)/dq) > 0.05$ on $\widehat{K}$. Since $db/dq = ib'(\theta)$, this is $\operatorname{Re} b'(\theta) > 0.05$ there. Shrink the convex $U$ so that $\operatorname{Re} b' > 0$ on it. For distinct $\theta_1, \theta_2 \in U$, the integral of $b'$ along their joining segment has positive real part, so $(b(\theta_2) - b(\theta_1))/(\theta_2 - \theta_1) \ne 0$. Thus $b$ is injective on $U$ as claimed. □

**Proposition 10.9** (The real seam anchor and an elliptic crossing test)**.** *For every real $p \in I$ of Proposition 10.7, the normalised seam function*

$$g_{p,q}(t) = \log |Q_{p,q}(\gamma_{p+\mathrm{i}q}(t))|, \qquad -1 < t < 1,$$

*has exactly one zero at $q = 0$, namely $t = 0$, and that zero is simple with $\partial_t g_{p,0}(0) < 0$. For fixed $p$, a change in the number of seam crossings on $0 \le q \le q_c(p)$ requires a pair $(q, t)$ with*

$$g_{p,q}(t) = \partial_t g_{p,q}(t) = 0.$$

*On the symmetric wide parameter box, the seam functions also obey $g_{p,-q}(-t) = -g_{p,q}(t)$. In particular, for $p = p_c$, exclusion of such pairs on this compact parameter path proves the hypothesis of Proposition 10.20 at its prescribed endpoint.*

*There is an equivalent scalar elliptic test on the fixed flat cylinder. Write $Y = 20 \operatorname{artanh} t$, let $\Psi_{p,q}$ denote the parameterisation of Proposition 10.6 in the physical $w$-coordinate, and set*

$$v_{p,q}(x, Y) = -\frac{1}{2\pi} \log |Q_{p,q}(\Psi_{p,q}(x + \mathrm{i}Y))|.$$

*With $\mu = \mu_{p+\mathrm{i}q}$ from that proposition, $v$ is periodic in $x$ and satisfies, weakly across the seam,*

$$\nabla \cdot (\mathsf{A}_\mu \nabla v) = 0, \qquad \mathsf{A}_\mu = \frac{1}{1 - |\mu|^2} \begin{pmatrix} |1 - \mu|^2 & -2 \operatorname{Im} \mu \\ -2 \operatorname{Im} \mu & |1 + \mu|^2 \end{pmatrix}.$$

*The two equations in the displayed crossing condition are exactly $v_{p,q}(0, Y) = \partial_Y v_{p,q}(0, Y) = 0$.*

*Proof.* At a real parameter, complex conjugation preserves the initial region and commutes with $E_b$. It descends to an antiholomorphic involution of the sewn sphere that exchanges its two punctures and fixes the real marked point. In the three-point coordinate it is $Q \mapsto 1/\overline{Q}$; its fixed locus is $|Q| = 1$. Conjugation sends the seam point labelled $t$ to the one labelled $-t$. The quotient seam is embedded, so a point on it is fixed only when $t = 0$. Near this point the real fixed locus has the horizontal physical tangent, while the chord seam has a nonzero vertical tangent. Conformality of the quotient chart makes their intersection transverse. Since $g$ tends from $+\infty$ to $-\infty$ as $t$ increases, its derivative at the unique zero is negative. For complex-conjugate parameters, the same involution interchanges the two punctures and preserves the marked normalisation, giving the displayed symmetry for $g$.

The uniform quasiconformal bound and compactness of the parameter region give uniform puncture tails for the normalised $Q_{p,q}$. Thus all zeros of $g_{p,q}$ lie in one compact subinterval of $(-1, 1)$. On that subinterval, the physical seam charts and the normalised uniformisation vary continuously in $C^1$ with $q$. If no double equation in the statement occurs, the implicit function theorem continues every simple zero locally in $q$; compactness and the uniform tails keep their number constant.

For the elliptic formula, choose a local lift $h = (2\pi\mathrm{i})^{-1} \log(Q \circ \Psi)$ on the flat cylinder. Its imaginary part is $v$, and it satisfies $h_{\bar{z}} = \mu h_z$. Writing $h = u + \mathrm{i}v$ and separating real and imaginary parts gives $\mathsf{A}_\mu \nabla v = (-u_Y, u_x)$. Taking the divergence proves the weak equation, including the periodic seam matching. Finally, $g_{p,q}(t) = -2\pi v_{p,q}(0, 20 \operatorname{artanh} t)$ and $dY/dt = 20/(1 - t^2)$, which gives the stated equivalence. □

**Lemma 10.10** (Uniform positive slopes at the two punctures)**.** *On the wide rectangle $\widehat{K}$ of Proposition 10.8, let $v_\theta$ be the elliptic height function of Proposition 10.9. Put $\lambda_\pm = \theta\cot\theta \pm \mathrm{i}\theta$ and take the principal logarithms $\ell_\pm = \operatorname{Log}\lambda_\pm$. Uniformly for $\theta \in \widehat{K}$,*

$$\partial_Y v_\theta(0,Y) \longrightarrow \operatorname{Im}\!\left(-\frac{1}{10\ell_+}\right) > 0.98 \quad (Y \to +\infty),$$
$$\partial_Y v_\theta(0,Y) \longrightarrow \operatorname{Im}\!\left(\frac{1}{10\ell_-}\right) > 0.98 \qquad (Y \to -\infty).$$

*Thus some finite $Y_0$ works uniformly on $\widehat{K}$: $\partial_Y v_\theta(0,Y) > 0$ for $|Y| > Y_0$. Every tangency in Proposition 10.9 therefore lies in $|Y| \le Y_0$.*

*Proof.* The wide interval certificate also verifies $\operatorname{Re}\lambda_\pm > 0.9$, $\operatorname{Im}\lambda_+ > 0.09$, and $\operatorname{Im}\lambda_- < -0.09$ throughout $\widehat{K}$, as well as the two 0.98 slope bounds above. In particular the stated logarithms vary continuously, have imaginary parts of the indicated signs, and never vanish. The factored formula for $\mu_\theta(x,Y)$ is analytic in $t = \tanh(Y/20)$ at $t = \pm 1$. It therefore tends, uniformly in $\theta$ and piecewise smoothly in $x$, to periodic coefficients $\mu_{\theta,\pm}(x)$ as $Y \to \pm\infty$.

Choose the uniformising lift $h_\theta$ on the universal cover of the cylinder so that $h_\theta(z+1) = h_\theta(z)+1$ and $h_\theta(0) = 0$. It is a quasiconformal homeomorphism of the plane with coefficient $\mu_\theta$ and dilatation below $11/9$; its imaginary part differs from $v_\theta$ by a constant. For any $\theta_j \to \theta_\infty$ in $\widehat{K}$ and $Y_j \to +\infty$, the maps

$$h_{\theta_j}(z + \mathrm{i}Y_j) - h_{\theta_j}(\mathrm{i}Y_j)$$

fix $0, 1, \infty$ and have a locally uniformly convergent subsequence by quasiconformal compactness. The limit is the unique normalised solution for $\mu_{\theta_\infty,+}$.

This limit is explicit. Since $1-\tanh(Y/20)$ is asymptotic to $2e^{-Y/10}$, the physical interpolation near $L_+$, after a nonzero conformal rescaling, tends to $e^{-Y/10}[1 + x(\lambda_+ - 1)]$. The normalised straight-sector solution for its coefficient is

$$h_{\theta,+}(x + \mathrm{i}Y) = \frac{\log[1 + x(\lambda_+ - 1)] - Y/10}{\ell_+}, \qquad 0 \le x \le 1,$$

extended by $h_{\theta,+}(z+1) = h_{\theta,+}(z)+1$. The logarithm along the segment from 1 to $\lambda_+$ is chosen to run from 0 to $\ell_+$. The segment remains in the right half-plane, and its argument increases strictly; this formula is the normalised quasiconformal homeomorphism with coefficient $\mu_{\theta,+}$. Uniqueness identifies it with every translated limit. At the lower end, $1 + \tanh(Y/20) \sim 2e^{Y/10}$ and the same argument gives

$$h_{\theta,-}(x + \mathrm{i}Y) = \frac{\log[1 + x(\lambda_- - 1)] + Y/10}{\ell_-}.$$

The preceding convergence also holds for the tangential $Y$-derivatives at the seam. Indeed, after dividing the physical coordinate $w - L_\pm$ by the corresponding factor $e^{\mp Y_j/10}$, the analytic gluing maps converge on fixed seam disks to multiplication by $\lambda_\pm$, uniformly in $\theta_j$. In these rescaled physical charts the lifted maps are holomorphic. Local uniform convergence followed by Cauchy's formula gives convergence of their physical derivatives, and the explicit rescaled interpolation gives convergence of their $Y$-derivatives. Hence the limits of $\partial_Y v$ are the imaginary parts of $\partial_Y h_{\theta,\pm}$ displayed above. Compactness of $\widehat{K}$ makes the convergence uniform. The two positive continuous limiting slopes have a positive minimum there, proving the last assertion. □

**Proposition 10.11** (A certified frozen Abel coordinate)**.** *Use the chord data $D_\theta(t)$ and $A_\theta(t,x)$ of Proposition 10.6, with $t = \tanh(Y/20)$. Set*

$$L_\theta(Y) = \operatorname{Log}\left(\frac{\theta}{\sin\theta}\right) + \mathrm{i}\theta t, \quad B_\theta = e^{L_\theta} = s'_\theta(t), \quad P_\theta(x,Y) = \frac{A_\theta(t,x)}{20 D_\theta(t)}.$$

*On the full marked path, including its boundary endpoint, $|P_\theta - \mathrm{i}| < 0.09$. On the wider rectangle $\widehat{K}$, the explicit frozen vertical slope*

$$c_\theta(Y) = \frac{e^{L_\theta(Y)} - 1}{20 D_\theta(t) L_\theta(Y)}$$

*has $\operatorname{Im} c_\theta(Y) > 0.98$ for every real $Y$. The quotient at $L_\theta = 0$ is interpreted by analytic continuation. Define*

$$F_\theta(x,Y) = \frac{\operatorname{Log}[1 + x(e^{L_\theta(Y)} - 1)]}{L_\theta(Y)}, \qquad h^0_\theta(x + \mathrm{i}Y) = F_\theta(x,Y) + \int_0^Y c_\theta(s)\, ds.$$

*Extending $h^0_\theta$ by $h^0_\theta(z+1) = h^0_\theta(z) + 1$ gives an explicit approximate Abel lift. Its exact residual in the pulled-back Beltrami equation is*

$$\partial_Y h^0_\theta - P_\theta \partial_x h^0_\theta = \partial_Y F_\theta, \qquad |\partial_Y F_\theta(x,Y)| < 10^{-3} \operatorname{sech}^2(Y/20).$$

*Its residual has $L^2$ norm below $0.005$ on the marked path. At the seam the approximate height derivative is greater than $0.98$. The residual and norm bounds do not yet give pointwise control of the exact uniformising derivative.*

*Proof.* The Beltrami formula of Proposition 10.6 rearranges exactly to $h_Y = P_\theta h_x$. Since $A_\theta = 1 + x(B_\theta - 1)$, differentiation gives

$$\partial_x F_\theta = \frac{B_\theta - 1}{L_\theta A_\theta}, \qquad P_\theta \partial_x F_\theta = c_\theta.$$

The displayed residual follows. The factor $F_\theta$ equals $0$ at $x = 0$ and $1$ at $x = 1$, so the periodic extension is continuous and its seam height derivative is $c_\theta$.

For a bound without cancellation at $L = 0$, write

$$F(L,x) = \int_0^1 \frac{x e^{sL}}{1 + x(e^{sL} - 1)}\, ds.$$

Its $L$-derivative has integrand

$$\frac{sx(1-x)e^{sL}}{[1 + x(e^{sL} - 1)]^2}.$$

The accompanying Arb script `certify_exterior_frozen_abel.py` verifies $|L| < 0.12$ on $\widehat{K}$. Consequently,

$$|F_L| \le \frac{e^{0.12}}{8(2 - e^{0.12})^2}, \qquad |L_Y| \le \frac{0.102}{20} \operatorname{sech}^2(Y/20).$$

Their product is less than $0.000945 \operatorname{sech}^2(Y/20)$. Squaring and integrating uses

$$\int_{\mathbb{R}} \operatorname{sech}^4(Y/20)\, dY = 80/3$$

and gives the stated $L^2$ bound. The same outward-rounded script encloses $P_\theta$ on the marked path and $c_\theta$ on the wide box; it evaluates $(e^L - 1)/L$ by a Taylor polynomial with a rigorous tail, so intervals containing $L = 0$ cause no division problem. □

**Proposition 10.12** (An $L^4$ correction to the frozen coordinate)**.** *On the full marked path let $h_\theta$ be the exact uniformising lift with $h_\theta(0)=0$ and $h_\theta(z+1)=h_\theta(z)+1$. The coordinate $h_\theta^0$ of Proposition 10.11 satisfies, on the infinite flat cylinder,*

$$\|\partial_x(h_\theta-h_\theta^0)\|_{L^4}<0.032,\qquad \|\partial_Y(h_\theta-h_\theta^0)-\mathrm{i}\partial_x(h_\theta-h_\theta^0)\|_{L^4}<0.005.$$

*In particular $\|\partial_Y(h_\theta-h_\theta^0)\|_{L^4}<0.037$. These are global integral bounds; they do not assert a pointwise sign for the exact seam derivative.*

*Proof.* Write $\mathcal{L}=\partial_Y-\mathrm{i}\partial_x$ and $f=\partial_Y F_\theta$. The preceding proposition gives

$$\|P_\theta-\mathrm{i}\|_\infty<0.09,\qquad \|f\|_4<0.000945\left(20\int_{\mathbb{R}}\operatorname{sech}^8 s\,ds\right)^{1/4}=0.000945(640/35)^{1/4}<0.002.$$

An exact correction $e$ must solve $\mathcal{L}e=(P_\theta-\mathrm{i})e_x-f$. On the cylinder let $B_{\mathrm{per}}$ be the periodised Beurling transform. The planar bound of [1, Thm. 1.2] at exponent 4 gives $\|B\|_4\le 4.725<12$. The same bound $\|B_{\mathrm{per}}\|_4<12$ follows by cutting a smooth periodic input of compact vertical support to $M$ horizontal periods, applying the planar estimate, and averaging the output over the interior $(1-\varepsilon)M$ periods. On each fixed-height output strip the omitted kernel tails tend uniformly to zero, since the kernel is $O(|x|^{-2})$ away from the input; first let $M\to\infty$, then $\varepsilon\downarrow 0$, then release the output-height cutoff. Density gives the bound on cylinder $L^4$.

Since $\mathcal{L}=-2\mathrm{i}\partial_{\bar z}$, the cylinder Cauchy operator has

$$T:=\partial_x\mathcal{L}^{-1}=-\frac{I+B_{\mathrm{per}}}{2\mathrm{i}},\qquad \|T\|_4<6.5.$$

Solve for $u$ as the unique fixed point in cylinder $L^4$ of $u=T[(P_\theta-\mathrm{i})u-f]$, because $6.5(0.09)=0.585<1$. It obeys

$$\|u\|_4<\frac{6.5(0.002)}{1-0.585}<0.032,\qquad \|g\|_4\le 0.09\|u\|_4+\|f\|_4<0.005.$$

For completeness, construct $e$ from this fixed point by applying the periodic Cauchy potential to $g=(P_\theta-\mathrm{i})u-f$. For compactly supported smooth inputs, its gradient is the usual Cauchy and Beurling pair. Approximation in $L^4$, the bounds above, and local Morrey compactness give a periodic $e\in W^{1,4}_{\mathrm{loc}}$ with $\nabla e\in L^4$ and $e(0)=0$. The identity $e_x=u$ passes to the limit; uniqueness modulo constants follows because a periodic entire holomorphic function with derivative in $L^4$ is constant.

Set $H=h_\theta^0+e$. The exact equation $H_Y=P_\theta H_x$ says that $H$ has the same Beltrami coefficient as $h_\theta$, of modulus below $1/10$. The positive bound $\operatorname{Im} c_\theta>0.98$ and boundedness of $F_\theta$ show that $\operatorname{Im} h_\theta^0$ tends to $+\infty$ and $-\infty$ at the corresponding cylinder ends. The $L^4$ gradient bound makes $e=o(|Y|)$ uniformly in $0\le x\le 1$: its horizontal mean grows at most $O(|Y|^{3/4})$ by Hölder's inequality, while the oscillation on each unit-height strip is uniformly bounded by Morrey's inequality. Hence $H$ is proper. The homotopy $(1-s)z+sH(z)$ is proper for $0\le s\le 1$ and has degree one. The weak quasiregular factorisation (or the quasiconformal chain rule followed by Weyl's lemma) makes $H\circ h_\theta^{-1}$ an entire holomorphic map. It is proper of degree one, so affine; the period relation forces its slope to be 1, and the value at 0 removes the constant. Therefore $H=h_\theta$, proving the claimed bounds for the exact lift. □

**Proposition 10.13** (Certified pointwise seam monotonicity)**.** *On the closed parameter rectangle $|\operatorname{Re}\theta-0.1000435842|\le 10^{-8}$, $|\operatorname{Im}\theta|\le 0.01$, including both sides of its non-cusp boundary subarc, the exact uniformising lift satisfies*

$$\sup_{z\in\mathbb{C}}|\partial_Y(h_\theta-h_\theta^0)(z)|<0.041,\qquad \partial_Y\operatorname{Im} h_\theta(\mathrm{i}Y)>0.93\quad(Y\in\mathbb{R}).$$

*Consequently the image of the chord seam under the three-point-normalised coordinate $Q_\theta$ crosses $|Q| = 1$ exactly once and transversely, for every parameter in this rectangle.*

*Proof.* Let $e = h_\theta - h_\theta^0$ and $f = \partial_Y F_\theta$. The same outward-rounded interval computation as in Proposition 10.11, now differentiating the factored interpolation remainder in $t$, gives throughout the wider rectangle $\|P_\theta - \mathrm{i}\|_\infty < 0.18$ and $\|\partial_Y P_\theta\|_\infty < 0.004$. Here the derivative of its order-$n$ polynomial has modulus at most $n^2/4$ on $[-1,1]$; starting at $n = 12$, its Taylor tail is bounded by $36(0.102)^{10}/[12!(1-0.102/10)]$. For the residual, put $E = e^{0.12}$ and $a = 2-E$. Differentiating the integral formula for $F_L$ in the preceding proof gives

$$|F_L| \le \frac{E}{8a^2}, \qquad |F_{LL}| \le \frac{E}{12a^2} + \frac{E^2}{6a^3}.$$

Since $|L_Y| \le (0.102/20)\operatorname{sech}^2(Y/20)$ and $|L_{YY}| \le (0.102/200)\operatorname{sech}^2(Y/20)$, the interval evaluation yields

$$|f_Y| < 0.00012\operatorname{sech}^2(Y/20), \qquad \|f_Y\|_4 < 0.00025.$$

The planar Beurling bound $\|B\|_4 \le 4.725 < 6$ quoted in Proposition 10.12 transfers to the periodic cylinder by the same cutoff-and-averaging argument used there. Hence the cylinder operator $T = \partial_x \mathcal{L}^{-1}$ has norm below 3.5. The fixed-point construction of that proposition now applies on the wider rectangle because $3.5(0.18) = 0.63 < 1$. It gives

$$\|e_x\|_4 < \frac{3.5(0.002)}{1-0.63} < 0.019, \quad \|\mathcal{L}e\|_4 < 0.18(0.019) + 0.002 < 0.0055, \quad \|e_Y\|_4 < 0.025.$$

The same properness and degree-one argument identifies this correction with the exact uniformising lift.

Take a vertical difference quotient $e^\delta(x,Y) = [e(x,Y+\delta) - e(x,Y)]/\delta$. Its $x$ derivative belongs to cylinder $L^4$ for each $\delta \neq 0$, and it satisfies

$$\mathcal{L}e^\delta = (P_\theta - \mathrm{i})(e^\delta)_x + g^\delta, \quad g^\delta = P_\theta^\delta e_x(x, Y+\delta) - f^\delta.$$

The bounds above give $\|g^\delta\|_4 < 0.004(0.019) + 0.00025 < 0.00033$. The same cylinder operator $T$ as before therefore gives

$$\|(e^\delta)_x\|_4 < \frac{3.5(0.00033)}{1-0.63} < 0.0032, \qquad \|(e^\delta)_Y\|_4 < 1.18(0.0032) + 0.00033 < 0.0042.$$

Passing to weak limits of the difference quotients proves that $v = e_Y \in W^{1,4}(\mathbb{C}/\mathbb{Z})$ and $\|v_x\|_4 < 0.0032$, $\|v_Y\|_4 < 0.0042$; the preceding bound gives $\|v\|_4 < 0.025$.

For clarity, a disk of radius $r < 1/2$ on the flat cylinder obeys the following pointwise Sobolev estimate, obtained by averaging along radial segments and applying Hölder's inequality:

$$|v(z_0)| \le (\pi r^2)^{-1/4}\|v\|_{L^4(B_r(z_0))} + (3/2)^{3/4}(2\pi)^{-1/4} r^{1/2}\|\nabla v\|_{L^4(B_r(z_0))}.$$

At $r = 1/4$ its two coefficients are below 1.51 and 0.43. Hence $|e_Y(z_0)| < 1.51(0.025) + 0.43(0.0032 + 0.0042) < 0.041$ for every $z_0$. The $W^{1,4}$ representative is continuous; on the seam it equals the classical tangential derivative supplied by the analytic physical chart. Since $h_Y^0(\mathrm{i}Y) = c_\theta(Y)$ and $\operatorname{Im} c_\theta(Y) > 0.98$, the stated strict derivative bound follows.

The seam height $\operatorname{Im} h_\theta(\mathrm{i}Y)$ tends to the opposite infinite limits at the two ends by the proof of Proposition 10.12. It is therefore strictly increasing through every real level. The three-point-normalised $Q_\theta$ differs from $\exp(2\pi\mathrm{i}h_\theta)$ by a nonzero constant factor, so $|Q_\theta| = 1$ is one such height level. It is crossed exactly once, with nonzero derivative. □

**Proposition 10.14** (An attracting overlap of the exterior chord)**.** *At the exact interior parameter $\theta_0 = 0.1000435842+0.01\mathrm{i}$, the upper fixed point $L_+(\theta_0)$ is attracting and the lower fixed point is repelling. The entire physical chord subarc $\gamma_{\theta_0}([-13/100, 1])$ lies in the immediate attracting basin of $L_+(\theta_0)$. This supplies a common physical chart region for the exterior quotient and the upper attracting dynamics; it does not identify the exterior solution with the inner sewn solution. Moreover, the unique point of the chord mapped by the marked quotient coordinate to $|Q_{\theta_0}| = 1$ lies on this certified attracting subarc, at a parameter $t_* > 0$. Consequently there is a noninteger real height $z_* \in (0, 1)$ and product neighbourhoods of $(\theta_0, z_*)$ on which the continued exterior superfunction takes values in the attracting basin and admits a jointly holomorphic upper Koenigs coordinate.*

*Proof.* The outward-rounded Arb certificate `certify_interior_chord_overlap.py` uses exact rational coordinates for $\theta_0$ and partitions the indicated real $t$-interval into 113 cells of width $1/100$. Put $a = \log b(\theta_0)$, $L = L_+(\theta_0)$, and $\delta = 1/100$. It verifies $|aL|e^{|a|\delta} < 1$ and $|aL_-| > 1$. For a cell centred at $t_j$, let $c_0 = \gamma_{\theta_0}(t_j)$ and $\rho_0 = |\mathrm{i}r \sin\theta_0|/200$. The rigorous recurrence

$$c_{n+1} = e^{ac_n}, \qquad \rho_{n+1} = |c_{n+1}|(e^{|a|\rho_n} - 1)$$

encloses the whole cell image under $E_b^{\circ n}$ in $\overline{B(c_n, \rho_n)}$. For each of the 113 cells the certificate finds an $n \le 505$ with $|c_n - L| + \rho_n < \delta$. The disk $B(L, \delta)$ is strictly forward invariant by the first verified inequality, because $|E_b(L + u) - L| \le |aL|e^{|a|\delta}|u|$. Thus the whole subarc belongs to the attracting basin. It is connected and contains $L$, so it lies in the immediate component.

For the final assertion, write the marked iterate in the flat interpolation coordinate as $\zeta_{28} = t_m + x_m(s_{\theta_0}(t_m) - t_m)$, $0 < x_m < 1$, and put $Y_m = 20 \operatorname{artanh} t_m$. The same Arb certificate checks $\operatorname{Re}\zeta_{28} > 0.15$ and $\operatorname{Re}(s_{\theta_0}(t) - t) < 0.01$ for every $-1 \le t \le 1$; hence $t_m > 0.14$ and $Y_m > 2.8$. The formula for $F_x$ in Proposition 10.11, together with $|L_{\theta_0}(Y)| < 0.12$, gives

$$|F_x - 1| \le \frac{(e^{0.12} - 1 - 0.12)/0.12 + (e^{0.12} - 1)}{2 - e^{0.12}} < 0.23.$$

The exact equation $e_Y = P_\theta e_x - f$, the wide-box bounds $|P_\theta - \mathrm{i}| < 0.18$, $|f| < 0.001$, and $|e_Y| < 0.041$ give $|e_x| < 0.052$. Thus $|h_{\theta_0,x} - 1| < 0.3$ on the flat cylinder. Since $h_{\theta_0}(0) = 0$ and its seam-height slope exceeds 0.93, we have

$$\operatorname{Im} h_{\theta_0}(x_m + \mathrm{i}Y_m) > 0.93Y_m - 0.3 > 0.$$

Three-point normalisation at the mark divides $\exp(2\pi\mathrm{i}h_{\theta_0})$ by its value there. Its unit-circle level on the seam therefore has positive height. Strict seam monotonicity makes its unique crossing satisfy $Y_* > 0$, hence $t_* = \tanh(Y_*/20) > 0$. The seam point represents a noninteger height $z_* \in (0, 1)$, because the marked point is in the interior of the Jordan cell. The developed value at $z_*$ is a preimage under $E_b^{\circ 28}$ of either $\gamma_{\theta_0}(t_*)$ or its forward image. Both lie in the immediate attracting basin, so that value lies in the full attracting basin. Finite entry into the invariant disk persists on product neighbourhoods of $(\theta_0, z_*)$; Proposition 10.35 then gives the stated joint Koenigs coordinate. □

**Proposition 10.15** (A certified interior sewing circle)**.** *At $\theta_0 = 0.1000435842 + 0.01\mathrm{i}$, the entire lower regular curve $S_{\theta_0}([0,1])$ lies in the full attracting basin of $L_+(\theta_0)$ and avoids every backward preimage of $L_+$. Let $\sigma_+$ be the upper Koenigs coordinate, normalised by $\sigma'_+(L_+) = 1$, and let $\ell_+ = \operatorname{Log}\lambda_+$ be the branch continued from the inner real axis. The upper time*

$$T(x) = \frac{1}{\ell_+}\log\frac{\sigma_+(S_{\theta_0}(x))}{\sigma_+(1)}$$

*has a branch along $[0,1]$ for which*

$$0.94 < \operatorname{Re} T'(x) < 1.06, \qquad |\operatorname{Im} T'(x)| < 0.056, \qquad T(1) - T(0) = 1.$$

*Consequently $T([0,1])/\mathbb{Z}$ is an embedded analytic Jordan circle in the upper cylinder. With $\varpi = 2\pi\mathrm{i}/\ell_+$, its shifted image $(T+\varpi)([0,1])$ lies in $\operatorname{Im} w < -2.5$. It therefore defines an analytic two-chart sewing near this strictly interior parameter, with the normalising upper point $w = 0$ above the seam. The resulting locally normalised sewn solution is jointly holomorphic in the base and height on its regular domains. This local family has not been identified with the continuation of the real inner sewn family or with the exterior family.*

*Proof.* The accompanying outward-rounded Arb certificates use the exact rational coordinates of $\theta_0$. Write $a = \log b$, $L_- = L_-(\theta_0)$, $\mu = aL_-$, $M = |\mu|$, $A = |a|$, and $f(u) = L_-(e^{au} - 1)$. They verify $1.01 < M < 1.02$, $A < 0.38$, and, with $\rho = 1/3000$, $\exp(4A\rho/(M-1)) < 2$. The normalised lower Poincaré function is the limit of

$$P_N(t) = L_- + f^{\circ N}(t\mu^{-N}).$$

For $|t| \le \rho$, elementary exponential bounds give $|f^{\circ j}(v)| \le 4\rho M^{j-N}$ and $|(f^{\circ N})'(v)| \le 2M^N$ on the line segment joining $t\mu^{-N}$ to $f(t\mu^{-(N+1)})$. Indeed the latter endpoint differs from the former by at most $\frac{1}{2}A\rho^2 M^{-2N-1}e^{A\rho}$, and both have modulus below $2\rho M^{-N}$. Summing the resulting telescoping differences gives

$$\sup_{|t|\le\rho}|P(t) - P_N(t)| \le \frac{A\rho^2 e^{A\rho}}{M-1}M^{-N} < 39\rho^2(100/101)^N.$$

At $N = 1000$ this last bound is below $2.1 \cdot 10^{-10}$.

Put $m = 700$ and $t(z) = -\exp((z-m)\operatorname{Log}\mu)$. The certificates check $|t(z)| < \rho$ on $0 \le z \le 1$ and use $S(z) = E_b^{\circ m}(P(t(z)))$. They partition $[0,1]$ into 32 rational cells. On each cell the initial $t$-disk has radius $|t(z_j)|(e^{|\operatorname{Log}\mu|/64} - 1)$. For a disk of radius $r$ centred at $u$ under $f$, the new radius is at most $|L_- e^{au}|(e^{Ar} - 1)$; under $E_b$ the analogous radius is $|E_b(u)|(e^{Ar} - 1)$. Adding the above uniform Poincaré tail after 1000 steps, the certificates find that every cell enters $B(L_+, 1/100)$ within a further 427 iterates after the 700 shifts. The disk is strictly invariant since $|aL_+|e^{A/100} < 1$. At entry each enclosure also excludes $L_+$; since $|a|/100 < 2\pi$, no later point of the invariant disk can map exactly to $L_+$. Thus $\sigma_+(S(x)) \ne 0$ throughout the interval.

For the slope, the second certificate propagates derivative disks on a finer 64-cell partition, using Cauchy's estimate for the Poincaré tail on $|t| < \rho$. Alongside a value disk $B(c,r)$ and a derivative disk $B(d,s)$, one exponential step uses

$$c_{\mathrm{new}} = e^{ac}, \qquad d_{\mathrm{new}} = ac_{\mathrm{new}}d, \qquad s_{\mathrm{new}} = |ac_{\mathrm{new}}|((e^{Ar} - 1)|d| + e^{Ar}s).$$

The lower Poincaré steps have the analogous formula with $c_{\mathrm{new}} = f(c)$. If $u_n = E_b^{\circ n}(S(z)) - L_+$ and $d_n = (E_b^{\circ n} \circ S)'(z)$, then

$$D_n = \frac{d_n}{u_n\ell_+} \longrightarrow T'(z), \qquad \frac{D_{n+1}}{D_n} = \frac{au_n e^{au_n}}{e^{au_n} - 1}.$$

On $|u| < 1/100$, the last factor differs from 1 by at most $A|u|$; subsequent displacements contract by $q = |aL_+|e^{A/100} < 1$. Thus the multiplicative tail after any certified iterate is bounded by $\exp(A|u_n|/(1-q)) - 1$. The outward-rounded derivative and tail disks after 1200 upper iterates give the stated slope inequalities.

Since $S(1) = E_b(S(0))$, logarithm continuation gives $T(1) - T(0) = 1 + k\varpi$ for some integer $k$. The certified bound $\operatorname{Re}\varpi > 50$ and the integral of the slope over $[0,1]$ force $k = 0$. For the seam location, compare the same-depth ratio $\mathcal{R}_n = (E_b^{\circ n}(S(0)) - L_+)/(E_b^{\circ n}(1) - L_+)$. On the invariant disk, the factor $h(u) = (e^{au} - 1)/(au)$ gives the ratio-tail bound

$$\left|\frac{\mathcal{R}_\infty}{\mathcal{R}_n} - 1\right| \le \exp\left(\frac{A(|u_n^{(0)}| + |u_n^{(1)}|)}{1-q}\right) - 1.$$

The certificate encloses the principal logarithm of this ratio, including its tail, and locates $\operatorname{Im}(T(0) + \varpi) < -2.5 - 0.056$. The imaginary-slope bound then places the entire shifted circle below $-2.5$. Positive real slope and degree one make it an embedded circle. A thin analytic collar persists for parameters near $\theta_0$, and the three-section compactification and normalisation argument of Proposition 3.5 produces the stated local holomorphic sewn family. □

**Proposition 10.16** (A transverse crossing on the straight interior path)**.** *Set* $\theta_\dagger = 23541/2560000 + 0.01\mathrm{i}$ *and let* $S = S_{\theta_\dagger}$ *be the lower regular superfunction. There are distinct real heights*

$$x \in 0.045633494813841561325455 13 + [-10^{-4}, 10^{-4}],$$
$$y \in 0.276750034386440064562133 2 + [-10^{-4}, 10^{-4}]$$

*for which* $S(x) = S(y)$*. This crossing is transverse. The physical subarc* $S([0.04, 0.28])$ *lies in a simply connected disc contained in the upper attracting chart and avoiding* $L_+$*. Hence the same two heights have equal upper Koenigs times on one logarithmic branch. Any upper-time sewing curve containing this subarc therefore self-intersects at* $\theta_\dagger$*. Moreover,* $\operatorname{Re} T'(0) < -0.1$*. In particular, the straight path* $\theta = p + 0.01\mathrm{i}$ *cannot transport the real inner sewing by maintaining this lower regular circle as an embedded, positive-slope seam at every parameter. This does not rule out continuation by a different seam or parameter path.*

*Proof.* Both accompanying certificates use the exact rational parameter and the lower Poincaré limit from the preceding proposition. For the crossing, `certify_interior_self_intersection.py` takes $\rho = 1/2000$, Poincaré depth 4500, and 800 further exponential iterates. Its uniform Poincaré value tail is bounded by

$$\frac{A\rho^2 e^{A\rho}}{M-1} M^{-4500}, \qquad A = |\log b|, \quad M = |E_b'(L_-)|,$$

and Cauchy's estimate bounds the derivative tail. It propagates value and derivative discs over the two stated height intervals. For $F(x,y) = S(x) - S(y)$, a rational preconditioner of the real $2 \times 2$ Jacobian gives a Krawczyk image within $[-1.3 \cdot 10^{-6}, 1.3 \cdot 10^{-6}] \times [-10^{-6}, 10^{-6}]$ relative to the box centre. The two Jacobian defect row sums are $< 0.1$. Thus the Krawczyk map contracts the box strictly into itself; its unique zero is transverse, and the two height intervals are disjoint.

Eight more rational height cells enclose $S([0.04, 0.28])$ in $B(2.69185 + 0.019515\mathrm{i}, 0.001)$; the maximum certified distance from its centre is $< 1.1 \cdot 10^{-6}$. This disc lies in $B(L_+, 0.02)$ and avoids $L_+$. The latter disc is invariant since $|E_b'(L_+)|e^{A/50} < 1$; the upper Koenigs coordinate has no zero there except $L_+$, by the local linearisation, the absence of other preimages in this small disc, and Hurwitz's theorem. Its logarithm therefore has a single branch on the smaller disc containing the whole subarc. Because $S(x) = S(y)$, that branch gives $T(x) = T(y)$; the nonzero conformal multiplier from $S'$ to $T'$ preserves transversality.

Independently, `certify_interior_path_fold.py` propagates the value and derivative discs at height 0 to the invariant upper disc and bounds the remaining Koenigs product. It gives $\operatorname{Re} T'(0) < -0.11272 < -0.1$. □

**Proposition 10.17** (The lower sewing circle in the exterior quotient)**.** *At $\theta_0 = 0.1000435842 + 0.01\mathrm{i}$, let $H_{\theta_0}$ be the exterior initial Jordan cell and set $w_n(x) = E_{b(\theta_0)}^{\circ n}(S_{\theta_0}(x))$. There is a unique $x_* \in (0,1)$ such that $w_{76}(x_*)$ lies on the chord boundary of $H_{\theta_0}$. For $0 \le x < x_*$ one has $w_{77}(x) \in H_{\theta_0}$, and for $x_* < x \le 1$ one has $w_{76}(x) \in H_{\theta_0}$. At $x_*$ these two representatives are the boundary pair $w_{77}(x_*) = E_b(w_{76}(x_*))$; at the other join, $w_{77}(0) = w_{76}(1)$. The resulting closed curve is an embedded essential analytic circle in the exterior quotient cylinder. Its annular physical chart extends jointly holomorphically to parameters near $\theta_0$. Thus the local lower sewing circle and the continued exterior quotient have a common annular physical chart. This does not identify their global uniformising coordinates.*

*Proof.* Use the chord coordinate $\zeta(w) = (w/r - \cos\theta_0)/(\mathrm{i}\sin\theta_0)$. The chord is $[-1,1]$, its exponential image is $s([-1,1])$, and direct substitution gives the exact identity

$$\zeta(E_b(w)) = s(\zeta(w)), \qquad s(t) = \frac{e^{\mathrm{i}\theta_0 t} - \cos\theta_0}{\mathrm{i}\sin\theta_0}.$$

On the central rectangle $\mathcal{R} = [0.45, 0.55] + \mathrm{i}[-0.05, 0.10]$, the outward-rounded certificate verifies $\operatorname{Re} s' > 0.98$ and $\operatorname{Im} s(t) < -0.03$ for real $t \in [0.45, 0.55]$, while $\operatorname{Re} s(0.45) < 0.48 < 0.52 < \operatorname{Re} s(0.55)$. Thus $s$ is injective on this convex rectangle, its real arc lies below the chord, and for $\zeta \in \mathcal{R} \cap s(\mathcal{R})$

$$\zeta \in \zeta(H_{\theta_0}) \quad \Longleftrightarrow \quad \operatorname{Im}\zeta < 0 < \operatorname{Im} s^{-1}(\zeta).$$

The inverse here is the branch on $s(\mathcal{R})$.

The same certificate uses the Poincaré approximation and explicit tail from Proposition 10.15, followed by 75–77 exponential iterates. It partitions $[0,1]$ into 64 rational cells and propagates both value and derivative disks. Throughout the interval it verifies that $\zeta(w_{75}), \zeta(w_{76}), \zeta(w_{77})$ belong to $\mathcal{R}$, all with real parts in $(0.48, 0.52)$, and

$$\operatorname{Im}\zeta(w_{75}) > 0.03, \qquad \operatorname{Im}\zeta(w_{77}) < -0.005, \qquad \frac{d}{dx}\operatorname{Im}\zeta(w_{76}) < -0.03.$$

Its endpoint disks put $\operatorname{Im}\zeta(w_{76}(0)) > 0$ and $\operatorname{Im}\zeta(w_{76}(1)) < 0$, so the middle curve crosses the chord exactly once and transversely. The criterion above, applied to $w_{77} = E_b(w_{76})$ before the crossing and to $w_{76} = E_b(w_{75})$ afterwards, proves the asserted membership in $H_{\theta_0}$.

For both $n = 76, 77$ the certificate also gives $d\operatorname{Re}\zeta(w_n(x))/dx > 10^{-4}$. Since $w_{77}(0) = w_{76}(1)$, the two interior arcs have disjoint real-coordinate ranges except at this common endpoint. The only other join is the single identified boundary pair at $x_*$. Hence the quotient curve is simple and analytic, and crossing the glued boundary once makes it essential. All inequalities are strict and all arcs avoid the two fixed endpoints, so compactness gives a jointly holomorphic annular collar for nearby parameters. □

**Corollary 10.18** (Two distinct return circles in the exterior quotient)**.** *At the same parameter $\theta_0$, the lower regular circle has a second embedded essential return circle in the exterior quotient, represented by $w_{15}(x)$ before a unique chord crossing of $w_{14}(x)$ and by $w_{14}(x)$ after it. This circle is disjoint from the return circle of Proposition 10.17. In particular, a forward return of the lower circle to the exterior initial cell does not by itself specify a canonical comparison map between the two quotient cylinders.*

*Proof.* The independent outward-rounded certificate `certify_interior_early_return.py` uses the same Poincaré tail and 64 rational cells as the preceding proposition. On $[0.25, 0.33] +$

i$[-0.10, 0.08]$ it verifies $\operatorname{Re} s' > 0.98$ and $\operatorname{Im} s(t) < -0.03$ for real $t \in [0.25, 0.33]$. All three curves $\zeta(w_{13})$, $\zeta(w_{14})$, and $\zeta(w_{15})$ lie in this rectangle, with real parts in $(0.26, 0.31)$, and the certificate gives

$$\operatorname{Im}\zeta(w_{13}) > 0.005, \qquad \operatorname{Im}\zeta(w_{15}) < -0.03, \qquad \frac{d}{dx}\operatorname{Im}\zeta(w_{14}) < -0.03.$$

The endpoint signs of $\operatorname{Im}\zeta(w_{14})$ are positive and negative. Thus $w_{14}$ crosses the chord once, and the same initial-cell criterion as above puts $w_{15}$ before the crossing and $w_{14}$ afterwards inside $H_{\theta_0}$. For $n = 14, 15$, $d \operatorname{Re}\zeta(w_n)/dx > 10^{-4}$; the endpoint identity $w_{15}(0) = w_{14}(1)$ and the boundary gluing make a simple essential circle. Its real-coordinate strip $(0.26, 0.31)$ is disjoint from the strip $(0.48, 0.52)$ certified for the earlier proposition, even after boundary identification. □

**Corollary 10.19** (Degree-one regular germs at the exterior ends)**.** *Let $Q_{\theta_0}$ be the three-point-normalised coordinate of the exterior quotient at $\theta_0$. On the corresponding quotient end germs, the lower regular superfunction $S$ and the upper regular superfunction $R$ satisfy*

$$Q_{\theta_0}(S(z))^{-1} = C_- e^{-2\pi i z}(1 + O(e^{-2\pi i z})) \quad (\operatorname{Im} z \to -\infty),$$

*and*

$$Q_{\theta_0}(R(z)) = C_+ e^{2\pi i z}(1 + O(e^{2\pi i z})) \quad (\operatorname{Im} z \to +\infty),$$

*where $C_\pm \neq 0$. These are degree-one local puncture maps. Together with a chosen return circle from Proposition 10.17 or Corollary 10.18, this locates the remaining chart-identification problem in the compact middle bands between that circle and the two puncture germs. It does not assert that the two local identifications continue through those bands or agree with one another.*

*Proof.* Let $\tau_-$ be the Koenigs coordinate at the repelling lower fixed point, normalised by $\tau_-'(L_-) = 1$, and let $\mu = E_b'(L_-)$. On a deep lower time half-cylinder, $\tau_-(S(z)) = -\mu^z$ exactly. The lower exterior puncture coordinate is, on its logarithmic sector quotient,

$$\xi_- = \exp\left(-\frac{2\pi i}{\log \mu}\log(-\tau_-(w))\right).$$

The choice of logarithmic branch is fixed by the lower regular curve; multiplication by $\mu$ changes its exponent by an integer multiple of $2\pi i$, so $\xi_-$ descends to the glued end. Consequently $\xi_-(S(z)) = e^{-2\pi i z}$. The exterior quotient was compactified by this simple puncture coordinate, so its global coordinate has the expansion $Q_{\theta_0}^{-1} = C_-\xi_-(1 + O(\xi_-))$ with $C_- \neq 0$.

At the upper end let $\sigma_+$ be the attracting Koenigs coordinate and $\lambda_+ = E_b'(L_+)$. On a deep upper time half-cylinder, $\sigma_+(R(z)) = \sigma_+(1)\lambda_+^z$. The upper exterior puncture coordinate

$$\xi_+ = \exp\left(\frac{2\pi i}{\log \lambda_+}\log\frac{\sigma_+(w)}{\sigma_+(1)}\right)$$

therefore obeys $\xi_+(R(z)) = e^{2\pi i z}$. The global coordinate has a simple zero there, giving $Q_{\theta_0} = C_+\xi_+(1 + O(\xi_+))$ with $C_+ \neq 0$. The local end degrees are one. The statements concern germs in the respective quotient end charts; they do not supply analytic continuation from the middle circle to those germs. □

**Proposition 10.20** (A single-crossing criterion for a full height segment)**.** *Fix the prescribed parameter $p_c$ of Proposition 10.4 and abbreviate $Q_q = Q_{p_c,q}$ for the three-point-normalised coordinates of Proposition 10.7. Suppose the curve $t \mapsto Q_{q_c}(\gamma_{q_c}(t))$, $-1 < t < 1$, meets the unit circle exactly once, and the meeting is transverse; equivalently,*

$$\log |Q_{q_c}(\gamma_{q_c}(t))| = 0$$

*has exactly one zero $t_*$ and its derivative there is nonzero. Then there are a complex parameter neighbourhood $U_*$ of $\theta_c = p_c + \mathrm{i}q_c$ and a connected open height neighbourhood $W \supset [0,1]$ such that the exterior germ of Proposition 10.8 extends jointly holomorphically to $U_* \times W$. In particular, it crosses a non-cusp boundary subarc and takes the values $F_\theta(0) = 1$, $F_\theta(1) = b(\theta)$ there. The single-crossing hypothesis is verified by Proposition 10.13.*

*Proof.* The normalised coordinates $Q_\theta$ and their inverses vary continuously in $C^1$ on compact collars away from the two punctures by Propositions 10.7 and 10.8. In puncture coordinates, $Q_\theta$ tends uniformly to 0 at the upper end and to $\infty$ at the lower end. Thus the unique transverse intersection with $|Q_{\theta_c}| = 1$ persists, and remains the only such intersection, throughout a sufficiently small complex neighbourhood $U_*$ of $\theta_c$.

The marked point $Q_\theta = 1$ and the seam point split $|Q_\theta| = 1$ into two compact subarcs. On each subarc the inverse of $Q_\theta$ has a representative in $H_\theta$. Starting at the interior marked point $w_{28}(\theta)$, pull the first representative back by the conformal branch $\Phi_\theta$ of Proposition 10.5. At the unique seam crossing the representative passes from the image arc to the chord; this is the orientation of the positive Abel deck generator. Continue on the second subarc using $E_{b(\theta)} \circ \Phi_\theta$. The seam identity in that proposition makes the two physical charts agree on their common analytic collar. They therefore define a holomorphic developing map at every height $z \in [0,1]$ with $Q_\theta = e^{2\pi \mathrm{i} z}$, taking the values 1 and $E_{b(\theta)}(1) = b(\theta)$ at its endpoints.

The path is compact and avoids both punctures. Finitely many of the collars just described cover it. Their domains, transition maps and physical developing maps vary jointly holomorphically in the parameter by Proposition 10.8 and the finite-log construction of Proposition 10.5. The seam identity of that proposition extends to the complex parameter neighbourhood by the identity theorem, since both sides are holomorphic in $\theta$ there and agree along the real $q$ path. After shrinking $U_*$ and these finitely many charts, they cover one connected complex neighbourhood $W$ of $[0,1]$ for every $\theta \in U_*$. The analytic overlap identities provide a jointly holomorphic family on $U_* \times W$. Near $z = 0$ it is the previously transported normalised Kneser germ, so the identity theorem identifies the family throughout $W$. □

**Corollary 10.21** (A full exterior height corridor)**.** *At the prescribed non-cusp Brjuno parameter $\theta_c$ there are a complex parameter neighbourhood $U_*$ and a connected open height neighbourhood $W \supset [0,1]$ on which the exterior Kneser family continued from real $b > \eta$ is jointly holomorphic and satisfies $F_\theta(0) = 1$ and $F_\theta(1) = b(\theta)$. The parameter neighbourhood crosses the Shell–Thron boundary. No comparison with the inner sewn family $K^W$ is asserted.*

*Proof.* Proposition 10.13 verifies the unique transverse seam crossing at $\theta_c$. Apply Proposition 10.20. □

**Corollary 10.22** (Full exterior height along an interior-reaching path)**.** *Put $p_0 = 0.1000435842$ and $J = \{p_0 + \mathrm{i}q : 0 \le q \le 0.01\}$. There are a connected complex parameter neighbourhood $\Omega \supset J$ and a connected height neighbourhood $W_J \supset [0,1]$ such that the exterior Kneser branch continued from $q = 0$ is jointly holomorphic on $\Omega \times W_J$, the map $\theta \mapsto b(\theta)$ is injective on $\Omega$, and the family satisfies $F_\theta(0) = 1$, $F_\theta(1) = b(\theta)$. In particular its full-height branch is defined at the strictly interior parameter $\theta_0 = p_0 + 0.01\mathrm{i}$ of Proposition 10.14.*

*Proof.* The wide-box certificate of Proposition 10.8 supplies the marked pullback cell, its analytic seam identity, and a jointly holomorphic normalised germ throughout a neighbourhood of $J$. Proposition 10.13 supplies a unique transverse seam crossing at every point of $J$. The compact-arc development in the proof of Proposition 10.20 therefore applies locally at each such parameter; it uses the Jordan cell and its puncture tails, not the sign of $|\lambda_+| - 1$. Finitely many resulting parameter charts cover $J$. Their full-height families agree on overlaps because all extend the same jointly holomorphic germ at $z = 0$; the identity theorem then extends the agreement along $[0, 1]$. Choose a thin connected parameter neighbourhood $\Omega$ of $J$ inside their union and inside the injectivity domain $U$ of Proposition 10.8, and a common thin connected tube $W_J$ about $[0, 1]$ inside their finitely many height domains. Patching gives the assertion. □

**Corollary 10.23** (The exterior zero label at the boundary)**.** *After shrinking $U_*$ in Corollary 10.21, the exterior family extends jointly holomorphically to $U_* \times W_-$ for a connected height neighbourhood $W_- \supset [-1, 1]$. Throughout this parameter neighbourhood,*

$$F_\theta(-1) = 0, \qquad \partial_z F_\theta(-1) = \frac{\partial_z F_\theta(0)}{\log b(\theta)} \neq 0.$$

*Its next local backward branch at $z = -2$ has the logarithmic monodromy $2\pi\mathrm{i}/\log b(\theta)$ of Proposition 10.41. Thus the exterior boundary germ has the same zero backward label as the real inner sewn family; this label alone does not identify the two families.*

*Proof.* The proof of Proposition 10.20 applies locally at every point of the compact marked path, since Proposition 10.13 gives the same strict crossing throughout it. The resulting full-height charts agree on overlaps: they contain the transported germ at $z = 0$, and the identity theorem extends that agreement along $[0, 1]$. They therefore give a continuous family of height paths $F_\theta([0, 1])$ from the real seed to $\theta_c$.

These paths avoid 0. Indeed, each developing representative lies either in $\Phi_\theta(H_\theta)$ or in $E_{b(\theta)}(\Phi_\theta(H_\theta))$. The second set omits 0 because the exponential map does. If $\Phi_\theta(w) = 0$ for $w \in \overline{H_\theta}$, then $w = E^{\circ 28}_{b(\theta)}(0)$, contrary to the certified singular-orbit clearance of Proposition 10.5. The compact height paths and the strict clearance allow a common small simply connected tube $V \supset [0, 1]$, after restricting the parameter to a neighbourhood of $\theta_c$, on which $F_\theta$ has no zeros.

Choose its jointly holomorphic logarithm $L_\theta(z)$ on $U_* \times V$ by $L_\theta(0) = 0$, and choose the local holomorphic branch $\ell(\theta) = \log b(\theta)$ from the real exterior seed. Since $F_\theta(1) = b(\theta)$, $[L_\theta(1) - \ell(\theta)]/(2\pi\mathrm{i})$ is an integer, constant along the marked path. It is zero at the real seed, where $F_\theta([0, 1])$ is positive. Hence $L_\theta(1) = \ell(\theta)$ near $\theta_c$.

For $z \in V - 1$ put $\widetilde{F}_\theta(z) = L_\theta(z + 1)/\ell(\theta)$. On the connected overlap with $V$, the Abel equation and the choice at $z = 0$ give $\widetilde{F}_\theta = F_\theta$; shrink the tube if needed so this overlap is connected. They glue on $W_- = V \cup (V - 1)$ and give $F_\theta(-1) = L_\theta(0)/\ell(\theta) = 0$. Differentiation gives the displayed nonzero derivative because the exterior height chart is locally conformal at $z = 0$. The final assertion is Proposition 10.41. □

**Corollary 10.24** (Exterior root isolation and local normality)**.** *There are a parameter disc $U_0 \Subset U_*$ about $\theta_c$, a radius $r > 0$, and constants $0 < c < C$ such that, for every $\theta \in U_0$ and $|z| \leq r$,*

$$c|z| \leq |F_\theta(z) - 1| \leq C|z|, \qquad c \leq |F_\theta(z)| \leq C.$$

*After reducing $r$, the same constants can be chosen so that $c|z + 1| \leq |F_\theta(z)| \leq C|z + 1|$ for $|z + 1| \leq r$. Joint holomorphy also gives the exterior half of condition (P1) on every $V \Subset W_-$ across this boundary subarc, while the displayed estimates isolate its normalised root uniformly. No corresponding inner estimate is asserted.*

*Proof.* The jointly holomorphic quotient $A(\theta, z) = (F_\theta(z) - 1)/z$ extends across $z = 0$ with $A(\theta_c, 0) = \partial_z F_{\theta_c}(0) \neq 0$. Shrink $U_0$ and $r$ so that $A$ is bounded above and away from zero on $\overline{U_0} \times \overline{B(0, r)}$. The same compactness argument applies to $F_\theta$ near $F_{\theta_c}(0) = 1$ and to $F_\theta(z)/(z+1)$ near $z = -1$, using Corollary 10.23 for its nonzero value there. Enlarge $C$ and decrease $c$ to cover all three inequalities. □

**Lemma 10.25** (The integer label at either vertical end)**.** *Let $E_b(w) = b^w$ have a fixed point $L$ admitting a local Koenigs coordinate $\sigma$, with multiplier $\lambda$. Choose $\varepsilon \in \{+1, -1\}$ and $\ell = \log \lambda$ so that $0 < \varepsilon \operatorname{Im} \ell < 2\pi$. Suppose $F$ is nonconstant and holomorphic on a sufficiently high $\varepsilon$-half-plane, satisfies $F(z+1) = E_b(F(z))$, and $F(x + \varepsilon i t) \to L$ uniformly for $0 \leq x \leq 1$ as $t \to +\infty$. Then, deep in that end and in a fundamental vertical strip, there are an integer $m \geq 0$ and a holomorphic function $h$ near $q = 0$, with $h(0) \neq 0$, such that*

$$\sigma(F(z)) = e^{\ell z} q^m h(q), \qquad q = e^{\varepsilon 2\pi i z}.$$

*The Koenigs logarithm consequently has the form*

$$\frac{\log \sigma(F(z))}{\ell} = \left(1 + \frac{\varepsilon 2\pi i m}{\ell}\right) z + \frac{\log h(q)}{\ell}.$$

*The degree-one form $z+$ a bounded holomorphic 1-periodic correction at this end is equivalent to $m = 0$. For a jointly holomorphic parameter family with a common such end and holomorphically chosen $L$, $\sigma$ and $\ell$, the constant Laurent coefficient $a_0(b)$ is holomorphic in $b$ (and equals $h_b(0)$ when $m = 0$). On a connected parameter domain, if nonzero at one base, its zeros are isolated.*

*Proof.* For large $t$, the values $F(x + \varepsilon i t)$, $0 \leq x \leq 1$, lie in the Koenigs chart. There $g(z) = e^{-\ell z} \sigma(F(z))$ is 1-periodic, so it descends to a holomorphic function on a punctured $q$-disk. Its Laurent coefficient at index $n$ is

$$a_n = \int_0^1 e^{-\ell(x+\varepsilon i t)} \sigma(F(x + \varepsilon i t)) e^{-\varepsilon 2\pi i n (x+\varepsilon i t)}\, dx.$$

For $n < 0$ its modulus is bounded by a constant times $\sup_{0 \leq x \leq 1} |\sigma(F(x+\varepsilon i t))| e^{(\varepsilon \operatorname{Im} \ell + 2\pi n) t}$, which tends to zero. Thus every negative Laurent coefficient vanishes and $g$ extends holomorphically to $q = 0$. It is not identically zero, so it has finite order $m \geq 0$ there and $g = q^m h(q)$ with $h(0) \neq 0$. Taking a logarithm on the deep end gives the displayed formula. For the parameter assertion, use the same coefficient integral on a fixed sufficiently deep line locally in $b$; it is holomorphic in $b$. The identity theorem makes its zeros isolated unless it vanishes identically, which the nonzero seed excludes. □

A fixed-point limit at either end therefore does not, by itself, select the degree-one time used in Definition 8.1. This is an actual ambiguity for a repelling fixed point: the inverse Koenigs map $P$ extends to an entire Poincaré function, and for every $m \geq 0$ and $c \neq 0$ the entire solution $F_m(z) = P(c e^{(\ell + \varepsilon 2\pi i m) z})$ satisfies the functional equation and tends to $L$ uniformly on $0 \leq \operatorname{Re} z \leq 1$ as $\varepsilon \operatorname{Im} z \to +\infty$, with exactly the label $m$. These examples impose one end condition; they do not assert the normalisation or the limit at the other end. The two integer labels and their nonvanishing leading coefficients are branch data for a two-regular-chart identification.

**Corollary 10.26** (The exterior branch has degree-one time at both ends)**.** *At every real $b > \eta$, Kneser's classical solution has $m = 0$ in Lemma 10.25 at both conjugate fixed-point ends. Its two leading coefficients are nonzero and vary holomorphically and without zeros in the local parameter families of Proposition 10.1.*

*Proof.* In the compactified cylinder of that proposition, the local puncture coordinates are $\xi_+ = \exp(2\pi\mathrm{i}\log\sigma_+/\ell_+)$ and $\xi_- = \exp(-2\pi\mathrm{i}\log\sigma_-/\ell_-)$. The global sphere coordinate $q_b$ has a simple zero at the upper end and a simple pole at the lower end. Consequently $q_b = c_+(b)\xi_+(1+O(\xi_+))$ and $q_b^{-1} = c_-(b)\xi_-(1+O(\xi_-))$, with $c_\pm(b)$ holomorphic and nonzero. Taking logarithms and using $z = (2\pi\mathrm{i})^{-1}\log q_b$ shows at either end that $\log\sigma_\pm/\ell_\pm = z + C_\pm(b) + O(e^{\varepsilon 2\pi\mathrm{i}z})$, where $\varepsilon = +1$ at the upper end and $-1$ at the lower end. Thus the coefficient of $z$ is one, so the label is zero. The same local expansions give nonvanishing holomorphic leading coefficients. □

**Corollary 10.27** (Parameter-uniform exterior tails)**.** *Fix $b_0 > \eta$. There are a complex disc $\Delta$ about $b_0$, $\delta, T, \rho > 0$, and functions $h_\pm \in \mathcal{O}(\Delta \times B(0,\rho))$ which are nowhere zero after shrinking the disc and radius, such that the local exterior family satisfies*

$$\sigma_{\pm,b}(\kappa_b(z)) = \mathrm{e}^{\ell_{\pm,b}z}h_\pm(b, \mathrm{e}^{\pm 2\pi\mathrm{i}z}) \quad \text{for } -\delta < \operatorname{Re} z < 1+\delta, \quad \pm\operatorname{Im} z > T.$$

*For every smaller closed base disc $\Delta' \Subset \Delta$ and every $0 < c < \inf_{b\in\Delta'}\{\operatorname{Im}\ell_{+,b}, -\operatorname{Im}\ell_{-,b}\}$, a constant $C$ gives*

$$\sup_{\substack{b\in\Delta'\\ 0\le x\le 1}} |\kappa_b(x \pm \mathrm{i}t) - L_\pm(b)| \le C\mathrm{e}^{-ct} \qquad (t > T).$$

*The same estimate holds uniformly over a complex neighbourhood of any compact subinterval of $(\eta,\infty)$, with a possibly smaller common $c > 0$.*

*Proof.* In the proof of Corollary 10.26, the global coordinate and the upper puncture coordinate satisfy $q_b = \phi_+(b,\xi_+)$, where $\phi_+(b,0) = 0$ and $\partial_\xi\phi_+(b,0) \neq 0$. The joint inverse function theorem therefore gives $\xi_+ = q_b\psi_+(b,q_b)$ with $\psi_+$ holomorphic and nonzero near $q_b = 0$. Since $q_b = \mathrm{e}^{2\pi\mathrm{i}z}$, taking a holomorphic logarithm of $\psi_+$ and then exponentiating the relation between $\xi_+$ and $\sigma_{+,b}$ yields the asserted $h_+$; the constant integer choice of the lift is absorbed into it. At the lower end apply the same argument to $q_b^{-1} = \phi_-(b,\xi_-)$, with $\xi_- = \mathrm{e}^{-2\pi\mathrm{i}\log\sigma_{-,b}/\ell_{-,b}}$. Choose $\rho$ and $T$ uniformly after shrinking $\Delta$. On $\Delta'$, both $h_\pm$ and their reciprocals are bounded on a smaller closed $q$-disc, and $|\mathrm{e}^{\ell_{\pm,b}(x\pm\mathrm{i}t)}| \le C_0\mathrm{e}^{-c_0 t}$ for $0 \le x \le 1$ and some $c_0 > c$. The inverse Koenigs maps satisfy $\sigma_{\pm,b}^{-1}(\zeta) = L_\pm(b) + \zeta + O(\zeta^2)$ uniformly there, giving the estimate. A finite subcover of a compact real base interval gives common parameters and a positive minimum decay rate. □

**Proposition 10.28** (A normal-family criterion for base continuation)**.** *Let $\Omega$ and $W$ be connected open sets in the base and height planes, respectively, and suppose $I = \Omega\cap(\eta,\infty)$ contains a nonempty interval. Let $D \subset W$ have an accumulation point in $W$. Suppose $A_N \in \mathcal{O}(\Omega\times W)$ is locally uniformly bounded in $N$ and, for every $(b,z) \in I \times D$, $A_N(b,z) \to \kappa_b(z)$, where $\kappa_b$ is Kneser's classical solution on $W$. Then $A_N$ converges locally uniformly on $\Omega\times W$ to a jointly holomorphic family $A$, and $A_b = \kappa_b$ on $W$ for every $b \in I$. If, in addition, $U_0 \subset \Omega$ is a nonempty open set where the sewn family $K^W$ is jointly holomorphic on $U_0 \times W$, and $A_N(b,z) \to K_b^W(z)$ for $(b,z) \in U_0 \times D$, then $A = K^W$ on $U_0 \times W$.*

*Proof.* Local boundedness makes the sequence normal on $\Omega\times W$. Any two subsequential limits agree on $I \times D$ by the prescribed pointwise convergence. For each $b \in I$ the identity theorem in $z$ makes them agree on $\{b\}\times W$; for each $z \in W$ the identity theorem in $b$ then makes them agree throughout $\Omega\times\{z\}$. Hence the normal sequence has a unique cluster limit and converges locally uniformly to it. Applying the identity theorem in $z$ once more identifies $A_b$ with $\kappa_b$ for $b \in I$ and, under the additional assumption, with $K_b^W$ for $b \in U_0$. □

This gives an alternative to checking (P0)–(P2) one boundary point at a time: a single locally bounded sequence of jointly holomorphic approximants on a corridor crossing the boundary would provide the continuation and, if its inner limit were verified, the branch identification. At

each fixed finite stage, the quadrature formulas are locally holomorphic in $b$ on a cell where all logarithm and inverse-chart branches are fixed and the normalising denominator is nonzero. A common branch-safe corridor for every stage, and a bound uniform in quadrature order and iteration count, have not been established for the approximants of [24, §5]. Real-base convergence by itself cannot supply the bound: for real $b_0$ the entire functions $a_N(b) = \exp[-N - N^3(b-b_0)^2]$ tend to zero at every real $b$, but are unbounded along every vertical line $b = b_0 + iy$ with $y \neq 0$.

10.7. **Reduction of the continuation to Brjuno boundary points.** Near the cusp the continuation is established in Section 9. For a boundary arc far from $\eta$ we record how continuation across it reduces to statements at boundary points with Brjuno rotation number. Let $\Gamma$ be an open arc of the Shell–Thron boundary in the upper half-plane. Parametrise it by $\lambda_c = e^{2\pi i\alpha}$, where $\lambda_c$ is the multiplier of the fixed point $L_1$. Let $U = U_{\text{out}} \cup \Gamma \cup U_{\text{in}}$ be a neighbourhood of $\Gamma$ split by it.

**Lemma 10.29** (Radial small divisors)**.** *For $\lambda = r\lambda_c$ with $r > 0$ and every $m \geq 1$, $|\lambda^m - 1| \geq \min\big(\frac{1}{\sqrt{2}}|\lambda_c^m - 1|,\ \frac{1}{2}\big)$.*

*Proof.* Write $\lambda_c^m = e^{i\theta}$ with $|\theta| \leq \pi$. The distance from 1 to the ray $\{te^{i\theta} : t > 0\}$ equals $|\sin\theta|$ if $|\theta| \leq \pi/2$ and 1 otherwise. In the first case, $|\sin\theta| = |e^{i\theta} - 1|\cos(\theta/2) \geq |e^{i\theta} - 1|/\sqrt{2}$. In the second, $|e^{i\theta} - 1| \leq 2$. □

The local germ has a useful exact form. Since $b = \exp(\lambda e^{-\lambda})$ and $L = e^{\lambda}$, the coordinate $v = (\log b)(w - L)$ transforms $w \mapsto b^w$ into

$$f_\lambda(v) = \lambda(e^v - 1). \tag{19}$$

**Lemma 10.30** (Radial convergence of local charts)**.** *Let $\lambda_c = e^{2\pi i\alpha}$ with $\alpha$ Brjuno, and put $\lambda = r\lambda_c$. The normalised inverse Schröder charts $H_\lambda$, defined by $f_\lambda \circ H_\lambda = H_\lambda \circ (\lambda\,\cdot)$ and $H_\lambda(0) = 0$, $H_\lambda'(0) = 1$, are holomorphic and uniformly bounded on one fixed disk $|v| < \rho$ for $r$ sufficiently close to 1 from either side. They converge locally uniformly there to the Siegel chart $H_{\lambda_c}$. On a possibly smaller fixed disk, their inverses have the same convergence property.*

*Proof.* The map $v \mapsto e^v - 1$ is univalent on $|v| < 1$. Thus the cone-uniform Brjuno majorant estimate of [17] applies directly to (19); the radial paths lie in such a cone. Lemma 10.29 gives the requisite uniform lower bounds for the small divisors. Each Taylor coefficient of $H_\lambda$ is rational in $\lambda$ with poles only at roots of unity, hence converges to its value at the irrational $\lambda_c$. The uniform majorant makes this coefficientwise convergence locally uniform on a fixed disk. Since $H_\lambda'(0) = 1$, Cauchy estimates on that disk give a smaller disk on which the $H_\lambda$ are uniformly close to the identity and univalent; their inverses then converge locally uniformly as well. □

This lemma concerns germs at $L$. Neither the Siegel disk nor its finite-forward-pullback domain supplies a complete sewing line (Proposition 10.42).

**Lemma 10.31** (The other fixed points are repelling)**.** *For $b = \exp(\lambda e^{-\lambda})$ with $0 < |\lambda| \leq 1$ and $\lambda \neq 1$, every fixed point of $E_b$ other than $L_1 = e^{\lambda}$ is simple and has a multiplier of modulus greater than 1.*

*Proof.* Put $a = \log b = \lambda e^{-\lambda}$ and $g(\mu) = \mu e^{-\mu}$. Fixed points $w = E_b(w)$ correspond to $g(\mu) = a$ through $\mu = aw$, $w = e^{\mu}$; $\mu$ is their multiplier. The chosen logarithm is indeed $a$, since $|a| \leq e < \pi$. We show that $g$ is injective on the closed unit disk. In polar coordinates,

$$\arg g(re^{i\theta}) = \theta - r\sin\theta, \qquad |g(re^{i\theta})| = re^{-r\cos\theta}.$$

For each $0 < r \leq 1$, the first expression is strictly increasing in $\theta$ modulo $2\pi$. Holding its value fixed and varying $r < 1$, implicit differentiation gives

$$\frac{d}{dr} \log |g(re^{i\theta(r)})| = \frac{|1 - re^{i\theta(r)}|^2}{r(1 - r\cos\theta(r))} > 0.$$

Thus different radii cannot have the same image, including by continuity at $r = 1$. Since $\lambda$ is already a preimage of $a$ in the closed unit disk, every other preimage has modulus greater than 1. Finally $g'(\mu) = e^{-\mu}(1 - \mu)$ vanishes only at $\mu = 1$, which cannot be another preimage when $\lambda \neq 1$. □

**Proposition 10.32** (Transport of the lower fixed point)**.** *Fix $\lambda_0 \in (0, 1)$ and its real second multiplier $\mu_0 > 1$, where $\mu_0 e^{-\mu_0} = \lambda_0 e^{-\lambda_0}$. Let $\gamma : [0, 1] \to \overline{\mathbb{D}} \setminus \{0, 1\}$ be a piecewise $C^1$ path with $\gamma(0) = \lambda_0$, $\gamma([0, 1)) \subset \mathbb{D}$, and $\gamma(1) = \lambda_c \in \partial\mathbb{D} \setminus \{1\}$. Then $\mu_0$ continues analytically along $\gamma$ to a multiplier $\mu_c$ with $|\mu_c| > 1$. The associated lower fixed point and its regular superfunction continue along a narrow parameter corridor and extend locally jointly holomorphically in the base and $z$ through $b_c = \exp(\lambda_c e^{-\lambda_c})$, with a transported branch of $\log \mu$.*

*Proof.* Put $g(t) = te^{-t}$. At every point of the path, the implicit equation $g(\mu) = g(\lambda)$ has a locally unique continuation of the second root as long as $\mu \neq 1$. Lemma 10.31 keeps it outside the closed unit disk, so it cannot meet the root $\lambda$. On the compact trace $K = \gamma([0, 1])$, which avoids 1, there is a $\delta > 0$ such that every such second root obeys $|\mu - 1| \geq \delta$: otherwise $\mu_j \to 1$ and $\lambda_j \to \lambda_* \in K$ would give $g(\lambda_*) = g(1)$, contradicting injectivity of $g$ on the closed unit disk. Along each smooth piece of $\gamma$,

$$\frac{d\mu}{d\lambda} = \frac{g'(\lambda)}{g'(\mu)} = \frac{1 - \lambda}{\lambda} \frac{\mu}{1 - \mu}.$$

Here $|\lambda|$ is bounded below on $K$, and $|\mu/(1 - \mu)| \leq 1 + 1/\delta$. The derivative is therefore bounded by a constant times the path speed. At any hypothetical first stopping point, finite path length gives a finite limit $\mu \neq 1$; the implicit-function theorem would extend the branch. Thus it reaches $\lambda_c$. It remains outside the unit disk by Lemma 10.31: it cannot merge with $\lambda_c$, because $g'(\lambda_c) \neq 0$ would make the local root uniquely equal to $\lambda$ near $\lambda_c$, contrary to $|\mu| > 1$ inside the disk. The lower Koenigs construction and the proof of Lemma 10.38 apply in neighbourhoods along the path and at its endpoint. Their normalisation and the continued branch of $\log \mu$ patch these local superfunctions. Finally $d \exp(g(\lambda))/d\lambda \neq 0$ at $\lambda_c \neq 1$, so the endpoint chart may be expressed in the base $b$. □

This transports the lower regular chart only. It does not transport the sewing correction or the solution $K^W$ to the boundary arc.

**Corollary 10.33** (A global lower regular family in the inner parameter disk)**.** *Let $\Omega = \mathbb{D} \setminus (-1, 0]$. The real second multiplier $\mu_2 > 1$ extends to a single-valued holomorphic function on $\Omega$ with $|\mu_2(\lambda)| > 1$. It has a holomorphic logarithm fixed by the positive real value on $(0, 1)$. With this choice, the lower regular superfunction $S_\lambda(z)$ is jointly holomorphic on $\Omega \times \mathbb{C}$ and extends locally across every point of $\partial\mathbb{D} \setminus \{1, -1\}$.*

*Proof.* The continuation argument of Proposition 10.32 works along every finite path in $\Omega$. Since $\Omega$ is simply connected, the monodromy theorem makes the continuation single-valued. Lemma 10.31 gives $|\mu_2| > 1$, hence a global holomorphic logarithm. At each parameter, the Koenigs coordinate at $L_2 = e^{\mu_2}$, normalised by derivative 1, is unique and varies locally holomorphically; so the local constructions of Lemma 10.38 patch. Its forward-iterate formula makes $S_\lambda$ entire in $z$ and jointly holomorphic locally in $(\lambda, z)$. Proposition 10.32 gives the stated local boundary extensions. □

For $0 < |\lambda| < 1$ the normalised attracting Koenigs germ also varies locally holomorphically in $\lambda$, since its coefficient equations have no interior resonances. Thus both local fixed-point charts can be transported through $\Omega$; persistence of the overlap needed for their sewing remains unproved.

**Proposition 10.34** (Infinite monodromy around $b = 1$)**.** *The map $\lambda \mapsto b = \exp(\lambda e^{-\lambda})$ is injective on $\mathbb{D}$. On the universal cover of $0 < |\lambda| < 1$, the lower multiplier is single-valued and holomorphic, and its regular superfunction is jointly holomorphic in the parameter and height. The lower multiplier branch of Corollary 10.33 has infinite-order monodromy around $b = 1$: continuing it around any positively oriented loop that winds once around $b = 1$ in the image of $0 < |\lambda| < 1$ changes its value, and its values after different integer winding numbers are distinct. Consequently neither the lower fixed point nor its normalised lower regular superfunction is single-valued on that punctured base domain.*

*Proof.* Injectivity of $g(\lambda) = \lambda e^{-\lambda}$ on $\mathbb{D}$ was proved in Lemma 10.31. Since $|g(\lambda)| < e < \pi$, the exponential is injective on $g(\mathbb{D})$ as well. Moreover $b'(\lambda) = b(\lambda)e^{-\lambda}(1 - \lambda) \neq 0$ in $\mathbb{D}$, so this is a conformal parameterisation. The continuation proof of Proposition 10.32 applies to every finite path in the punctured disk. Monodromy on its simply connected universal cover gives a holomorphic root $\mu$. Since $|\mu| > 1$, it has a global holomorphic logarithm there. The local construction of Lemma 10.38 then patches to give the asserted superfunction.

Choose $0 < \varepsilon < 1$ so small that $R := -\log \varepsilon > \varepsilon$, and continue the real lower root $\mu_0 > 1$ along $\lambda(t) = \varepsilon e^{it}$, with $t$ running from 0 to $2\pi n$ for any integer $n$. The path-continuation proof of Proposition 10.32 applies to every finite number of turns. All continued roots obey $|\mu(t)| > 1$. Their common modulus equation gives

$$\operatorname{Re} \mu(t) = R + \operatorname{Re} \lambda(t) + \log |\mu(t)| > R - \varepsilon > 0.$$

Thus the principal $\operatorname{Log} \mu(t)$ remains defined throughout. Taking continuous logarithms in $g(\mu(t)) = g(\lambda(t))$ and using the real equation at $t = 0$ yields

$$\mu(t) - \operatorname{Log} \mu(t) = R - it + \varepsilon e^{it}.$$

At the end of $n$ turns, the right side is $R + \varepsilon - 2\pi i n$. Since the same single-valued function $\mu \mapsto \mu - \operatorname{Log} \mu$ is used at every endpoint, roots obtained after different winding numbers cannot coincide. A radial path from any real $\lambda_0 \in (0, 1)$ to $\varepsilon$ conjugates this monodromy to the loop based at $\lambda_0$.

Distinct endpoint roots give distinct fixed points $L_2 = e^{\mu}$, because $\mu = (\log b)L_2$, and distinct regular superfunctions, because $S(x) \to L_2$ as $x \to -\infty$. Homotopy invariance extends the conclusion to every loop of the same winding number. □

**Proposition 10.35** (The upper chart at the normalising orbit)**.** *Let $0 < |\lambda| < 1$, $a = \lambda e^{-\lambda}$, $b = e^a$, and $L_1 = e^{\lambda}$. Then $E_b^{\circ n}(1) \to L_1$ locally uniformly in $\lambda$. The attracting Koenigs coordinate $\sigma_\lambda$, normalised by $\sigma_\lambda(L_1) = 0$ and $\sigma'_\lambda(L_1) = 1$, extends to the attracting basin of $L_1$ and depends locally jointly holomorphically on $(\lambda, w)$ there. In particular $c(\lambda) = \sigma_\lambda(1)$ is holomorphic and nonzero on the punctured unit disk, and $\sigma'_\lambda(1) \neq 0$. It extends across $\lambda = 0$ with*

$$c(\lambda) = -\lambda - \tfrac{1}{2}\lambda^2 + O(\lambda^3).$$

*Consequently there is a unique holomorphic function $h$ on $|\lambda| < 1$ with $h(0) = 0$ and $c(\lambda) = -\lambda \exp h(\lambda)$; in particular $h(\lambda) = \lambda/2 + O(\lambda^2)$.*

*On the universal cover of $0 < |\lambda| < 1$, the normalised upper regular superfunction germ*

$$R_\lambda(z) = \sigma_\lambda^{-1}(c(\lambda) \exp(z \log \lambda)), \qquad R_\lambda(0) = 1,$$

*is jointly holomorphic near each parameter and $z = 0$. Along every compact parameter path avoiding 0 it has a common positive height radius on a sufficiently narrow corridor. On a*

*possibly smaller common disk, $R_\lambda(z) = 1$ only at $z = 0$, and $R_\lambda$ and $1/R_\lambda$ are uniformly bounded along the path. The deck transformation around $\lambda = 0$ acts with infinite order on these normalised upper germs.*

*Proof.* Under the change of coordinate $v = aw$, the map $E_b$ becomes $v \mapsto ae^v$, whose attracting fixed point is $v = \lambda$ and whose only finite singular value is 0. The singular-value theorem for exponential maps places 0 in the immediate attracting basin of this fixed point [28, §2]. Since $0 \mapsto a$ in the $v$ coordinate, $w = 1$ belongs to the same basin. Attraction is locally uniform in the parameter: at any fixed $\lambda_0$, a finite iterate of 1 enters a small invariant contraction disk about $L_1(\lambda_0)$, and this persists for nearby parameters.

The local Koenigs coordinate varies holomorphically with $\lambda$. For a basin point $w$, choose $N$ with $E_b^{\circ N}(w)$ in its local domain and define

$$\sigma_\lambda(w) = \lambda^{-N}\sigma_\lambda(E_b^{\circ N}(w)).$$

This formula is independent of larger $N$ and locally jointly holomorphic. Its $w$ derivative never vanishes, since $E_b$ has no critical points and the local Koenigs coordinate is univalent. The local coordinate has only the zero $L_1$, so $c(\lambda) = 0$ exactly when a finite iterate of 1 lands on $L_1$. In that case the singular orbit of $v \mapsto ae^v$ would be finite. For a postsingularly finite exponential map, a preperiodic dynamic ray lands at its singular value [15, Thm. 3.1]; its points escape, so it cannot land at a point in the open attracting basin. This contradicts the membership of 0 in that basin proved above. Thus $c$ never vanishes.

To analyse the removable endpoint $\lambda = 0$, write $u = w - L_1$ and

$$G_\lambda(u) = E_b(L_1 + u) - L_1 = e^\lambda(\exp(\lambda e^{-\lambda}u) - 1) = \lambda u + O(\lambda^2 u^2).$$

On a fixed small $u$-disk and for sufficiently small $|\lambda|$, $|G_\lambda(u)| \le C|\lambda||u|$ with $C^2|\lambda| < 1$. Put $u_n = G_\lambda^{\circ n}(u)$. Each $\lambda^{-n}u_n$ has a removable singularity at $\lambda = 0$, and

$$\left|\lambda^{-n-1}u_{n+1} - \lambda^{-n}u_n\right| \le C'|\lambda|(C^2|\lambda|)^n|u|^2.$$

The telescoping limit is the normalised Koenigs coordinate and extends jointly holomorphically through $\lambda = 0$ with $\sigma_\lambda(L_1 + u) = u + O(\lambda u^2)$. Since $1 - L_1 = -\lambda - \lambda^2/2 + O(\lambda^3)$, this gives the stated expansion of $c$. The function $-c(\lambda)/\lambda$ is now holomorphic and zero-free on the whole unit disk, with value 1 at the origin. It therefore has the asserted normalised holomorphic logarithm.

Choose a holomorphic branch of $\log\lambda$ on the universal cover. The inverse-function theorem at $w = 1$, applied to $\sigma'_\lambda(1) \ne 0$, gives the displayed germ and its joint holomorphy. A finite cover of a compact lifted parameter path gives one positive height radius after narrowing its corridor. Since $R'_\lambda(0) = c(\lambda)\log\lambda/\sigma'_\lambda(1) \ne 0$, the same compactness argument gives a smaller common root-isolation disk. Shrink it once more to avoid zeros of $R_\lambda$ and obtain the stated bounds. On one positive turn around 0, $c(\lambda)$ and $\sigma'_\lambda(1)$ return unchanged, whereas $\log\lambda$ gains $2\pi\mathrm{i}$. After $n$ turns, $R'_\lambda(0)$ therefore changes by $2\pi\mathrm{i}nc(\lambda)/\sigma'_\lambda(1) \ne 0$ for $n \ne 0$. □

**Corollary 10.36** (Diverging chart-entry depth at a Brjuno boundary point)**.** *Let $\lambda_c = e^{2\pi\mathrm{i}\alpha} \ne 1$ with $\alpha$ Brjuno, $\lambda = r\lambda_c$, and $b(r) = \exp(\lambda e^{-\lambda})$. Choose $\rho > 0$ so that the charts of Lemma 10.30 are univalent near $\{|\zeta| \le \rho\}$ for $r$ close to 1 from either side, and let $0 < s < \rho$. Put*

$$\mathcal{D}_{r,s} = L_1(b(r)) + (\log b(r))^{-1}H_{r\lambda_c}(B(0,s)).$$

*For every fixed $N \ge 0$, the point $E_{b(r)}^{\circ N}(1)$ lies outside $\mathcal{D}_{r,s}$ for all $r \ne 1$ sufficiently close to 1. For $r < 1$ the first-entry time*

$$m_s(r) = \min\{n \ge 0 : E_{b(r)}^{\circ n}(1) \in \mathcal{D}_{r,s}\}$$

*is finite and tends to infinity as $r \uparrow 1$.*

*Proof.* For the boundary map $f_{\lambda_c}(v) = \lambda_c(e^v - 1)$, the singular value $-\lambda_c$ belongs to the Julia set whenever the map has a Siegel disk [26, Introduction, remark after Thm. 2]. Under the coordinate $v = (\log b_c)(w - L_1(b_c))$, this singular value is $w = 0$. The Julia set is forward invariant, and $E_{b_c}(0) = 1$; hence every $E_{b_c}^{\circ N}(1)$ belongs to the Julia set and lies outside the boundary Siegel chart $\mathcal{D}_{1,\rho}$.

If the first assertion failed, there would be $r_j \to 1$, $r_j \neq 1$, and $\zeta_j \in B(0,s)$ with

$$E_{b(r_j)}^{\circ N}(1) = L_1(b(r_j)) + (\log b(r_j))^{-1} H_{r_j\lambda_c}(\zeta_j).$$

After taking a subsequence, $\zeta_j \to \zeta$ with $|\zeta| \le s < \rho$. Lemma 10.30 and finite-iterate continuity put $E_{b_c}^{\circ N}(1)$ in $\mathcal{D}_{1,\rho}$, a contradiction. For each $r < 1$, Proposition 10.35 says the orbit of 1 converges to $L_1(b(r))$, so it eventually enters $\mathcal{D}_{r,s}$. The first assertion excludes every bounded entry time as $r \uparrow 1$. □

This concerns finite pullbacks of a fixed local upper chart at the normalising orbit. It does not preclude continuation of the sewn solution by a different mechanism.

**Proposition 10.37** (Collapse of the attracting basin at a Brjuno boundary point)**.** *Let $\lambda_c = e^{2\pi i\alpha} \neq 1$ with $\alpha$ Brjuno, $b(\lambda) = \exp(\lambda e^{-\lambda})$, and let $\mathcal{A}_\lambda$ be the immediate attracting basin of $L_1(\lambda) = e^\lambda$ for $0 < |\lambda| < 1$. Then $1 \in \mathcal{A}_\lambda$ and*

$$d(\lambda) := \operatorname{dist}(1, \mathbb{C} \setminus \mathcal{A}_\lambda) \longrightarrow 0 \qquad (\lambda \to \lambda_c,\ |\lambda| < 1).$$

*The Poincaré distance within the immediate basin also satisfies*

$$\operatorname{dist}^{\mathrm{hyp}}_{\mathcal{A}_\lambda}(1, b(\lambda)) \longrightarrow \infty \qquad (\lambda \to \lambda_c,\ |\lambda| < 1).$$

*Moreover, let $\lambda_j \to \lambda_c$ from the disk and let $V$ be a connected height domain containing 0. If holomorphic maps $G_j : V \to \mathcal{A}_{\lambda_j}$ satisfy $G_j(0) = 1$, then $G_j \to 1$ locally uniformly on $V$, without a boundedness assumption. Consequently, if $1 \in V$, there is a neighbourhood $U$ of $\lambda_c$ such that no $\lambda \in U \cap \{0 < |\lambda| < 1\}$ admits a holomorphic $G : V \to \mathcal{A}_\lambda$ with $G(0) = 1$ and $G(1) = b(\lambda)$.*

*Proof.* The proof of Proposition 10.35 puts 0 in the immediate attracting basin for every inner parameter; since $E_{b(\lambda)}(0) = 1$, this basin also contains 1. The proof of Corollary 10.36 puts 1 in the Julia set of the boundary map $E_{b(\lambda_c)}$. Repelling periodic points are dense in this Julia set [2, Thm. 4]. Given $\epsilon > 0$, choose such a point $p_c$ with $|p_c - 1| < \epsilon/2$. If its period is $m$, then $|(E_{b(\lambda_c)}^{\circ m})'(p_c)| > 1$, so the implicit-function theorem continues it to repelling periodic points $p_\lambda$ for all $\lambda$ near $\lambda_c$. They lie outside $\mathcal{A}_\lambda$ and satisfy $|p_\lambda - 1| < \epsilon$. Thus $d(\lambda) < \epsilon$ near $\lambda_c$.

For the second assertion, first note that $b(\lambda_c) \neq 1$: indeed $0 < |\lambda_c e^{-\lambda_c}| \le e < 2\pi$, so its exponential cannot equal 1. Both 1 and $b(\lambda_c) = E_{b(\lambda_c)}(1)$ belong to the Julia set. Choose a repelling periodic point $q_c \neq 1$ near $b(\lambda_c)$; it continues to repelling periodic points $q_j \to q_c$ of $E_{b(\lambda_j)}$. By the first assertion, choose $p_j \in \mathbb{C} \setminus \mathcal{A}_{\lambda_j}$ with $p_j \to 1$. For all large $j$ the functions

$$H_j(z) = \frac{G_j(z) - p_j}{q_j - p_j}$$

omit both 0 and 1, while $H_j(0) \to 0$. Montel's theorem gives spherical normality of this family. Every subsequential limit takes the value 0 at the origin; Hurwitz's theorem for the zero-free $H_j$ forces that limit to be identically zero. Hence $H_j \to 0$ and $G_j \to 1$ locally uniformly on $V$. Since $\mathcal{A}_{\lambda_j}$ omits the two distinct points $p_j, q_j$, it is a hyperbolic domain. If the asserted Poincaré distances stayed bounded along a sequence, take universal covering maps $\phi_j : \mathbb{D} \to \mathcal{A}_{\lambda_j}$ with $\phi_j(0) = 1$. The lifts of $b(\lambda_j)$ can then be chosen as points $t_j$ in a fixed compact subdisk of $\mathbb{D}$. Applying the preceding collapse to $\phi_j$ gives $b(\lambda_j) = \phi_j(t_j) \to 1$, contrary to $b(\lambda_c) \neq 1$. If $1 \in V$ and $G_j(1) = b(\lambda_j)$, this gives $b(\lambda_c) = 1$, a contradiction. The neighbourhood assertion follows by choosing a contrary sequence $\lambda_j \to \lambda_c$. □

This collapse rules out a fixed basin-valued upper chart carrying both normalising heights 0 and 1 near the boundary. If a normalised holomorphic solution exists on such a fixed connected height domain, its image must meet $\partial\mathcal{A}_\lambda \subset J(E_{b(\lambda)})$ for all sufficiently close inner parameters. This does not decide whether the sewn family reaches such a fixed height domain near $\lambda_c$; continuation through another chart remains possible.

**Lemma 10.38** (Parameter stability of the lower chart)**.** *Let $b_c$ be a point of the boundary arc at which the chosen lower fixed point $L_2(b_c)$ is simple and its multiplier $\mu_c = E_{b_c}'(L_2(b_c))$ satisfies $|\mu_c| > 1$. Choose the local branches of $\log b$ and $\log\mu$ used to define $E_b$ and the lower regular superfunction. Near $b_c$, the normalised Koenigs coordinate $\tau_b$ at $L_2(b)$ and its local inverse depend jointly holomorphically on the base and their arguments on fixed disks. Moreover the entire superfunction*

$$S_b(z) = E_b^{\circ m}\Big(\tau_b^{-1}\big(-\exp((z-m)\log\mu(b))\big)\Big)$$

*is independent of every sufficiently large integer $m$, locally jointly holomorphic in $(b,z)$, and converges with all $z$-derivatives locally uniformly to $S_{b_c}$ as $b \to b_c$ from either side.*

*Proof.* The implicit-function theorem gives holomorphic $L_2(b)$ and $\mu(b)$ near $b_c$. In the coordinate $u = w - L_2(b)$, set $G_b(u) = E_b(L_2(b)+u) - L_2(b)$ and let $h_b$ be its local inverse. Then $h_b'(0) = \mu(b)^{-1}$. Choose $|\mu_c|^{-1} < q < |\mu_c|^{-1/2}$. After shrinking the parameter neighbourhood and a fixed disk $|u| < r$, we have $|h_b(u)| \le q|u|$, $|\mu(b)h_b(u) - u| \le C|u|^2$, and $Mq^2 < 1$, where $M = \sup_b|\mu(b)|$. Hence

$$\tau_b(L_2(b)+u) := \lim_{n\to\infty} \mu(b)^n h_b^{\circ n}(u)$$

converges normally in $(b,u)$: successive terms differ by at most $C(Mq^2)^n|u|^2$. It has derivative 1 at the fixed point and obeys $\tau_b(E_b(w)) = \mu(b)\tau_b(w)$ wherever both sides are local. The inverse-function theorem, uniformly after one more shrinking, gives jointly holomorphic inverses on a fixed disk about 0.

Since $\operatorname{Re}\log\mu(b) = \log|\mu(b)|$ stays positive, for any compact set of $(b,z)$ one integer $m$ puts $-\exp((z-m)\log\mu(b))$ in that inverse disk. The local conjugacy makes the displayed formula independent of increasing $m$. These formulas therefore define a locally jointly holomorphic function for all $z \in \mathbb{C}$. Holomorphy and Cauchy's formula give the asserted locally uniform convergence, including all $z$-derivatives. □

**Proposition 10.39** (Backward-value label of the lower regular family)**.** *In the parameter neighbourhood of Lemma 10.38, put $a(b) = \log b \neq 0$. For every $k \in \mathbb{Z}\setminus\{0\}$ there is a simple root $c$ of $S_{b_c}(c) = 1$ with $S_{b_c}(c-1) = 2\pi\mathrm{i}k/a(b_c)$. Each such root continues locally holomorphically as $c(b)$, and $G_b(z) := S_b(z + c(b))$ is a jointly holomorphic, entire-in-$z$ solution with $G_b(0) = 1$ and*

$$G_b(-1) = \frac{2\pi\mathrm{i}k}{\log b}.$$

*Conversely every normalised translate of $S_b$ has one of these nonzero integer labels, which is constant along local parameter continuation. In fact every entire-in-$z$ solution with $F(0) = 1$ has a nonzero backward-value label. For real $1 < b < \eta$, the geometric sewn solution satisfies $K_b^W(-1) = 0$; this identity persists on its local complex parameter discs. Consequently a normalised lower regular translate cannot agree with the sewn solution on a common connected regular $z$-domain containing $-1$.*

*Proof.* The functional equation gives $S_b(z) = E_b(S_b(z-1)) \neq 0$, so the nonconstant entire function $S_b$ omits 0. Picard's theorem makes it attain every nonzero value, including $2\pi\mathrm{i}k/a(b_c)$. Choose $t$ with that value and put $c = t+1$; then $S_{b_c}(c) = 1$. The formula in Lemma 10.38, differentiated after choosing a sufficiently large $m$, shows $S_b'(z) \neq 0$: the derivatives of $E_b^{\circ m}$

and of the local inverse $\tau_b^{-1}$, and the exponential factor, are all nonzero. Thus the root $c$ is simple and the implicit-function theorem continues it holomorphically in $b$. The displayed value at $-1$ persists by the functional equation and continuity. Conversely $S_b(c) = 1$ forces $a(b)S_b(c-1) \in 2\pi\mathrm{i}\mathbb{Z}$; omission of 0 excludes the zero integer. The integer is locally constant because it depends holomorphically on $b$. For an arbitrary entire solution $F$, the same argument applies: $F(z) = E_b(F(z-1))$ makes $F$ zero-free, so $F(0) = 1$ forces $F(-1) = 2\pi\mathrm{i}k/a(b)$ with $k \neq 0$.

For real $b \in (1, \eta)$, the marked upper point $w = 0$ of Proposition 3.4 corresponds to $z = 0$, and its deck translate $w = -1$ corresponds to $z = -1$. The real attracting regular branch has $R_b(-1) = 0$, since $E_b(0) = 1 = R_b(0)$ and this inverse branch is real. Hence $K_b^W(-1) = 0$. Proposition 3.5 and the identity theorem preserve the equality on each local parameter disc. The two values at $-1$ exclude the asserted agreement. □

**Corollary 10.40** (Rigidity of the backward branch)**.** *Let $U$ be a connected base domain on which $a(b) = \log b$ is holomorphic and nonzero. Suppose $F_b(z)$ is jointly holomorphic in $(b, z)$ near $U \times \{-1, 0\}$, satisfies $F_b(0) = 1$, and obeys the tetration equation near $z = -1$. Then*

$$k(b) := \frac{a(b)F_b(-1)}{2\pi\mathrm{i}} \in \mathbb{Z}$$

*is constant on $U$. If $k = 0$, none of the $F_b$ can extend holomorphically through $z = -2$ while preserving the tetration equation; in particular they cannot be entire solutions. The inner sewn family has $k = 0$ on its real-axis parameter neighbourhood, as does any continuation of Kneser's real-normalised family from $b > \eta$ that remains regular at $z = -1$. Thus their values and all base derivatives at $z = -1$ agree automatically wherever both branches are defined.*

*Proof.* The equation at $z = -1$ gives $\exp(a(b)F_b(-1)) = 1$. The displayed function is holomorphic and integer-valued, hence constant. If $k = 0$ and $F_b$ extended regularly through $-2$, the equation would give $E_b(F_b(-2)) = F_b(-1) = 0$, impossible because an exponential never vanishes. Proposition 10.39 gives $k = 0$ for the sewn family. Kneser's family has $F_b(-1) = 0$ for real $b > \eta$ by its real normalisation; the identity theorem transports this value on any stated parameter continuation. □

**Proposition 10.41** (Logarithmic monodromy at the first backward singularity)**.** *Let $U$ be a simply connected parameter disc on which $a(b) = \log b \neq 0$. Suppose $F_b(z)$ is jointly holomorphic near $U \times \{-1\}$, $F_b(-1) = 0$, and $F_b'(-1) \neq 0$. Locally in $(b, t)$ write $F_b(-1 + t) = t\,u_b(t)$ with $u_b(0) \neq 0$. On any simply connected sector $0 < |t| < r$, all holomorphic backward branches satisfying $\exp(a(b)G_b(-2 + t)) = F_b(-1 + t)$ have the form*

$$G_b(-2 + t) = \frac{\operatorname{Log} t + \log u_b(t) + 2\pi\mathrm{i}k}{a(b)}, \qquad k \in \mathbb{Z}.$$

*Continuation once around $t = 0$ adds $2\pi\mathrm{i}/a(b)$. Hence no single-valued holomorphic or meromorphic backward branch satisfying the equation exists on a full punctured disc about $z = -2$. For every real $b \in (1, \eta)$, the sewn solution $K_b^W$ has a simple zero at $z = -1$, so this description applies to it and to its local holomorphic parameter family.*

*Proof.* The quotient $u_b(t)$ extends jointly holomorphically through $t = 0$. After shrinking $U$ and $r$, it has a joint holomorphic logarithm. Two logarithms of $tu_b(t)$ on a connected sector differ by a constant multiple of $2\pi\mathrm{i}$, giving the formula. A turn about $t = 0$ changes $\operatorname{Log} t$ by $2\pi\mathrm{i}$. If a single-valued backward branch existed on the punctured disc, integrating the logarithmic derivative of $\exp(aG) = tu$ around a small circle would give both 0 and $2\pi\mathrm{i}$. A meromorphic branch has no pole away from the centre, since its exponential would have an essential singularity there; it is therefore subject to the same contradiction.

For the sewn family, the upper chart of Proposition 3.4 contains $w = -1$ and its inverse uniformising coordinate $w(z)$ is conformal, with $w(-1) = -1$. Proposition 10.35 gives $R_b'(0) \neq 0$, and differentiating the functional equation at $z = -1$ gives $R_b'(-1) = R_b'(0)/a(b) \neq 0$. Thus $(K_b^W)'(-1) = R_b'(-1)w'(-1) \neq 0$. Proposition 3.5 preserves this simple zero on a sufficiently small complex parameter disc about each real base. □

This monodromy is shared by every simple-zero branch with $F(-1) = 0$; it does not by itself identify the two families in (P2).

The families $G_b$ show that the functional equation and the normalisation $F(0) = 1$ allow entire solutions jointly holomorphic across a non-cusp boundary point. The integer at $z = -1$ distinguishes these families from the inner sewn branch. Thus (P0) still requires continuation of the *specified* branches with backward value 0.

**Proposition 10.42** (Siegel pullbacks have no complete sewing band)**.** *Let $\lambda_c = \mathrm{e}^{2\pi\mathrm{i}\alpha}$ be Brjuno and different from 1, and let $b_c = \exp(\lambda_c \mathrm{e}^{-\lambda_c})$. Take $\rho > 0$ small enough that the chart of Lemma 10.30 is univalent on a neighbourhood of $\{|\zeta| \le \rho\}$ and its image lies in that lemma's common inverse-chart disk, and put*

$$\mathcal{D}_\rho = L_1(b_c) + (\log b_c)^{-1} H_{\lambda_c}(\{|\zeta| < \rho\}), \qquad \chi_c(w) = H_{\lambda_c}^{-1}\big((\log b_c)(w - L_1(b_c))\big).$$

*Then $E_{b_c}(\mathcal{D}_\rho) = \mathcal{D}_\rho$ and $\chi_c \circ E_{b_c} = \lambda_c \chi_c$ there. Define*

$$\Omega_\rho = \{z \in \mathbb{C} : \ S_{b_c}(z+n) \in \mathcal{D}_\rho \text{ for some } n \in \mathbb{Z}_{\ge 0}\}, \qquad V_c(z) = \lambda_c^{-n} \chi_c(S_{b_c}(z+n)).$$

*The following assertions hold.*

1. *$\Omega_\rho$ is nonempty, open and invariant under $z \mapsto z+1$ and $z \mapsto z-1$. The formula for $V_c$ is independent of the admissible integer $n$ and defines a holomorphic function with $V_c(z+1) = \lambda_c V_c(z)$. It has zeros, while $S_{b_c}^{-1}(1) \neq \varnothing$ and $S_{b_c}^{-1}(1) \cap \Omega_\rho = \varnothing$.*
2. *$\Omega_\rho$ contains no nonempty path-connected set $C$ with $C + m = C$ for any nonzero integer $m$. In particular it contains no complete horizontal line, horizontal strip, or lift of an essential closed curve in $\Omega_\rho/\mathbb{Z}$. The same statement holds if $\mathcal{D}_\rho$ is replaced by the maximal Siegel disk.*
3. *For every compact $K \subset \Omega_\rho$ there is one integer $N$ such that the pulled-back upper coordinates*

$$V_{b,N}(z) = \lambda(b)^{-N} H_{\lambda(b)}^{-1}\big((\log b)(S_b(z+N) - L_1(b))\big)$$

*are defined near $K$ for $b = \exp(\lambda \mathrm{e}^{-\lambda})$, $\lambda = r\lambda_c$, and $r$ sufficiently close to 1 from either side. They converge to $V_c$ locally uniformly near $K$, together with all $z$-derivatives.*

*Proof.* The Schröder identity and $|\lambda_c| = 1$ make $\mathcal{D}_\rho$ invariant in both directions. The admissible sets $\{z : S_{b_c}(z+n) \in \mathcal{D}_\rho\}$ increase with $n$. If $m \ge n$ are both admissible, the Schröder identity gives $\chi_c(S_{b_c}(z+m)) = \lambda_c^{m-n} \chi_c(S_{b_c}(z+n))$. This proves independence, holomorphy and the translation identity; admissibility after either integer translation follows by increasing the pullback depth when needed. The nonconstant entire function $S_{b_c}$ omits 0, since its defining formula has at least one exponential iterate. Little Picard's theorem therefore makes it attain $L_1(b_c) \neq 0$; any such preimage lies in $\Omega_\rho$ and is a zero of $V_c$. The same argument shows that $S_{b_c}$ attains 1. By Corollary 10.36, 1 belongs to the Julia set of $E_{b_c}$. Every point in its forward orbit remains in that set, whereas $\mathcal{D}_\rho$ lies in the Fatou set. Hence no $z$ with $S_{b_c}(z) = 1$ can satisfy the defining condition of $\Omega_\rho$.

Put $a = \log b_c$ and $\mathcal{B} = \bigcup_{n \ge 0} E_{b_c}^{-n}(\mathcal{D}_\rho)$. Because $E_{b_c}$ maps $\mathcal{D}_\rho$ bijectively onto itself,

$$E_{b_c}^{-1}(\mathcal{D}_\rho) = \bigcup_{k \in \mathbb{Z}} (\mathcal{D}_\rho + 2\pi\mathrm{i}k/a),$$

a disjoint union. Thus $\mathcal{D}_\rho$ is a component of its first preimage. Inductively it is a component of every $E_{b_c}^{-n}(\mathcal{D}_\rho)$: a path at level $n$ starting in $\mathcal{D}_\rho$ maps to a path at level $n-1$ starting there, so the induction hypothesis and the first-preimage description keep it in $\mathcal{D}_\rho$. It is also a component of $\mathcal{B}$, since any path in the increasing union has compact image and is contained in one finite level.

Suppose $C \subset \Omega_\rho$ were nonempty, path-connected and $C + m = C$ for some $m > 0$. The set $S_{b_c}(C)$ is path-connected in $\mathcal{B}$ and meets $\mathcal{D}_\rho$: for any $z \in C$, choose an admissible depth $n$ and then a multiple $N$ of $m$ with $N \geq n$; $z + N \in C$ and $S_{b_c}(z+N) \in \mathcal{D}_\rho$. Hence $S_{b_c}(C) \subset \mathcal{D}_\rho$. But for a fixed $z \in C$, $S_{b_c}(z - jm) \to L_2(b_c)$ as $j \to \infty$, whereas $\chi_c(S_{b_c}(z-jm)) = \lambda_c^{-jm}\chi_c(S_{b_c}(z))$ lies on a compact circle inside the chart disk. Its limit would be a fixed point inside $\mathcal{D}_\rho$, necessarily $L_1(b_c)$, not $L_2(b_c)$. The same argument uses the maximal Siegel disk in place of $\mathcal{D}_\rho$.

Compactness of $K$ and the increasing admissible sets give one $N$ with $S_{b_c}(K+N) \Subset \mathcal{D}_\rho$. Lemma 10.30 and Lemma 10.38 give the convergence in (3); Cauchy's formula gives all $z$-derivatives. □

**Proposition 10.43** (No bounded periodic sewing collar in the attracting basin)**.** *Let $\lambda_c = e^{2\pi i\alpha} \neq 1$ with $\alpha$ Brjuno, and continue the chosen lower regular family $S_\lambda$ to a neighbourhood of $\lambda_c$ as in Lemma 10.38. For $0 < |\lambda| < 1$ let $\mathcal{A}_\lambda^{\mathrm{all}} = \bigcup_{n\geq 0} E_{b(\lambda)}^{-n}(\mathcal{A}_\lambda)$ be the full attracting basin of $L_1(\lambda)$. Let $V \subset \mathbb{C}$ be any nonempty connected open set with $V + 1 = V$, and let $\lambda_j \to \lambda_c$ from $0 < |\lambda| < 1$ on the selected lower branch. There is no locally bounded sequence of holomorphic 1-periodic functions $Q_j$ on $V$ such that*

$$S_{\lambda_j}(z + Q_j(z)) \in \mathcal{A}_{\lambda_j}^{\mathrm{all}} \qquad (z \in V,\ j \geq 1).$$

*In particular, taking $Q_j = 0$ excludes any fixed lower-time collar whose entire image stays in the attracting basin up to this boundary point.*

*Proof.* Suppose the sequence existed. Montel's theorem gives a subsequence $Q_j \to Q$ locally uniformly on $V$. Set $\phi_j(z) = z + Q_j(z)$ and $\phi(z) = z + Q(z)$. The identity $\phi(z+1) = \phi(z) + 1$ makes $\phi$ nonconstant and $W = \phi(V)$ a nonempty connected open set with $W + 1 = W$. Lemma 10.38 gives $H_j := S_{\lambda_j} \circ \phi_j \to H := S_{\lambda_c} \circ \phi$ locally uniformly on $V$.

We claim $H(V)$ avoids the Julia set. Otherwise its open image contains a repelling periodic point $p_c$, by the density theorem [2, Thm. 4]. Continue it to repelling periodic points $p_j$ of the nearby maps. Choose a closed disk $\overline{D} \Subset V$ containing a zero of $H - p_c$ and none on its boundary. Rouché's theorem gives a zero of $H_j - p_j$ in $D$ for large $j$, contrary to $H_j(V) \subset \mathcal{A}_{\lambda_j}^{\mathrm{all}}$.

Thus the connected set $H(V) = S_{\lambda_c}(W)$ lies in one Fatou component $U$. The identity $W + 1 = W$ and the functional equation make $U$ invariant. The Fatou-component classification for exponential maps [25, §1, Remarks on notation] says that, in the presence of this Siegel disk, every Fatou component is an iterated preimage of it. It applies here because, with $a = \log b(\lambda_c) \neq 0$, the affine map $w \mapsto aw + \log a$ conjugates $E_{b(\lambda_c)}$ to $u \mapsto e^u + \log a$. Hence $U$ is the maximal Siegel disk. Thus $W$ lies in the eventual maximal-Siegel domain of Proposition 10.42(2), whose maximal-disk version forbids a nonempty connected translation-invariant set. □

**Corollary 10.44** (No uniformly controlled boundary sewing)**.** *Keep the Brjuno parameter and disk $\mathcal{D}_\rho$ of Proposition 10.42. Let $r_j \to 1$ with $r_j \neq 1$, $\lambda_j = r_j\lambda_c$, and $b_j = \exp(\lambda_j e^{-\lambda_j})$. For $0 < s \leq \rho$ put*

$$\mathcal{D}_{j,s} = L_1(b_j) + (\log b_j)^{-1} H_{\lambda_j}(B(0,s)).$$

*On a complete horizontal strip $\Sigma$, there is no locally bounded sequence of holomorphic 1-periodic functions $Q_j$ with the following property: for every compact $K \subset \Sigma$ there are an integer $N_K \geq 0$ and a radius $s_K < \rho$ such that, for all large $j$,*

$$S_{b_j}(z + Q_j(z) + N_K) \in \mathcal{D}_{j,s_K} \qquad (z \in K).$$

*Proof.* By Montel's theorem a subsequence of $Q_j$ converges locally uniformly on $\Sigma$ to a holomorphic 1-periodic $Q$. The lower-chart lemma and the radial-chart lemma imply, for each compact $K$, that $S_{b_c}(z + Q(z) + N_K)$ belongs to the image of the closed disk $\{|\zeta| \le s_K\}$ under the boundary chart. This image is contained in $\mathcal{D}_\rho$, since $s_K < \rho$. Hence $z + Q(z) \in \Omega_\rho$ for every $z \in \Sigma$. The image $(\mathrm{id} + Q)(\Sigma)$ is nonempty and path-connected, and periodicity gives $(\mathrm{id} + Q)(\Sigma) + 1 = (\mathrm{id} + Q)(\Sigma)$. This contradicts Proposition 10.42(2). □

In particular, a boundary solution cannot have overlapping upper-Siegel and lower-repelling regular representations on a complete horizontal line with 1-periodic time corrections: the lower times $z + Q(z)$ would form a path-connected, translation-invariant subset of the corresponding eventual Siegel domain, contradicting part (2).

**Proposition 10.45** (The local chart has a natural parameter boundary)**.** *The family $H_\lambda$ for (19) has no jointly holomorphic extension in $(\lambda, v)$ from either side of the unit circle through any root of unity, on any fixed nonzero disk in $v$. Consequently it has no such extension across any open arc of the unit circle.*

*Proof.* If an extension existed at a primitive $q$th root $\lambda_0$, the identity $f_\lambda \circ H_\lambda = H_\lambda \circ (\lambda\,\cdot)$ and $H'_\lambda(0) = 1$ would persist at $\lambda_0$. Iterating $q$ times would give $f_{\lambda_0}^{\circ q} \circ H_{\lambda_0} = H_{\lambda_0}$. Local invertibility of $H_{\lambda_0}$ would make $f_{\lambda_0}^{\circ q}$ the identity near 0, hence everywhere by the identity theorem for entire functions. This would make $f_{\lambda_0}$ injective, contrary to its $2\pi\mathrm{i}$-periodicity. Roots of unity are dense on the unit circle. □

This obstruction concerns the chart, not the sewn Kneser-type function: parameter singularities in the separate charts might cancel in the sewing. The monogenic parameter regularity of [17] is therefore the relevant comparison for the local step.

**Lemma 10.46** (Root isolation implies normality)**.** *Let $\mathcal{F} \subset \mathcal{O}(B(0,R))$ satisfy $F(0) = 1$, $F(z) \neq 0$ throughout $B(0,R)$, and $F^{-1}(1) \cap B(0,R) = \{0\}$ for every $F \in \mathcal{F}$. Then the normalised logarithms $g_F$ with $g_F(0) = 0$, the functions $F$, and their reciprocals $1/F$ are uniformly bounded on every $|z| \le r < R$, with bounds depending only on $r/R$. In particular $|F'(0)| \le C/R$ for an absolute constant $C$. If $F(-1) = 0$ and $F(z) = \exp(aF(z-1))$ on $B(0,R)$, then $g_F(z) = aF(z-1)$; when $|a|$ is bounded below, the family is also uniformly bounded on every smaller disk $B(-1,r)$. For these solutions the zero-free hypothesis is automatic.*

*Proof.* Choose the holomorphic logarithm $g_F$ with $g_F(0) = 0$. Root isolation makes $h_F = \frac{1}{2} + g_F/(4\pi\mathrm{i})$ omit 0 and 1, while $h_F(0) = \frac{1}{2}$. Schottky's theorem bounds $h_F$, hence $g_F$, $F = \exp g_F$, and $1/F$, uniformly on compact subdisks. Cauchy's estimate at radius $R/2$ gives the derivative bound. The functional equation makes $aF(z-1)$ a logarithm of $F(z)$ normalised at 0, so it equals $g_F$. □

For a zero-free $F \in \mathcal{O}(B(0,R_0))$ with $F(0) = 1$ and $F \not\equiv 1$, let $d_F$ be the distance from 0 to the nearest other preimage of 1, or $\infty$ if there is none. Applying the lemma on every disk of radius $s < \min(R_0, d_F)$ gives $|F'(0)| \min(R_0, d_F) \le C$. Thus derivative blow-up on a fixed disk forces another preimage of 1 to approach 0.

**Proposition 10.47** (Reduction)**.** *Let $W$ be a nonempty open subset of a common regular $z$-domain for both families, with $\operatorname{Re} z > -2$ on $W$. Suppose the following hold.*

(P0) *The branch $\kappa$ from $b > \eta$ has a jointly holomorphic continuation on a connected exterior parameter corridor reaching $U_{\mathrm{out}} \times W$. The sewing family of Theorem 7.1 has a jointly holomorphic continuation on a connected interior parameter corridor reaching $U_{\mathrm{in}} \times W$. Both continuations use the stated branches and have $W$ in their common regular $z$-domain.*

(P1) *The base-continued family $\kappa_b(z)$ on $U_{\text{out}} \times W$ and the two-sided regular family $K_b^W(z)$ on $U_{\text{in}} \times W$ are locally uniformly bounded up to $\Gamma$: for every compact subarc $\Gamma' \subset \Gamma$ and $V \Subset W$, they are bounded on a neighbourhood of $\Gamma'$ on their respective sides, uniformly for $z \in V$.*

(P2) *Let $D \subset W$ be a countable set with an accumulation point in $W$. For each $z \in D$, the two one-sided radial limits exist and are equal at almost every point of $\Gamma$ (in arc length).*

*Then $\kappa$ continues jointly holomorphically in $(b, z)$ across $\Gamma$ on $W$ and coincides with the continued $K^W$ on $U_{\text{in}} \times W$. The equality extends to any common connected regular $z$-domain by the identity theorem.*

*Proof.* Intersect the full-measure subsets in (P2) over the countable set $D$. At a point $b_c$ in the intersection, (P1) makes each radial family normal as a family of holomorphic functions on $W$. Every subsequential limit has the values prescribed on $D$; the identity theorem makes it unique. Thus the two radial families converge locally uniformly on $W$ and have the same limit. Fix $z \in W$. On each side, bounded holomorphic functions have non-tangential limits almost everywhere (Fatou); the radial limits just obtained agree with these. The $H^\infty$ gluing theorem across an analytic arc therefore gives a holomorphic function of $b$ through $\Gamma$. The bounds in (P1) are locally uniform in $z$. At a point of $\Gamma$, approach from $U_{\text{in}}$: the functions of $z$ form a locally bounded family and converge pointwise to the glued values. Montel's theorem shows that the limit is holomorphic in $z$. The glued function is thus separately holomorphic and locally bounded, so Hartogs' theorem makes it jointly holomorphic. It equals $\kappa$ on the outer side and $K^W$ on the inner side; the identity theorem extends the equality in $z$ wherever both are defined on a common connected domain. □

**Proposition 10.48** ($H^1$ alternative to (P1))**.** *In Proposition 10.47, (P1) may be replaced by the following condition. In every local conformal coordinate $\xi = s + \mathrm{i}t$ straightening $\Gamma$, for each compact interval $I$ on the straightened arc and each $V \Subset W$, there are $\epsilon > 0$ and $C < \infty$ such that on both sides*

$$\sup_{0<|t|<\epsilon} \int_I \sup_{z \in V} |F_\pm(s + \mathrm{i}t, z)|\, ds \le C, \tag{20}$$

*where $F_+$ and $F_-$ denote the respective families from (P0). The conclusion of Proposition 10.47 still holds.*

*Proof.* Shrink $I$ slightly to avoid the ends of the coordinate rectangle. By Fubini and (20), the two vertical sides of a smaller rectangle can be chosen with integrable $\sup_{z \in V} |F_\pm|$; its opposite horizontal edge is away from the arc. The local $H^1$ boundary theorem (or the disk theorem after conformally mapping this rectangle) [7] applied for each fixed $z$ gives $L^1(I)$ traces $f_\pm(\cdot, z)$, with convergence in $L^1$ on the smaller interval as $t \to 0^\pm$. The radial parameter paths in (P2) meet the analytic arc transversely, so their limits agree almost everywhere with these Hardy traces. The bound (20) and Cauchy's formula in $z$ show that $z \mapsto f_\pm(\cdot, z)$ are holomorphic as $L^1(I)$-valued maps: on each compact $z$-circle the $L^1$ norms are uniformly bounded, so Bochner dominated convergence passes the boundary limit through the Cauchy integral. By (P2), their difference vanishes in $L^1(I)$ at every $z \in D$. The Banach-valued identity theorem makes the difference zero on $W$.

The two piecewise defined families are locally integrable in $(s, t, z)$ by (20). Equality of their boundary traces removes the jump term in the distributional $\bar\partial_\xi$ derivative. The distributional $\bar\partial_z$ derivative also vanishes, since the families are jointly holomorphic away from the measure-zero arc. Weyl's lemma for $\bar\partial$ therefore extends the glued family jointly holomorphically across the arc. The identity theorem gives the same conclusion on the common connected regular $z$-domain as before. □

**Remark 10.49.** The exponent 1 is sharp for this abstract boundary-gluing argument. Indeed, on opposite sides of the real $\xi$-axis let $F_\pm(\xi, z) = 1 + z/\xi$. The two boundary values agree almost everywhere and $F_\pm(\xi, 0) = 1$. For every $0 < p < 1$ and compact $I, V$, the integrals $\int_I \sup_{z\in V} |F_\pm(s + \mathrm{i}t, z)|^p \, ds$ stay uniformly bounded as $t \to 0^\pm$, but the common function has a pole at $\xi = 0$ for $z \neq 0$. Thus replacing (20) by a bare $H^p$ bound with $p < 1$ would not prove the gluing statement; this example does not address additional constraints of the tetration equation.

Lemma 10.46 gives a geometric sufficient condition for (P1) when the common $z$-domain contains $B(0, R) \cup B(-1, R)$: if both families have no preimage of 1 in $B(0, R)$ other than 0 for every base in the two collars, then (P1) holds on every smaller disk $W = B(0, r)$. Along a connected parameter corridor, the argument principle transports this root count from one base where 0 is the only, simple root, provided $F_b$ is defined near the closed disk $\overline{B(0, R)}$ and $F_b(z) \neq 1$ on the fixed circle $|z| = R$ throughout the corridor. This contour exclusion near $\Gamma$ has not been proved. Corollary 6.5 supplies a starting base and a fixed root-isolation disk for the inner sewing branch near the real cusp; transport to $\Gamma$ still requires the contour exclusion.

The open set $W$ is essential. At every $n \in \mathbb{Z}_{\geq 0}$ both families equal $E_b^{\circ n}(1)$, together with all base derivatives. Corollary 10.40 also makes their values and base derivatives agree at $z = -1$ when both retain the zero backward label; this point can still exclude entire lower regular translates. Thus these integer heights cannot test (P2). Corollary 3.9 gives two locally holomorphic families sharing these values and even the logarithmic monodromy at $z = -2$, yet differing at fractional real heights by order $\Lambda$. An accumulating set of noninteger heights is needed.

The monodromy in Proposition 10.35 gives an explicit illustration: after any nonzero number of parameter turns, the upper regular germs still take the same values $E_b^{\circ n}(1)$ at every nonnegative integer $n$, although their derivatives at $z = 0$ differ. A pointwise version of (P1)–(P2) would be vacuous and could not establish continuation of the family.

Even finitely many $z$-derivatives at every integer do not remove this ambiguity. For any $m \geq 0$ and any solution $F$ regular near the integers, the local solution $F_\epsilon(z) = F(z + \epsilon(\mathrm{e}^{2\pi \mathrm{i} z} - 1)^{m+1})$ obeys the same functional equation and has the same $z$-jets through order $m$ at each integer where both are defined. For small nonzero $\epsilon$ it generally differs at noninteger heights; this observation makes no claim that $F_\epsilon$ remains in the two-chart sewing class. Condition (P0) is also substantive: Proposition 3.5 constructs $K^W$ on discs along the whole real segment $(1, \eta)$, and Theorem 7.1 adds quantitative bounds near $\eta$; neither reaches an arbitrary Shell–Thron boundary arc. Proposition 10.1 supplies a jointly holomorphic exterior germ along the entire real ray $b > \eta$. Proposition 10.2 gives an explicit *geometric* base corridor on which its unmarked initial curve and two repelling ends persist up to a non-cusp boundary arc. Proposition 10.3 certifies transport of the normalising point $w = 1$ along one path in that corridor, hence of the exterior germ at $z = 0$; Proposition 10.4 places its endpoint at a prescribed Brjuno parameter. Proposition 10.5 pulls the whole initial region back through 28 iterates to a Jordan cell containing $w = 1$, even at that endpoint. Proposition 10.6 gives a uniform quasiconformal model of these quotient cylinders, and Proposition 10.7 gives one-sided limits on a fixed small height disk about $z = 0$ along a short boundary subarc, without an arithmetic restriction. Proposition 10.8 upgrades these limits to a jointly holomorphic germ across the subarc in a genuine complex base corridor, certified on a rectangle with $|\operatorname{Im} \theta| \leq 0.01$. Proposition 10.9 proves one transverse seam crossing at the real seed and reduces its possible loss along the path to a scalar elliptic tangency test. Lemma 10.10 confines any such tangency to a finite-height cylinder band, without yet giving its explicit width. Propositions 10.11 and 10.12 give an explicit positive-slope approximate Abel coordinate and a global $L^4$ bound on its correction to the exact uniformisation. Proposition 10.13 differentiates the equation vertically and upgrades this to a pointwise derivative bound, excluding all seam tangencies on a wider box that also contains interior bases. Proposition 10.14 certifies

at one such base that a nontrivial subarc of the exterior chord lies in the immediate attracting basin. This physical overlap does not identify the two quotient coordinates. The single-crossing criterion of Proposition 10.20 is therefore met. Corollary 10.21 supplies a common regular height neighbourhood of $[0, 1]$ across a non-cusp boundary subarc for the exterior family. This proves the exterior full-height substep of (P0) at that subarc; the inner sewn family has not been transported to the same subarc. (Near the cusp the whole question is settled differently by Theorem 9.13.) Corollary 10.22 carries that same full-height branch along the certified path to a strictly interior base where Proposition 10.14 supplies a physical attracting overlap. Proposition 10.15 separately certifies an embedded lower-to-upper sewing circle and a locally holomorphic sewn family at this interior base. Its parameter neighbourhood has not been connected to the real inner sewn family, and the two local sewing constructions have not been identified. Proposition 10.16 certifies a transverse self-intersection of that lower regular curve at another interior base on the proposed straight parameter path. Thus the same circle cannot remain an embedded positive-slope seam throughout that path; a parameter detour or a change of seam remains possible. Proposition 10.17 now places the entire lower sewing circle, after 76–77 forward iterates, as one embedded essential loop in the exterior quotient at that same base. This supplies a common annular physical chart, but does not yet identify the compactified quotient surfaces or their normalised coordinates. Corollary 10.18 certifies another, disjoint return loop after 14–15 iterates, so the physical return alone does not select a unique quotient comparison. Corollary 10.19 also matches both regular charts to degree-one puncture germs of the exterior quotient. For either chosen return loop, the unproved part is its analytic connection through the two compact middle bands to those germs. Corollary 10.23 continues this exterior family through height $z = -1$, where it has a simple zero and the same backward label as the inner sewn family. That label does not by itself identify the two branches. Corollary 10.24 gives a fixed disc of uniform root isolation and the exterior local bounds required by (P1) on this boundary subarc. The inner bounds remain open. This exterior substep is not established by the parameter argument of [24, Prop. 3 and §5]. Proposition 10.35 transports the upper normalised germ along every compact inner path avoiding $\lambda = 0$, but gives neither a boundary-uniform height disk nor the sewing correction. Corollary 10.36 shows that no fixed finite pullback depth at the normalising orbit can supply such a boundary chart: the required depth diverges at every Brjuno point. Proposition 10.37 further shows that any chart on a fixed connected height domain whose image stays in the immediate attracting basin and whose value at $z = 0$ is 1 must collapse to the constant 1 there, even without a prior bound. Such a chart cannot carry the normalised value at $z = 1$ near a Brjuno boundary point. Proposition 10.43 also rules out a locally bounded periodic lower-time correction on a fixed translation-invariant collar whose image remains in the full attracting basin up to that point.

Non-Brjuno numbers have measure zero, so (P2) need only be tested at Brjuno points. Lemma 10.30 gives two-sided local convergence to the Siegel chart at $L_1$; Lemmas 10.31 and 10.38 give locally uniform convergence of the lower superfunction on non-cusp $\Gamma$.

Yet Proposition 10.42(2) excludes a complete sewing band even after finite pullbacks, and Corollary 10.44 excludes uniform finite-depth chart entry from a fixed strip under bounded periodic lower times. The convergence in part (3) is local on $\Omega_\rho$ and supplies no essential sewing curve. Thus (P2) still requires direct control of the sewn solutions. The samples in Section 10.6 are compatible with a smooth join at $\alpha = (\sqrt{2} - 1)/5$ but do not identify the specified families. At that non-cusp arc we have not proved (P0), (P1), its Hardy alternative (20), or (P2); these are not needed for the continuation around the cusp of Section 9.

## 11. Open problems

(1) *Global single-valuedness of the Kneser family.* Theorem E settles the continuation along paths near the cusp. Is the continuation of Kneser's solution single-valued on the whole upper half of the Shell–Thron region, for instance along paths that cross the boundary far from $\eta$? Section 10.6 gives pointwise numerical compatibility at one such arc. One possible approach is the following. The non-Brjuno rotation numbers have measure zero, so the corresponding set $E$ on the boundary curve has zero length. By Painlevé's theorem, sets of zero length are removable for bounded holomorphic functions. It would therefore suffice to show two things on a nonempty open set of $z$-values, locally uniformly in $z$: the two families stay bounded as $b$ approaches the boundary from their respective sides, and their one-sided limits exist and agree at every point of $\Gamma \setminus E$, where $\Gamma$ is a boundary arc. The latter is a statement about Siegel linearisations at Brjuno multipliers approached from inside and outside the unit circle. The question is related to a global uniqueness theorem for Kneser-type solutions inside the Shell–Thron region. The condition by limits at $\pm\mathrm{i}\infty$ alone admits solutions of the form $R \circ (z + 2\pi\mathrm{i}mz/\log\lambda + \cdots)$.

(2) *The first-order coefficients.* Theorem 5.6 expresses $\kappa^{(n)}$ as a convergent series over parabolic orbits. Is there a closed form, for instance in terms of the horn-map data $B_n$, $a$ and the iterative residue? Is the expansion in $p$ convergent, or only Gevrey, as $\frac{1}{2}$-summability [6] suggests in a different normalisation?

(3) *Certification.* Give interval enclosures of $B_1$, $B_2$ and $a$, and of the ladder values. This would turn the numerical sections into computer-assisted statements.

## Code and data availability

All computations use the Python library `kneser` (mpmath). The transition-map invariants are computed by `transition_map.py`, `eta_approach.py`, `eta_sweep.py`, `deficit_param.py` and `deficit_modes.py`. The parabolic data come from `parabolic_horn.py` and `parabolic_horn_inverse.py`. The first-order series and the second-order check are `mrr_derivative.py` and `mrr_second.py`. The boundary comparison is reproduced by `boundary_multiheight.py` from the unmodified `fatou.gp`. The confluence check is `confluence_check.py`, and the sewing solution is computed by `weld_solve.py`. The complex-base sewing solution and its comparison with `fatou.gp` are computed by `weld_complex.py`. The ladder computations of $K_b$ use the complex-base builder of the library together with `c1_invariant.py` and `first_mode_check.py`. The exterior interval bounds use `certify_exterior_wide_corridor.py` and `certify_exterior_frozen_abel.py`. The interior interval certificates are `certify_interior_chord_overlap.py`, `certify_interior_lower_overlap.py`, `certify_interior_transition_slope.py`, `certify_interior_path_fold.py`, `certify_interior_self_intersection.py`, `certify_interior_quotient_loop.py`, and `certify_interior_early_return.py`. The signs and branches used in Section 9 (normal component of the step, the first crossing of the base orbit, and the seam height $-h/2$ through the lower half-plane) are checked numerically by `cusp_tilted_check.py`.

Code and data are available from the author on request.

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